

# A First-Principles Dynamic Model of a Rubber V-Belt CVT for General Transient Simulation

With Mechanical Actuation and Traction-Limited Torque Transfer

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# Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt geometric closure, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the CVT kinematics, the rotational dynamics of the primary and secondary pulleys, the axial equation of motion governing shift, and the torque-transfer closure required to distinguish sticking from traction-limited slip. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is also derived to account for the reduction in effective normal force at speed. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document in which each major modeling step remains physically interpretable. Simulation results are used to examine whether the completed model behaves in mechanically sensible ways under transient launch conditions. Comparisons of launch shift trajectories, pulley axial-force balance, and torque admissibility across multiple tuning cases show a consistent mechanism chain from tuning changes, to clamping-force evolution, to stick-slip behavior, to overall launch outcome. The default tuning produces the most balanced launch response, combining strong early admissible coupling torque, controlled low-ratio retention, and a smoother main shift. More aggressive cases reveal renewed slip and visibly underdamped oscillatory transients during shift onset, highlighting both the usefulness of the model and the limitations introduced by idealized loss assumptions.

Overall, the formulation provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behavior during launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behavior remains traceable back to clear mechanical causes.

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# Nomenclature and Notation

All quantities are expressed in SI units unless otherwise stated. Angles are expressed in radians. Angular velocities are expressed in rad/s. Torques are expressed in N m, forces in N, lengths in m, and masses in kg.

## Symbol Summary

Symbols used throughout the derivation are summarized in [Table 1](#).

| Symbol                                         | Meaning                                                               |
|------------------------------------------------|-----------------------------------------------------------------------|
| <b>State and kinematics</b>                    |                                                                       |
| $\mathbf{x}(t)$                                | state vector                                                          |
| $t$                                            | time                                                                  |
| $\omega_p, \omega_s$                           | primary and secondary angular speed                                   |
| $s, \dot{s}, \ddot{s}$                         | shift position, velocity, and acceleration                            |
| $R, \dot{R}$                                   | CVT ratio and ratio rate                                              |
| $v_b$                                          | belt transport speed                                                  |
| $v_b^*, T_b$                                   | slip-branch target belt speed and belt-speed relaxation time constant |
| $v_\Delta$                                     | slip metric: primary-imposed minus secondary-imposed belt-line speed  |
| <b>Geometry</b>                                |                                                                       |
| $r_{p,\text{outer}}(s), r_{s,\text{outer}}(s)$ | primary and secondary outer radius                                    |
| $r_{p,\text{eff}}(s), r_{s,\text{eff}}(s)$     | primary and secondary effective torque radius                         |
| $r_{p,\text{outer},\text{min}}$                | minimum primary outer radius                                          |
| $r_{s,\text{outer},\text{max}}$                | maximum secondary outer radius                                        |
| $s_{\text{dz}}, s_{\text{max}}$                | deadzone threshold and maximum shift                                  |
| $\beta$                                        | sheave face half-angle (radial reference)                             |
| $C$                                            | pulley center distance                                                |
| $L_b$                                          | total belt length                                                     |
| $\phi$                                         | belt wrap angle used in distributed contact model                     |
| $r, \dot{r}$                                   | local belt path radius and its time derivative                        |
| $r_{\text{cm}}$                                | belt element center-of-mass radius                                    |
| <b>Constraint and mapping functions</b>        |                                                                       |
| $f(r_p, r_s, C)$                               | belt-length compatibility root function ( $f = 0$ )                   |
| $\text{root}_{r_s}(\cdot)$                     | numerical root solve for secondary radius                             |
| $r_f(s), \frac{dr_f}{ds}$                      | primary flyweight radius map and gradient                             |
| $\theta_s(s), \frac{d\theta_s}{ds}$            | secondary helix angle map and gradient                                |

| Symbol                                           | Meaning                                                                  |
|--------------------------------------------------|--------------------------------------------------------------------------|
| <b>Forces and torques</b>                        |                                                                          |
| $F_{p,ax}, F_{s,ax}$                             | primary and secondary axial mechanism force                              |
| $F_{c,p}, F_{c,s}$                               | primary and secondary belt centrifugal axial contribution                |
| $F_{ax}$                                         | generic axial clamp force at one pulley contact                          |
| $\tau_{eng}, \tau_{load}$                        | engine input torque and reflected load torque                            |
| $\tau_p, \tau_s$                                 | coupling torque seen at primary and secondary                            |
| $\tau, \tau_c$                                   | generic interface torque and applied coupling/friction torque            |
| $\tau_{ns}$                                      | no-slip requested coupling torque                                        |
| $\tau_{-}^{stick}, \tau_{+}^{stick}$             | aggregate lower and upper coupling-torque bounds for stick admissibility |
| $\tau_{-}^{slip}, \tau_{+}^{slip}$               | aggregate lower and upper coupling-torque bounds on slip branches        |
| $\tau_{c,p,\pm}^{stick}, \tau_{c,s,\pm}^{stick}$ | primary/secondary pulley-wise stick-branch lower/upper bounds            |
| $\tau_{c,p,\pm}^{slip}, \tau_{c,s,\pm}^{slip}$   | primary/secondary pulley-wise slip-branch lower/upper bounds             |
| <b>Mass/inertia and material parameters</b>      |                                                                          |
| $I_p, I_s$                                       | primary and secondary rotational inertia                                 |
| $m_f$                                            | effective flyweight mass                                                 |
| $m_{p,m}, m_{s,m}$                               | equivalent primary and secondary translating mechanism mass              |
| $\rho_b$                                         | belt material density                                                    |
| $A_b$                                            | belt cross-sectional area                                                |
| $\mu$                                            | belt-pulley traction coefficient                                         |
| $k_p, x_{p,0}$                                   | primary spring stiffness and preload displacement                        |
| $k_{s,\theta}, \theta_{s,0}$                     | secondary torsional stiffness and preload angle                          |
| $k_{s,x}, x_{s,0}$                               | secondary axial stiffness and preload displacement                       |

Table 1: Core symbols used throughout the derivation.

## Constants and Parameters

All parameters in [Table 2](#) are treated as constant values that do not vary through the model. They are either physical constants, fixed by the physical design of the CVT and vehicle or else are tuning parameters that are set and held constant during a given simulation run.

| Symbol                   | Name                           | Unit | Description                                      |
|--------------------------|--------------------------------|------|--------------------------------------------------|
| <b>Geometry and belt</b> |                                |      |                                                  |
| $L_b$                    | Belt length                    | m    | Fixed total belt length                          |
| $\beta$                  | Sheave half-angle              | rad  | Cone half-angle used for radial/axial conversion |
| $h_b$                    | Belt thickness                 | m    | Radial belt thickness                            |
| $b_{out}$                | Belt outer width               | m    | Belt width at outer face                         |
| $b_{in}$                 | Belt inner width               | m    | Belt width at inner face                         |
| $r_{p,outer,min}$        | Primary minimum outer radius   | m    | Radius at fully open primary                     |
| $r_{s,outer,max}$        | Secondary maximum outer radius | m    | Radius at fully closed secondary                 |
| $s_{dz}$                 | Shift deadzone threshold       | m    | Onset of effective radial motion                 |

| Symbol                             | Name                                     | Unit              | Description                                        |
|------------------------------------|------------------------------------------|-------------------|----------------------------------------------------|
| $s_{\max}$                         | Maximum shift                            | m                 | Physical upper travel bound                        |
| <b>Primary/secondary actuation</b> |                                          |                   |                                                    |
| $m_f$                              | Flyweight mass                           | kg                | Effective flyweight mass                           |
| $r_{f,0}$                          | Initial flyweight radius                 | m                 | Flyweight radius offset at $s = 0$                 |
| $k_p$                              | Primary spring stiffness                 | N/m               | Primary compression spring rate                    |
| $x_{p,0}$                          | Primary spring preload                   | m                 | Initial compression of primary spring              |
| $k_{s,\theta}$                     | Secondary torsional stiffness            | N m/rad           | Torsional spring rate in helix assembly            |
| $\theta_{s,0}$                     | Secondary torsional preload              | rad               | Initial torsional deflection preload               |
| $r_h$                              | Helix reference radius                   | m                 | Effective cylindrical radius in helix kinematics   |
| $k_{s,x}$                          | Secondary axial stiffness                | N/m               | Secondary axial compression spring rate            |
| $x_{s,0}$                          | Secondary axial preload                  | m                 | Initial compression of secondary axial spring      |
| <b>Inertia and drivetrain</b>      |                                          |                   |                                                    |
| $I_{\text{eng}}$                   | Engine inertia                           | kg m <sup>2</sup> | Engine rotational inertia                          |
| $I_{\text{prim}}$                  | Primary pulley inertia                   | kg m <sup>2</sup> | Primary CVT pulley inertia                         |
| $I_p$                              | Effective primary inertia                | kg m <sup>2</sup> | Implemented primary-side inertia in current setup  |
| $I_{\text{sec}}$                   | Secondary pulley inertia                 | kg m <sup>2</sup> | Secondary CVT pulley inertia                       |
| $I_{\text{gb}}$                    | Gearbox inertia                          | kg m <sup>2</sup> | Gearbox/intermediate shaft inertia                 |
| $I_{\text{wheel}}$                 | Wheel inertia                            | kg m <sup>2</sup> | Wheel/hub inertia about wheel axis                 |
| $I_s$                              | Effective secondary inertia              | kg m <sup>2</sup> | Lumped secondary-side inertia about secondary axis |
| $G$                                | Final drive ratio                        | 1                 | Fixed ratio defined by $\omega_s = G \omega_w$     |
| $m$                                | Vehicle mass                             | kg                | Total vehicle mass                                 |
| $r_w$                              | Wheel rolling radius                     | m                 | Effective tire rolling radius                      |
| <b>Contact and environment</b>     |                                          |                   |                                                    |
| $\rho_b$                           | Belt density                             | kg/m <sup>3</sup> | Belt material density                              |
| $\mu_s$                            | Belt-pulley static friction coefficient  | 1                 | Static traction coefficient used at impending slip |
| $\mu_k$                            | Belt-pulley kinetic friction coefficient | 1                 | Kinetic traction coefficient used on slip branches |
| $C_{\text{rr}}$                    | Rolling resistance coefficient           | 1                 | Tire-road rolling resistance coefficient           |
| $C_d$                              | Drag coefficient                         | 1                 | Aerodynamic drag coefficient                       |
| $A_f$                              | Frontal area                             | m <sup>2</sup>    | Vehicle aerodynamic reference area                 |
| $\rho$                             | Air density                              | kg/m <sup>3</sup> | Ambient air density used in drag model             |
| $g$                                | Gravitational acceleration               | m/s <sup>2</sup>  | Gravity constant                                   |

Table 2: Canonical non-varying parameters used by the CVT model equations.

# Chapter 1

## Introduction & Background

### 1.1 Problem and Motivation

Mechanically actuated rubber V-belt continuously variable transmissions (CVTs) are widely used in compact powersport drivetrains, including snowmobiles, all-terrain vehicles (ATVs), and utility terrain vehicles (UTVs) (Messick, 2018), because they can vary the speed ratio automatically while transmitting power through a relatively simple mechanical assembly. Their visible layout is straightforward: two variable-radius pulleys are connected by a belt. Their transient behaviour, however, is not. The operating state of the transmission emerges from the interaction of its internal mechanisms, the belt–pulley contacts, and the mechanical conditions under which it operates.

Several coupled effects determine that response. Engine speed influences the centrifugal actuation of the primary pulley, while transmitted torque influences the torque-reactive response of the secondary pulley. Together, these mechanisms generate the clamping forces that press the belt into the sheave grooves. Clamping force determines the traction available at the belt–pulley contacts, which in turn determines how torque is carried and whether the belt and pulleys can remain coupled without gross slip. At the same time, pulley motion changes the belt radii and therefore the transmission ratio, which changes the speed and torque relationships between the two pulleys. These effects do not occur independently; each changes the conditions under which the others act.

This coupling makes the behaviour of a mechanically actuated CVT difficult to interpret from external measurements alone. A shift curve, pulley-speed trace, or observed loss of traction may reflect the combined effect of primary actuation, secondary reaction, belt traction, and shift motion. The underlying states responsible for that behaviour are largely internal to the transmission and are not usually available directly during testing. This is especially consequential because mechanically actuated CVTs are tuned by changing physical components such as springs, flyweights, ramps, and helices; determining the effect of a change can therefore require repeated hardware changes and testing (Skinner, 2020).

Addressing this problem requires a physically interpretable dynamic framework that preserves the connection between the internal CVT mechanisms and the behaviour they produce. A useful model should do more than reproduce an output trace: it should make it possible to follow how actuation, clamping, belt traction, shift motion, and inertia combine as the transmission responds to changing operating conditions. If an undesirable response can be traced to its mechanical cause, changes to the hardware or operating conditions can be made deliberately rather than through trial and

error. The motivation is therefore not only to predict transient CVT behaviour, but to make that behaviour understandable enough to diagnose and improve.

The remainder of this chapter develops the context for that framework. [Section 1.2](#) introduces the physical layout and operating principles of the belt–sheave CVT, including ratio change, traction, and mechanical actuation. [Section 1.3](#) contains a literature review that organizes prior work by modelling emphasis and identifies the foundations available for mechanical actuation, belt contact, and transient CVT dynamics. Given this context, [Section 1.4](#) then states the aim, scope, assumptions, and contributions of the present formulation.

## 1.2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the operating principles of the continuously variable transmission (CVT) considered in this work. The goal is not yet to derive the model equations, but to build the mechanical intuition needed to understand them. In particular, this section explains why a CVT is useful, how a rubber V-belt CVT changes ratio, why clamping and traction matter, how the primary and secondary pulleys generate their forces, and how these mechanisms interact to determine the transmission operating state.

### 1.2.1 Why Use a Continuously Variable Transmission?

A transmission is responsible for matching the operating speed of an engine to the speed required at the vehicle wheels. In a conventional stepped gearbox, this is achieved using a fixed set of discrete gear ratios. Each gear imposes a fixed relationship between engine speed and wheel speed. As the vehicle accelerates, the engine speed rises through one gear, drops when the transmission shifts, and then rises again in the next gear. This repeated sweep through the engine speed range is a direct consequence of having only a small number of available ratios.

This matters because an internal combustion engine (ICE) does not produce the same torque and power at every speed. Instead, the engine has a useful operating region where it can produce power more effectively, as shown conceptually in [Figure 1.1](#). A stepped gearbox can pass through this region, but it cannot generally stay there while vehicle speed continues to increase. [Figure 1.2](#) contrasts this stepped behavior with the smoother operating path made possible by a continuously variable transmission.

A continuously variable transmission (CVT) addresses this limitation by allowing the transmission ratio to vary smoothly within a finite range rather than changing through a fixed set of gear steps. Since the ratio can change continuously, the engine can remain closer to a desired operating region while the vehicle accelerates. This is the main benefit of a CVT: it gives the drivetrain more freedom to match engine behavior to vehicle load.

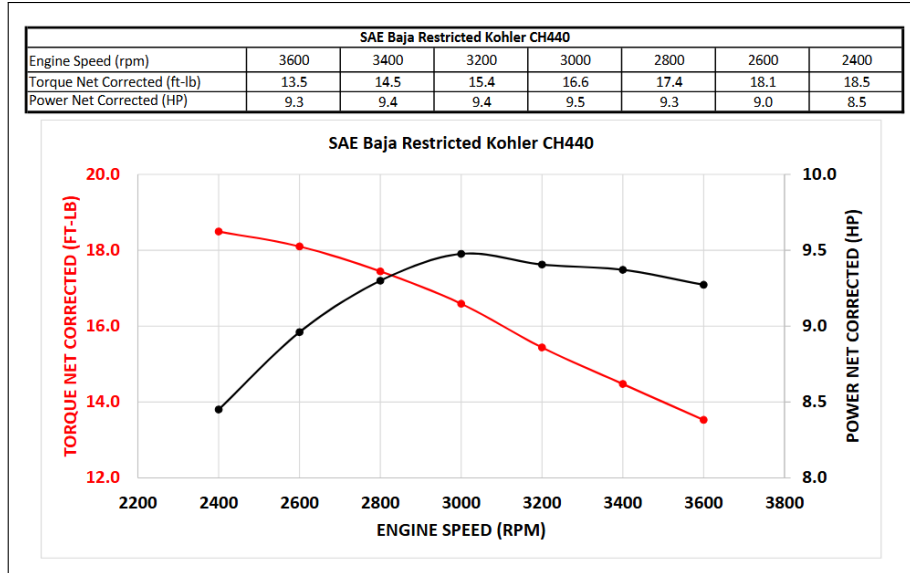


Figure 1.1: Torque and power of the Kohler CH440 engine as functions of engine speed. Both quantities vary substantially across the operating range, with peak torque occurring at a lower engine speed than peak power. (Baja SAE, 2022)

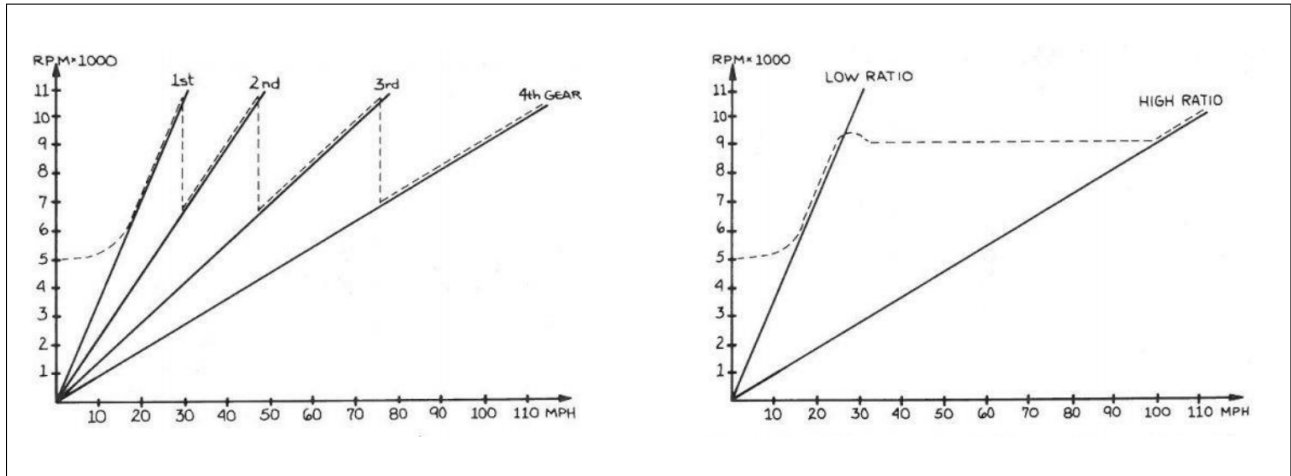


Figure 1.2: Conceptual comparison of engine-speed behavior during acceleration. Solid lines show the engine-speed–vehicle-speed relationship imposed by the available transmission ratios, while the dotted line illustrates a representative engine-speed trajectory during acceleration. With a stepped gearbox (left), the engine repeatedly sweeps through a speed range and drops in speed at each gear change. With a CVT (right), engine speed can rise to a desired operating speed and remain approximately constant while the transmission continuously changes ratio, increasing again only after the high-ratio limit is reached. (Aaen, 2007)

### 1.2.2 Physical Layout and Single-Pulley Geometry

There are several forms of continuously variable transmission, including different mechanical layouts and different ways of producing the variable speed relationship. This work focuses on a belt-sheave CVT, shown conceptually in Figure 1.3. In this arrangement, two variable-radius pulleys are

connected by a rubber V-belt. The pulley connected to the engine side is called the primary (or driving) pulley. The pulley connected to the downstream drivetrain is called the secondary (or driven) pulley. The belt transmits motion and torque between these two pulleys.

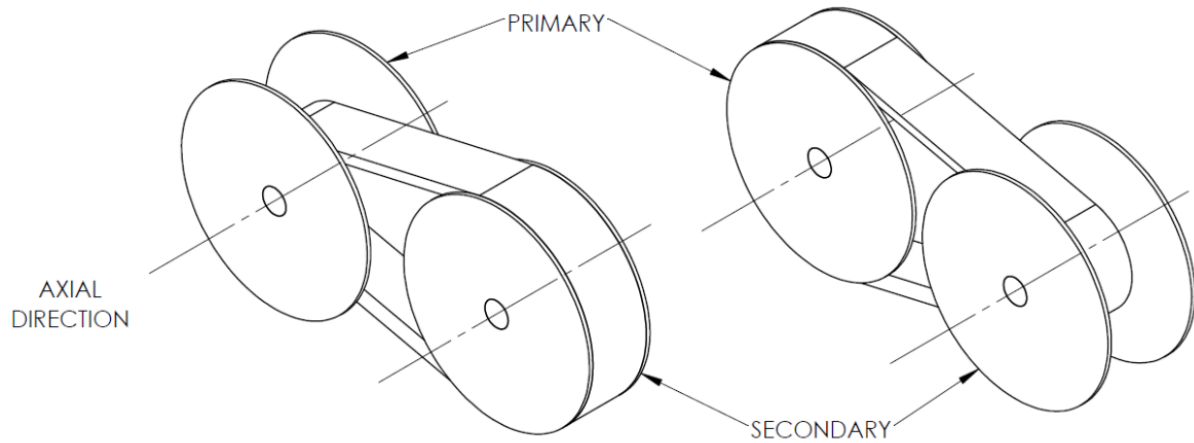


Figure 1.3: Two sample configurations of the belt-sheave CVT considered in this work. The primary pulley is connected to the engine side, the secondary pulley is connected to the downstream drivetrain, and the rubber V-belt transmits motion and torque between them. (Messick, 2018)

Before considering the full two-pulley system, it is helpful to look at the geometry of a single pulley. Both the primary and secondary pulleys share the same basic anatomy. Each pulley is made from two opposing conical sheaves that form a V-shaped groove. One sheave is fixed axially to the shaft, while the other can move along the shaft. The rubber V-belt sits between the sheave faces. This basic layout is shown in Figure 1.4.

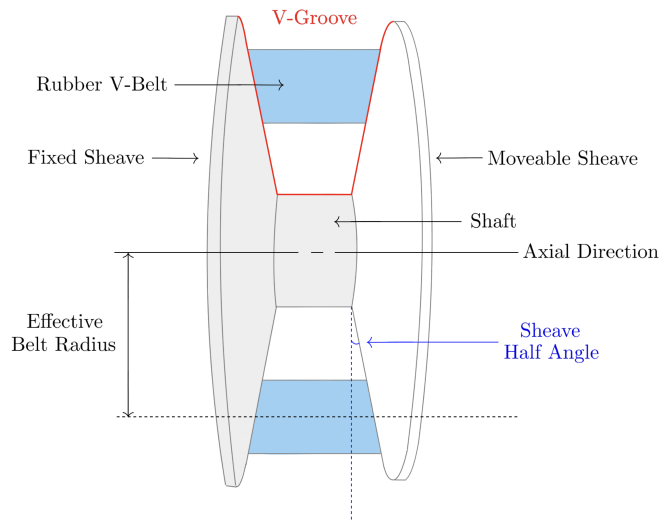


Figure 1.4: Generic anatomy of a belt-sheave CVT pulley. Each pulley consists of opposing conical sheaves that form a V-groove. Motion of the movable sheave changes the groove width and therefore changes the radius at which the belt rides.

The key geometric idea is simple: changing the axial separation between the sheaves changes the effective radius of the belt. If the movable sheave moves toward the fixed sheave, the groove becomes narrower. Since the belt has a finite width, it cannot occupy the same radial position in the narrower groove, so it is pushed outward along the conical sheave faces. The effective belt radius therefore increases.

If the movable sheave moves away from the fixed sheave, the groove becomes wider and the belt can fall deeper between the sheaves. This reduces the effective belt radius. At this level, the mechanism is just wedge geometry: axial motion of the sheave becomes radial motion of the belt. Closing the sheaves makes the belt radius larger, while opening the sheaves makes the belt radius smaller, as shown in [Figure 1.5](#).

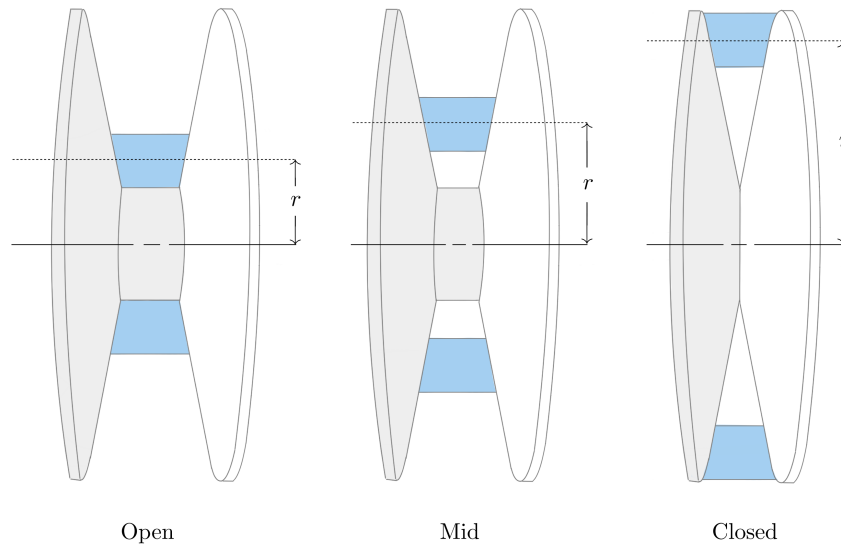


Figure 1.5: Single-pulley radius change. As the sheaves close, the V-groove narrows and the belt is pushed outward to a larger effective radius. As the sheaves open, the belt can fall deeper in the groove and the effective radius decreases.

### 1.2.3 Two-Pulley Coupling and Ratio Change

The full CVT contains two of the variable-radius pulleys described above. The important difference is that the two pulleys are connected by the same belt. Because the belt length is approximately fixed, the effective radii of the two pulleys cannot change independently.

If the belt rides outward on the primary pulley, it must ride inward on the secondary pulley. Likewise, if the primary radius decreases, the secondary radius must increase. This opposite motion is the core geometric behavior that allows the CVT ratio to change. In the physical mechanism, the actual motion of the sheaves occurs through contact forces, springs, actuator mechanisms, and the axial force balance of the pulleys.

The CVT begins launch in a low-speed, high-reduction configuration. In this state, the primary pulley has a small effective radius and the secondary pulley has a large effective radius. This behaves like a low gear in a conventional drivetrain: the engine-side pulley turns relatively quickly while the downstream side turns more slowly, amplifying torque through the drivetrain.

As the CVT upshifts, the primary sheaves close and the primary effective radius increases. At the same time, the secondary effective radius decreases. The transmission therefore moves from its launch configuration toward a higher-speed configuration. By the end of the shift, the primary radius is larger and the secondary radius is smaller. This coupled motion is shown in [Figure 1.6](#).

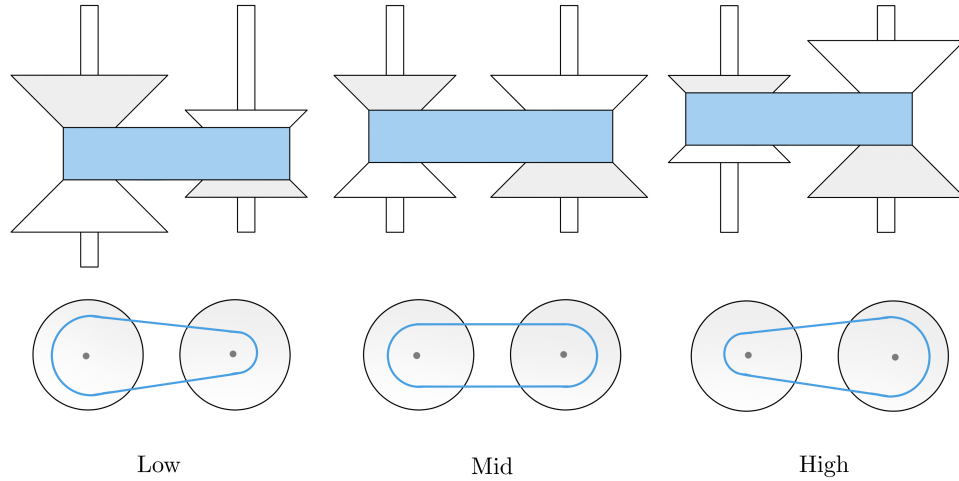


Figure 1.6: Low-, intermediate-, and high-ratio states of a belt-sheave CVT. As the belt moves outward on the primary pulley and inward on the secondary pulley, the effective radii vary in opposite directions while the belt length remains approximately constant. The lower views highlight the corresponding change in transmission ratio.

#### 1.2.4 Torque Transfer, Clamping Force, and Slip

The geometry described so far determines the speed relationship between the primary and secondary pulleys. However, a speed relationship alone does not guarantee that torque can be transmitted. Unlike a gear pair, the belt and pulley are not mechanically locked together by teeth. Torque is transmitted through frictional traction between the belt and the sheave faces.

For traction to develop, the belt must be physically squeezed between the sheave faces. [Figure 1.7](#) shows how this happens at a small portion of the belt-pulley contact. The circled region on the pulley wrap is enlarged to show the belt cross-section within the pulley V-groove. At that location, the movable sheave presses the belt toward the fixed sheave. This clamping action produces contact normal forces on the two sides of the belt, creating the contact pressure needed for frictional traction.

The same local interaction occurs continuously around the wrapped portion of the belt. At each position along the wrap, the normal forces between the belt and sheave faces allow a tangential friction force to act. Together, these small tangential traction forces transfer torque between the pulley and the belt. In practical terms, transferring more torque through the CVT requires enough clamping force to create enough available traction at the belt-pulley interface. More clamping force increases the normal contact force, which in turn increases the tangential force that friction can support.

If the available traction is not sufficient for the torque being transferred, the belt slips relative to the pulley. Slip reduces useful torque transfer and can produce heat, wear, and efficiency loss. A

working CVT must therefore do two things at once: it must allow the pulley radii to change so the ratio can shift, and it must maintain enough clamping force for the belt to transmit torque without excessive slip.

This distinction is central to the rest of the model. The pulley geometry determines how the speed ratio changes. The contact mechanics determine how much torque can actually be carried through the belt–pulley contacts. A CVT is therefore not only a variable-geometry mechanism; it is also a clamping and traction problem.

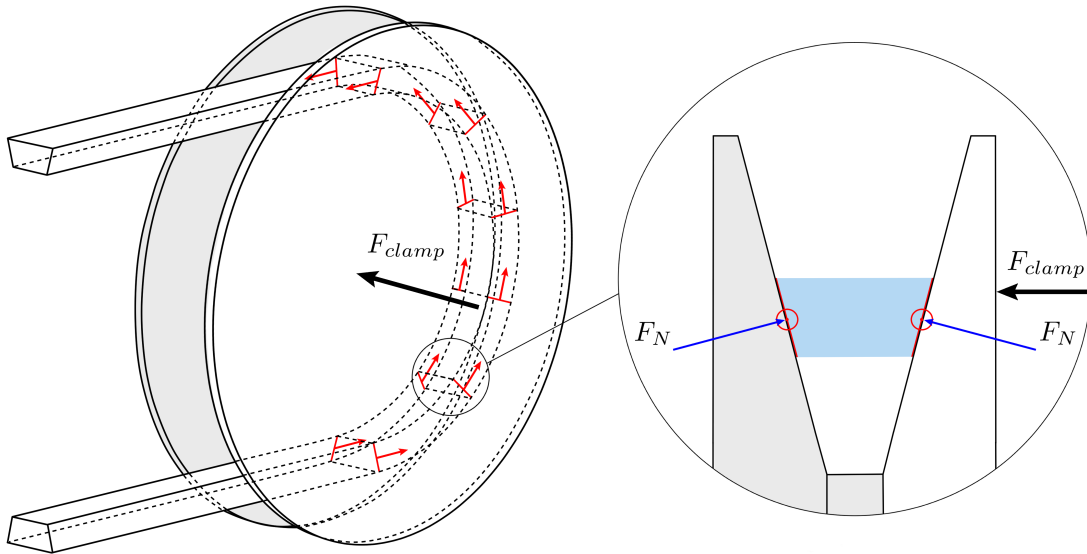


Figure 1.7: Clamping and traction in a rubber V-belt CVT. A representative location on the wrapped belt–pulley contact is circled and enlarged to show the local belt cross-section within the V-groove. Motion of the movable sheave toward the fixed sheave creates normal contact forces on the belt faces. These normal forces allow tangential frictional traction to act continuously along the wrap, collectively transmitting torque between the pulley and belt. The indicated tangential forces are representative local contributions to this distributed traction.

### 1.2.5 Mechanical Actuation in the Present CVT

Different CVTs generate radius change and clamping force in different ways. Some systems use electronic or hydraulic actuation to command pulley motion or clamping pressure directly. The system studied here is mechanically actuated: its pulley forces arise from mechanisms inside the primary and secondary pulley assemblies rather than from an externally commanded ratio.

These mechanisms are important because they determine both how the belt moves through the ratio range and how much clamping force is available for torque transfer. The primary pulley is mainly speed-sensitive, while the secondary pulley is mainly torque-reactive.

The primary pulley uses flyweights, ramps, and a spring. As engine speed increases, the flyweights move outward under centrifugal effects. The ramp surfaces convert this outward flyweight motion into a closing force on the primary sheaves. The spring resists this motion and helps set the speed at

which the primary begins to close. As the primary closes, the belt is pushed outward, the primary effective radius increases, and the CVT shifts in the upshift direction introduced in [Figure 1.6](#). The primary actuation mechanism is shown conceptually on the left side of [Figure 1.8](#).

The secondary pulley uses a helix mechanism and spring loading. The helix can be understood like a screw-like interface: when torque is transmitted through the secondary, the angled helix surfaces convert part of that torque reaction into a clamping force between the sheaves. In this way, torque through the secondary helps create the clamping needed to carry torque through the belt. The spring provides preload and helps set the baseline clamping behavior.

The helix also means that secondary shift is not purely axial. As the secondary moves through its shift range, axial motion is coupled with relative rotation between secondary sheave components. This detail becomes important later when the secondary force model is developed, but the basic intuition is simply that the helix converts torque and relative rotation into axial clamping. The secondary actuation mechanism is shown conceptually on the right side of [Figure 1.8](#).

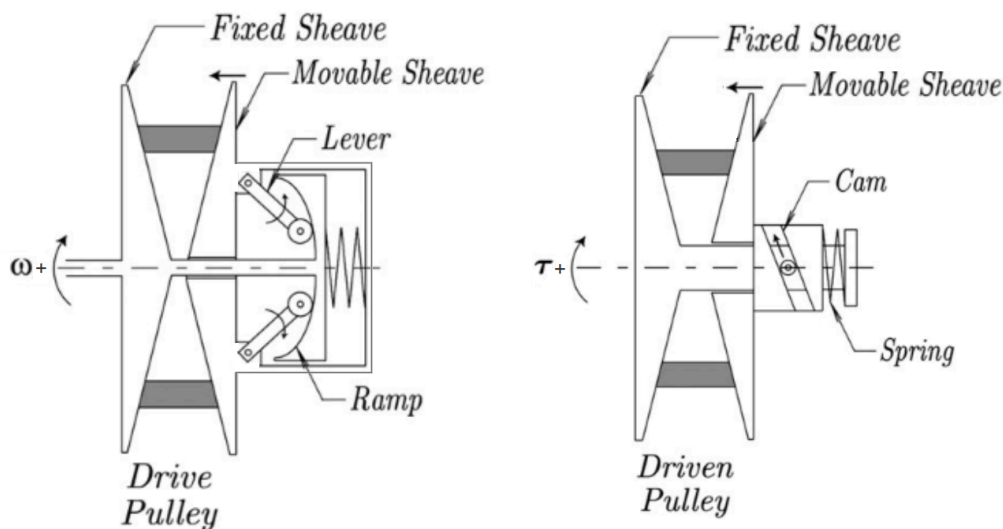


Figure 1.8: Conceptual illustration of the mechanical actuation mechanisms in the present CVT. Left: the primary (drive) pulley, where increasing rotational speed moves the flyweights outward and the ramp geometry converts that motion into a closing force on the sheaves, opposed by a spring. Right: the secondary (driven) pulley, where transmitted torque loads a cam or helix mechanism that converts part of the torque reaction into sheave clamping force, together with spring preload. In this way, the primary is mainly speed-sensitive and the secondary is mainly torque-reactive. ([Julió and Plante, 2011](#))

Although the primary and secondary mechanisms respond most strongly to different physical quantities, they act on the same belt and the same shift process. Primary speed affects centrifugal actuation, transmitted torque affects the torque-reactive secondary, the resulting clamping forces determine the available belt traction, and the belt geometry couples motion of the two pulleys. As the operating conditions change, the balance among these effects can change as well, allowing the transmission to shift in either direction or to enter a different traction state. These coupled relationships form the physical foundation for the dynamic behaviour developed later in this work.

## 1.3 Motivation and Literature Review

CVT modelling is not a single, uniform problem. A model developed to size a variator, estimate belt efficiency, design a control strategy, or predict transient behaviour may involve the same physical components, but it will usually make different choices about which quantities are prescribed and which are solved. Some models are primarily design tools. Some focus on mechanical actuation and force balance. Some study belt forces, axial thrust, slip, or efficiency at fixed operating points. Others attempt to resolve transient shift behaviour or detailed belt–pulley contact mechanics.

The progression followed here is from design intuition, to component force balance, to mechanical actuation, to belt contact, to ratio change, and finally to general transient CVT behaviour. This ordering mirrors the way the modelling problem becomes difficult. A belt–sheave CVT can first be understood as a variable-radius transmission, then as a collection of mechanical actuators that generate axial force, then as a friction-limited belt contact problem, and finally as a coupled dynamic system in which pulley rotation, shift motion, belt behaviour, and contact state evolve together. The literature contains strong foundations for each of these pieces, but those foundations are distributed across several modelling traditions.

### 1.3.1 Framework for Classifying CVT Models

A useful way to compare CVT models is to ask what part of the system is being allowed to move, and what part is being held fixed. The same physical transmission can be treated as a geometry problem, a force-balance problem, a belt-contact problem, a shift-dynamics problem, or a coupled system-dynamics problem. Each viewpoint can be useful, but each one answers a different question. This is why the literature is best read by tracking the modelling choices that define each formulation, rather than by treating all CVT models as attempts at the same final system.

One of the first choices is how the ratio is represented. In the simplest case, the transmission is evaluated at a fixed ratio. A slightly broader approach computes the ratio from a quasi-static force balance at each operating point. Other models prescribe a ratio trajectory, or use an empirical relationship for the rate of ratio change. More detailed shift models treat ratio change as a mechanical process in its own right, involving radial belt motion, pulley motion, and contact forces (Kim et al., 2002; Sorge, 2008; Cammalleri and Sorge, 2009; Sorge and Cammalleri, 2011). These approaches are not interchangeable. A fixed-ratio model can explain forces at a known condition, while a shift model can explain how the system moves from one condition to another. Even then, a formulation that describes the mechanics of an imposed or already active shift does not necessarily determine when a shift begins, which direction it takes, or what operating condition follows it.

A second choice is how axial force is produced. Some models take pulley clamping forces as known inputs. This is natural when the goal is to study belt response under specified loading, or when the variator is externally controlled. Other models include the mechanism that generates those forces, such as centrifugal weights, springs, torque-sensitive cams, hydraulic pressure, or electronic actuation (Cammalleri, 2005; Kim et al., 2002; Skinner, 2020; Sorge and Cammalleri, 2011). The important distinction is not only the actuator type, but how completely its mechanical action is resolved. A clamping force may be prescribed directly, obtained from an instantaneous force balance or empirical relation, or produced by a mechanism whose own motion is treated dynamically. Each choice answers a different question about how the pulley force arises.

A third choice is how much of the belt–pulley contact is represented. At low fidelity, the belt may be treated through a capstan-style tension relation, often adjusted for the wedge action of the V-groove.

More detailed models include radial and tangential friction, sliding angle, belt penetration, bending stiffness, local normal pressure, tension variation, slip mechanisms, and efficiency losses (Gerbert, 1996; Kim and Kim, 1989; Sorge, 1996; Kong and Parker, 2005, 2006; Chen et al., 1998; Bertini et al., 2014; Messick, 2018). At still higher fidelity, the belt may be discretized into nodes or elements so that local belt motion and contact behaviour are solved directly (Julió and Plante, 2011; Ballew, 2015). Increasing contact fidelity can make the local mechanics more realistic, but it does not by itself determine how much of the complete transmission response is being solved. Local mechanical fidelity and global dynamic closure are therefore separate modelling choices.

A fourth choice is how the rotating system and its interaction with the surroundings are treated. Many models prescribe one or both pulley speeds, or impose the speed ratio algebraically. This is often appropriate for steady contact analysis, efficiency prediction, or controlled operating-point studies. Other models solve rotational equations of motion so that pulley speeds and transmitted torques evolve dynamically (Srivastava and Haque, 2007; Ballew, 2015; Duan et al., 2016). Any physical transmission must, of course, interact with whatever machinery is attached to it. The important distinction is whether those external systems supply only the mechanical loading associated with that interaction, or whether quantities belonging to the internal CVT mechanics—such as pulley-speed histories, clamping-force histories, axial pulley motion, or ratio trajectories—must also be prescribed. This distinction determines how much of the transmission operating state is actually predicted by the model.

A further choice is the operating region over which the same formulation remains valid. Some models are intended for steady or fully engaged operation. Others describe transient shifting once the variator is already engaged, focus specifically on clutching or engagement, or move through a predefined sequence of operating phases. A broader formulation must determine which mechanical regime is active as conditions change and remain valid as the transmission enters or leaves those regimes. A model may therefore reproduce behaviour accurately within one operating phase without necessarily providing the mechanics that determine how the transmission reaches, leaves, or reverses that phase.

These questions form a useful progression for understanding the modelling problem, but the associated modelling choices are partly independent. A formulation may resolve belt contact in great detail while prescribing global pulley motion. Another may solve pulley rotation dynamically while receiving clamping force as an external input. Another may model mechanical actuation carefully while treating ratio change through quasi-static or empirical relations. None of these choices is inherently wrong; they reflect the intended modelling objective. The important point is that model fidelity in one part of the CVT does not by itself imply closure of the complete transmission response.

The following review therefore considers both what each formulation was developed to predict and what quantities or operating conditions remain prescribed. It proceeds from practical and quasi-static design models, through mechanical actuation and belt-contact mechanics, to transient rubber-belt formulations and related dynamic models in metal-belt and chain CVTs. This provides a consistent basis for identifying which parts of the CVT dynamics are already well established and where the remaining modelling gaps lie.

### 1.3.2 Practical Design and Tuning Foundations

The practical understanding of mechanically actuated rubber V-belt CVTs developed largely through machine design and tuning practice. General machine design references such as Budynas and Nisbett

(2015) provide the background needed to treat the CVT as a real mechanical assembly involving belts, springs, clutches, shafts, friction interfaces, uncertainty, and design iteration. Not all of these effects are included in a compact model, but they provide useful context for deciding which behaviours are retained and which are idealized.

Practical tuning literature provides a more application-specific foundation. [Aaen \(2007\)](#) describes powersport CVTs using the language of engagement speed, shift speed, flyweights, primary springs, secondary springs, torque sensing, backshift, and speed variance. Its value is not that it provides a complete dynamic model, but that it builds component-level intuition from the perspective of a tuner or racer. It explains how changes to weights, springs, ramps, and helices are expected to affect behaviour, while leaving much of the coupled system response to experience and testing.

These sources are valuable because they describe the physical machine before it is reduced to equations. They also show why prediction is difficult: the same hardware change can affect engagement, shift rate, belt grip, backshift, engine speed, and vehicle acceleration at the same time. Qualitative tuning rules are therefore useful starting points, but they are not enough to determine the complete transient response of the CVT under changing operating conditions.

### 1.3.3 Design-Oriented and Quasi-Static CVT Models

Design-oriented CVT models commonly evaluate the transmission at a fixed operating condition or through a sequence of quasi-static operating points. Rather than asking how every transient state evolves in time, these models usually ask which geometry, actuator, belt, or tuning parameters should be selected to satisfy a desired operating requirement. In a fixed-operating-point analysis, quantities such as ratio and pulley speed may be specified while the corresponding forces, tensions, or capacities are calculated. In a quasi-static model, the operating condition may instead be obtained from an instantaneous force balance, while inertia and the time required to move between neighbouring equilibria are neglected. These approaches commonly relate vehicle requirements, pulley geometry, belt length, wrap angle, speed ratio, flyweight force, spring preload, helix force, belt tension, and torque capacity. Their main value is that they connect tunable hardware to expected operating behaviour.

A useful starting point is the numerical design method of [Arango and Muñoz Alzate \(2020\)](#), which begins from vehicle requirements, road specifications, and desired performance before working toward the CVT design. This is valuable because it places component selection in its proper context: the belt, pulley geometry, speed-ratio range, engine or motor selection, and mechanical centrifugal actuation are not independent choices, but responses to the requirements imposed by the vehicle and its operating route. In this sense, the work provides a design-level framework where system requirements drive CVT component choices.

Other design-oriented works focus more narrowly on particular parts of the variator design. [Cammalleri \(2005\)](#) develops a speed-torque-controlled rubber V-belt variator design using equilibrium relationships for a centrifugal driver and torque-sensitive driven pulley. [La Battaglia et al. \(2022\)](#) focuses on the kinematic design of motorcycle rubber V-belt CVTs, emphasizing the roller sliding profile of the front pulley as a critical design variable. Both works show how simplified mathematical models can be used to choose actuator or geometry parameters without requiring a complete transient simulation of the transmission.

[Skinner \(2020\)](#) provides a practical force-balance and tuning-oriented model. The work emphasizes quantifiable tuning objectives such as engagement speed, shift-out speed, static RPM shifting, and

efficient power transfer, and relates those objectives to the primary and secondary subsystems. It is useful as an applied tuning reference because it connects the variables adjusted by teams to predicted operating behaviour, while remaining primarily an operating-point and force-balance model.

The common strength of these works is that they make CVT design less arbitrary. They identify the parameters that matter and show how geometry, springs, flyweights, helices, belt forces, and vehicle requirements can be related. They also highlight a recurring tension in CVT modelling: models that are simple enough to support early design may omit important coupled behaviour, while models that resolve more detailed mechanics can become difficult to apply directly as design tools. In this sense, design-oriented and quasi-static models occupy an important middle ground. They provide usable relationships for component selection and tuning, but they do not generally determine the complete time evolution of engagement, traction, shift motion, and pulley response.

### 1.3.4 Mechanical Actuation and Clamping Force Generation

After the basic design variables are identified, the next modelling question is how the CVT generates the axial forces that clamp the belt and drive ratio change. In externally controlled CVTs, these forces may be treated as prescribed hydraulic or electronic inputs. In mechanically actuated rubber V-belt CVTs, however, the forces are produced by the hardware itself. The primary force is typically speed-sensitive, while the secondary force is torque-sensitive.

[Cammalleri \(2005\)](#) provides an important design treatment of this problem through a speed-torque-controlled rubber V-belt variator. The centrifugal driver and torque-sensitive driven pulley are treated as part of the variator design, with equilibrium relationships connecting actuator geometry, spring loading, pulley torque, and the intended operating behaviour of the CVT. This establishes a quantitative link between the mechanical hardware and the clamping forces it is designed to produce.

[Kim et al. \(2002\)](#) also study a rubber belt CVT with mechanical actuators, including both steady-state and transient characteristics. Their driver and driven thrusts are obtained from the mechanical actuator geometry and instantaneous force balances, while the transient ratio response is represented separately by an experimentally identified first-order relationship. The work therefore connects mechanical primary and secondary actuation to transient CVT behaviour, but the motion of the actuator hardware itself is not what determines the shift acceleration.

[Sorge and Cammalleri \(2011\)](#) address the secondary side in substantially greater mechanical detail through the helical shift mechanics of rubber V-belt variators. Their formulation allows the fixed and movable driven faces to rotate differently and resolves how the belt-contact state determines the torque carried by the movable face. The resulting helix response therefore depends on spring loading, helix geometry, transmitted torque, and the internal relative motion of the driven pulley rather than on a simple assumed torque split. The instantaneous pulley radius and shift speed are nevertheless supplied to the contact calculation, so the resulting actuator relation determines the forces associated with a specified shift state rather than the acceleration of the movable secondary assembly itself.

Together, these works establish much of the mechanical foundation needed to model passive CVT actuation. They show how centrifugal primary mechanisms and torque-reactive secondaries generate configuration-dependent axial forces, and how detailed actuator geometry can materially change those forces. They also illustrate an important distinction introduced in [Section 1.3.1](#):

including the actuator mechanism does not necessarily mean that the actuator is itself a dynamically propagated subsystem. In these formulations, the actuator force is generally closed configuration by configuration, while transient ratio motion is either outside the model or supplied through a separate shift description. This distinction becomes important when the actuator hardware has inertia and its motion is coupled directly to the shift and shaft dynamics.

### 1.3.5 Rubber V-Belt Contact, Slip, and Efficiency Models

Once axial force is generated by the actuation mechanisms, the belt-pulley interface determines what that force actually does. The sheaves may clamp the belt, but the resulting torque capacity, axial reaction, radial belt motion, slip, and loss all depend on how the rubber belt interacts with the pulley groove. A rubber V-belt therefore cannot be treated as an ideal kinematic constraint. It wedges into the sheaves, develops a tension distribution around the wrap, experiences radial and tangential friction, deforms under load, and dissipates energy through several mechanisms.

Classical belt theory gives the starting point: friction and wrap angle determine the maximum tension ratio that can be supported before gross sliding occurs. For V-belts, this picture is modified by the groove angle and the associated wedge effect. However, real belt slip is more complicated than a single capstan-style traction limit. [Gerbert \(1996\)](#) presents a unified treatment of belt slip in which creep, radial compliance, shear deflection, and seating/unseating effects all contribute to speed loss. The important point is not that every model must include all of these effects, but that “slip” is not one physical mechanism. A model should be clear about which slip and loss mechanisms it represents and which it leaves outside its scope.

The contact problem also determines the axial reaction felt by the sheaves. [Kim and Kim \(1989\)](#) developed a theoretical analysis of axial forces in a rubber V-belt CVT using simplified assumptions about the contact regions and belt force distribution. [Sorge \(1996\)](#) developed a compact model for axial thrust in V-belt drives, emphasizing radial penetration and the connection between belt forces and pulley thrust. These works are useful because they connect the belt contact state back to the sheave force balance: axial force is not only generated by the actuator, but also shaped by how the belt seats, slides, and transmits load in the groove.

A more detailed treatment is given by [Kong and Parker \(2005\)](#), who formulate the steady mechanics of belt-pulley systems using distributed belt equations. [Kong and Parker \(2006\)](#) extend this framework to pulley grooves and sliding friction. This line of work resolves local belt mechanics at much greater fidelity than simple force-balance models, including the distribution of forces and sliding behaviour around the contact arc.

[Messick \(2018\)](#) extends this contact-mechanics tradition for rubber V-belt CVTs by building on the Kong and Parker framework and validating axial-force and efficiency predictions experimentally. Messick’s work is valuable for its contact-region formulation, belt property measurement, operating-range analysis, and dynamometer validation. It is also clear about its modelling scope: shifting is treated as a quasi-equilibrium process rather than as a transient response-time problem. In this sense, it is a strong reference for belt force and efficiency modelling, while leaving the time evolution of the complete CVT operating state to a different class of model.

Efficiency-focused studies provide another view of the same interface. [Chen et al. \(1998\)](#) experimentally investigated speed and torque losses in a rubber V-belt CVT, while [Bertini et al. \(2014\)](#) developed an analytical model for rubber V-belt CVT power losses including sliding, hysteresis, and entry/exit effects. These works show that transmission losses arise from several interacting

mechanisms whose relative importance depends on the operating condition.

The contact and efficiency literature therefore provides the foundation for understanding how belt force, axial thrust, slip, and loss arise. It also reinforces the distinction between local fidelity and global dynamic closure. A high-fidelity contact model may be necessary for efficiency prediction, axial-force prediction, or detailed local belt mechanics. A lower-dimensional contact model may instead be appropriate when the modelling objective requires the belt to remain coupled transparently to actuator dynamics, pulley rotation, shift motion, and the broader transient operating state of the CVT.

### 1.3.6 Transient Rubber-Belt Models and Dynamic Closure

The contact and force-balance literature establishes how a rubber belt CVT behaves at a specified operating condition. Transient models extend this picture by asking how the transmission moves between operating conditions and which parts of that motion must be resolved in time. In the rubber-belt literature, this has been approached through models of shift mechanics, distributed belt motion, pulley rotational dynamics, and mechanically actuated vehicle response. A useful comparison among these works is therefore which parts of the CVT state are advanced dynamically and which quantities are supplied as operating conditions or inputs.

[Sorge \(2008\)](#) provides an important bridge from quasi-static analysis to transient shift mechanics. Rather than representing ratio change as a sequence of neighbouring equilibria, the formulation treats the radial migration of the belt through the pulley grooves as a time-dependent process coupled to circumferential belt motion, mass conservation, and sliding. [Cammalleri and Sorge \(2009\)](#) later develop approximate closed-form solutions for the same class of shift problem. These works establish a mechanical basis for transient ratio change, while taking the pulley loading and active shift condition as part of the problem definition.

A more detailed transient treatment is given by [Julió and Plante \(2011\)](#), who discretize the rubber belt and apply Newton’s law to the individual belt elements. Their model predicts the time response of transmission ratio under specified drive-pulley axial force, drive-pulley speed, and load torque, while resolving belt elasticity, bending, damping, and frictional contact directly. The formulation is validated for steady and shifting conditions and provides substantially greater local belt detail than reduced force-balance models. The authors also report difficulty obtaining a stabilized prediction for the low-torque start-up condition in which belt clamping itself acts as the clutch, indicating that the operating range of the formulation is narrower than the full sequence of CVT engagement and shift behaviour.

[Ballew \(2015\)](#) extends this discretized-belt model family by removing the prescribed input-pulley speed used in earlier formulations. Applied torques, pulley inertias, and belt forces instead determine the pulley angular accelerations, allowing both shaft speeds to evolve dynamically. Engine and vehicle submodels are also included so that changes in external loading can feed back through the transmission rather than being represented by a prescribed pulley-speed history.

Axial actuation in [Ballew \(2015\)](#) remains a separate input to the belt–pulley model. The thesis notes that the axial-force inputs could be supplied either by a control system or by a model of a mechanically governed actuator. In the simulations presented, a PI controller with feed-forward gain generates the primary axial force, while the secondary axial force is held fixed. The formulation therefore extends the dynamic closure of the rotating system substantially, while leaving the source of the pulley clamping forces to a separate actuation model.

da Silva et al. (2023) present a reduced mechanically actuated rubber-belt CVT model coupled to a longitudinal vehicle simulation. Axial equations of motion are written for the movable sheaves, with flyweight loading on the primary and a torque-dependent secondary contribution. The operating sequence is divided into a clutching phase followed by a fully engaged phase, and the model is validated against a full-throttle acceleration. The work therefore provides an example of movable-sheave dynamics and mechanical actuation being retained in a transient application, while using a simplified, phase-specific representation of the CVT behaviour.

Taken together, these works show a clear progression in transient rubber-belt modelling. Ratio change has been treated as a mechanical process, belt motion and contact have been propagated directly, pulley rotational motion has been allowed to respond to applied torques, and movable-sheave inertia has been introduced in mechanically actuated applications. The formulations differ, however, in which parts of the transmission remain prescribed or are restricted to a particular operating phase. This makes the broader CVT literature useful for examining how other variator architectures have divided the boundary between prescribed actuation and dynamically resolved transmission behaviour.

### 1.3.7 Adjacent Metal-Belt and Chain CVT Dynamic Models

The broader CVT literature contains an extensive modelling tradition for metal push-belt and chain variators, particularly in automotive applications. These systems are not direct equivalents of dry rubber V-belt CVTs, since their load-carrying mechanisms, deformation, lubrication, and actuation systems differ. They nevertheless provide useful comparisons for transient ratio change, frictional contact, pulley dynamics, and the treatment of applied clamping forces.

Srivastava and Haque (2009) review the dynamics and control of belt and chain CVTs and summarize a wide range of steady and transient formulations, including slip and traction models, control-oriented shift laws, and detailed belt or chain mechanics. The review illustrates both the maturity of dynamic CVT modelling outside the rubber-belt literature and the importance of keeping the underlying transmission architecture in view when comparing models.

The Carbone–Mangialardi–Mantriota (CMM) family provides a well-developed example of mechanically derived shift dynamics for metal-belt and chain variators. Carbone et al. (2001, 2005) derive ratio-changing behaviour from belt or chain kinematics, mass conservation, friction, and pulley deformation, including creep-mode and slip-mode shifting. Experimental work by Yildiz et al. (2018) validates the resulting CMM shift dynamics under load. In these tests, the predicted quantity is the speed-ratio response to a prescribed change in clamping-force ratio, while drive-pulley angular speed, driven-side clamping force, and load torque are specified operating quantities. The formulation is therefore directed at the variator shift response under known actuation and loading.

Other metal-belt studies similarly treat transient ratio change and traction behaviour directly. Srivastava and Haque (2005, 2008) develop dynamic formulations that retain transient belt motion and examine the effects of friction characteristics and band-pack slip. Bonsen et al. (2003) study variator slip and its relationship to transmitted torque and clamping. These works extend the treatment of shift and traction beyond steady operating-point analysis while retaining the controlled clamping architecture typical of metal-belt automotive CVTs.

Schindler et al. (2012) develop a spatial transient multibody model of a pushbelt variator with detailed representation of the pulley sheaves, ring packages, friction, and unilateral contact. The model is used to study the internal transient response of the pushbelt system, including deformation

and three-dimensional contact behaviour. In the simulations presented, the output pulley receives prescribed clamping force and load torque, while the input pulley is assigned angular-velocity and clamping-velocity histories. The detailed belt and pulley response is therefore resolved under a specified global variator excitation.

Chain CVTs provide another example of strongly coupled transient mechanics. [Duan et al. \(2016\)](#) include pulley rotational dynamics, movable-pulley axial dynamics, chain radial motion, chain and pulley deformation, and a speed-dependent friction model spanning micro-slip and macro-slip. The axial and rotational pulley dynamics are coupled directly through the chain forces. The movable-pulley equation is driven by an applied clamping force, however, so the generation of that clamping force remains outside the variator model.

These adjacent models show how transient shift mechanics, detailed contact behaviour, pulley rotation, and movable-pulley motion can be represented at substantially different levels of detail. They also illustrate a recurring modelling boundary: in hydraulically or otherwise externally actuated variators, clamping force and pulley actuation are naturally supplied as inputs to the transmission model. For a mechanically actuated rubber V-belt CVT, those same functions are produced by hardware contained within the transmission, making the treatment of the actuation mechanism itself an important part of how the model boundary is defined.

### 1.3.8 Synthesis of Prior Work

The literature does not point to a single missing ingredient in CVT modelling. Instead, several mature lines of work resolve different parts of the same machine. Practical and design references establish the component relationships used to size and tune rubber V-belt CVTs. Mechanical-actuation studies explain how centrifugal and torque-reactive hardware generate pulley forces. Contact and efficiency models resolve the belt-pulley interface at progressively greater local detail. Transient formulations add radial belt motion, ratio change, pulley rotation, movable-sheave dynamics, or distributed belt and chain motion. The resulting body of work provides strong foundations for the individual mechanisms required in a dynamic transmission model.

A recurring distinction among these formulations is the location of the model boundary. Any CVT must receive mechanical interaction from the systems attached to it, but the quantities treated as external inputs differ substantially among models. Depending on the modelling objective, pulley speed, axial force, clamping pressure, ratio motion, or actuator commands may be prescribed while the remaining transmission response is solved. These choices are appropriate for many component, control, and contact studies, but they also determine how much of the CVT itself is contained within the dynamic system.

For a mechanically actuated rubber V-belt CVT, the mechanism that generates clamping is itself part of the transmission. Its motion therefore cannot be separated completely from the response that it produces: the actuation, pulley, belt, and rotational mechanics form one coupled physical system. No formulation identified in the reviewed literature was found to close that complete mechanically actuated system dynamically from the loading applied at the primary and secondary shafts. Treating the clamping mechanism dynamically is one part of this gap, but the broader requirement is that the internal CVT response be solved as a coupled whole rather than assembled from separately prescribed or separately closed pieces.

A separate gap concerns the range of operating behaviour covered by the same formulation. Much of the literature is developed for an already engaged variator, a specified shift, or a prescribed

sequence of operating phases. Individual works address clutching, upshift and backshift, ratio limits, or traction behaviour, but no formulation identified in the reviewed literature was found to remain mechanically valid across the complete operating sequence of disengagement and engagement, seated ratio limits, free shifting, and traction loss and recovery without imposing a predefined operating phase or trajectory. This is a different requirement from accurately modelling any one of those operating conditions: it requires the same physical formulation to remain valid as the active mechanical constraints change.

The remaining modelling opportunity is therefore defined by two independent forms of closure. The first is system-level closure of the complete mechanically actuated CVT, so that its internal state evolves as one coupled mechanical system in response to the loading applied at the primary and secondary shafts. The second is operating-domain closure, so that the same model remains valid as the transmission moves between its major mechanical regimes rather than being constructed around one specified phase or trajectory.

Table 1.1 summarizes selected formulations that most directly establish this progression. The table is not a ranking of model quality; each formulation reflects a different modelling objective. It instead shows which parts of the transmission are resolved within each model and which quantities remain prescribed as operating conditions or inputs.

| <b>Formulation</b>                                                | <b>Principal behaviour resolved</b>                                               | <b>Actuation / clamping</b>                                                   | <b>Quantities supplied or separately closed</b>                                    | <b>Primary operating scope</b>                          |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------|
| <a href="#">Cammalleri (2005)</a>                                 | Mechanical variator design and equilibrium force relationships                    | Centrifugal primary and torque-sensitive secondary represented mechanically   | Operating speed, torque, and equilibrium condition                                 | Steady / quasi-static design                            |
| <a href="#">Kim et al. (2002)</a>                                 | Steady mechanical response with an empirical first-order ratio transient          | Mechanical primary and secondary thrust relations                             | Operating conditions and experimentally identified transient shift relation        | Engaged upshift and downshift response                  |
| <a href="#">Sorge (2008)</a>                                      | Transient radial belt migration, sliding, and ratio-change mechanics              | Pulley loading forms part of the specified shift problem                      | Pulley loading and active shift condition                                          | Engaged transient shift                                 |
| <a href="#">Sorge and Cammalleri (2011)</a>                       | Torque-reactive secondary helix mechanics with differential sheave rotation       | Helix and spring represented mechanically                                     | Instantaneous pulley radius and shift motion supplied to the secondary calculation | Engaged secondary shift mechanics                       |
| <a href="#">Messick (2018)</a>                                    | Distributed rubber-belt contact, axial thrust, and efficiency                     | Axial loading prescribed                                                      | Pulley speed, torque, clamping, and operating point                                | Steady / quasi-equilibrium contact and efficiency       |
| <a href="#">Julió and Plante (2011)</a>                           | Transient discretized rubber-belt motion and contact                              | Drive-pulley axial force prescribed                                           | Drive-pulley speed, axial force, and load torque                                   | Engaged steady and shifting operation                   |
| <a href="#">Ballew (2015)</a>                                     | Discretized belt dynamics with pulley rotational response to applied torques      | Primary axial force controlled; secondary axial force fixed in reported cases | Axial-force inputs; actuator mechanism remains separate                            | Engaged transient operation with dynamic shaft response |
| <a href="#">Carbone et al. (2001, 2005); Yildiz et al. (2018)</a> | Mechanically derived metal-variator ratio dynamics including creep and slip modes | Clamping-force ratio prescribed                                               | Drive speed, clamping quantities, and load torque                                  | Engaged underdrive / overdrive shifting                 |
| <a href="#">Schindler et al. (2012)</a>                           | Detailed spatial pushbelt, pulley, friction, and unilateral-contact dynamics      | Global pulley actuation prescribed                                            | Input angular and clamping velocities; output clamping force and load torque       | Detailed transient response of an engaged variator      |
| <a href="#">Duan et al. (2016)</a>                                | Coupled chain, pulley axial and rotational dynamics, deformation, and slip        | Applied clamping force drives movable-pulley motion                           | Applied clamping and external loading                                              | Engaged chain-CVT shift dynamics                        |

Table 1.1: Selected CVT formulations illustrating the progression from operating-point and component models to increasingly coupled transient descriptions. The comparison emphasizes which behaviour is resolved within each formulation and which quantities remain prescribed according to its modelling objective.

## 1.4 Aim, Scope, and Contributions

The literature review identifies two remaining modelling needs for mechanically actuated rubber V-belt CVTs: closure of the transmission as one coupled dynamic system, and a formulation that remains valid as the CVT moves between its different operating regimes. The present work addresses these needs with a first-principles transient model of the CVT itself.

### 1.4.1 Aim

The aim of this work is to develop a general transient model of a mechanically actuated dry rubber V-belt CVT whose behaviour follows from its internal mechanics and from the mechanical loading applied at the primary and secondary shafts. The model is intended to determine its own ratio motion, clamping response, pulley dynamics, belt behaviour, and traction state rather than requiring those quantities to be prescribed as an operating history. External systems may therefore be coupled to either shaft without changing the internal CVT formulation.

The same physical model is also intended to remain applicable as the transmission moves between its different operating conditions. Engagement, operation at the ratio limits, free shifting, and changes in traction state are treated as different admissible states of the same CVT rather than as separate scenario-specific models. The direction and sequence of those transitions are determined by the evolving mechanical conditions.

A further aim is physical interpretability. The derivation is organized so that predicted behaviour can be traced back to the geometry, actuation, inertial response, belt mechanics, and active constraints that produced it. The assumptions, numerical implementation, and reported simulation cases are also documented so that the formulation can be reproduced and independently examined.

### 1.4.2 Scope

The modeled system is the mechanically actuated CVT itself. It includes the primary and secondary pulley assemblies, the mechanical hardware that produces their clamping forces, and the complete rubber V-belt loop. The model is deliberately independent of any particular engine, motor, dynamometer, vehicle driveline, or other surrounding system. Instead, those systems may be coupled to the CVT through the primary and secondary shafts without changing the internal transmission formulation.

The formulation retains the rotational motion of both pulley assemblies, the axial motions required for ratio change, the motion of the retained actuation hardware, and the transport motion of the belt. The retained component masses and inertias participate in the corresponding dynamic balances, while the actuation mechanisms and belt–pulley contacts provide the forces and reactions through which these motions influence one another.

The belt and contact mechanics are intentionally represented at lower order than in distributed deformable-belt formulations. The model does not attempt to resolve the complete spatial deformation of the rubber belt, local creep and shear fields, detailed seating and unseating behaviour, full actuator tribology, flexible pulley and shaft deformation, thermal evolution, wear, or a complete transmission-loss model. The principal consequences of these and the other idealizations are stated explicitly in [Section 2.2](#). These choices define the physical resolution of the present formulation; they do not prevent more detailed component laws from being introduced in future work.

### 1.4.3 Contributions

The principal contribution of this work is a first-principles transient formulation of the mechanically actuated rubber V-belt CVT as a complete dynamic component. Its internal response is solved as one coupled mechanical system under the loading applied at the primary and secondary shafts, without requiring prescribed histories of ratio, clamping force, pulley speed, or shift direction. The same formulation is constructed to remain valid across the major operating regimes of the transmission rather than around one predefined shift or operating sequence.

Realizing this overall capability requires several specific mechanical and formulation developments. These are contributions in their own right, while also providing the physical closure needed by the complete CVT model:

1. **Dynamic primary centrifugal actuation.** The primary flyweight mechanism is treated as a moving inertial subsystem rather than as an instantaneous centrifugal-force relation. The derivation retains the effect of flyweight acceleration in the axial force balance and the configuration-dependent contribution of the flyweights to primary shaft angular momentum. The same actuator motion therefore participates consistently in both the shift and rotational dynamics.
2. **Dynamic axial-rotational coupling of the torque-reactive secondary.** The movable secondary sheave is allowed to rotate relative to the shaft as required by the helix geometry while translating axially. Its retained rotational inertia consequently enters both the axial dynamics of the movable assembly and the angular-momentum balance of the secondary shaft. This produces the reciprocal coupling required between shift motion, helix reaction, movable-sheave rotation, and shaft dynamics.
3. **Reduced transient belt, geometry, and traction closure.** The pulley geometry and fixed belt length are carried through position, velocity, and acceleration level so that the evolving pulley radii enter the dynamic balances consistently. The belt retains an independent transport motion, and the reduced wrap mechanics preserve the radial and tangential acceleration terms implied by the selected kinematics. Belt reactions and traction admissibility are then coupled directly to the pulley and shift dynamics.
4. **Common mechanics across CVT operating regimes.** The formulation includes the mechanical constraints associated with disengaged or deadzone motion, engagement, seated ratio limits, free shift, and traction-limited stick-slip operation within one regime-dependent model. Transitions are selected from the admissibility of the current mechanical state rather than from a scripted operating sequence. When activation of a rigid constraint changes the allowable velocity, the post-transition motion is obtained from the retained inertia and impulse-momentum balance rather than an arbitrary state reset.

Together, these developments provide the system-level and operating-domain closure identified as missing in [Section 1.3.8](#). They also preserve a direct mechanical interpretation of the model: changes in the predicted CVT response can be traced to specific actuator, inertial, geometric, belt, or contact effects within the formulation.

## 1.5 Document Roadmap

The remainder of this work develops the model from system definition through simulation results. **Chapter 2** formalizes the CVT as a simulation problem by defining the modeled transmission, its interaction with surrounding systems, the retained assumptions and coordinates, the state variables, and the mechanical regimes through which the model evolves. **Chapter 3** then derives the governing mechanics, proceeding from pulley and belt geometry to axial actuation, pulley rotational dynamics, belt transport and contact, and finally the coupled transient model and its regime transitions.

**Chapter 4** presents the simulation results and discussion using representative launch cases to examine the model behaviour and its response to targeted CVT tuning changes. It also documents the implementation and reproducibility artifacts and concludes the present study. The appendices collect supporting derivations and numerical details that are useful for reproducing or extending the formulation but would interrupt the main development.

## Chapter 2

# Model Definition and Simulation Problem

With the aim and scope established in [Section 1.4](#), the next step is to define the CVT as a precise simulation problem. This requires a clear account of what belongs to the modeled transmission, how surrounding systems interact with it mechanically, and what information describes the CVT at a given instant.

The definition proceeds from the outside inward. [Section 2.1](#) first establishes the modeled CVT and its shaft interfaces, after which [Section 2.2](#) states the physical idealizations that set the resolution of the formulation. [Section 2.3](#) then defines the coordinates and sign conventions used to describe the retained motions. With those choices fixed, [Section 2.4](#) identifies the internal state carried from one instant to the next, while [Section 2.5](#) introduces the active mechanical constraints that distinguish the admissible operating regimes. Finally, [Section 2.6](#) separates the roles of fixed data, current state, shaft-interface loading, and instantaneous model results, completing the simulation problem used by the derivation in Chapter 3.

These definitions provide the common problem statement used by the derivation in [Chapter 3](#). At any instant, the fixed CVT data, current internal state, and mechanical loading presented at the two shafts are known. The governing mechanics determine the corresponding instantaneous response and state derivatives, which may then be advanced through time by numerical integration.

### 2.1 Modeled CVT and Shaft Interfaces

The CVT is connected to the surrounding machinery through its primary and secondary shafts. These two connections form the external mechanical interfaces of the modeled transmission. Everything that physically belongs to the CVT between them is retained inside the model boundary.

The modeled CVT includes the primary and secondary pulley assemblies, the complete belt loop, and both belt–pulley contacts. It also includes the primary flyweights, ramp, and spring and the secondary spring and helix. Any masses and inertias retained for these components are included with them. Their forces and inertias therefore act together as parts of the same coupled mechanism.

Placing the model boundary at the two shafts keeps the CVT description independent of any particular surrounding system. An attached system supplies the mechanical loading that it presents

at the corresponding shaft, so changing what is connected to the CVT changes the loading seen by the model rather than the definition of the transmission itself.

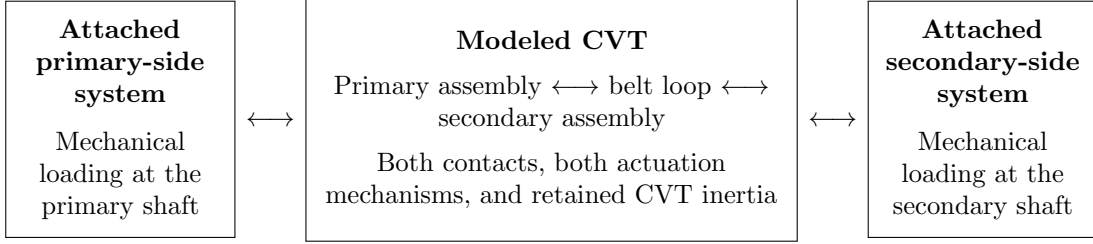


Figure 2.1: The modeled CVT and its external mechanical interfaces. Everything in the centre box is resolved as part of the transmission. An attached system acts on the CVT through the corresponding shaft, while the shaft motion remains part of the modeled response.

The same interface description applies at either shaft. Let  $j \in \{p, s\}$  identify the primary or secondary side. The attached system contributes two mechanical quantities:

$$\tau_{B,j}, \quad I_{B,j}.$$

The applied torque  $\tau_{B,j}$  is the signed torque exerted on the CVT through shaft  $j$ . The equivalent inertia  $I_{B,j}$  is the inertia of the attached system as seen at that same shaft. A rotor connected directly to the shaft contributes its own inertia; a wheel, drivetrain, or other mechanism connected through gearing contributes the corresponding inertia reflected to the shaft.

These quantities describe how the attached system loads the CVT, not how the shaft must move. The shaft speed remains part of the modeled motion. An attached system may use the current shaft speed, time, and its own internal state to determine the torque and equivalent inertia that it presents at the interface. Those quantities then act together with the inertia and mechanics inside the CVT to determine the shaft acceleration.

A positive applied torque acts in the positive rotational direction defined for the corresponding shaft, and the equivalent inertia is expressed about that same coordinate. The equivalent inertia contains only inertia belonging to the machinery attached beyond the shaft interface. Any pulley, sheave, flyweight, helix, or belt inertia retained inside the CVT is therefore excluded from  $I_{B,j}$ , so no component inertia is counted twice.

The terms *primary-side interface* and *secondary-side interface* identify the two shaft connections rather than a required direction of power flow. Torque may act through either interface in either rotational direction without changing which components belong to the CVT. Changing the attached driver or load therefore changes the loading at the interface, not the definition of the modeled CVT.

## 2.2 Modeling Assumptions

The assumptions below define the physical resolution of the formulation rather than a particular operating case. Each assumption is stated together with its principal consequence for the mechanics that can and cannot be represented. The consequence summaries are not intended to enumerate every secondary effect of an idealization; they identify the main physical resolution removed by that choice so that later results can be interpreted against a clear model boundary.

### 2.2.1 Material and Mechanism Idealizations

#### Assumption 1 (Time-Invariant Material and Surface State)

##### Model statement

The material properties and surface condition assigned to the CVT remain unchanged during a simulation. Intrinsic component dimensions are likewise not allowed to evolve through wear or thermal growth. The selected properties describe the hardware for the duration of the simulated event rather than evolving with its operating history.

##### Consequences for model fidelity

Temperature-dependent stiffness or friction, thermal state evolution, wear, glazing, conditioning, degradation, and other feedback from accumulated contact history are not represented. The formulation therefore describes short-term mechanical response for a specified material and surface state rather than the progressive evolution of that state.

#### Assumption 2 (Rigid Pulley and Actuation Hardware)

##### Model statement

The pulleys, shafts, actuation hardware, guides, and supporting CVT structure are represented as rigid bodies. Their prescribed geometry does not deform under the loads developed during operation.

##### Consequences for model fidelity

Pulley bending, shaft bending or torsion, guide and housing deflection, load-dependent groove geometry, and structural vibration generated by those compliances are outside the model. More detailed CVT formulations show that pulley deformation can alter local belt mechanics and axial loading; those effects would require compliant component models rather than changes to the rigid-body balances alone ([Carbone et al., 2005](#); [Duan et al., 2016](#)).

#### Assumption 3 (Idealized Actuator Mechanics)

##### Model statement

The primary ramp and flyweight mechanism, secondary helix, guides, and springs follow their prescribed kinematic and force laws without additional contact friction, backlash, or local compliance unless such an effect is introduced explicitly by the selected component law.

##### Consequences for model fidelity

Ramp or roller friction, pin–helix friction, roller or pin spin, guide friction, backlash, helix-flank transitions, contact compliance, and unmodelled spring hysteresis or internal damping are

omitted. Relaxing this assumption would add actuator-specific tribology and compliance to the same mechanism free bodies; for example, contact friction appears explicitly in more detailed centrifugal and helix treatments ([Kim et al., 2002](#); [Sorge and Cammalleri, 2011](#)).

## Assumption 4 (Fixed, Uniform Belt Cross-Section)

### Model statement

The same prescribed cross-section is assigned everywhere along the belt and is carried unchanged as the belt runs and shifts. This description fixes the cross-sectional shape and dimensions and treats mass density as uniform through the section and along the belt. The belt therefore has a constant mass per unit length, and its cross-sectional centre of mass coincides with the geometric centroid. The cord layer is retained only as a prescribed tension-carrying line at the fixed depth  $d_{\text{cord}}$ ; its distinct density and contribution to the through-section mass distribution are not resolved. As the section follows the curved modeled belt path, it does not compress, bulge, shear, warp, or grow under centrifugal or sheave loading.

### Consequences for model fidelity

Accordingly, neither spatial variations caused by cogs, ribs, joints, local reinforcement, manufacturing variation, or nonuniform wear nor load-induced changes caused by lateral and radial compression, bulging, cross-sectional shear and warping, centrifugal growth, or local cog and rib deformation are represented. Concentrated cord mass and differences between the densities of the belt layers cannot shift the modeled centre of mass away from the geometric centroid. Changes in belt width, height, contact geometry, mass distribution, and internal strain energy caused by these omitted effects therefore lie outside the present formulation. Detailed belt models retain some of these spatial and compliance effects explicitly ([Gerbert, 1996](#); [Kong and Parker, 2005](#); [Messick, 2018](#)).

## 2.2.2 Belt Geometry and Kinematics

## Assumption 5 (Aligned Pulley Axes and Fixed Center Distance)

### Model statement

The primary and secondary pulley axes remain parallel at a fixed spacing. The pulley centers and shaft directions therefore do not change as the CVT shifts.

### Consequences for model fidelity

Shaft misalignment, eccentricity, geometric runout, bearing-center motion, and the associated changes in belt path and face loading are not represented. Removing this assumption would require the belt geometry and contact loading to respond to the additional shaft and bearing motion.

## Assumption 6 (Fixed Belt Length)

---

### Model statement

The modeled belt has a fixed total length. Longitudinal extension is not treated as an independent deformation of the belt, so every admissible pulley configuration must be compatible with the same installed length.

---

### Consequences for model fidelity

Tension-dependent elongation, longitudinal strain energy, extension waves, permanent elongation, and the longitudinal component of elastic creep are not represented. Detailed traction-belt models can retain these effects through a longitudinal compliance field; the present formulation instead assigns ratio change to the pulley and contact motions of a fixed-length belt ([Gerbert, 1996](#); [Kong and Parker, 2005](#); [Messick, 2018](#)).

## Assumption 7 (Single Belt Transport Coordinate)

---

### Model statement

The circumferential motion of the complete belt is represented by one global transport coordinate. Individual material points are not tracked around the loop, and the local tangential belt speed is not given an independent spatial field.

---

### Consequences for model fidelity

The geometry-induced redistribution of local tangential material speed during shift, material-point migration through the wraps, and the additional momentum transport associated with that spatially varying velocity field are not resolved. The model still retains acceleration of the global belt transport and the motion of the prescribed wrap geometry. Shift-mechanics and discretized belt formulations resolve the omitted local transport in greater detail ([Sorge, 2008](#); [Cammalleri and Sorge, 2009](#); [Julió and Plante, 2011](#)).

## Assumption 8 (Prescribed Planar Belt Path)

---

### Model statement

At each instant, the belt is taken to wrap each pulley along a circular arc and to pass between the pulleys along two straight tangent spans in one common plane. The belt path is therefore set by the current pulley geometry rather than solved as a flexible dynamic curve.

While both pulleys are engaged, the belt is treated as taut and perfectly flexible. It carries load along its length through a single longitudinal tension, but it carries no compression, transverse shear force, or bending moment. The belt also remains in continuous contact with the sheave faces over each prescribed wrap. This engaged description is physically admissible only while the belt tension and distributed normal contact loading remain nonnegative.

---

**Consequences for model fidelity**

Slack or buckled spans within the engaged CVT, local belt lift-off, partial wrap, and contact reattachment are not represented. Bending stiffness, flexural strain energy, and bending hysteresis are also absent even though the prescribed belt path is curved. A calculated negative tension or normal loading therefore indicates that the assumed engaged path is no longer physically admissible; it does not represent a physical negative belt load.

Progressive or spiral radial migration within a wrap, seating and unseating trajectories, span sag, transverse belt vibration, and transverse inertia associated with straight-span reorientation are omitted. The same planar description also excludes the out-of-plane span skew, twist, and axial belt inertia that would arise when the two physical groove centerplanes move differently during shift. More detailed belt and shift-mechanics models resolve subsets of these path and contact effects explicitly (Sorge, 2008; Kong and Parker, 2005; Messick, 2018).

---

**Assumption 9 (Reduced Belt Motion While the Primary Is Disengaged)**

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**Model statement**

While the primary sheaves are separated from the belt, no primary sheave normal resultant, tangential traction, or belt torque is retained. Any radial support provided at the minimum primary radius is not assigned a tangential contact law. Belt pretension and the changing shape of the slack spans are not retained as additional mechanical states.

The belt is instead taken to remain seated on the fully closed secondary and to be carried without slip at the secondary effective radius. The belt transport speed and secondary speed are therefore joined by one kinematic constraint throughout the disengaged-primary regime.

---

**Consequences for model fidelity**

The disengaged belt and secondary do not have independent transport coordinates. Slack-span motion, sag, belt whip, intermittent primary rubbing, secondary slip, and tension waves created as slack is taken up cannot be predicted. The model also cannot determine deadzone belt pretension or the transient contact loads produced by a detailed re-engagement history. Instead, the entire belt moves at the speed imposed by the secondary contact.

### 2.2.3 Belt–Pulley Contact Representation

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**Assumption 10 (Gross Tangential Coulomb Traction)**

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**Model statement**

During continuous motion, each belt–pulley interface is represented by a dry gross stick–slip law with fixed static and kinetic traction capacities for the selected surface state. The frictional contact action retained by the model acts in the circumferential belt-transport direction.

---

**Consequences for model fidelity**

The formulation can distinguish torque-carrying stick from gross kinetic slip, but it does not resolve radial friction, local creep or microslip, detailed presliding behaviour, lubrication or Stribeck effects, or friction variation with pressure, temperature, conditioning, or local sliding speed. Friction associated specifically with radial migration, seating, and unseating is also outside the retained contact direction. These mechanisms are treated separately in higher-fidelity rubber-belt contact models (Gerbert, 1996; Kong and Parker, 2006).

The static and kinetic traction limits are force-level laws for continuous motion. They do not supply a separate impulse-level friction law during an instantaneous change in mechanical constraint.

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**Assumption 11 (Wrap-Level Contact Resolution)**

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**Model statement**

The model may resolve how belt tension and normal loading vary around a pulley wrap, but each wrap is assigned one overall traction state at an instant. The contact is not resolved through the belt width or depth, and no separate local traction state is carried for different portions of the same wrap.

---

**Consequences for model fidelity**

Local adhesion and sliding subregions, spatial traction reversals, detailed pressure concentrations, edge loading, and cross-sectional contact-stress fields cannot be predicted. Resolving those effects requires a distributed or discretized contact description of the type used in higher-fidelity belt models (Kong and Parker, 2006; Messick, 2018; Julió and Plante, 2011).

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**Assumption 12 (Symmetric Sheave-Face Sharing)**

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**Model statement**

The two sheave faces at a pulley are assumed to share the total normal contact load equally. On the secondary pulley, the belt torque is also divided equally between the fixed and movable faces for the purpose of the movable-sheave balance. The primary sheave faces rotate together with their shaft and therefore have one common angular speed. On the secondary, helix motion gives the movable face a different instantaneous angular speed from the fixed face. Under **Assumption 11 (Wrap-Level Contact Resolution)**, the wrap-level tangential law does not retain those two face contacts separately. It therefore uses one representative secondary contact speed, defined so that the aggregate contact retains the pulley-side tangential power implied by the two equal face-torque paths.

---

**Consequences for model fidelity**

Face-to-face differences in normal loading, transmitted torque, and local sliding state are not resolved. The power-equivalent representative speed preserves the pulley-side torque power

implied by the equal-share reduction, but it does not make the two physical face speeds equal or recover their separate contact states. The helix-induced rotation of the movable face remains in the actuation and rotational balances and also enters the representative contact speed. Secondary sticking therefore denotes gross sticking of this aggregate contact, rather than simultaneous pointwise adhesion to both physical faces during helix motion. Face-resolved shift models show that the two secondary face torques and sliding conditions can depart substantially from this aggregate description during shifting ([Sorge and Cammalleri, 2011](#)).

#### 2.2.4 Boundary and Transition Idealizations

##### **Assumption 13 (Instantaneous, Perfectly Inelastic Constraint Activation)**

###### **Model statement**

When a mechanical stop or a new kinematic constraint becomes active at finite relative velocity, its finite contact duration and local compliance are replaced by an instantaneous, perfectly inelastic event. This treatment applies to arrival at a mechanical travel stop, first belt engagement, and activation of the deadzone belt lock.

The generalized positions remain continuous. The generalized velocities may change, and their post-event values are determined by impulse–momentum balance over all retained inertial coordinates subject to the newly active kinematic constraints. Applied torques, spring forces, damping forces, and other bounded forces act over no resolved time during the event and therefore contribute no finite impulse. All masses and inertias retained by the continuous mechanics remain part of the momentum balance. A constraint release produces no impulse unless another constraint activates at the same instant.

###### **Consequences for model fidelity**

The event model does not resolve contact duration, local indentation, peak impact force, stress waves, rebound, chatter, or the high-frequency vibration created by compliant contact. Individual velocities are neither preserved nor set independently; incoming momentum is redistributed among the velocities that remain compatible with the new mechanical topology. Because the event is perfectly inelastic, kinetic energy need not be conserved. The decrease in kinetic energy is retained as the impact or topology-capture loss, but the model cannot divide that loss among local belt deformation, material damping, sound, heat, or structural vibration. Predicting those details would require a finite-duration compliant-contact model. An impulsive load applied through an external shaft boundary would likewise require an explicit boundary impulse rather than the continuous torque description.

##### **Assumption 14 (Tangential Contact Through Boundary Events)**

###### **Model statement**

A belt–pulley contact that is sticking immediately before an instantaneous boundary event remains constrained against tangential slip through the event. Contacts that are sliding, together with contacts created by the event, are not assigned a new impulsive no-slip constraint. Their relative tangential motion after the event, together with the ordinary continuous traction limits, determines the stick–slip state that follows.

---

**Consequences for model fidelity**

An existing sticking contact may transmit whatever tangential constraint impulse is required to preserve no slip; that impulse is not checked against a separate impulse-level friction bound. The model therefore cannot predict impact-induced loss of an existing stick, brief impulsive slip followed by readhesion, or a transition caused by exhausting an impulse-level friction capacity.

Conversely, normal contact or kinematic capture does not automatically glue a new belt–pulley interface tangentially. In particular, a newly formed primary contact may retain tangential relative motion and begin its continuous motion in slip. Resolving the omitted impulsive friction behaviour would require an impulse-level traction law coupled to the normal constraint impulse.

## 2.2.5 Other Omitted Losses

### Assumption 15 (Negligible Unmodeled Parasitic Losses)

---

**Model statement**

Bearing, seal, windage, and auxiliary-hardware losses are neglected unless they are introduced explicitly by a component or external boundary model. No global empirical efficiency factor is inserted to represent these omitted mechanisms.

---

**Consequences for model fidelity**

Energy loss and damping arise only from mechanisms represented explicitly by the formulation, including gross kinetic slip and perfectly inelastic boundary or topology-capture events. Absolute efficiency may therefore be optimistic, and real hardware may exhibit additional damping or resisting torque when the omitted parasitic losses are significant. Adding such losses does not require changing the kinematic structure of the CVT; it requires supplying the corresponding physical torque or force laws.

**Resulting fidelity** Taken together, these assumptions retain the dominant rigid-body actuation, pulley rotation, sheave-shift motion, global belt transport, wrap tension and contact resultants along an admissible taut belt path, clamping, traction limitation, gross stick–slip mechanics, and momentum redistribution when a new mechanical constraint activates. They also retain the kinetic-energy loss associated with gross slip and perfectly inelastic capture.

The formulation does not reproduce the complete deformable belt and local contact problem, spatial variation within the belt construction, slack or buckled engaged spans, partial-wrap lift-off and reattachment, belt bending mechanics, slack-belt dynamics while the primary is disengaged, actuator

tribology, flexible-structure response, finite-duration impact and rebound, impulse-level friction limits, thermal or wear evolution, or miscellaneous parasitic losses. The assumptions are separated by physical mechanism so that a higher-fidelity model can replace one closure without obscuring which additional behaviour that change is intended to recover.

## 2.3 Coordinate and Sign Conventions

This section defines the coordinates and sign conventions used throughout the formulation.

### 2.3.1 Rotational and Belt-Transport Directions

Figure 2.2 shows the two pulley assemblies face-on, with their parallel shafts extending away from the viewer. Let  $\omega_p$  and  $\omega_s$  denote the angular velocities of the primary and secondary pulley assemblies, respectively. In this view, counterclockwise is positive for both pulleys, as shown by the arrows in the figure. Thus,  $\omega_p > 0$  and  $\omega_s > 0$  denote counterclockwise rotation, while negative values denote clockwise rotation.

Let  $v_b$  denote the signed transport speed of belt material around the loop. The positive direction is the direction shown in Figure 2.2: from the secondary pulley to the primary pulley along the upper span, and from the primary pulley to the secondary pulley along the lower span. Thus,  $v_b > 0$  denotes transport in this direction, while  $v_b < 0$  denotes transport in the opposite direction.

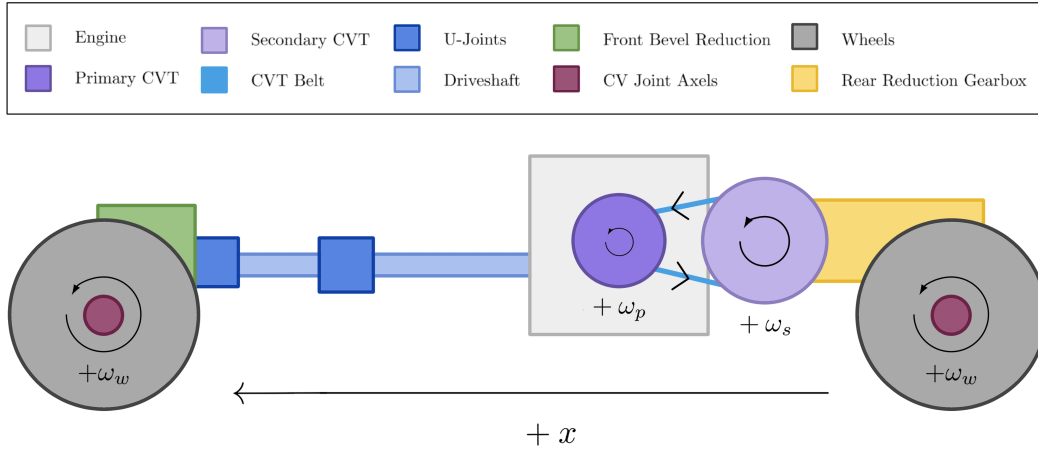


Figure 2.2: Rotational and belt-transport sign convention viewed from the outboard pulley faces, looking inward along the parallel shaft axes. Counterclockwise rotation is positive for both pulleys. With the primary pulley on the left and the secondary pulley on the right, positive belt transport runs from secondary to primary along the upper span and from primary to secondary along the lower span.

Torque follows the corresponding shaft convention. A torque acting on shaft  $j \in \{p, s\}$  is positive when it acts in the rotational sense defined by  $\omega_j > 0$ . In particular, the interface torque  $\tau_{B,j} > 0$  acts on the CVT in this positive direction. A torque that opposes positive shaft rotation is negative, but the torque coordinate itself does not change if the shaft reverses.

An attached engine, vehicle, dynamometer, or other system may use its own coordinates internally. The shaft speed and applied torque exchanged at its CVT interface are expressed using the convention defined here.

### 2.3.2 Local Axial Sheave Coordinates

Each movable sheave is assigned a local axial coordinate. Let  $x_p$  denote the position of the primary movable sheave and  $x_s$  the position of the secondary movable sheave. More generally,  $x_j$  denotes the corresponding coordinate for pulley  $j \in \{p, s\}$ .

For either pulley,  $x_j = 0$  denotes the fully open position. Increasing  $x_j$  moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing  $x_j$  opens it. If  $x_{j,\max}$  denotes the available travel, then

$$x_j = 0 \quad \text{is fully open,} \quad x_j = x_{j,\max} \quad \text{is fully closed.}$$

The velocity follows the same convention:  $\dot{x}_j > 0$  denotes closing motion and  $\dot{x}_j < 0$  denotes opening motion. The fully open position is only the geometric origin of the coordinate; it does not prescribe the configuration from which a simulation must begin.

The two coordinates are local to their respective sheave pairs. Their positive directions may therefore point in opposite physical directions along a common global axis. What they share is their mechanical meaning: positive axial motion closes the corresponding pulley.

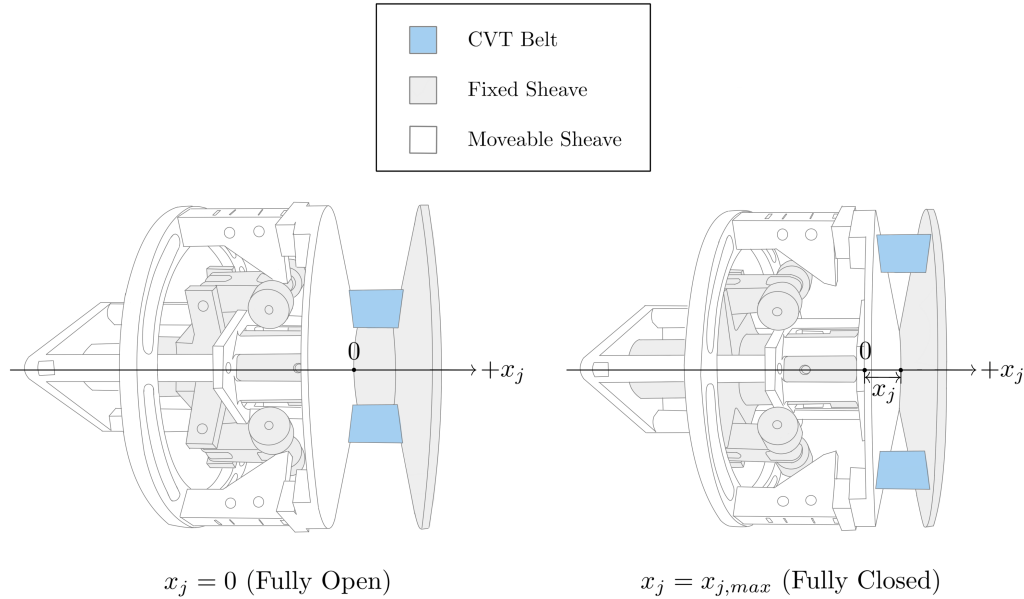


Figure 2.3: Local axial convention for the primary and secondary movable sheaves. For either pulley  $j$ ,  $x_j = 0$  denotes the fully open position and increasing  $x_j$  moves the movable sheave toward the fixed sheave. The fully closed position is  $x_j = x_{j,\max}$ . Because the coordinates are local, the positive primary and secondary directions need not point in the same global direction.

### 2.3.3 Local Radial, Tangential, and Axial Directions

At either pulley, the radial and tangential directions lie in the plane of pulley rotation. At any point around the pulley, the radial unit vector  $\hat{e}_r$  points outward from the pulley axis. The tangential unit vector  $\hat{e}_\theta$  is perpendicular to  $\hat{e}_r$  and points in the direction produced by the positive rotation defined in [Section 2.3.1](#). A radial or tangential component is positive when it points along its corresponding unit vector.

These vectors form a local polar basis rather than a fixed Cartesian pair. Their directions therefore change with the point around the pulley, while  $\hat{e}_r$  remains outward and  $\hat{e}_\theta$  remains tangent in the positive rotational direction. The inset in [Figure 2.4](#) shows this local basis.

The positive axial direction for pulley  $j$  is the direction of increasing  $x_j$  defined in [Section 2.3.2](#). It lies along the pulley shaft and is perpendicular to the radial–tangential plane. In the face-on view of [Figure 2.4](#), the shaft axis therefore passes through the page. That figure illustrates only the radial and tangential directions; the axial sign remains the local sheave-closing direction shown in [Figure 2.3](#).

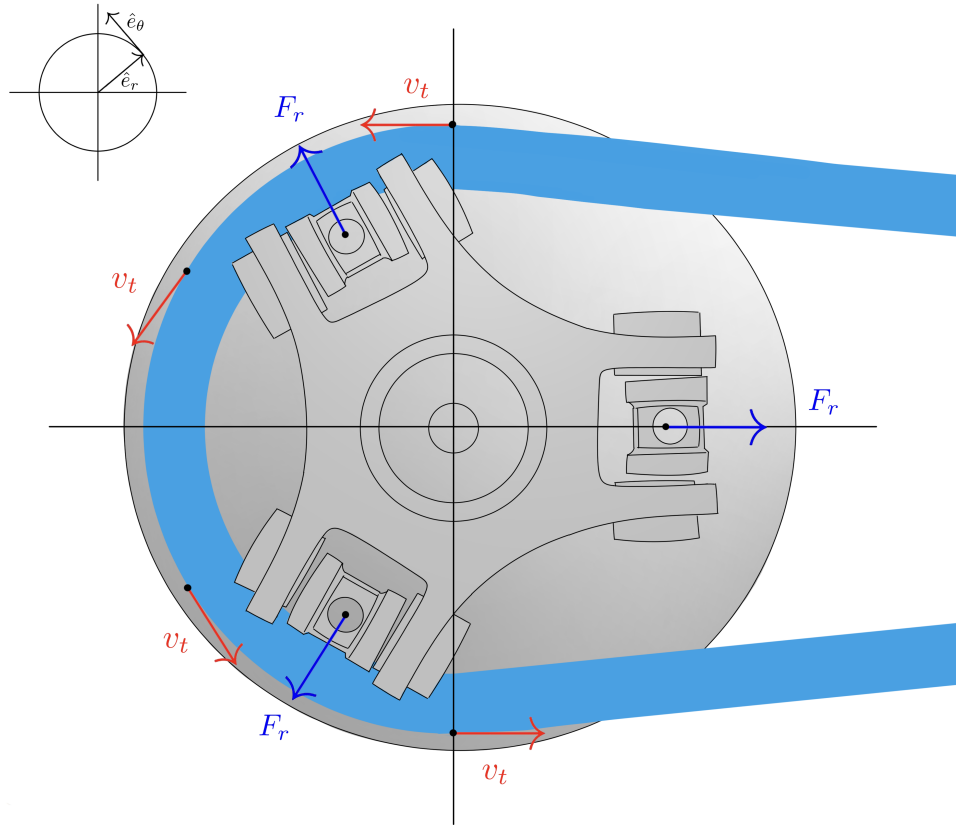


Figure 2.4: Radial and tangential directions in the plane of pulley rotation. The inset shows the local polar basis:  $\hat{e}_r$  points outward from the pulley axis, while  $\hat{e}_\theta$  is tangent in the positive rotational direction defined in [Section 2.3.1](#). The basis changes orientation with the point around the pulley rather than remaining fixed like a Cartesian pair.

## 2.4 CVT State Variables

At any instant, the modeled CVT's motion and configuration are described by five internal numerical quantities. These quantities are collected in the state vector,

|                                                                                             |                |
|---------------------------------------------------------------------------------------------|----------------|
| <b>State Vector</b>                                                                         | <b>(2.4.1)</b> |
| $\mathbf{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ v_b \end{bmatrix}$ |                |

Here,  $\omega_p$  and  $\omega_s$  are the primary and secondary shaft speeds. The shift coordinate  $s$  uniquely identifies the positions of both movable sheaves, while  $\dot{s}$  is the corresponding shift velocity. The geometric relation between  $s$  and the local coordinates  $x_p$  and  $x_s$  is established in [Section 3.1.3](#). The remaining entry,  $v_b$ , is the belt transport speed.

The primary and secondary pulley assemblies rotate about separate shafts. Although the belt couples their motion, neither shaft speed can generally be reconstructed from the other. Both  $\omega_p$  and  $\omega_s$  must therefore be retained. Absolute shaft angles are not required because the internal CVT mechanics depend on the current angular velocities, but not on the angular orientation of either pulley about its shaft.

The current pulley geometry and the configurations of the retained actuation mechanisms depend on the shift position, so  $s$  must also be retained. Shift motion has inertia and is governed by a second-order balance. Its present velocity  $\dot{s}$  is therefore required alongside its position.

The belt transport speed is retained independently because belt motion need not equal the surface speed of either pulley when tangential slip is present. Under the uniform-belt and single-transport-coordinate assumptions, the mechanics do not depend on which material point occupies a particular location around the loop. No belt-phase or accumulated belt-displacement coordinate is therefore required.

Together, these five values contain all of the internal position and velocity information that must be carried from one instant to the next. For the selected CVT, these values, the current shaft-interface conditions, and the active mechanical constraints are sufficient for the formulation to determine

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{\omega}_p \\ \dot{\omega}_s \\ \dot{s} \\ \ddot{s} \\ \dot{v}_b \end{bmatrix}. \quad (2.4.2)$$

This derivative gives the time rate of change of every state quantity. While the active constraints remain unchanged, advancing the state using these rates produces its value at the next instant, where the same evaluation is repeated. No other internal position or velocity variable must be retained.

The vector in [Equation \(2.4.1\)](#) describes only the modeled CVT. An engine, vehicle, dynamometer, or other attached system may carry additional states in its own model and use them when determining the torque and equivalent inertia presented at a shaft interface.

## 2.5 Mechanical Regimes and State Evolution

Consider the shift coordinate at one of its travel limits with  $\dot{s} = 0$ . The same position and velocity can describe two different mechanical conditions. If the surrounding forces press the movable sheave against the limit, a stop reaction is required and further motion into the stop is prevented. If the unconstrained tendency is away from the limit, the stop reaction vanishes and the sheave is free to accelerate away. The values stored in  $\mathbf{x}$  are identical in both cases, but the motion that is allowed is not.

The missing information is which contacts and mechanical constraints are currently active. Their active set is called the *mechanical regime*. For the modeled CVT, a regime identifies which belt–pulley contacts exist, whether the shift coordinate is free or supported at a unilateral boundary, and whether each existing tangential contact is sticking or sliding. Let  $q(t)$  denote the current regime. This label is not another position or velocity and does not introduce another moving part; it records the mechanical arrangement under which the state is evolving.

The complete internal condition needed to continue the model is therefore the pair  $(\mathbf{x}(t), q(t))$ . The state vector records the current motion and configuration, while the regime identifies the constraints that determine which motion is admissible. For the selected CVT and a fixed regime, the current state and shaft-interface conditions determine the state derivative. At this level, that relation may be written as

$$\dot{\mathbf{x}} = \mathcal{F}_q(t, \mathbf{x}, \tau_{B,p}, I_{B,p}, \tau_{B,s}, I_{B,s}), \quad (2.5.1)$$

The function  $\mathcal{F}_q$  represents the mechanics of the selected CVT under the current mechanical arrangement. Its subscript indicates that the active contacts and constraints select the applicable form of the mechanical balances. While that regime remains admissible, the state advances continuously according to [Equation \(2.5.1\)](#).

Every regime is defined by physical conditions that must remain satisfied. An event occurs when the evolving motion reaches the boundary of those conditions: a contact forms or separates, a travel limit is reached or released, or an existing tangential contact can no longer maintain its current stick–slip condition. At that instant, the old mechanical arrangement no longer provides the admissible continuation of the motion and the regime must change.

Not every regime change alters the state instantaneously. The simplest case is a nonimpulsive regime change: the active force or constraint description changes without producing an impulse, so the positions and velocities pass through the event unchanged,

$$\mathbf{x}^+ = \mathbf{x}^-, \quad (2.5.2)$$

where the superscripts  $(-)$  and  $(+)$  denote the values immediately before and after the event. The new regime changes the derivative obtained from [Equation \(2.5.1\)](#), even though the state itself is continuous.

A different transition is required when a rigid kinematic constraint becomes active with a nonzero velocity into that constraint. The configuration remains continuous, but one or more velocity components may change over the effectively instantaneous event. In general, the compatible post-event state may be written as

$$\mathbf{x}^+ = \Delta_{q^- \rightarrow q^+}(\mathbf{x}^-). \quad (2.5.3)$$

The transition law enforces the newly active kinematic constraint together with impulse–momentum balance for the retained inertia. It does not require each velocity or each coordinate momentum to remain unchanged. A constraint impulse can change the velocity components, and kinetic energy may decrease when the capture is inelastic. Once the transition is complete, continuous evolution resumes under the new regime.

The resulting motion has the repeated structure

$$\text{continuous evolution} \longrightarrow \text{event} \longrightarrow \text{constraint-consistent transition} \longrightarrow \text{new continuous evolution}. \quad (2.5.4)$$

Because the CVT combines continuous state evolution with discrete changes in its active mechanical arrangement, its motion forms a *hybrid initial-value problem*. A complete initial condition specifies both an initial state and an initial regime,

$$\mathbf{x}(t_0) = \mathbf{x}_0, \quad q(t_0) = q_0, \quad (2.5.5)$$

with  $\mathbf{x}_0$  satisfying the contacts and constraints represented by  $q_0$ . The regimes are therefore not separate CVT models. They are different admissible arrangements of the same pulley assemblies, belt, actuation mechanisms, and shaft interfaces. The state vector records how that mechanism is moving; the regime records which motion is presently allowed.

## 2.6 Model Quantities and Evaluation Flow

Advancing the CVT from one instant to the next requires more than the stored internal condition alone. The calculation also uses data that define the selected CVT and loading supplied through the two shaft interfaces, and it produces quantities that are needed only for the current evaluation. These quantities serve different roles in the model and should not all be treated as state.

**Table 2.1** separates the four roles used throughout the formulation. The fixed data represented by its first row are collected in ??, while the instantaneous quantities represented by its final row are listed by mechanical regime in ??. Fixed CVT data may be specified directly or calculated once from the selected hardware during initialization. In either case, they define the CVT being modeled and remain unchanged while that simulation is advanced. The time-varying internal condition is instead carried by the state and regime introduced in the preceding sections.

Table 2.1: Roles of the quantities used during a CVT model evaluation.

| Quantity role             | Contents                                                                                                          | Use during evaluation                                                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed CVT data            | Physical and geometric data for the selected hardware, including any values calculated once during initialization | Define the particular CVT and the mechanical relations used to represent it; they are not advanced through time.                                                        |
| Current CVT condition     | State vector $\mathbf{x}(t)$ and mechanical regime $q(t)$                                                         | Record the time-varying internal information retained from the preceding instant and identify the currently active constraints.                                         |
| Shaft-interface data      | Applied torque $\tau_{B,j}$ and equivalent inertia $I_{B,j}$ at each shaft                                        | Supply the current mechanical effect of the systems attached to the primary and secondary shafts.                                                                       |
| Instantaneous CVT results | State derivative, current internal mechanical quantities, and regime admissibility measures                       | Determine how the state can advance from the current instant and whether the active regime can continue; they are recomputed rather than retained as independent state. |

The pair  $(\mathbf{x}, q)$  is therefore the only time-varying internal CVT condition passed from one model evaluation to the next. The shaft-interface quantities are supplied for the current evaluation. An attached system may use time, shaft speed, and states belonging to its own model to determine those quantities, but those external states do not become states of the CVT.

Using these roles, one evaluation proceeds as follows:

1. The current time, state  $\mathbf{x}(t)$ , and regime  $q(t)$  identify the internal CVT condition at the start of the evaluation.
2. The systems attached to the primary and secondary shafts supply the torque and equivalent inertia presented at their respective interfaces under the current shaft conditions.
3. The CVT mechanics use the fixed CVT data, current internal condition, and shaft-interface data to determine the instantaneous mechanical solution. This solution provides  $\dot{\mathbf{x}}(t)$  together with the conditions used to test whether the current regime remains admissible.
4. If the regime remains admissible, the state is advanced using  $\dot{\mathbf{x}}$  and the regime is retained. If an event is reached, the event state is established, the applicable transition is applied when required, and the new regime is selected.
5. The resulting state and regime become the current internal condition for the next evaluation, and the cycle repeats.

# Chapter 3

## Derivation

**Chapter 2** established the CVT as a hybrid initial-value problem: at a given instant, the current state, mechanical regime, and shaft-interface loading are known, and the model must determine the corresponding instantaneous response. This chapter derives the mechanics required to perform that evaluation. The development begins with the most general continuous configuration, in which the belt is fully engaged and the CVT is free to shift, and then returns to the mechanical boundaries and regime changes that connect successive intervals of continuous motion.

**Section 3.1** first develops the pulley and belt-loop kinematics, reducing the motion of both sheaves and both pulley radii to the shared shift coordinate. Using that kinematic structure, **Section 3.2** derives the axial actuation and movable-sheave dynamics, while **Section 3.3** develops the corresponding rotational dynamics of the two pulley assemblies. **Section 3.4** then closes the mechanics of the belt itself, relating its transport motion, tension, pulley loading, and transmitted torque. These pieces are assembled in **Section 3.5**, where the admissible belt–pulley contact state completes the instantaneous dynamics of the freely shifting engaged CVT. Finally, **Section 3.6** extends this result to the remaining mechanical regimes, showing how the continuous equations reduce when additional constraints become active and how the state is carried consistently through transitions between them.

### 3.1 Pulley Geometry

The conceptual overview introduced the CVT as two variable-radius pulleys connected by a single belt. Moving the pulley sheaves changes where the belt sits on each pulley and therefore changes the transmission ratio.

Both pulleys contain a movable sheave. Let  $x_p$  denote the axial position of the movable primary sheave and  $x_s$  the axial position of the movable secondary sheave. The geometric problem is therefore twofold: how does the axial position of each movable sheave set the belt radius on its own pulley, and how does the shared belt constrain the two pulley positions to move together?

The first question can be answered from the local geometry of each pulley. The second requires following the belt around the complete loop. Once these position relationships are known, they can be differentiated to obtain the corresponding radius rates and accelerations.

The natural place to begin is the geometry of a single pulley.

### 3.1.1 Local Pulley Geometry

The first part of the geometric question can be answered on a single pulley: how does the axial position of its movable sheave determine the pulley radius? The same conical geometry appears on both pulleys, so the relationship can be derived once and then applied to the primary and secondary.

The axial direction and primary-sheave convention were introduced in [Section 2.3.2](#) and illustrated in [Figure 2.3](#). The local coordinates  $x_p$  and  $x_s$  follow the same convention on their respective pulleys.

For either pulley  $j \in \{p, s\}$ ,  $x_j = 0$  denotes the fully open configuration. Increasing  $x_j$  moves the movable sheave toward the fixed sheave and closes the pulley, while decreasing  $x_j$  opens it. The corresponding velocity follows the same sign convention:  $\dot{x}_j > 0$  denotes closing motion and  $\dot{x}_j < 0$  denotes opening motion.

The primary therefore begins at  $x_p = 0$  and moves in the positive direction as it closes. The secondary begins in a closed position, with  $x_s > 0$ , and moves toward smaller values of  $x_s$  as it opens.

Consider a belt seated between two straight sheave faces with half-angle  $\beta$ , as shown in [Figure 3.1](#). Let  $b_{\text{in}}$  denote the width of the belt at its inner surface. As assumed in [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#), this width remains constant.

Suppose the movable sheave closes the pulley by a distance  $\Delta x_j$ . At the original belt radius, the available space between the sheaves is therefore reduced from  $b_{\text{in}}$  to

$$b_{\text{in}} - \Delta x_j.$$

The belt responds by moving outward through a radial distance  $\Delta r_j$ . At its new position, the angled sheave faces provide an additional axial distance  $d$  on each side.

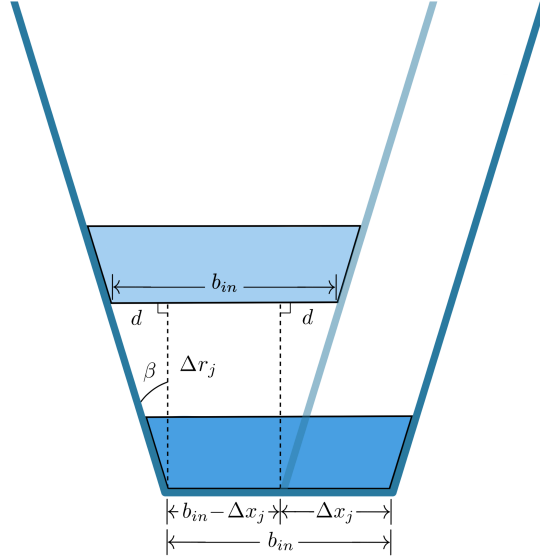


Figure 3.1: Local geometry of a belt moving outward between the sheave faces of pulley  $j$ . Closing the pulley by  $\Delta x$  reduces the available width at the original belt position to  $b_{\text{in}} - \Delta x$ . Moving outward by  $\Delta r$  provides an additional axial distance  $d$  at each sheave face.

At the new radius, the total space between the sheave faces must again equal the fixed belt width. Reading directly from [Figure 3.1](#),

$$b_{\text{in}} = 2d + (b_{\text{in}} - \Delta x_j). \quad (3.1.1)$$

The right triangle beside either sheave face relates  $d$  to the radial movement. For the sheave half-angle  $\beta$ ,

$$\tan \beta = \frac{d}{\Delta r_j}, \quad (3.1.2)$$

and therefore

$$d = \Delta r_j \tan \beta. \quad (3.1.3)$$

Substituting [Equation \(3.1.3\)](#) into [Equation \(3.1.1\)](#) gives

$$b_{\text{in}} = 2\Delta r_j \tan \beta + (b_{\text{in}} - \Delta x_j). \quad (3.1.4)$$

The constant belt-width terms cancel, leaving

$$\Delta r_j = \frac{\Delta x_j}{2 \tan \beta}. \quad (3.1.5)$$

This is the basic axial-to-radial relationship of the pulley. Closing the pulley moves the belt outward, while opening it moves the belt inward. A smaller sheave angle produces more radial movement for the same axial motion, whereas a larger sheave angle produces less.

Taking the infinitesimal limit of Equation (3.1.5) gives

$$\frac{dr_j}{dx_j} = \frac{1}{2 \tan \beta}. \quad (3.1.6)$$

Because the sheave faces are straight, this slope is constant.

**Radius convention** The derivation above follows the inner surface of the belt, where  $b_{\text{in}}$  is defined. The belt also has the finite radial height  $h_b$ , so its inner and outer surfaces occupy different radii. Let  $r_{j,\text{inner}}$  and  $r_{j,\text{outer}}$  denote those radii. They are related by

$$r_{j,\text{outer}} = r_{j,\text{inner}} + h_b. \quad (3.1.7)$$

The fixed-cross-section assumption in Assumption 4 (Fixed, Uniform Belt Cross-Section) makes  $h_b$  constant, so the axial-to-radial displacement derived above can be read from either surface. An absolute pulley radius must nevertheless identify which surface is being measured. The outer surface is used here because it provides a convenient and directly measurable reference. Throughout the geometric development,

$$r_j \equiv r_{j,\text{outer}}. \quad (3.1.8)$$

Using the outer surface gives a clear measure of where the belt is seated. It does not mean that every physical property of the belt acts at that radius.

The belt tension is carried along the cords embedded within the belt, so it acts through the centre of the cord layer. The distance from the pulley shaft to this line is therefore the lever arm through which belt tension transmits torque, making it the effective radius for gearing. This cord-line radius is commonly called the pitch radius (Gerbert, 1996).

Let  $d_{\text{cord}}$  denote the radial depth of the cord-layer centre from the outer surface, with  $0 < d_{\text{cord}} < h_b$ . Because the cord line lies this far inside the outer surface, its effective radius is

$$r_{j,\text{eff}} = r_{j,\text{outer}} - d_{\text{cord}}. \quad (3.1.9)$$

The belt mass is spread throughout the whole cross-section rather than concentrated at the outer surface or the cord line. To describe where this mass is with one radius, we use its centre of mass and denote the radius to that point by  $r_{j,\text{cm}}$ . Under the uniform-density approximation, the centre of mass is the geometric centroid of the trapezoid.

Let  $y$  measure radially inward from the outer surface, with  $y = 0$  at the outer surface and  $y = h_b$  at the inner surface, and let  $y_{\text{cm}}$  denote the depth of the centre of mass. These two depths and their corresponding radii are shown in Figure 3.2.

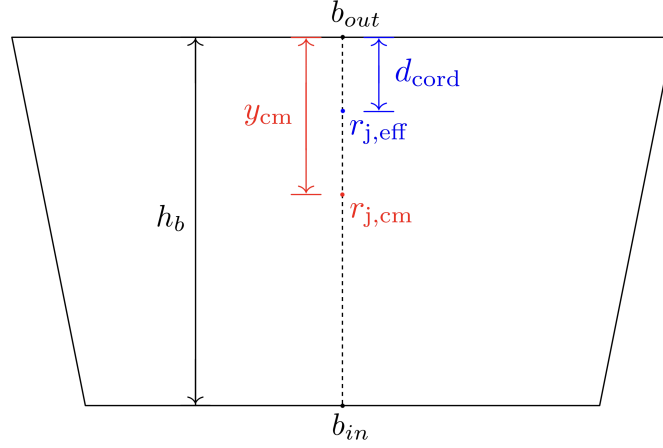


Figure 3.2: Radial reference lines within the belt cross-section. The cord-line radius  $r_{j,\text{eff}}$  is the lever arm through which belt tension transmits torque, while  $r_{j,\text{cm}}$  locates the belt's centre of mass. The depths  $d_{\text{cord}}$  and  $y_{\text{cm}}$  are measured radially inward from the outer surface.

If  $b_{\text{out}}$  and  $b_{\text{in}}$  are the outer and inner widths, respectively, the straight sides of the trapezoid give

$$b(y) = b_{\text{out}} - \frac{b_{\text{out}} - b_{\text{in}}}{h_b} y. \quad (3.1.10)$$

Because the density is uniform, a thin strip at depth  $y$  contains mass in proportion to its area  $b(y) dy$ . The centroid depth is therefore the weighted average of  $y$  over the cross-section:

$$y_{\text{cm}} = \frac{\int_0^{h_b} y b(y) dy}{\int_0^{h_b} b(y) dy}. \quad (3.1.11)$$

The denominator is the belt cross-sectional area,

$$A_b = \int_0^{h_b} b(y) dy = \frac{h_b}{2} (b_{\text{out}} + b_{\text{in}}). \quad (3.1.12)$$

Using this area in the weighted average gives

$$\begin{aligned} y_{\text{cm}} &= \frac{1}{A_b} \int_0^{h_b} y b(y) dy \\ &= h_b \frac{b_{\text{out}} + 2b_{\text{in}}}{3(b_{\text{out}} + b_{\text{in}})}. \end{aligned} \quad (3.1.13)$$

The result has the expected limiting behaviour. If  $b_{\text{out}} = b_{\text{in}}$ , the cross-section is rectangular and  $y_{\text{cm}} = h_b/2$ . For the usual trapezoid with  $b_{\text{out}} > b_{\text{in}}$ , the additional area near the outer surface

places the centroid slightly closer to that surface, so  $y_{\text{cm}} < h_b/2$ . The belt mass therefore follows the radius

$$r_{j,\text{cm}} = r_{j,\text{outer}} - y_{\text{cm}}. \quad (3.1.14)$$

The outer-surface radius map still requires one known configuration to establish its absolute placement. Let  $x_{j,\text{ref}}$  denote a known axial position at which the belt is seated, and let  $r_{j,\text{ref}}$  denote the outer radius measured at that same position. The outer radius at any other seated position is then

$$r_j(x_j) = r_{j,\text{ref}} + \frac{x_j - x_{j,\text{ref}}}{2 \tan \beta}. \quad (3.1.15)$$

The pair  $(x_{j,\text{ref}}, r_{j,\text{ref}})$  simply anchors the linear relationship to a known physical configuration.

For the secondary pulley, a convenient reference is its known closed starting position. The secondary begins with the belt seated at a large radius and opens as the CVT shifts. Since opening corresponds to decreasing  $x_s$ , [Equation \(3.1.15\)](#) produces the corresponding decrease in  $r_s$ .

The primary introduces an important geometric asymmetry: it can open farther than the width of the belt. At  $x_p = 0$ , the distance between the primary sheaves can therefore exceed  $b_{\text{in}}$ . Initial positive motion of the movable primary sheave only removes this clearance, leaving the belt at the same radius. The primary radius begins to increase only once the movable sheave reaches the belt.

This interval is the *primary deadzone*. Let  $x_{p,\text{dz}}$  denote the primary position at which the movable sheave first reaches the belt. Within the deadzone,

$$0 \leq x_p < x_{p,\text{dz}},$$

the primary coordinate changes, but the belt remains at its minimum primary radius. At  $x_p = x_{p,\text{dz}}$ , the clearance has been removed and the belt becomes seated between the sheave faces. Further closure then moves the belt outward according to the local geometry derived above.

The first-contact position provides the natural reference for the seated primary branch:

$$x_{p,\text{ref}} = x_{p,\text{dz}}, \quad r_{p,\text{ref}} = r_{p,\text{min}}.$$

The complete primary radius map is therefore

$$r_p(x_p) = \begin{cases} r_{p,\text{min}}, & 0 \leq x_p < x_{p,\text{dz}}, \\ r_{p,\text{min}} + \frac{x_p - x_{p,\text{dz}}}{2 \tan \beta}, & x_{p,\text{dz}} \leq x_p \leq x_{p,\text{max}}. \end{cases} \quad (3.1.16)$$

The linear axial-to-radial relationship in [Equation \(3.1.15\)](#) is standard in reduced CVT geometry models ([Duan et al., 2016](#); [Ballew, 2015](#)). It applies to both pulleys whenever the belt is seated, while [Equation \(3.1.16\)](#) extends the primary description through the deadzone before contact.

The local geometry now tells us the outer-surface radius associated with either movable-sheave position. The two local descriptions belong to the same closed belt, so only certain primary and secondary positions are geometrically compatible.

### 3.1.2 Complete Belt-Loop Geometry

The local pulley geometry describes how axial motion changes the belt's outer-surface radius on each pulley. The belt, however, passes around both pulleys. To connect those radii, the belt must now be followed around the complete loop.

Under **Assumption 8 (Prescribed Planar Belt Path)**, the modeled belt path consists of a circular wrap on each pulley joined by two straight tangent spans. The aligned, fixed-center geometry of **Assumption 5 (Aligned Pulley Axes and Fixed Center Distance)** makes the two spans equal. If each straight span has length  $l$ , and the primary and secondary wrap angles are  $\phi_p$  and  $\phi_s$ , the total geometric length of the loop is

$$L_{\text{geo}} = r_p \phi_p + r_s \phi_s + 2l. \quad (3.1.17)$$

The quantities  $l$ ,  $\phi_p$ , and  $\phi_s$  follow directly from the geometry shown in **Figure 3.3**.

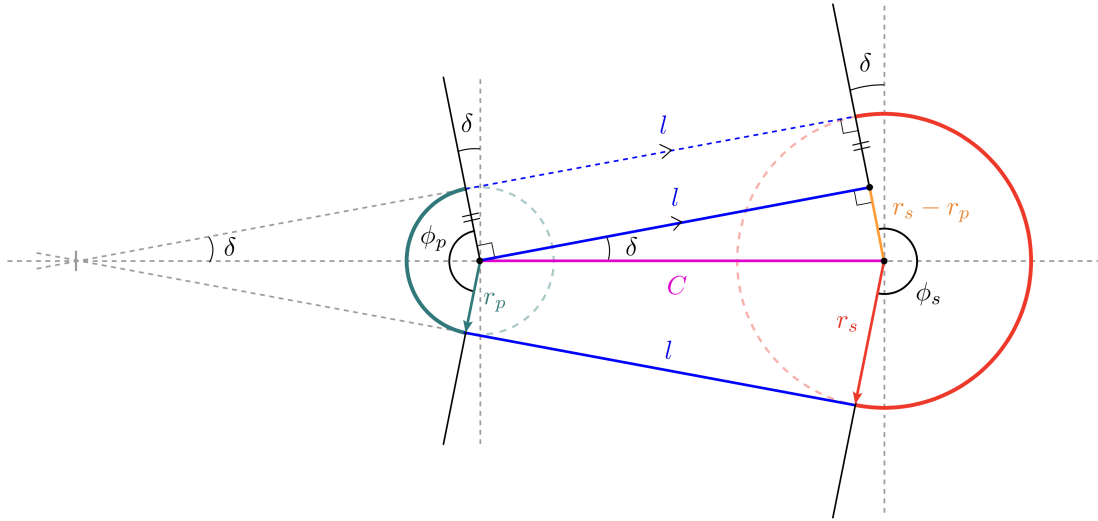


Figure 3.3: Geometry of the complete belt loop. The belt path consists of two wrapped arcs and two equal straight spans of length  $l$ . Unequal pulley radii incline the straight spans by the angle  $\delta$ , producing the primary and secondary wrap angles  $\phi_p$  and  $\phi_s$ .

Consider the right triangle in the upper span in **Figure 3.3**. The line joining the pulley centres forms its hypotenuse and has length  $C$ . One side lies along the straight belt span and therefore has length  $l$ .

The other side is perpendicular to the span. The primary and secondary radii drawn to their respective contact points are also perpendicular to this span and are therefore parallel. As shown

by the orange construction in [Figure 3.3](#), the difference between their lengths gives the remaining side of the triangle,  $r_s - r_p$ . The triangle therefore satisfies

$$l^2 + (r_s - r_p)^2 = C^2, \quad (3.1.18)$$

and the length of either straight span is

$$l = \sqrt{C^2 - (r_s - r_p)^2}. \quad (3.1.19)$$

This expression requires

$$|r_s - r_p| < C. \quad (3.1.20)$$

This condition is already enforced by the physical separation of the pulleys. The pulley circles must remain apart, requiring  $C > r_s + r_p$ , and  $r_s + r_p$  is necessarily greater than  $|r_s - r_p|$ .

The straight-span length accounts for only two parts of the belt path. The two wrapped portions still depend on how the straight spans meet the pulleys. If the pulley radii are equal, the spans are parallel to the line joining the pulley centres and the belt wraps halfway around each pulley. Both wrap angles are then equal to  $\pi$ .

When the radii differ, the straight spans rotate through the angle  $\delta$  shown in [Figure 3.3](#). The same right triangle gives

$$\sin \delta = \frac{r_s - r_p}{C}, \quad (3.1.21)$$

and therefore

$$\delta = \sin^{-1} \left( \frac{r_s - r_p}{C} \right). \quad (3.1.22)$$

The angle  $\delta$  is signed. For the configuration shown,  $r_s > r_p$ , so  $\delta > 0$ . The primary wrap loses an angle  $\delta$  at each end of its contact arc, while the secondary wrap gains the same amount at each end. Their wrap angles are therefore

$$\phi_p = \pi - 2\delta, \quad \phi_s = \pi + 2\delta. \quad (3.1.23)$$

If  $r_p > r_s$ , then  $\delta < 0$ , and the same equations automatically reverse which pulley has the larger wrap angle.

The complete belt length follows by adding the two wrapped arcs and the two straight spans. Substituting [Equations \(3.1.19\), \(3.1.22\) and \(3.1.23\)](#) into [Equation \(3.1.17\)](#) gives

**Exact geometric belt length****(3.1.24)**

$$\begin{aligned}
L_{\text{geo}}(r_p, r_s, C) = & \pi (r_p + r_s) \\
& + 2 (r_s - r_p) \sin^{-1} \left( \frac{r_s - r_p}{C} \right) \\
& + 2 \sqrt{C^2 - (r_s - r_p)^2}
\end{aligned}$$

Each term still corresponds directly to part of the belt path. The first is the combined half-circle contribution from the two pulley wraps. The second accounts for the unequal wrap angles, and the third is the combined length of the two straight spans.

For any chosen values of  $r_p$ ,  $r_s$ , and  $C$ , [Equation \(3.1.24\)](#) tells us how much belt is required to follow that path. The installed belt cannot change its total length as the pulleys shift. Under [Assumption 6 \(Fixed Belt Length\)](#), every possible pulley position must therefore satisfy

$$L_{\text{geo}}(r_p, r_s, C) = L_b, \quad (3.1.25)$$

where  $L_b$  is the fixed length of the same outer-surface path described by  $r_p$  and  $r_s$ , consistent with the convention in [Equation \(3.1.8\)](#). This is the geometric compatibility condition for the belt loop. Once any two of  $r_p$ ,  $r_s$ , and  $C$  are known, the condition determines the third.

For a particular CVT, the first quantity to determine is the centre distance  $C$ . The pulley radii change as the sheaves move, but the pulley shafts do not move with them. Once the CVT is assembled,  $C$  remains fixed throughout operation.

The fixed centre distance can be found from any installed pulley position that can be measured. At that position, the belt length and the two pulley radii are known. Denoting the measured radii by  $r_{p,\text{ref}}$  and  $r_{s,\text{ref}}$ , the only unknown in [Equation \(3.1.25\)](#) is  $C$ .

In the general unequal-radius case,  $C$  appears both inside the square root and inside the inverse sine. The exact relation therefore cannot be rearranged into a useful explicit expression for  $C$ . Instead, the centre distance is found with a one-dimensional root solve.

For a trial centre distance, the difference between the belt length required by the geometry and the belt length actually available is

$$\mathcal{R}_L(r_p, r_s, C) = L_{\text{geo}}(r_p, r_s, C) - L_b. \quad (3.1.26)$$

A positive residual means that the trial geometry requires more belt than is available. A negative residual means that it requires less. The correct centre distance is the value for which the required and available lengths match:

$$\mathcal{R}_L(r_{p,\text{ref}}, r_{s,\text{ref}}, C) = 0. \quad (3.1.27)$$

The root solver varies  $C$  until this condition is satisfied.

### Remark (Simplified Belt-Length Relation)

The exact root solve is required because changing pulley radii alter both the straight-span lengths and the wrap angles. When the radius difference is small relative to the centre distance,

$$\frac{|r_s - r_p|}{C} \ll 1,$$

the straight-span angle is also small:

$$\delta = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

A common simplification is to neglect the resulting wrap-angle correction while retaining the exact straight-span length. The compatibility relation then becomes

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}.$$

Unlike the exact relation, this approximation can be rearranged directly. For example, the centre distance is

$$C \approx \sqrt{(r_s - r_p)^2 + \frac{1}{4}[L_b - \pi(r_p + r_s)]^2}.$$

The present model retains the exact relation. The effect of using the simplified form is examined in [Appendix A](#).

Once the initial root solve has determined  $C$ , both the centre distance and the belt length are known constants. The compatibility condition can then be used to connect the two pulley radii at every subsequent position.

Suppose the primary radius  $r_p$  is known. Substituting  $r_p$ ,  $C$ , and  $L_b$  into [Equation \(3.1.26\)](#) leaves the secondary radius as the only unknown. Its compatible value is found from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \quad \text{solved for } r_s. \quad (3.1.28)$$

The resulting  $r_s$  is the secondary radius required for the same fixed belt to pass around both pulleys. The calculation also works in the opposite direction. If  $r_s$  is known instead, the compatible primary radius follows from

$$\mathcal{R}_L(r_p, r_s, C) = 0 \quad \text{solved for } r_p. \quad (3.1.29)$$

Thus, once  $C$  has been established, either pulley radius can be used to determine the other. The two radii remain local geometric quantities, but they cannot vary independently while belonging to the same fixed-length belt loop.

The arc-and-tangent construction used above is standard in belt-drive geometry. [Budynas and Nisbett \(2015\)](#) presents the same underlying construction for relating pulley radii, straight spans, wrap angles, and total belt length. The exact geometry is retained here because it provides the compatibility relation between the two shifting pulleys.

The local relations  $r_p(x_p)$  and  $r_s(x_s)$  now describe how each pulley produces its radius, while [Equation \(3.1.25\)](#) describes which pairs of radii can belong to the same belt loop. Combining these results gives the relationship between the two local axial coordinates.

### 3.1.3 From Local Pulley Coordinates to the Shift Coordinate

The preceding geometry gives a local relation for each pulley:  $x_p$  determines  $r_p(x_p)$ , and  $x_s$  determines  $r_s(x_s)$ . A complete CVT configuration, however, cannot be formed by choosing these positions independently. Both radii must belong to the same belt loop of length  $L_b$ . The remaining problem is therefore to identify the pairs  $(x_p, x_s)$  that satisfy this shared geometry.

For any proposed pair, the local pulley relations first give the two radii. Substituting those radii into the belt-length residual gives

$$\mathcal{R}_x(x_p, x_s) \equiv \mathcal{R}_L(r_p(x_p), r_s(x_s), C) = 0. \quad (3.1.30)$$

Every pair that makes [Equation \(3.1.30\)](#) equal to zero is a possible position of the two pulleys. Together, these pairs form the compatibility curve.

To reveal the shape of this curve, either local coordinate could be chosen as a starting point. Take  $x_p$  simply as an arbitrary choice. For each value of  $x_p$ , the primary relation gives  $r_p(x_p)$ . The shaft spacing fixes  $C$ , and  $L_b$  remains fixed by [Assumption 6 \(Fixed Belt Length\)](#), so the belt-length residual can then be solved for the required secondary radius  $r_s$ . The local secondary relation finally gives the corresponding position  $x_s$ :

$$x_p \longrightarrow r_p(x_p) \longrightarrow r_s \longrightarrow x_s.$$

Repeating these steps across the primary travel produces the representative compatibility curve in [Figure 3.4](#).

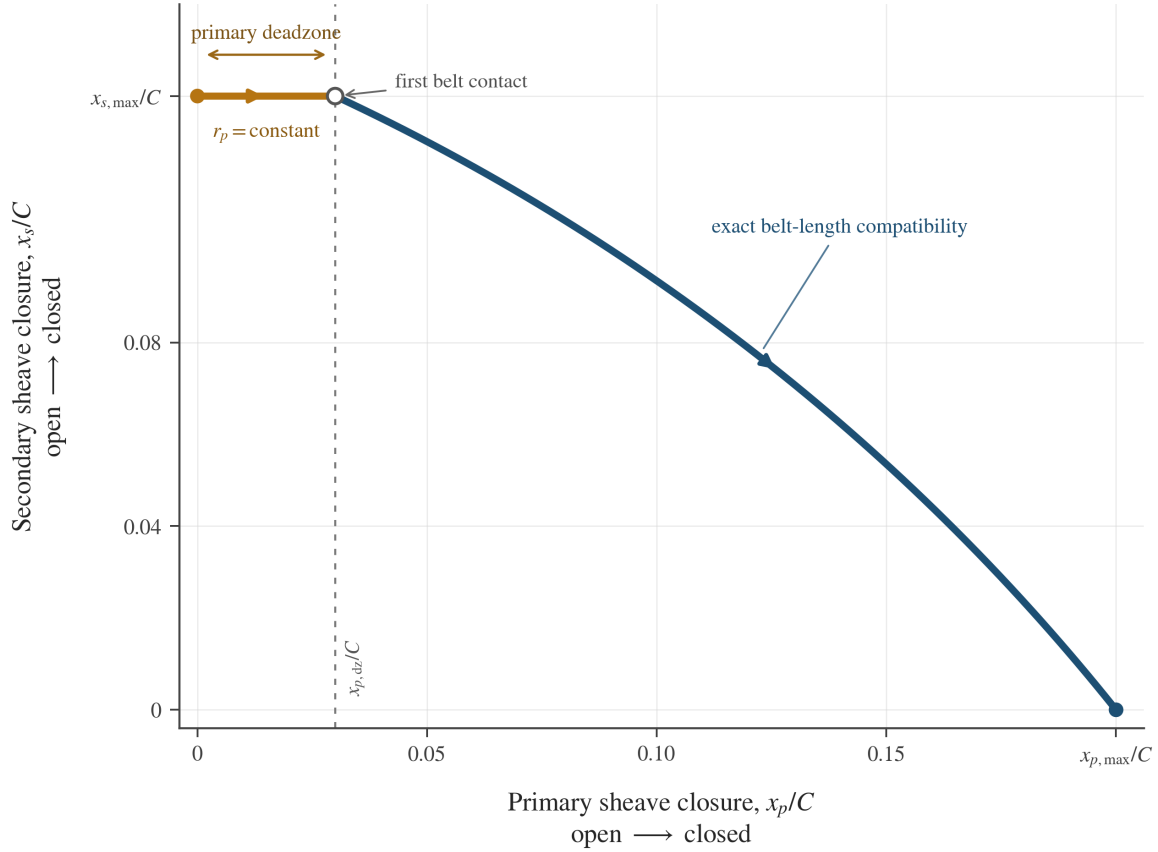


Figure 3.4: Representative compatibility curve between the local primary and secondary axial coordinates for an admissible belt–pulley geometry. Both coordinates are normalized by the same centre distance  $C$ , so their relative travel and the changing slope of the engaged branch are shown on a common scale. The dimensions are illustrative and do not represent a particular CVT.

The horizontal portion is created by the primary deadzone. While  $0 \leq x_p < x_{p,dz}$ , the primary sheave closes only the clearance to the belt. The primary radius remains fixed, so the compatible secondary radius and position remain fixed as well.

Once the primary reaches the belt, further closure increases  $r_p$ . The fixed belt loop then requires  $r_s$  to decrease, which means that the secondary opens and  $x_s$  decreases. The two sheaves do not move by equal amounts, and the resulting branch is not linear. Changing the radii alters both wrap lengths and both straight spans, so the compatible secondary displacement changes throughout the shift.

The curve also reveals which local coordinate can represent the complete geometry. A vertical line at any  $x_p$  meets the curve once, giving one compatible  $x_s$ . At the deadzone height, however, a horizontal line meets every  $x_p$  between 0 and  $x_{p,dz}$ . Thus  $x_s$  is a unique function of  $x_p$  over the full travel, whereas  $x_p$  is not a unique function of  $x_s$ .

The primary coordinate is therefore adopted as the system shift coordinate. This is the same  $s$  included in the state vector of ??:

**System shift coordinate**

**(3.1.31)**

$$s \equiv x_p, \quad x_p(s) = s, \quad x_s = x_s(s)$$

The secondary mapping is then

$$x_s(s) = \begin{cases} x_{s,\max}, & 0 \leq s < x_{p,\text{dz}}, \\ \text{root}_{0 \leq x_s \leq x_{s,\max}} [\mathcal{R}_x(s, x_s) = 0], & x_{p,\text{dz}} \leq s \leq x_{p,\max}. \end{cases} \quad (3.1.32)$$

The first branch holds the secondary fixed while the primary closes its clearance; the second returns the unique compatible position after belt contact. Both give  $x_{s,\max}$  at  $s = x_{p,\text{dz}}$ , so the mapping is continuous. The shift coordinate therefore determines both movable-sheave positions and, through the local pulley relations, both pulley radii. The position geometry is now complete. What remains is to determine how these positions and radii move when  $s$  changes with time.

### 3.1.4 Pulley-Radius Rates

The shift coordinate now determines the positions of both movable sheaves and the corresponding pulley radii. Its rate form asks the same geometric question over a small motion: if the primary closes by an amount  $ds$ , how far must the secondary open, and how quickly do the two pulley radii change?

We first find how each local coordinate and pulley radius changes per unit  $s$ . Throughout this subsection, a prime denotes differentiation with respect to  $s$ . For either pulley  $j \in \{p, s\}$ , the corresponding time rates are then

$$\dot{x}_j = x'_j(s)\dot{s}, \quad \dot{r}_j = r'_j(s)\dot{s}. \quad (3.1.33)$$

**Primary radius rate** The primary side follows immediately from the definition  $x_p(s) = s$ :

$$x'_p(s) = 1, \quad \dot{x}_p = \dot{s}. \quad (3.1.34)$$

The corresponding radius map is constant through the primary deadzone and linear after the movable sheave reaches the belt. Differentiating [Equation \(3.1.16\)](#) therefore gives

$$r'_p(s) = \begin{cases} 0, & 0 \leq s < x_{p,\text{dz}}, \\ \frac{1}{2 \tan \beta}, & x_{p,\text{dz}} < s \leq x_{p,\max}. \end{cases} \quad (3.1.35)$$

Multiplying this slope by  $\dot{s}$  gives

### Primary Radius Rate

(3.1.36)

$$\dot{r}_p = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ \frac{\dot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

Thus, the primary sheave can move while crossing the deadzone even though the primary belt radius does not. After contact, the sheave angle fixes the radial change produced by each unit of primary closure.

**Secondary coordinate and radius rates** The secondary requires one additional step. Its local pulley relation tells us how  $r_s$  changes when  $x_s$  changes, but it does not tell us how far the secondary must move. That motion is set by the fixed belt loop.

At every compatible position,

$$L_{\text{geo}}(r_p(s), r_s(s), C) = L_b. \quad (3.1.37)$$

The centre distance  $C$  is fixed by the pulley shafts, while  $L_b$  is constant under **Assumption 6 (Fixed Belt Length)**. A small change in  $s$  therefore cannot change the geometric length of the belt path:

$$\frac{dL_{\text{geo}}}{ds} = 0. \quad (3.1.38)$$

Both radii change during this motion. The change caused by  $r_p$  is the sensitivity of the belt length to  $r_p$ , multiplied by the actual rate  $r'_p$ . The secondary contributes in the same way. Applying the chain rule to **Equation (3.1.37)** gives

$$\frac{\partial L_{\text{geo}}}{\partial r_p} r'_p + \frac{\partial L_{\text{geo}}}{\partial r_s} r'_s = 0. \quad (3.1.39)$$

The two partial derivatives have a direct purpose: each tells us how much additional belt path would be required if one pulley radius increased while the other radius remained fixed. The actual pulley motion must make these two changes cancel. We therefore calculate the two sensitivities next.

The span angle and straight-span length entering the belt-length relation were derived in **Equations (3.1.19)** and **(3.1.22)**. Writing them together in the form needed for differentiation,

$$\delta = \sin^{-1} \left( \frac{r_s - r_p}{C} \right), \quad l = \sqrt{C^2 - (r_s - r_p)^2}, \quad (3.1.40)$$

Here,  $\delta$  sets the wrap angles through **Equation (3.1.23)**, while  $l$  is the length of either straight span. Using these quantities, the exact path length from **Equation (3.1.24)** is

$$L_{\text{geo}} = \pi(r_p + r_s) + 2(r_s - r_p)\delta + 2l. \quad (3.1.41)$$

First, hold  $r_s$  and  $C$  fixed and increase  $r_p$ . The radius difference  $r_s - r_p$  then decreases at the same rate:

$$\frac{\partial(r_s - r_p)}{\partial r_p} = -1. \quad (3.1.42)$$

Using this result in the definitions of  $\delta$  and  $l$  gives

$$\begin{aligned} \frac{\partial \delta}{\partial r_p} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(-\frac{1}{C}\right) \\ &= -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}} = -\frac{1}{l}, \end{aligned} \quad (3.1.43)$$

$$\begin{aligned} \frac{\partial l}{\partial r_p} &= \frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}} [2(r_s - r_p)] \\ &= \frac{r_s - r_p}{l}. \end{aligned} \quad (3.1.44)$$

We can now differentiate the complete path length in [Equation \(3.1.41\)](#). The first term differentiates directly, the wrap correction requires the product rule, and the final term contains the changing straight spans:

$$\begin{aligned} \frac{\partial L_{\text{geo}}}{\partial r_p} &= \pi + 2 \left[ (-1)\delta + (r_s - r_p) \left(-\frac{1}{l}\right) \right] + 2 \left( \frac{r_s - r_p}{l} \right) \\ &= \pi - 2\delta \\ &= \phi_p. \end{aligned} \quad (3.1.45)$$

The two terms containing  $(r_s - r_p)/l$  cancel. The remaining sensitivity is the primary wrap angle. The secondary calculation follows the same path, but increasing  $r_s$  increases the radius difference:

$$\frac{\partial(r_s - r_p)}{\partial r_s} = 1. \quad (3.1.46)$$

The corresponding derivatives of  $\delta$  and  $l$  are

$$\begin{aligned}\frac{\partial \delta}{\partial r_s} &= \frac{1}{\sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2}} \left(\frac{1}{C}\right) \\ &= \frac{1}{l},\end{aligned}\tag{3.1.47}$$

$$\begin{aligned}\frac{\partial l}{\partial r_s} &= \frac{1}{2\sqrt{C^2 - (r_s - r_p)^2}} [-2(r_s - r_p)] \\ &= -\frac{r_s - r_p}{l}.\end{aligned}\tag{3.1.48}$$

Differentiating the full path length with these signs gives

$$\begin{aligned}\frac{\partial L_{\text{geo}}}{\partial r_s} &= \pi + 2 \left[ (1)\delta + (r_s - r_p) \left(\frac{1}{l}\right) \right] + 2 \left( -\frac{r_s - r_p}{l} \right) \\ &= \pi + 2\delta \\ &= \phi_s.\end{aligned}\tag{3.1.49}$$

Again, the terms involving the changing straight-span length cancel. The sensitivity to the secondary radius is the secondary wrap angle. Together,

$$\frac{\partial L_{\text{geo}}}{\partial r_p} = \phi_p, \quad \frac{\partial L_{\text{geo}}}{\partial r_s} = \phi_s.\tag{3.1.50}$$

This result also has a simple geometric interpretation. If one pulley radius increases by a small amount while the other is held fixed, the required belt path increases by that radial change multiplied by the angle over which the belt wraps the pulley. The remaining changes in the tangent geometry cancel one another.

Returning to the fixed-length condition [Equation \(3.1.39\)](#), the two radial changes must therefore satisfy

$$\phi_p r'_p + \phi_s r'_s = 0.\tag{3.1.51}$$

After primary contact, the local pulley relation gives the same axial-to-radial slope on both pulleys. Since  $x'_p = 1$ ,

$$r'_p = \frac{1}{2 \tan \beta}, \quad r'_s = \frac{1}{2 \tan \beta} x'_s.\tag{3.1.52}$$

Substituting these two slopes into [Equation \(3.1.51\)](#) gives

$$\phi_p \left( \frac{1}{2 \tan \beta} \right) + \phi_s \left( \frac{1}{2 \tan \beta} x'_s \right) = 0.\tag{3.1.53}$$

The common local pulley slope cancels, leaving

$$\phi_p + \phi_s x'_s = 0. \quad (3.1.54)$$

Solving for the slope of the secondary mapping gives

|                                                                                                                           |                 |
|---------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Secondary Coordinate Slope</b>                                                                                         | <b>(3.1.55)</b> |
| $x'_s(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |                 |

Within the deadzone, the primary radius does not change, so neither the compatible secondary radius nor  $x_s$  changes. After contact,  $x'_s < 0$ : primary closure requires the secondary to open. Its magnitude is not generally one and changes with the two wrap angles, which is why the engaged branch in [Figure 3.4](#) is curved rather than straight.

The time rate of the secondary coordinate is now obtained by multiplying this slope by  $\dot{s}$ :

$$\dot{x}_s = x'_s(s)\dot{s} = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}\dot{s}, & x_{p,dz} < s \leq x_{p,max}. \end{cases} \quad (3.1.56)$$

Applying the local axial-to-radial slope once more gives the secondary radius rate:

|                                                                                                                                                          |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Secondary Radius Rate</b>                                                                                                                             | <b>(3.1.57)</b> |
| $\dot{r}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s} \frac{\dot{s}}{2 \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |                 |

The rates above were written for  $r_p$  and  $r_s$ , which denote the outer-surface radii. The inner surface, cord line, and mass centre remain at fixed depths within the belt cross-section. When the belt moves radially on either pulley, all four reference radii therefore change by exactly the same amount. For either pulley  $j \in \{p, s\}$ ,

$$r'_j = r'_{j,inner} = r'_{j,eff} = r'_{j,cm}, \quad \dot{r}_j = \dot{r}_{j,inner} = \dot{r}_{j,eff} = \dot{r}_{j,cm}. \quad (3.1.58)$$

The choice of reference changes the absolute radius, but not its rate. The two pulley-radius rates are now determined by  $s$  and  $\dot{s}$ . However, the secondary multiplier  $-\phi_p/\phi_s$  changes as the radii and wrap angles change. Even if  $\dot{s}$  is constant, the secondary rate is not. Its acceleration must therefore include the changing slope of the compatibility curve.

### 3.1.5 Pulley-Radius Accelerations

The pulley-radius rates in [Equations \(3.1.36\)](#) and [\(3.1.57\)](#) were obtained by multiplying the slope of each geometric mapping by the shift velocity  $\dot{s}$ . Their accelerations follow by asking how those rates change. On the primary, the mapping slope is constant within each branch. On the secondary, the slope in [Equation \(3.1.55\)](#) changes as the pulley radii and wrap angles change. The secondary acceleration must therefore account for both the imposed shift acceleration and the changing geometric slope.

**Primary radius acceleration** The primary coordinate requires no additional geometry. The system-coordinate definition in [Equation \(3.1.31\)](#) gives  $x_p(s) = s$ , so

$$x'_p(s) = 1, \quad x''_p(s) = 0, \quad (3.1.59)$$

and therefore

$$\dot{x}_p = \dot{s}, \quad \ddot{x}_p = \ddot{s}. \quad (3.1.60)$$

The general shift-to-time relation in [Equation \(3.1.33\)](#) gives the primary radius rate in the form

$$\dot{r}_p = r'_p(s)\dot{s}. \quad (3.1.61)$$

Differentiating this product with respect to time gives

$$\begin{aligned} \ddot{r}_p &= \frac{d}{dt} [r'_p(s)\dot{s}] \\ &= \frac{dr'_p}{dt}\dot{s} + r'_p(s)\ddot{s}. \end{aligned} \quad (3.1.62)$$

Because  $r'_p$  is itself a function of  $s$ ,

$$\frac{dr'_p}{dt} = r''_p(s)\dot{s}. \quad (3.1.63)$$

Substituting [Equation \(3.1.63\)](#) into [Equation \(3.1.62\)](#) gives

$$\ddot{r}_p = r''_p(s)\dot{s}^2 + r'_p(s)\ddot{s}. \quad (3.1.64)$$

Within either smooth branch of the primary radius map,  $r'_p(s)$  is constant. Hence

$$r''_p(s) = 0. \quad (3.1.65)$$

Using the primary slopes from [Equation \(3.1.35\)](#), the primary radius acceleration is

### Primary Radius Acceleration

(3.1.66)

$$\ddot{r}_p = \begin{cases} 0, & 0 \leq s < x_{p,\text{dz}}, \\ \frac{\ddot{s}}{2 \tan \beta}, & x_{p,\text{dz}} < s \leq x_{p,\text{max}}. \end{cases}$$

The result is direct: within the deadzone the primary radius remains fixed, while after contact the shift acceleration is converted into radial acceleration by the same constant sheave slope found in Equation (3.1.35).

**Secondary coordinate curvature** The secondary mapping is not linear after contact. Its first derivative was found from the fixed-length rate balance in Equation (3.1.54):

$$\phi_p + \phi_s x'_s = 0.$$

The balance in Equation (3.1.54) contains the information needed for the second derivative. We will differentiate it once more with respect to  $s$ , which first requires the rates at which the two wrap angles change with  $s$ .

The required geometric quantities were defined in Equations (3.1.22) and (3.1.23). Written together, they are

$$\delta = \sin^{-1} \left( \frac{r_s - r_p}{C} \right), \quad \phi_p = \pi - 2\delta, \quad \phi_s = \pi + 2\delta. \quad (3.1.67)$$

As  $s$  changes, both  $r_p$  and  $r_s$  change, so the radius difference inside  $\delta$  changes at the rate

$$\frac{d}{ds}(r_s - r_p) = r'_s - r'_p. \quad (3.1.68)$$

Applying the chain rule to  $\delta$ , and using the straight-span length from Equation (3.1.19) in the final line, gives

$$\begin{aligned} \delta' &= \frac{1}{\sqrt{1 - \left( \frac{r_s - r_p}{C} \right)^2}} \left( \frac{r'_s - r'_p}{C} \right) \\ &= \frac{r'_s - r'_p}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \frac{r'_s - r'_p}{l}. \end{aligned} \quad (3.1.69)$$

After primary contact, the local pulley slopes are given by Equation (3.1.52):

$$r'_p = \frac{1}{2 \tan \beta}, \quad r'_s = \frac{x'_s}{2 \tan \beta}.$$

Substituting these slopes into Equation (3.1.69) gives

$$\begin{aligned} \delta' &= \frac{\frac{x'_s}{2 \tan \beta} - \frac{1}{2 \tan \beta}}{l} \\ &= \frac{x'_s - 1}{2l \tan \beta}. \end{aligned} \tag{3.1.70}$$

The wrap-angle derivatives now follow directly from Equation (3.1.67):

$$\phi'_p = -2\delta', \quad \phi'_s = 2\delta'. \tag{3.1.71}$$

We can now differentiate the secondary slope balance in Equation (3.1.54). The primary term contributes  $\phi'_p$ , while the secondary term requires the product rule because both  $\phi_s$  and  $x'_s$  change with  $s$ :

$$\begin{aligned} 0 &= \frac{d}{ds} (\phi_p + \phi_s x'_s) \\ &= \phi'_p + \phi'_s x'_s + \phi_s x''_s. \end{aligned} \tag{3.1.72}$$

Substituting the wrap-angle derivatives from Equation (3.1.71) gives

$$-2\delta' + 2\delta' x'_s + \phi_s x''_s = 0. \tag{3.1.73}$$

Solving first for  $x''_s$ ,

$$\begin{aligned} \phi_s x''_s &= 2\delta'(1 - x'_s), \\ x''_s &= \frac{2\delta'(1 - x'_s)}{\phi_s}. \end{aligned} \tag{3.1.74}$$

Using the expression for  $\delta'$  from Equation (3.1.70),

$$\begin{aligned} x''_s &= \frac{2}{\phi_s} \left( \frac{x'_s - 1}{2l \tan \beta} \right) (1 - x'_s) \\ &= -\frac{(1 - x'_s)^2}{\phi_s l \tan \beta}. \end{aligned} \tag{3.1.75}$$

On the engaged branch, the first-derivative result in Equation (3.1.55) gives  $x'_s = -\phi_p/\phi_s$ . The remaining factor can therefore be simplified:

$$\begin{aligned} 1 - x'_s &= 1 + \frac{\phi_p}{\phi_s} \\ &= \frac{\phi_s + \phi_p}{\phi_s} \\ &= \frac{2\pi}{\phi_s}, \end{aligned} \tag{3.1.76}$$

where  $\phi_p + \phi_s = 2\pi$  follows from Equation (3.1.23). Substituting this result into Equation (3.1.75) gives the curvature of the secondary coordinate mapping:

|                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Secondary Coordinate Curvature</b></div> <div><b>(3.1.77)</b></div> </div> <hr style="margin: 10px 0;"/> $x''_s(s) = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{4\pi^2}{\phi_s^3 l \tan \beta}, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The deadzone branch in Equation (3.1.32) is horizontal, so both  $x'_s$  and  $x''_s$  are zero there. On the engaged branch,  $x''_s < 0$ : as the primary closes, the secondary mapping becomes progressively steeper in the opening direction.

**Secondary axial and radial accelerations** The curvature in Equation (3.1.77) describes how the slope of the compatibility curve changes with position. To convert it into acceleration, begin with the secondary axial velocity from Equation (3.1.56):

$$\dot{x}_s = x'_s(s)\dot{s}.$$

Differentiating with respect to time gives

$$\begin{aligned} \ddot{x}_s &= \frac{d}{dt} [x'_s(s)\dot{s}] \\ &= \frac{dx'_s}{dt} \dot{s} + x'_s(s)\ddot{s}. \end{aligned} \tag{3.1.78}$$

Because  $x'_s$  changes only through  $s$ ,

$$\frac{dx'_s}{dt} = x''_s(s)\dot{s}. \tag{3.1.79}$$

The secondary axial acceleration is therefore

$$\ddot{x}_s = x'_s(s)\ddot{s} + x''_s(s)\dot{s}^2. \quad (3.1.80)$$

Substituting the slope from Equation (3.1.55) and the curvature from Equation (3.1.77) gives

$$\text{Secondary Axial Acceleration} \quad (3.1.81)$$

$$\ddot{x}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s}\ddot{s} - \frac{4\pi^2}{\phi_s^3 l \tan \beta} \dot{s}^2, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The local radius derivative in Equation (3.1.6) is the constant  $1/(2 \tan \beta)$ . Applying it to the secondary gives

$$\ddot{r}_s = \frac{1}{2 \tan \beta} \ddot{x}_s. \quad (3.1.82)$$

Substituting Equation (3.1.81) into Equation (3.1.82) gives

$$\text{Secondary Radius Acceleration} \quad (3.1.83)$$

$$\ddot{r}_s = \begin{cases} 0, & 0 \leq s < x_{p,dz}, \\ -\frac{\phi_p}{\phi_s} \frac{\ddot{s}}{2 \tan \beta} - \frac{2\pi^2}{\phi_s^3 l \tan^2 \beta} \dot{s}^2, & x_{p,dz} < s \leq x_{p,max}. \end{cases}$$

The two terms have different physical origins. The first comes from the imposed shift acceleration, converted through the coordinate slope in Equation (3.1.55) and the local pulley slope in Equation (3.1.6). The second comes from the curvature in Equation (3.1.77): even when  $\ddot{s} = 0$ , moving through the engaged branch changes  $-\phi_p/\phi_s$ , so the secondary velocity continues to change.

As with the radius rates, the fixed offsets between the four radial references give, for either pulley  $j \in \{p, s\}$ ,

$$\ddot{r}_j = \ddot{r}_{j,inner} = \ddot{r}_{j,eff} = \ddot{r}_{j,cm}. \quad (3.1.84)$$

#### Remark (Deadzone Boundary)

The idealized maps in Equations (3.1.16) and (3.1.32) change slope when the primary first reaches the belt at  $s = x_{p,dz}$ . Their first derivatives therefore change discontinuously, and their ordinary second derivatives are not defined at that single point. The acceleration relations in Equations (3.1.66) and (3.1.83) apply within the deadzone and engaged branches, while first contact is treated as the transition between them.

#### Remark (Practical Consequences of the Pulley Geometry)

The radius, rate, and acceleration relations derived in this section describe how the pulley geometry evolves as the CVT shifts. [Appendix B](#) explores the practical consequences of these results in more detail. In particular, it examines the resulting geometric CVT ratio and ratio-change rate across the shift travel, together with how the current pulley geometry and sheave angle shape the available ratio range and ratio-change sensitivity.

### 3.1.6 Geometric Closure and Model Scope

Specifying the shift state now determines the complete modeled pulley geometry. The definition in [Equation \(3.1.31\)](#) sets the primary position directly, while [Equation \(3.1.32\)](#) gives the unique compatible secondary position. The local pulley maps in [Equations \(3.1.15\)](#) and [\(3.1.16\)](#) convert those two positions into the outer-surface radii used to describe the belt path. Finally, the rate and acceleration relations in [Equations \(3.1.36\)](#), [\(3.1.57\)](#), [\(3.1.66\)](#) and [\(3.1.83\)](#) determine how those radii move. Thus, for any admissible shift state,

$$(s, \dot{s}, \ddot{s}) \longmapsto (r_p, \dot{r}_p, \ddot{r}_p, r_s, \dot{r}_s, \ddot{r}_s).$$

Here  $r_p$  and  $r_s$  retain the outer-surface convention used throughout the geometric development. The corresponding inner-surface, effective, and centre-of-mass radii follow directly from their fixed positions within the belt cross-section. Nothing else has to be solved for.

These results follow from the geometry derived above under the idealizations stated in [Section 2.2](#). The belt cross-section and total length remain fixed under [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#) and [Assumption 6 \(Fixed Belt Length\)](#), while [Assumption 8 \(Prescribed Planar Belt Path\)](#) constrains the belt to circular wraps and straight tangent spans. Together with the aligned pulley axes and fixed centre distance of [Assumption 5 \(Aligned Pulley Axes and Fixed Center Distance\)](#), these conditions leave one modeled belt path for each admissible value of  $s$ .

**Relation to prior transient models** These choices also define the level of detail represented by the present geometry. It determines the complete belt path that is globally compatible with the imposed shift position, but it does not track the trajectory of each belt element through the pulley wraps.

[Duan et al. \(2016\)](#) begin from closely related open-belt arc-and-tangent geometry and the same wedge relationship between movable-sheave motion and nominal radius motion. Their chain-CVT formulation retains additional internal deformation. Chain extension and its rate contribute to the span tension, and pulley deformation allows the local radius followed by a chain element to differ from the nominal pulley radius. The present formulation instead fixes the belt length through [Assumption 6 \(Fixed Belt Length\)](#) and prescribes the arc-and-tangent belt path through [Assumption 8 \(Prescribed Planar Belt Path\)](#). Consequently, elastic path-length change and deformation-driven departure from that prescribed path are not represented in this geometric closure.

Rubber-belt transient models resolve the same shifting process at several levels of detail. Analytical shift-mechanics models, such as those of [Sorge \(2008\)](#) and [Cammalleri and Sorge \(2009\)](#), describe local radial belt migration through a pulley wrap and couple it to circumferential transport, sliding, and the mechanics of the contact region. Discretized rubber-belt models, such as those of [Julió and Plante \(2011\)](#) and [Ballew \(2015\)](#), instead obtain the evolving belt geometry from the motion and

contact state of individual belt elements. These approaches can resolve local deformation, tension redistribution, and contact migration in greater detail, but require additional spatial or nodal states and the contact laws that govern them.

The distinction is therefore one of model resolution. The present formulation uses the global compatibility condition in Equation (3.1.25) to find the representative pulley-radius pair and its derivatives directly from  $s$ ,  $\dot{s}$ , and  $\ddot{s}$ . The cited transient models additionally ask how the belt or chain deforms and migrates within that path. Those internal motions lie outside the present geometric layer; within the assumptions stated above, the global pulley geometry and its motion are fully determined.

This closure determines which sheave and radius motions are compatible with the modeled belt path, but it does not determine which compatible acceleration occurs. In particular, every acceleration relation above still contains the shift acceleration  $\ddot{s}$ . Its value must be consistent with the forces and inertias acting on the two movable pulley assemblies.

## 3.2 Axial Actuation and Sheave Balances

The preceding section established how the two movable sheaves must move together. It did not determine the motion itself. That requires the forces acting on each movable assembly and the inertia that those forces must accelerate.

The primary and secondary remain separate mechanisms even though the belt geometry couples their positions. Each has its own actuator, spring, moving hardware, and belt reaction, so each must satisfy its own axial equation of motion. We therefore complete the primary balance in its local coordinate  $x_p$ , then do the same for the secondary in  $x_s$ . Once each mechanism has been interpreted and every local force has been derived, the geometric maps developed above in Section 3.1 express both balances through the shared shift coordinate  $s$ . This final step shows how the two local free bodies act on one shift motion.

### 3.2.1 Primary Movable-Sheave Balance

The primary is easiest to understand by separating rotation about the shaft from motion along it. The entire primary rotates with the shaft at  $\omega_p$ , but its parts do not all move axially. The fixed sheave remains at one axial position. Opposite it, the movable sheave, ramp carrier, and the hardware attached rigidly to them slide together along the shaft. Motion in the closing direction narrows the pulley groove and clamps the belt; motion in the opening direction widens the groove.

The flyweights provide the closing drive. They are attached to the fixed sheave and carried around the shaft with it, but each can also pivot about its attachment to the fixed sheave. As the primary rotates, centrifugal loading tends to swing each flyweight radially outward until it presses against a ramp carried by the movable sheave. Because the ramp surface is inclined, its contact reaction has an axial component that drives the movable sheave in the closing direction. The primary spring pushes the movable sheave in the opening direction, and the belt pushes back through its normal reaction on the movable face.

This leaves two distinct moving bodies in the model. The movable sheave, ramp carrier, and attached hardware form one translating assembly with axial coordinate  $x_p$ . The flyweights rotate with the primary and pivot separately, but do not translate rigidly with  $x_p$ ; they deliver their closing action to the movable sheave through the ramp contacts. During a shift, the net axial force must

accelerate the translating assembly, while the flyweights require their own rotational balance. We therefore begin by isolating the translating assembly, then determine the force transmitted to it by the flyweights. This mechanism and the resulting axial free body are shown together in [Figure 3.5](#).

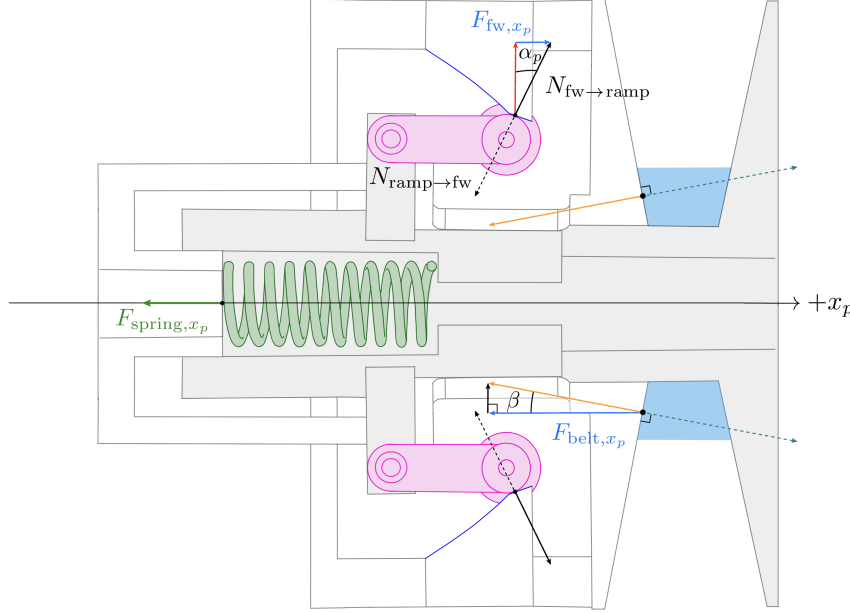


Figure 3.5: Axial free-body diagram of the primary movable assembly (unshaded). The symmetric sides of the cross-section are used to show the flyweight–ramp and belt–sheave force resolutions separately. Dotted arrows show the equal-and-opposite forces on the flyweight and belt, with  $F_{\text{clamp}}$  indicating the movable sheave’s squeeze on the belt.

As established in [Section 3.1](#), positive  $x_p$  closes the primary. Let  $m_p$  be the total mass of the movable sheave, ramp carrier, and every other component that translates rigidly with them. The three axial forces shown in [Figure 3.5](#) are:

- $F_{\text{fw}, x_p}$ , the closing force delivered to the ramp carrier by the flyweight contacts;
- $F_{\text{spring}, x_p}$ , the opening force exerted by the primary spring; and
- $F_{\text{belt}, x_p}$ , the opening force exerted by the belt on the movable face.

Newton’s second law along the shaft is therefore

$$m_p \ddot{x}_p = F_{\text{fw}, x_p} - F_{\text{spring}, x_p} - F_{\text{belt}, x_p}. \quad (3.2.1)$$

This free body tells us exactly what remains to be found. The spring force will follow from its compression, and the belt force from the normal reaction on the conical face. The flyweight force is harder. It is the axial result of a contact on a body that pivots rather than translating rigidly with the sheave. We therefore determine  $F_{\text{fw}, x_p}$  first.

## Force Transmitted by the Flyweights

**Flyweight free body** Figure 3.6 isolates one flyweight in the radial–axial plane fixed to the primary. The flyweight is pinned at  $P$  and pivots about the circumferential axis through that point. Its pose is described by  $q_f$ , measured from  $+x_p$  to the body-fixed reference line shown in the figure. At  $q_f = 0$ , this line is aligned with  $+x_p$ ; at  $q_f = \pi/2$ , it points radially outward. Increasing  $q_f$  therefore describes the outward-pivoting direction.

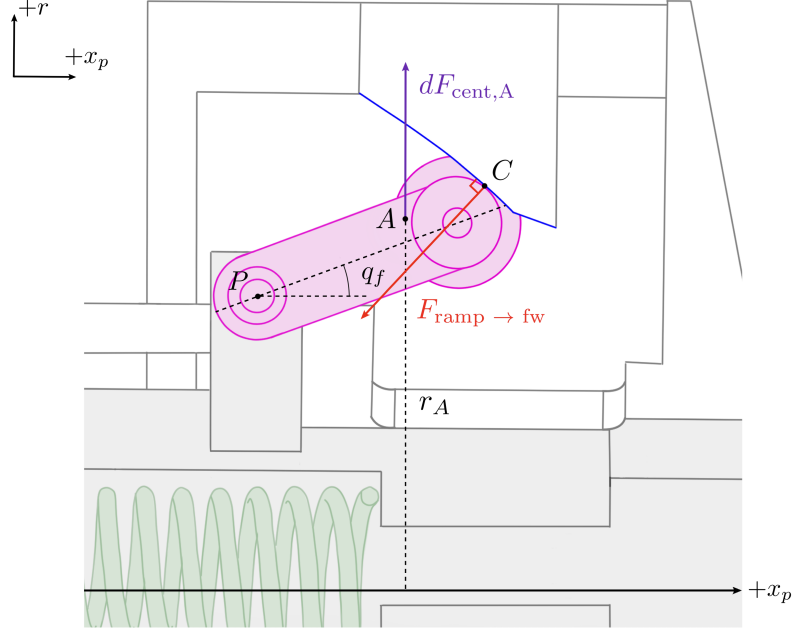


Figure 3.6: Free body of one primary flyweight in the frame rotating with the primary. Centrifugal loading acts throughout the flyweight mass; the marked element  $A$  identifies one infinitesimal mass  $dm$  at shaft radius  $r_A$  and its outward load. The ramp-contact normal at  $C$  and the pivot reaction at  $P$  complete the force system.

As the primary rotates at  $\omega_p$ , the flyweight experiences centrifugal loading throughout its mass. Each infinitesimal element is loaded radially outward according to its distance from the primary shaft. The marked element  $A$  is one example: for mass  $dm$  at shaft radius  $r_A$ , its outward load is

$$dF_{\text{cent},A} = \omega_p^2 r_A dm. \quad (3.2.2)$$

Every other element is loaded in the same way at its own shaft radius. Together, these forces form the distributed centrifugal loading of the flyweight.

Because the flyweight's relative motion is confined to the radial–axial plane, the Euler and Coriolis inertial forces are circumferential and have no moment about the circumferential pivot axis. The  $q_f$  balance therefore depends on  $\omega_p$ , but not on  $\dot{\omega}_p$ .

Taking moments about  $P$  separates the roles of these loads. The pivot reaction passes through  $P$  and therefore contributes no moment. The radial lines of action of the centrifugal loads generally do not pass through  $P$ , so together they produce a moment in the direction of increasing  $q_f$ . The ramp normal produces the opposing moment.

The figure shows one flyweight, whereas the primary contains a symmetric set whose members follow the same  $q_f$ . Let  $\tau_{f,\text{cent}}$  and  $\tau_{f,\text{contact}}$  denote the respective magnitudes of the centrifugal and contact moments summed over this complete set. Since their shared coordinate  $q_f$  is a rotation about the flyweight pivots, let  $I_f$  denote the sum of each flyweight's mass moment of inertia about its own pivot axis. With positive moment defined in the direction of increasing  $q_f$ , rotational Newton's second law gives

$$\tau_{f,\text{cent}} - \tau_{f,\text{contact}} = I_f \ddot{q}_f. \quad (3.2.3)$$

**Mechanism geometry** The moment balance is written in the flyweight coordinate  $q_f$ , whereas the force required by the movable-sheave balance must be expressed through the axial coordinate  $x_p$ . The ramp geometry connects the two: each sheave position fixes the corresponding flyweight pose.

The centrifugal moment in Equation (3.2.3) comes from the distributed loading shown in the figure, so its value depends on how the complete flyweight mass is distributed about the primary shaft. The marked element  $A$  contributes  $r_A^2 dm$  to the mass moment of inertia about that shaft. Adding the corresponding contributions from every element of every flyweight gives the complete shaft-axis inertia  $J_{f,p}$ . Since  $x_p$  fixes the flyweight pose, it fixes every element's shaft radius and therefore fixes  $J_{f,p}$ . This shaft-axis inertia should not be confused with the pivot-axis inertia  $I_f$  in Equation (3.2.3), which is constant for rigid flyweights.

The mechanism therefore supplies two configuration maps:

$$q_f = q_f(x_p), \quad J_{f,p} = J_{f,p}(x_p). \quad (3.2.4)$$

Together with the constant  $I_f$ , these maps abstract the physical geometry of the flyweight mechanism into the quantities required by the general force balance.

#### Remark (Constructing the flyweight maps)

The mechanism-specific construction is not carried out here. Appendix C.1 works through two common flyweight arrangements, showing how their geometry and mass distribution produce  $q_f(x_p)$ ,  $J_{f,p}(x_p)$ , and  $I_f$ .

With these mechanism properties specified, the remaining task is to express the two moments and  $\ddot{q}_f$  in Equation (3.2.3) through the primary motion. We begin with the moments.

Both could be found by resolving the forces and constructing their moment arms for the specific flyweight geometry. A more convenient approach is equivalent virtual work. Hold the mechanism at its current position and imagine a small compatible motion. During this motion, a moment  $\tau_f$  acting through  $dq_f$  performs the work

$$\tau_f dq_f = \delta W_f. \quad (3.2.5)$$

We can therefore find each moment by calculating the work of its underlying load during the same motion.

For the centrifugal loading, at fixed instantaneous angular speed, the work associated with a change  $dJ$  in shaft-axis inertia is  $\frac{1}{2}\omega^2 dJ$ . Applying this relation to the flyweights gives

$$\tau_{f,\text{cent}} dq_f = \delta W_{f,\text{cent}} = \frac{1}{2}\omega_p^2 dJ_{f,p}. \quad (3.2.6)$$

For the contact moment, the corresponding work is the force–displacement work  $F d\ell$ . The movable sheave moves axially through  $dx_p$ , so only the axial component of the ramp–contact force performs work. This component is  $F_{\text{fw},x_p}$ , the closing force introduced by the movable-sheave free body, and its work is therefore  $F_{\text{fw},x_p} dx_p$ . Under **Assumption 3 (Idealized Actuator Mechanics)**, the ramp contact is treated as an ideal frictionless constraint, so no work is dissipated at the contact. The work of the contact moment therefore appears entirely as axial work on the movable assembly. Using the positive magnitudes defined above,

$$\tau_{f,\text{contact}} dq_f = \delta W_{f,\text{contact}} = F_{\text{fw},x_p} dx_p. \quad (3.2.7)$$

**Flyweight acceleration** The two moment terms are now written in forms that can be returned to the rotational balance. Its remaining term is the flyweight angular acceleration. Ramp contact makes this acceleration a consequence of the sheave motion rather than an independent state. Since the contact geometry already gives  $q_f$  as a function of  $x_p$ , its angular velocity is

$$\dot{q}_f = \frac{dq_f}{dx_p} \dot{x}_p, \quad (3.2.8)$$

and one more derivative gives

$$\ddot{q}_f = \frac{dq_f}{dx_p} \ddot{x}_p + \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2. \quad (3.2.9)$$

The second term appears when the amount of flyweight rotation required per unit sheave travel changes along the ramp. Even at constant  $\dot{x}_p$ , the flyweight then changes angular speed as it moves.

**Axial closing force** There is now an expression for every term in the rotational balance. Multiplying **Equation (3.2.3)** by  $dq_f$  and substituting **Equations (3.2.6)**, **(3.2.7)** and **(3.2.9)** brings them together,

$$\frac{1}{2}\omega_p^2 dJ_{f,p} - F_{\text{fw},x_p} dx_p = I_f \left( \frac{dq_f}{dx_p} \ddot{x}_p + \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2 \right) dq_f. \quad (3.2.10)$$

Rearranging for the force transmitted to the movable sheave gives

$$F_{\text{fw},x_p} = \frac{1}{2}\omega_p^2 \frac{dJ_{f,p}}{dx_p} - I_f \left( \frac{dq_f}{dx_p} \right)^2 \ddot{x}_p - I_f \frac{dq_f}{dx_p} \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2. \quad (3.2.11)$$

The result now has the form required by the movable-sheave free body. The first term is the centrifugal drive and therefore uses the shaft-axis property  $J_{f,p}$ . The final two terms arise from acceleration about the flyweight pivots and therefore use the pivot-axis property  $I_f$ : the first reflects  $\ddot{x}_p$  through the motion ratio, while the second appears when that motion ratio changes with position.

#### Remark (Admissible flyweight maps)

The completed force law uses  $q_f''(x_p)$  and  $J_{f,p}'(x_p)$ , while the selected coordinate directions require primary closure to correspond to outward flyweight rotation. Over the admissible shift range, the supplied mechanism maps must therefore satisfy

$$q_f \in C^2, \quad J_{f,p} \in C^1, \quad 0 < \frac{dq_f}{dx_p} < \infty.$$

These are requirements on the fitted or constructed maps, not additional equations of motion.

The one-sided flyweight contact can transmit compression but not tension, so the force predicted by Equation (3.2.11) must satisfy  $F_{\text{fw},x_p} \geq 0$ .

### Spring and Belt Forces

Returning to the axial force balance, the two remaining forces in Equation (3.2.1) act directly on the translating assembly.

**Primary spring** Let  $k_p$  be the spring stiffness and  $x_{p,\text{pre}}$  its installed compression when the primary is open at  $x_p = 0$ . Closing the sheave by  $x_p$  adds the same amount of compression, so the spring opening force is

$$F_{\text{spring},x_p} = k_p (x_{p,\text{pre}} + x_p). \quad (3.2.12)$$

**Belt reaction** Let  $N_p$  be the total integrated normal load over both primary sheave faces. This load is coupled to the belt and the secondary balance, so it remains an unknown at this stage. Under Assumption 12 (Symmetric Sheave-Face Sharing), the movable face carries one half of this total load. Its normal is inclined by the sheave half-angle  $\beta$  from the axial direction, so the opening force on the movable face is

$$F_{\text{belt},x_p} = \frac{N_p}{2} \cos \beta. \quad (3.2.13)$$

### Completed Primary Balance

Each force in the original free body now has a mechanism-level expression. Substitute Equations (3.2.11) to (3.2.13) into Equation (3.2.1) and collect the terms multiplying  $\ddot{x}_p$ :

$$\begin{aligned}
\left[ m_p + I_f \left( \frac{dq_f}{dx_p} \right)^2 \right] \ddot{x}_p &= \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{dx_p} \\
&\quad - I_f \frac{dq_f}{dx_p} \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2 \\
&\quad - k_p (x_{p,\text{pre}} + x_p) - \frac{N_p}{2} \cos \beta.
\end{aligned} \tag{3.2.14}$$

The flyweights now appear through the same two axes introduced by their free body. Rotation about the primary shaft, described by  $J_{f,p}$ , provides the speed-dependent closing drive. Rotation about the flyweight pivots, described by  $I_f$ , adds to the inertia that the axial motion must accelerate. If the motion ratio  $dq_f/dx_p$  changes along the ramp, the flyweights also accelerate even when  $\dot{x}_p$  is constant, producing the term proportional to  $\dot{x}_p^2$ . The spring and belt forces act directly on the movable assembly and oppose closure exactly as shown in the first free body.

This is the primary axial equation of motion. It states how the flyweight action is divided among spring compression, belt loading, and acceleration of the translating and pivoting hardware.

### Point-Mass Limit and Relation to Existing Models

The general force law becomes the familiar classical actuator model when the distributed flyweight mass is collapsed to its centre of mass and the flyweight's pivot dynamics are neglected. Let  $m_f$  be the total mass of the symmetric flyweight set and let  $r_f(x_p)$  be the common instantaneous centre-of-mass radius. Under the point-mass assumption, the shaft-axis inertia reduces to

$$J_{f,p} = m_f r_f^2. \tag{3.2.15}$$

Substituting this point-mass inertia into the centrifugal term of [Equation \(3.2.11\)](#), while omitting the pivot-acceleration terms, gives

$$F_{\text{fw},x_p}^{\text{pm}} = m_f r_f \omega_p^2 \frac{dr_f}{dx_p}. \tag{3.2.16}$$

This is the familiar point-mass result. The factor  $m_f r_f \omega_p^2$  is the total centrifugal loading, while  $dr_f/dx_p$  is the local radial-to-axial motion ratio that converts that loading into sheave force. The path need not be straight, and its motion ratio need not be constant. A sliding mass on a straight ramp is one simple case; for a pivoted flyweight,  $r_f(x_p)$  instead follows from the flyweight pose and its mass geometry.

If the movable sheave is also in axial equilibrium, the point-mass actuator force balances the opening forces from the spring and belt:

$$m_f r_f \omega_p^2 \frac{dr_f}{dx_p} = k_p (x_{p,\text{pre}} + x_p) + \frac{N_p}{2} \cos \beta. \tag{3.2.17}$$

This reduction appears directly in the quasi-static model of [Messick \(2018\)](#). Their three wedge-shaped flyweights slide along linear ramps without changing orientation. Measured CAD geometry supplies

the centre-of-mass radius and the radial-to-axial displacement ratio, giving a force of the same form as Equation (3.2.16). Friction is neglected, and the resulting primary force is supplied to the belt model as a boundary condition; shifting is treated as a sequence of equilibrium configurations rather than a time-dependent response. Arango and Muñoz Alzate (2020) use the same instantaneous force conversion to construct both sliding-weight ramps and pivoted flyweight-cam profiles. Their contact geometry is designed so that the available centrifugal force supplies the required clamping force through the local mechanical advantage.

Other treatments retain more of the contact geometry while keeping the same instantaneous-equilibrium closure. Kim et al. (2002) resolve the two contacts of a centrifugal roller and include contact friction when converting roller centrifugal force into driver thrust. The current roller radius and contact angles vary with ratio, but no roller acceleration enters the thrust equation. Skinner (2020) resolve a flyweight, its supporting link, and the ramp reactions in a complete free body, including the centrifugal loading of both the flyweight and the link. In the present notation, those separate mass contributions could be collected in  $J_{f,p}(x_p)$ . Skinner’s derivation therefore retains a substantially richer equilibrium force path than a single fixed ramp factor, but still closes the flyweight system by force equilibrium at each configuration.

Despite their different levels of geometric detail, none of these actuator relations includes the acceleration of the flyweight mechanism. Messick does not model shift time, Skinner solves algebraically for the speed required at each configuration, and Kim places the time dependence in a separate experimentally identified first-order ratio law, using different fitted coefficients for upshift and downshift. Conversely, the transient belt model of Ballew (2015) imposes primary axial force through a controller rather than resolving the mechanical actuator. Across these examples, transient shift motion is therefore obtained without returning the flyweights’ own acceleration to the primary axial balance.

The distinction of the present formulation is therefore not that it can describe a curved ramp or a detailed flyweight shape; the equilibrium models above can already retain substantial geometric detail. Here, the complete mass distribution enters through  $J_{f,p}(x_p)$ , so the centrifugal drive is not restricted to a centre-of-mass point model. More importantly, the flyweight moment balance is not closed at instantaneous equilibrium. Mapping  $I_f \ddot{q}_f$  into  $x_p$  produces both the reflected pivot inertia and the changing-motion-ratio term in Equation (3.2.14). Consequently, the flyweight drive, spring force, belt reaction, translating mass, and pivoting mass all determine the same shift acceleration rather than being divided between an equilibrium actuator law and a separate shift-response model. No equivalent explicit reflection of the flyweights’ pivoting inertia was identified in the reviewed mechanically actuated rubber-belt CVT formulations.

This extension does not retain every effect represented in the earlier models. Structural deformation is excluded by Assumption 2 (Rigid Pulley and Actuation Hardware), while Assumption 3 (Idealized Actuator Mechanics) treats the ramp contact as an ideal constraint rather than resolving its tribology or local compliance. The roller-contact friction retained by Kim et al. (2002), for example, therefore lies outside the present force law; roller spin and contact compliance are likewise not resolved. The added fidelity is in the coupled mechanism dynamics, not in frictional or compliant contact detail.

### 3.2.2 Secondary Movable-Sheave Balance

While the primary closes mainly in response to rotational speed, the secondary is torque reactive: torque transmitted through the pulley can also increase the axial force clamping the belt.

To see how, begin with the two secondary sheaves rotationally disconnected. The fixed sheave rotates with the secondary shaft. The movable sheave can slide along that shaft and rotate relative to the fixed sheave. Because the belt contacts the two faces separately, it applies torque to each body. With no connection between them, the two bodies can accelerate differently and rotate relative to one another.

Now place a pin on the movable sheave in a straight slot running along the shaft. The movable sheave remains free to slide, but the pin prevents relative rotation. Whenever the two sheaves tend to rotate differently, the pin presses against the side of the slot and a normal reaction develops. Because the slot is axial, this normal acts entirely in the circumferential direction. It transfers torque between the sheaves but produces no axial force.

A helix is the same slot tilted around the shaft. The contact normal remains perpendicular to the slot surface, but now has both circumferential and axial components. The circumferential component continues to couple the rotational motion of the sheaves, while the axial component acts on the movable sheave. Relative rotation is therefore allowed only when accompanied by axial travel along the helix.

These three cases and their contact reactions are compared in [Figure 3.7](#).

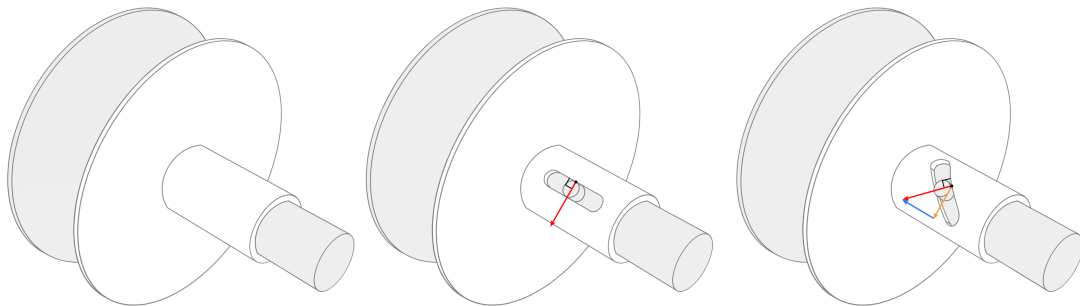


Figure 3.7: How the helix creates torque-reactive axial loading. With no connection, the two sheaves may rotate independently. A straight axial slot couples their rotation through a purely circumferential reaction; inclining the slot adds an axial component to that same reaction, coupling relative rotation to axial travel.

If the axial component of the helix reaction is balanced by the other axial loads, the movable sheave remains at its current position. Its relative angle then remains fixed, and the two sheaves rotate together with the shaft. If the axial loads do not balance, their resultant accelerates the movable sheave along the helix, and it rotates by the amount required by the slot geometry.

In the actual secondary, the belt and spring load this mechanism. The belt applies torque to the two sheave faces separately. The secondary spring is installed both compressed and twisted: its torsional preload tends to rotate the movable sheave relative to the fixed sheave, while its axial compression pushes the movable sheave toward closure. The torsional action loads the helix; the axial action enters the movable-sheave balance directly.

The axial free body gives the target for the derivation.

## Axial Free Body

Figure 3.8 isolates the movable assembly in the axial direction.

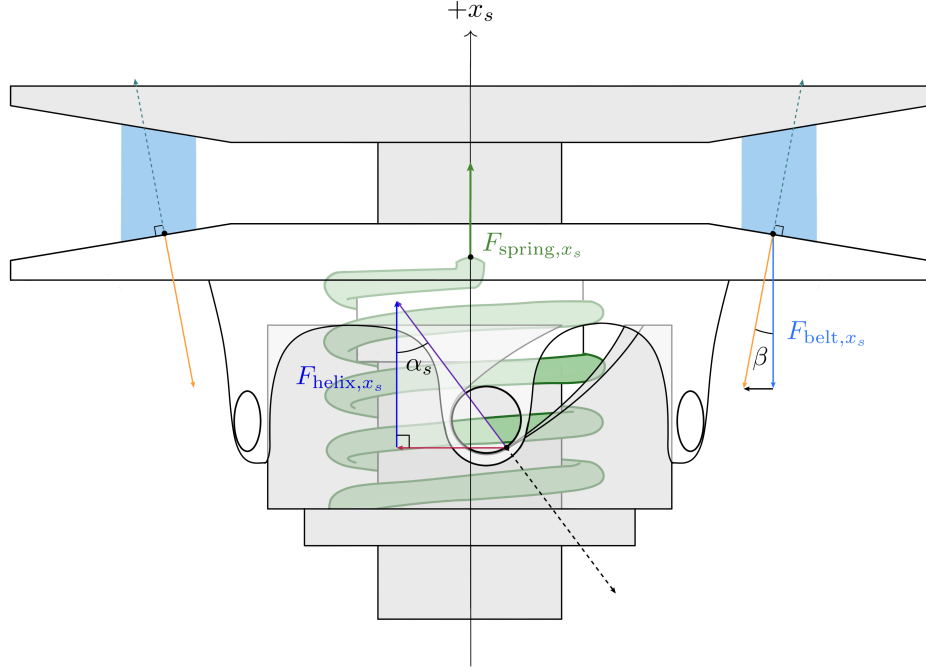


Figure 3.8: Axial free body of the complete movable secondary assembly. The axial spring force and belt reaction act directly on the translating assembly, while the secondary's torque-reactive behaviour enters through the pin-helix force  $F_{\text{helix}, x_s}$ .

Let  $m_s$  be the mass of the movable sheave and every component that translates rigidly with it. Three axial loads act on this assembly. The pin-helix contact contributes the signed force  $F_{\text{helix}, x_s}$ . The compressed spring pushes the movable sheave toward the fixed sheave in  $+x_s$ . The belt pushes the two sheave faces apart, so its reaction on the movable face acts in  $-x_s$ , just as on the primary. Newton's second law therefore gives

$$m_s \ddot{x}_s = F_{\text{helix}, x_s} + F_{\text{spring}, x_s} - F_{\text{belt}, x_s}. \quad (3.2.18)$$

This free body makes the remaining task clear. The direct spring and belt forces follow from compression and contact geometry. The helix force does not: the tilted slot has established that an axial reaction exists, but not its magnitude or sign. Because the same contact reaction must also carry the torque that constrains relative rotation, its rotational effect is resolved first. This sets the order of the derivation:

1. use the rotational balance to find the torque carried by the helix;
2. use the helix geometry to convert that torque into axial force; and
3. add this force to the direct spring and belt loads in the axial balance.

## Rotational Loading of the Movable Sheave

As established in [Section 3.1](#),  $x_s$  is measured from the open secondary toward the fixed sheave, so  $+x_s$  closes the pulley. Let  $\theta_s$  measure the rotation of the movable sheave relative to the fixed sheave in the same positive rotational sense already assigned to  $\omega_s$  (see [Section 2.3.1](#)). Only the conventional torque-reactive helix orientation is considered: as the secondary closes in  $+x_s$ , the movable sheave turns relative to the fixed sheave in  $+\theta_s$ .

With these directions set, the first question is what loads the helix circumferentially. The belt applies torque to the movable face, and the twisted spring applies torque between the two sheaves. The pin–helix contact supplies the torque that constrains the resulting relative motion to follow the slot.

Our goal is to find the torque reacted at the pin–helix contact. To do so, isolate the movable assembly and apply Newton’s second law for rotation. Use the following quantities:

- $I_{s,M}$ : rotational inertia of the movable sheave and every component that rotates rigidly with it;
- $\omega_{s,M}$ : absolute angular velocity of that assembly;
- $\tau_{s,M}$ : signed belt torque applied to the movable face;
- $\tau_{s,\text{spring}}$ : signed spring torque applied to the movable assembly; and
- $\tau_{s,\text{helix}}$ : the contact torque at the pin–helix connection, produced by the circumferential component of the contact normal. A positive  $\tau_{s,\text{helix}}$  acts on the movable assembly in  $-\theta_s$ .

The figures show one representative pin and track. The quantities listed above and later in this section are the totals over the complete set of loaded pin–track contacts, so no additional track multiplier is required.

The rotational balance is therefore

$$\tau_{s,M} + \tau_{s,\text{spring}} - \tau_{s,\text{helix}} = I_{s,M}\dot{\omega}_{s,M}. \quad (3.2.19)$$

Solving for the helix torque gives

$$\tau_{s,\text{helix}} = \tau_{s,M} + \tau_{s,\text{spring}} - I_{s,M}\dot{\omega}_{s,M}. \quad (3.2.20)$$

[Equation \(3.2.20\)](#) gives the contact torque we want, but its right-hand side still needs three relations:

- the share of belt torque applied to the movable face,  $\tau_{s,M}$ ;
- the torque supplied by the wound spring,  $\tau_{s,\text{spring}}$ ; and
- the absolute angular acceleration of the movable assembly,  $\dot{\omega}_{s,M}$ .

The belt term follows immediately from the face-torque assumption.

**Movable-face belt torque** Let  $\tau_s$  be the total signed torque exerted by the belt on the secondary, so  $\tau_s = \tau_{s,F} + \tau_{s,M}$ . Under [Assumption 12 \(Symmetric Sheave-Face Sharing\)](#), the two faces receive equal torque shares. Only the movable-face share appears in its free body:

$$\tau_{s,M} = \frac{\tau_s}{2}. \quad (3.2.21)$$

This equal split is a reduced-contact assumption. During shift, the movable face rotates slightly faster or slower than the fixed face. The two belt–face contacts can therefore have different sliding conditions, so their torques need not divide equally.

**Remark (Why the face-torque split can matter)**

Sorge and Cammalleri (2011) resolve the two contacts separately and show that the face torques can depart substantially from an equal split; distributed tangential forces on the two faces can even act in opposite directions over parts of the wrap. Since  $\tau_{s,M}$  enters Equation (3.2.20) directly, this is also a potential error in the torque reacted by the helix.

The remaining terms both depend on the helix kinematics:  $\dot{\omega}_{s,M}$  depends on how the movable sheave rotates as it shifts, and the spring torque depends on how far that motion unwinds the spring. We therefore derive the relation between  $x_s$  and  $\theta_s$  next.

**Motion Imposed by the Helix**

The helix geometry tells us how axial shift changes the relative angle of the movable sheave. Unwrap one track from its effective cylindrical radius  $r_h$ , as seen in Figure 3.9. A small relative rotation  $d\theta_s$  moves the pin by  $r_h d\theta_s$  circumferentially while the sheave moves by  $dx_s$  axially. Let  $\alpha_s$  be the local angle of the unwrapped track, measured from the positive circumferential direction toward the positive axial direction. For the conventional helix orientation considered here,

$$0 < \alpha_s < \frac{\pi}{2}.$$

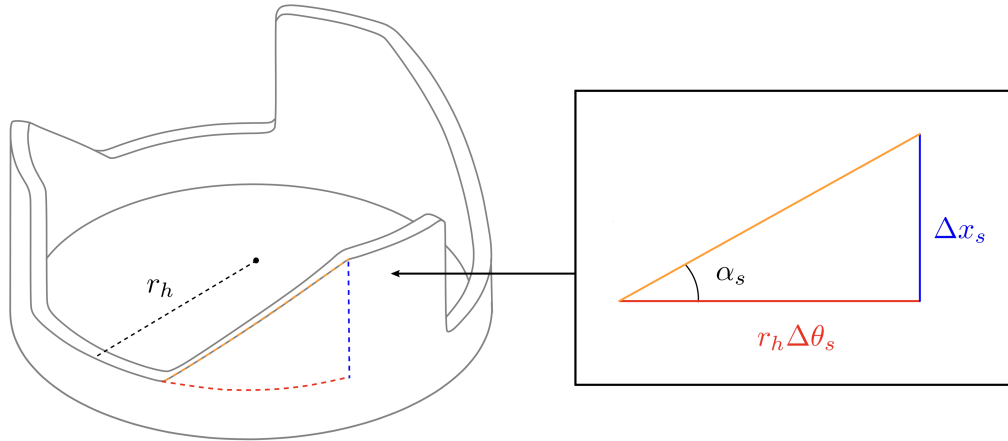


Figure 3.9: Unwrapped secondary-helix geometry. The same track angle relates relative rotation to axial travel and reaction torque to axial force.

Taking the infinitesimal limit of the triangle in Figure 3.9,

$$\tan \alpha_s = \frac{dx_s}{r_h d\theta_s}, \quad \frac{d\theta_s}{dx_s} = \frac{1}{r_h \tan \alpha_s}. \quad (3.2.22)$$

Thus,  $d\theta_s/dx_s$  is the relative rotation required per unit axial travel. It is positive for the conventional helix orientation assumed here. Returning to the straight axial slot from the opening example,  $\alpha_s \rightarrow \pi/2$  and  $d\theta_s/dx_s \rightarrow 0$ , as expected.

Taking  $\theta_s(0) = 0$  at the open secondary, integrate the local motion ratio to obtain

$$\theta_s(x_s) = \int_0^{x_s} \frac{1}{r_h \tan \alpha_s(\xi)} d\xi. \quad (3.2.23)$$

**Equation (3.2.23)** defines the helix as a function  $\theta_s(x_s)$ : for each axial position, it gives the relative angle enforced by the track. The function need not come from a constant helix angle. Any supplied function may be used provided it is single-valued and strictly increasing over the admissible shift range.

**Rotation of the movable sheave** We can now use this function to determine the two remaining quantities in the rotational balance. First consider the absolute motion of the movable sheave. The fixed sheave rotates with the secondary shaft at  $\omega_s$ , while the movable sheave also has the relative rotation  $\theta_s$ :

$$\omega_{s,M} = \omega_s + \dot{\theta}_s = \omega_s + \frac{d\theta_s}{dx_s} \dot{x}_s. \quad (3.2.24)$$

Because  $\theta_s$  was defined in the same positive rotational sense as  $\omega_s$ , the relative speed enters with the plus sign shown here. Differentiating gives the angular acceleration required by the rotational balance:

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2. \quad (3.2.25)$$

The shaft acceleration  $\dot{\omega}_s$  remains unknown here because  $\omega_s$  is a state of the coupled model. It is solved together with the axial motion rather than by the helix relation alone. The term proportional to  $\ddot{x}_s$  is the angular acceleration caused by axial acceleration. The term proportional to  $\dot{x}_s^2$  appears only when the motion ratio varies with position.

**Torsional Spring Loading** The same helix function tells us how much the spring has unwound. Let  $k_{s,\theta}$  be the spring's torsional stiffness and  $\theta_{s,\text{pre}}$  its signed twist at  $x_s = 0$ . Increasing  $\theta_s$  reduces this twist, leaving  $\theta_{s,\text{pre}} - \theta_s(x_s)$ . The resulting torque on the movable assembly is

$$\tau_{s,\text{spring}} = k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)]. \quad (3.2.26)$$

For the installed winding assumed here, this quantity is positive while the spring remains preloaded. The opposite winding direction can be represented by a signed  $\theta_{s,\text{pre}}$ .

We now have relations for all three terms on the right-hand side of **Equation (3.2.20)**. Substituting them gives the pin-helix contact torque:

$$\begin{aligned}\tau_{s,\text{helix}} = & \frac{\tau_s}{2} + k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)] \\ & - I_{s,M} \left[ \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right].\end{aligned}\tag{3.2.27}$$

This completes the rotational part of the derivation. The next step is to use the pin–helix contact geometry to find the axial force associated with this torque.

### Axial Component of the Helix Reaction

The torque above is the moment produced by the circumferential component of the contact normal. Let  $F_{\text{helix},x_s}$  denote the signed axial force produced by the helix reaction, positive in  $+x_s$ .

Under **Assumption 3 (Idealized Actuator Mechanics)**, the pin–helix contact is treated as an ideal frictionless constraint. The same compatible-work argument used for the primary therefore converts the rotational contact load into its axial effect without contact dissipation. Consider the motion of the movable sheave relative to the helix carrier. The contact normal does no net work in this relative motion. Its axial component contributes  $F_{\text{helix},x_s} dx_s$ , while its circumferential component contributes  $-\tau_{s,\text{helix}} d\theta_s$ . Their sum is zero, so

$$F_{\text{helix},x_s} dx_s = \tau_{s,\text{helix}} d\theta_s.\tag{3.2.28}$$

Using the motion ratio gives

$$F_{\text{helix},x_s} = \tau_{s,\text{helix}} \frac{d\theta_s}{dx_s}.\tag{3.2.29}$$

The motion ratio  $d\theta_s/dx_s$  is therefore the mathematical origin of the axial effect: it converts the contact torque into axial force. In the straight-slot limit used in the opening example, this ratio is zero, so the slot can react torque while the axial conversion vanishes.

Substituting **Equation (3.2.27)** gives the helix force in terms of the model variables:

$$\boxed{F_{\text{helix},x_s} = \frac{d\theta_s}{dx_s} \left\{ \frac{\tau_s}{2} + k_{s,\theta} [\theta_{s,\text{pre}} - \theta_s(x_s)] - I_{s,M} \left[ \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right] \right\}}.\tag{3.2.30}$$

This gives  $F_{\text{helix},x_s}$ . The belt and wound spring provide the driving torques inside the braces; part of that torque may be required to accelerate the movable assembly, and only the remainder is converted into axial thrust.

### Remark (Admissible helix maps)

The completed force uses the first two derivatives of  $\theta_s(x_s)$ . Over the admissible shift range, the supplied helix map must therefore be single-valued and twice continuously differentiable, with a finite, nonzero motion ratio:

$$\theta_s \in C^2, \quad 0 < \frac{d\theta_s}{dx_s} < \infty.$$

A manufactured multi-angle helix may still be represented, but nominal segments must be joined smoothly rather than by a mathematical corner.

The difficult force in the preliminary axial balance is now known. The two remaining forces act directly on the translating assembly.

### Direct Spring and Belt Forces

**Direct axial spring force** The spring force follows directly from its compression. Let  $k_{s,x}$  be the axial stiffness and  $x_{s,\text{pre}}$  the compression at  $x_s = 0$ . Closing the secondary by  $x_s$  lets the spring extend by the same amount, so its remaining compression is  $x_{s,\text{pre}} - x_s$ . Hence,

$$F_{\text{spring},x_s} = k_{s,x} (x_{s,\text{pre}} - x_s). \quad (3.2.31)$$

This force and the torque in Equation (3.2.26) come from the same installed spring. Following the reduced decomposition of Kim et al. (2002), the model represents its axial compression and torsional twist with separate linear laws. This use of prescribed, uncoupled spring laws is part of Assumption 3 (Idealized Actuator Mechanics); compression–twist coupling, hysteresis, and end-contact friction are not retained.

**Belt opening reaction** The belt opening force follows the same projection already used on the primary. Let  $N_s$  be the integrated normal load over both secondary faces. Under Assumption 12 (Symmetric Sheave-Face Sharing), the movable face carries one half of this total load, giving

$$F_{\text{belt},x_s} = \frac{N_s}{2} \cos \beta. \quad (3.2.32)$$

As on the primary,  $N_s$  is not determined by this local free body. It remains a coupled unknown supplied by the belt-contact and remaining system equations.

### Completed Local Secondary Balance

We now have a relation for every term in the axial free body. Substitute them into Equation (3.2.18) and collect the terms multiplying  $\ddot{x}_s$ :

$$\begin{aligned}
\left[ m_s + I_{s,M} \left( \frac{d\theta_s}{dx_s} \right)^2 \right] \ddot{x}_s = & \left[ \frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) - I_{s,M} \dot{\omega}_s \right] \frac{d\theta_s}{dx_s} \\
& - I_{s,M} \frac{d\theta_s}{dx_s} \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \\
& + k_{s,x} (x_{s,\text{pre}} - x_s) - \frac{N_s}{2} \cos \beta.
\end{aligned} \tag{3.2.33}$$

The completed equation separates into four physical pieces:

- $m_s + I_{s,M}(d\theta_s/dx_s)^2$  is the local axial inertia. Just as the flyweight pivot inertia appeared in the primary balance, the helix makes the secondary's relative-rotational inertia part of the load that axial motion must accelerate;
- the first bracket contains the belt and torsional-spring torques, less the portion required to accelerate the movable assembly with the secondary shaft;
- the term proportional to  $\dot{x}_s^2$  accounts for a helix motion ratio that varies with position; and
- the last two terms are the direct spring closing force and belt opening force.

This is the axial equation of motion we set out to obtain. If its right-hand side remains zero and the sheave is initially stationary,  $x_s$  and  $\theta_s$  remain fixed while the two sheaves rotate together. A nonzero right-hand side accelerates the sheave axially, and  $\theta_s(x_s)$  fixes the accompanying relative rotation.

### Classical Limit and Relation to Existing Models

As on the primary, the full dynamic balance should recover the established actuator equilibrium when the shift mechanism is stationary. Consider a fixed secondary configuration at constant shaft speed,  $\dot{x}_s = \ddot{x}_s = \dot{\omega}_s = 0$ . Under the equal face-torque split adopted here, the helix force reduces to

$$F_{\text{helix},x_s} = \left[ \frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) \right] \frac{d\theta_s}{dx_s}. \tag{3.2.34}$$

Adding the direct spring and belt loads gives the corresponding equilibrium condition for the complete movable assembly:

$$\left[ \frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) \right] \frac{d\theta_s}{dx_s} + k_{s,x} (x_{s,\text{pre}} - x_s) = \frac{N_s}{2} \cos \beta. \tag{3.2.35}$$

These equations expose the classical mechanism directly. The axial spring supplies one closing force, while the belt torque and torsional-spring torque are converted into a second closing force by the helix motion ratio. The factor  $1/2$  is not a property of the helix itself; it follows from the equal face-torque closure used here. A face-resolved contact model would instead use the torque  $\tau_{s,M}$  actually applied to the movable face.

The important comparison is how this equilibrium relation is used. In the present formulation, [Equations \(3.2.34\) and \(3.2.35\)](#) are stationary limits of a dynamic balance. In much of the mechanically actuated CVT literature, an equivalent algebraic relation remains the actuator closure while the transmission is shifting.

The formulation of Kim et al. (2002) is a direct example. Their driven thrust is the sum of a spring force and a torque-cam force obtained from instantaneous equilibrium. The cam term retains the applied torque, torsional-spring load, cam geometry, and cam friction. Suppressing that friction and adopting the equal face-torque split used here gives the same spring-plus-torque-cam structure as Equation (3.2.35). Their shift transient is not obtained from the movable secondary’s equation of motion, but from a separately identified first-order ratio law. Kim et al. therefore retain contact friction that is absent here, but do not use the secondary assembly’s inertia to determine its shift acceleration.

The design-oriented sources examined here apply the same configuration-by-configuration logic at different levels of detail. Messick (2018) calculates the secondary thrust required by a steady-state belt solution and uses it as a target for helix design, explicitly excluding the shift response time. Skinner (2020) calculates the compression-spring, torsional-spring, and torque-feedback contributions to secondary clamping algebraically. Arango and Muñoz Alzate (2020) construct the torque-sensing cam profile from the transmitted torque, spring force, required axial force, and axial displacement over the operating range. These treatments are useful for sizing and tuning the mechanism, but they do not determine its motion from a movable-sheave equation of motion.

Among the reviewed sources, Sorge and Cammalleri (2011) provide the most detailed treatment of the belt–face–helix interaction during shifting. Their distributed contact model allows the fixed and movable faces to rotate at different speeds and determines the torque  $M_{N,s}$  carried by the movable face. The resulting driven-side actuator relation combines the spring load with a helix contribution having the same torque-times-motion-ratio structure as Equation (3.2.34). It therefore replaces the assumed  $\tau_s/2$  with a face-resolved torque and can capture substantial departures from an equal split.

That contact calculation is shift-aware, but the secondary actuator itself is still closed quasi-statically. Sorge and Cammalleri prescribe the instantaneous radius and the dimensionless shift speed  $\rho = \dot{r}/(\omega_f r)$ , then solve for the corresponding belt-contact state. A nonzero  $\rho$  changes the relative face speeds, sliding directions, contact tractions, and face-torque split, but it is not produced by the forces on the movable secondary. Because no axial or relative-rotational equation of motion is written for the movable half-pulley, its translational inertia and helix-reflected rotational inertia are absent by construction.

The models therefore retain complementary parts of the physics. Kim et al. retain cam friction, while Sorge and Cammalleri resolve the two belt–face contacts and their unequal torques. The present reduced model retains neither capability. It instead carries the movable assembly’s rotational balance through the general constraint  $\theta_s(x_s)$  and solves its free axial motion. This introduces the shaft-acceleration coupling, the reflected inertia  $I_{s,M}(d\theta_s/dx_s)^2$ , and the velocity-squared term generated when the helix motion ratio varies with position. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent explicit projection of the movable side’s rotational inertia through a mechanical secondary helix was identified.

This distinction is deliberately narrow. The rigid-hardware, ideal-actuator, and symmetric-face idealizations are stated explicitly in Assumption 2 (Rigid Pulley and Actuation Hardware), Assumption 3 (Idealized Actuator Mechanics), and Assumption 12 (Symmetric Sheave-Face Sharing). They leave the compliant and frictional actuator details, and the face-resolved load redistribution of the models above, outside the present balance. The comparison is therefore not a claim of greater local contact fidelity; it isolates the added treatment of the movable assembly’s inertia and helix-constrained dynamics.

### 3.2.3 Coupling Through the Shared Shift Coordinate

The primary and secondary balances have been derived in their natural local coordinates,  $x_p$  and  $x_s$ . The fixed-length belt geometry makes those motions compatible. Choosing the primary displacement as the shared shift coordinate leaves the primary balance unchanged and supplies the secondary motion through the map  $x_s = x_s(s)$ .

#### Local-to-Shared Kinematics

Because the shared coordinate is the primary displacement,

$$x_p = s, \quad \dot{x}_p = \dot{s}, \quad \ddot{x}_p = \ddot{s}. \quad (3.2.36)$$

The secondary displacement follows from the fixed-length belt map. Its first two time derivatives are

$$x_s = x_s(s), \quad \dot{x}_s = \frac{dx_s}{ds} \dot{s}, \quad \ddot{x}_s = \frac{dx_s}{ds} \ddot{s} + \frac{d^2x_s}{ds^2} \dot{s}^2. \quad (3.2.37)$$

Positive  $x_p$  and  $x_s$  both denote sheave closing. Increasing  $s$  closes the primary but opens the secondary, so  $dx_s/ds < 0$ . If the belt geometry is nonlinear, this motion ratio changes through the shift and  $d^2x_s/ds^2 \neq 0$ . Both derivatives are determined by the belt-geometry relation established earlier.

#### Two Conditions on One Shift Motion

Substituting Equation (3.2.36) into Equation (3.2.14) gives the primary condition directly:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div> <b>Primary Condition on the Shared Shift Motion</b> </div> <div> <b>(3.2.38)</b> </div> </div> <hr style="margin: 10px 0;"/> $  \begin{aligned}  \left[ m_p + I_f \left( \frac{dq_f}{ds} \right)^2 \right] \ddot{s} = & \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{ds} \\  & - I_f \frac{dq_f}{ds} \frac{d^2q_f}{ds^2} \dot{s}^2 \\  & - k_p (x_{p,\text{pre}} + s) - \frac{N_p}{2} \cos \beta.  \end{aligned}  $ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Because  $x_p = s$ , this is simply the local primary balance written using the shared coordinate.

The secondary condition follows just as directly. Substituting Equation (3.2.37) into Equation (3.2.33) and collecting terms gives

$$\begin{aligned}
 & \left[ m_s + I_{s,M} \left( \frac{d\theta_s}{dx_s} \right)^2 \right] \frac{dx_s}{ds} \ddot{s} \\
 &= \frac{d\theta_s}{dx_s} \left[ \frac{\tau_s}{2} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) - I_{s,M} \dot{\omega}_s \right] \\
 & \quad + k_{s,x} (x_{s,\text{pre}} - x_s) - \frac{N_s}{2} \cos \beta \\
 & \quad - \left\{ \left[ m_s + I_{s,M} \left( \frac{d\theta_s}{dx_s} \right)^2 \right] \frac{d^2 x_s}{ds^2} \right. \\
 & \quad \quad \left. + I_{s,M} \frac{d\theta_s}{dx_s} \frac{d^2 \theta_s}{dx_s^2} \left( \frac{dx_s}{ds} \right)^2 \right\} \dot{s}^2.
 \end{aligned}$$

Here  $x_s$ ,  $\theta_s$ , and their local derivatives are evaluated at  $x_s = x_s(s)$ . Since  $dx_s/ds < 0$ , a force that closes the secondary drives the shift toward decreasing  $s$ , thereby opening the primary. Conversely, the secondary belt force acts in the opening direction and drives the shift toward increasing  $s$ . The kinematic map therefore carries the inverse coupling between the two pulleys directly into the acceleration balance.

The two terms proportional to  $\dot{s}^2$  have different geometric origins. The term containing  $d^2 x_s/ds^2$  appears because the belt geometry can require a changing amount of secondary travel per unit primary travel. The term containing  $d^2 \theta_s/dx_s^2$  is inherited from the local secondary balance and appears when the helix motion ratio varies with position; substituting  $\dot{x}_s = (dx_s/ds)\dot{s}$  supplies its squared local-to-shared motion ratio.

The two balances are not added into one axial free body. They remain separate conditions on the same shift acceleration: the same  $\ddot{s}$  must satisfy both movable assemblies. The belt reactions  $N_p$  and  $N_s$ , together with the coupled shaft and contact quantities, allow both conditions to hold. Those unknowns are supplied by the rotational, belt, and contact relations developed later.

### Relation to Existing Dynamic Mappings

Transient belt and pulley models provide broad precedent for dynamic ratio prediction. [Julió and Plante \(2011\)](#) discretize the rubber belt and integrate its motion, while [Ballew \(2015\)](#) allow both pulley speeds to evolve from the applied torques and inertias. In that model family, however, pulley axial force is prescribed and the radius follows from belt compression; a movable-sheave equation  $m_j \ddot{x}_j = \sum F_{j,x}$  is not integrated.

Movable-sheave dynamics and shared-coordinate reductions also predate the present model. [Duan et al. \(2016\)](#) retain movable-pulley mass in a chain CVT and couple the pulley motions through chain deformation and tension. Most directly, [da Silva et al. \(2023\)](#) write axial equations for both movable sheaves of a mechanically actuated rubber-belt CVT, impose a linear relation between their displacements, and reduce the two translations to one shift equation. Thus transient shifting, axial sheave inertia, and a shared shift coordinate are not individually new.

The distinction here lies in how these ingredients are combined. The fixed-length belt geometry supplies the nonlinear map  $x_s(s)$ , so both  $dx_s/ds$  and  $d^2 x_s/ds^2$  remain in the secondary acceleration.

At the same time, the flyweight pivot inertia and the movable secondary's relative-rotational inertia remain tied to their own mechanism maps,  $q_f(x_p)$  and  $\theta_s(x_s)$ . The result is a pair of local free-body balances expressed as simultaneous conditions on one shift acceleration.

This reduction also defines the tradeoff. It does not retain the spatial belt deformation fields of [Julió and Plante \(2011\)](#) and [Ballew \(2015\)](#), or the prescribed-shift, face-resolved torque distribution of [Sorge and Cammalleri \(2011\)](#). It instead keeps the actuator and movable-sheave inertias explicit while closing the nominal belt geometry without spatial discretization. Among the reviewed mechanically actuated rubber-belt formulations, no equivalent combination of these features was identified.

### 3.2.4 What the Axial Balances Establish

The primary and secondary now have complete axial equations of motion in their natural coordinates. On the primary, shaft speed produces a closing action because the flyweight centres of mass move outward as the movable sheave closes. Part of that action accelerates the flyweights about their pivots before the remainder is transmitted to the movable assembly. On the secondary, the belt and torsional-spring torques act through the helix. Part of their action changes the rotation of the movable sheave relative to the shaft-side assembly before the remainder appears as axial force.

Although the two actuators are physically different, their constrained motions produce the same underlying structure. The map  $q_f(x_p)$  reflects the flyweights' pivot inertia into the primary axial coordinate, while the helix map  $\theta_s(x_s)$  reflects the movable-sheave rotational inertia into the secondary coordinate. In each case, the local motion ratio contributes an effective axial inertia, and variation of that ratio with position produces a velocity-squared term. The primary closing action is additionally set by  $dr_f/dx_p$ , while the secondary balance couples to the shaft acceleration because its movable sheave rotates relative to an assembly that is itself rotating.

The shared belt geometry adds a second layer of constrained motion. The primary map  $x_p(s) = s$  carries its local balance directly into the shift coordinate, whereas the secondary map  $x_s(s)$  reflects its local motion into the same  $\ddot{s}$  and contributes its own velocity-squared curvature term. Both free-body relations must therefore be satisfied by one shift motion.

They do not, however, determine that motion in isolation. At this stage, the two axial equations contain the five unknown quantities

$$\ddot{s}, \quad N_p, \quad N_s, \quad \tau_s, \quad \dot{\omega}_s.$$

The belt normal loads, transmitted secondary torque, and secondary shaft acceleration follow only when the rotational and belt-contact mechanics are added. The two balances derived here therefore form the axial part of one simultaneous dynamic solution for the engaged CVT.

This completes the axial force description. Attention now turns from translation along the pulley shafts to rotation about them.

## 3.3 Rotational Dynamics of the Pulley Assemblies

The preceding section established how forces along the pulley shafts move the sheaves, but its equations still contain quantities belonging to rotation about those shafts. The primary flyweight force depends on the primary speed  $\omega_p$ , while the secondary balance contains the transmitted torque  $\tau_s$  and the secondary acceleration  $\dot{\omega}_s$ . The axial equations can use these quantities, but cannot supply them.

The primary pulley is carried around the primary shaft at  $\omega_p$ , while the secondary rotates about its own shaft at  $\omega_s$ . These are separate rotations, so their speeds need not change together. The mechanisms inside each pulley also move relative to these underlying shaft rotations. The primary flyweights pivot as the primary rotates, while the secondary movable sheave translates and rotates relative to its shaft through the helix. These internal motions are already tied to the shift coordinate, but they also change how angular momentum is distributed within each pulley as the CVT shifts.

The remaining question is therefore what governs the evolution of  $\omega_p$  and  $\omega_s$ , and consequently how shifting affects that evolution.

### 3.3.1 Following the Torque

To determine how  $\omega_p$  and  $\omega_s$  evolve, each pulley side requires a rotational equation of motion. That equation must account for the angular momentum carried about the corresponding shaft and the torques capable of changing it. The first step is therefore to identify the rotating systems represented by the two shaft speeds.

The primary-side system contains the primary shaft and the complete primary pulley assembly. The movable assembly and flyweights remain part of this system even while they move relative to the fixed sheave, since they continue to carry angular momentum about the primary axis. The secondary-side system is defined in the same way: it contains the secondary shaft and the complete secondary pulley assembly, including the helix-guided motion of the movable sheave. The belt passes between these two systems and is not included in either one.

The resulting division is shown in [Figure 3.10](#).

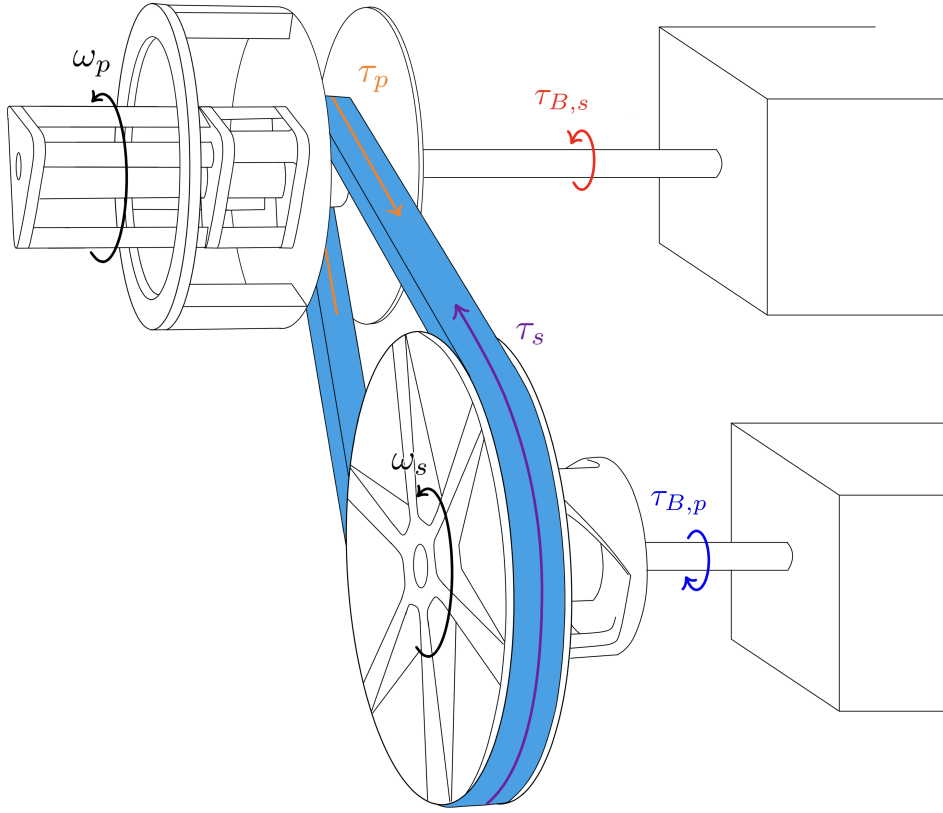


Figure 3.10: Rotating systems and torque path of the engaged CVT. The shaft-side boundaries represent the torque and equivalent inertia presented by the attached systems. The belt remains outside the two pulley-side systems and interacts with them through the contact torques  $\tau_p$  and  $\tau_s$ . The arrows show the signed directions associated with forward power transfer.

Begin with the primary. When its speed changes, the primary shaft must accelerate not only the pulley itself, but also the rotating hardware attached beyond the CVT boundary. As established in [Section 2.1](#), that external hardware is represented at the shaft by the equivalent inertia  $I_{B,p}$ , while  $\tau_{B,p}$  is the torque it applies to the primary side. The remaining external interaction is the belt, which exerts the torque  $\tau_p$  on the primary pulley.

The secondary follows the same construction. Its attached hardware contributes the equivalent inertia  $I_{B,s}$  and applies the boundary torque  $\tau_{B,s}$ , while the belt exerts  $\tau_s$  on the secondary pulley.

The signs follow directly from the convention established in [Section 2.3.1](#). During forward power transfer, the primary boundary drives the primary pulley and the belt resists it. The belt then drives the secondary pulley against the resisting secondary boundary. Thus,

$$\tau_{B,p} > 0, \quad \tau_p < 0, \quad \tau_s > 0, \quad \tau_{B,s} < 0.$$

Although the primary and secondary contain different mechanisms, their rotational balances now have the same outer structure. Let  $j \in \{p, s\}$  denote the pulley side under consideration.

The angular momentum on side  $j$  has two contributions. The fixed equivalent boundary inertia rotates at  $\omega_j$ , so it carries angular momentum  $I_{B,j}\omega_j$ . Let  $H_{\text{CVT},j}$  denote the angular momentum carried by the corresponding shaft and pulley assembly, including the internal motions of its components. The total is therefore

$$H_j = I_{B,j}\omega_j + H_{\text{CVT},j}. \quad (3.3.1)$$

Bearing, seal, windage, and other unmodeled resisting torques are omitted under [Assumption 15 \(Negligible Unmodeled Parasitic Losses\)](#). Within the resulting pulley-side free body, the applied boundary torque and belt torque are therefore the only external torques retained. Newton's second law for rotation gives

$$\tau_{B,j} + \tau_j = \frac{dH_j}{dt}.$$

Substituting [Equation \(3.3.1\)](#) writes this as

$$\tau_{B,j} + \tau_j = \frac{d}{dt}(I_{B,j}\omega_j + H_{\text{CVT},j}).$$

The equivalent boundary inertia is fixed, so differentiating yields the common pulley-side equation of motion:

|                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Common Pulley-Side Rotational Balance</b></div> <div><b>(3.3.2)</b></div> </div> <hr style="border: 0.5px solid black; margin: 10px 0;"/> $\tau_{B,j} + \tau_j = I_{B,j}\dot{\omega}_j + \frac{dH_{\text{CVT},j}}{dt}, \quad j \in \{p, s\}.$ |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The way shifting changes the angular momentum carried within a pulley is contained in  $H_{\text{CVT},j}$ . On the primary, the flyweights redistribute mass relative to the shaft as they pivot. On the secondary, the movable sheave rotates relative to the shaft-side assembly as it follows the helix. The two pulley sides therefore share the balance in [Equation \(3.3.2\)](#), but require separate expressions for their internal angular momentum. Beginning with the primary, each assembly can now be examined in turn.

### 3.3.2 Primary Pulley Assembly

The primary is the simpler of the two pulley assemblies because nearly all of its hardware shares the shaft speed  $\omega_p$ . The shaft, sheaves, ramp carrier, and other coaxial components may translate during a shift, but that translation is along the shaft and therefore does not change their distance from its axis. Their combined shaft-axis inertia can consequently be collected into one constant quantity,  $I_{p,\text{const}}$ .

The flyweights are different. They are carried around the primary shaft at  $\omega_p$ , but their mass moves radially as they pivot. The same motion was already used in [Section 3.2.1](#) to determine the centrifugal action that closes the primary. There, the flyweight geometry was summarized by the shaft-axis inertia map  $J_{f,p}(x_p)$ . Since  $x_p = s$ , that map is now  $J_{f,p}(s)$ .

It is useful to recall what this inertia represents. The quantity  $J_{f,p}$  measures the flyweight mass distribution about the *primary shaft*. It is therefore distinct from the constant pivot-axis inertia  $I_f$  that appeared in the flyweight moment balance of Equation (3.2.3). The two quantities describe the same flyweights about different axes and enter different equations of motion.

At a given shift position, the angular momentum carried inside the primary CVT boundary is therefore

$$H_{\text{CVT},p} = [I_{p,\text{const}} + J_{f,p}(s)] \omega_p. \quad (3.3.3)$$

If the shift position were fixed, differentiating this expression would simply multiply the current inertia by  $\dot{\omega}_p$ . During a shift, however,  $J_{f,p}$  changes as well. Because its only configuration variable is  $s$ , differentiating Equation (3.3.3) gives

$$\frac{dH_{\text{CVT},p}}{dt} = [I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p + \frac{dJ_{f,p}}{ds} \dot{s} \omega_p. \quad (3.3.4)$$

The two terms describe different ways of changing the same angular momentum. The first accelerates the primary with the flyweights held at their current configuration. The second changes the angular momentum because the flyweight mass itself moves toward or away from the shaft while already rotating.

Substituting this result into the common pulley-side balance from Equation (3.3.2) gives the primary equation of motion:

|                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div> <b>Primary Pulley Rotational Balance</b> </div> <div> <b>(3.3.5)</b> </div> </div> <hr style="margin: 10px 0;"/> $\begin{aligned} \tau_{B,p} + \tau_p = & [I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p \\ & + \frac{dJ_{f,p}}{ds} \dot{s} \omega_p. \end{aligned}$ |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The connection to the axial flyweight balance is now explicit. In Equation (3.2.38), the derivative  $dJ_{f,p}/ds$  determined how shaft rotation produces a centrifugal closing action. Here, the same derivative determines how that closing motion changes the shaft-axis angular momentum. The two appearances are not separate flyweight corrections: they are the axial and rotational consequences of the same radial redistribution of flyweight mass.

For the conventional primary mechanism, outward flyweight motion increases  $J_{f,p}$ . During an upshift,  $\dot{s} > 0$ , so the rotating primary must supply the angular momentum required by that outward motion in addition to accelerating the shaft. During a backshift the redistribution reverses, and the flyweights return angular momentum to the shaft-side motion.

If the shift is held fixed,  $\dot{s} = 0$ , and the flyweight mass distribution no longer changes. The additional term vanishes and the familiar fixed-inertia form is recovered:

$$\tau_{B,p} + \tau_p = [I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p, \quad \dot{s} = 0. \quad (3.3.6)$$

### 3.3.3 Secondary Pulley Assembly

The secondary differs from the primary in one important way. During a shift, the complete secondary remains one mechanical assembly, but not every part of that assembly has the same angular speed.

The secondary shaft, fixed sheave, and hardware locked to them rotate at  $\omega_s$ . Let their combined shaft-axis inertia inside the CVT boundary be  $I_{s,F}$ . The movable sheave and the components rigidly attached to it have inertia  $I_{s,M}$ . As established while deriving the secondary axial balance, the helix makes this movable assembly rotate relative to the shaft-side assembly as it translates. Its absolute angular speed is therefore

$$\omega_{s,M} = \omega_s + \frac{d\theta_s}{dx_s} \dot{x}_s, \quad (3.3.7)$$

which is the relation already obtained in Equation (3.2.24). The angular momentum carried by the secondary must therefore retain the two angular speeds separately:

$$H_{\text{CVT},s} = I_{s,F}\omega_s + I_{s,M}\omega_{s,M}. \quad (3.3.8)$$

The free body here is larger than the movable-sheave free body used in Section 3.2.2. There, only the movable assembly was isolated, so the spring and helix torques crossed the boundary and had to appear explicitly. Here, the shaft-side and movable-side hardware are enclosed together. The torque exerted by the helix on the movable side is accompanied by an equal-and-opposite torque on the shaft-side assembly, and both actions lie inside the same free body. The torsional spring produces the same kind of internal pair. These torques therefore cancel from the moment balance of the complete secondary. Their mechanical effect has not disappeared: the helix still allows the movable side to rotate at  $\omega_{s,M}$  rather than  $\omega_s$ , and that difference remains in Equation (3.3.8).

The common pulley-side balance in Equation (3.3.2) requires the time rate of change of the angular momentum inside this boundary. Because Equation (3.3.8) contains two angular velocities, differentiating it requires the corresponding two angular accelerations:

$$\frac{dH_{\text{CVT},s}}{dt} = I_{s,F}\dot{\omega}_s + I_{s,M}\dot{\omega}_{s,M}. \quad (3.3.9)$$

The first acceleration,  $\dot{\omega}_s$ , is the shaft acceleration governed by the present rotational balance. The second is not another independent degree of freedom. The helix already tied the movable-side rotation to the axial sheave motion in Section 3.2.2. In particular, differentiating the helix kinematics there gave Equation (3.2.25):

$$\dot{\omega}_{s,M} = \dot{\omega}_s + \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2. \quad (3.3.10)$$

The rotational balance has therefore led back to a kinematic relation already established by the axial mechanism. Substituting Equation (3.3.10) into Equation (3.3.9) gives

$$\begin{aligned} \frac{dH_{\text{CVT},s}}{dt} = & (I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \left[ \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right]. \end{aligned} \quad (3.3.11)$$

Substituting this rate into the common pulley-side rotational balance in Equation (3.3.2) produces the local secondary equation of motion:

|                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-between;"> <div><b>Secondary Pulley Rotational Balance</b></div> <div><b>(3.3.12)</b></div> </div> <hr style="border: 0.5px solid black; margin: 10px 0;"/> $\begin{aligned} \tau_{B,s} + \tau_s = & (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \left[ \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right]. \end{aligned}$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The first term accelerates the rotation shared by the complete secondary. The bracketed terms change only the movable side's rotation relative to the shaft. They are the rotational counterpart of the helix-reflected inertia already encountered in the secondary axial balance.

The complete CVT uses the shared shift coordinate rather than  $x_s$  as an independent coordinate. The required mapping was already established in Equation (3.2.37). Substituting

$$\dot{x}_s = \frac{dx_s}{ds} \dot{s}, \quad \ddot{x}_s = \frac{dx_s}{ds} \ddot{s} + \frac{d^2x_s}{ds^2} \dot{s}^2$$

into Equation (3.3.12) gives the same rotational balance in the coordinate used by the complete model:

$$\begin{aligned} \tau_{B,s} + \tau_s = & (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s \\ & + I_{s,M} \frac{d\theta_s}{dx_s} \frac{dx_s}{ds} \ddot{s} \\ & + I_{s,M} \left[ \frac{d\theta_s}{dx_s} \frac{d^2x_s}{ds^2} + \frac{d^2\theta_s}{dx_s^2} \left( \frac{dx_s}{ds} \right)^2 \right] \dot{s}^2. \end{aligned} \quad (3.3.13)$$

All local helix quantities in this expression are evaluated at  $x_s = x_s(s)$ . Its signs follow directly from the kinematics already chosen. For the conventional helix,  $d\theta_s/dx_s > 0$ , while the belt geometry gives  $dx_s/ds < 0$  over the engaged shift range. Hence

$$\frac{d\theta_s}{ds} = \frac{d\theta_s}{dx_s} \frac{dx_s}{ds} < 0.$$

During an upshift,  $\dot{s} > 0$ , so  $\dot{\theta}_s < 0$ : the secondary opens and the movable side rotates more slowly than the shaft-side assembly. The sign of the corresponding angular-momentum term is therefore carried by the mechanism maps rather than assigned separately in the rotational balance.

If the shift is held fixed,  $\dot{s} = \ddot{s} = 0$ . The helix then produces no relative motion, and the fixed and movable sides rotate together. The balance reduces to

$$\tau_{B,s} + \tau_s = (I_{B,s} + I_{s,F} + I_{s,M}) \dot{\omega}_s. \quad (3.3.14)$$

### 3.3.4 What the Rotational Balances Establish

The primary and secondary now have complete shaft-axis equations of motion. Both remain direct statements of  $\sum \tau = dH/dt$ . The additional terms appear because the hardware that actuates the pulleys continues to move while the pulley assemblies rotate, so shifting changes the angular momentum stored inside the assemblies themselves.

On the primary, this connection is carried by the same flyweight inertia map  $J_{f,p}(s)$  that appeared in the axial derivation. There,  $dJ_{f,p}/ds$  contributes to the centrifugal closing action because the flyweights move outward as the primary closes. Here,  $J_{f,p}(s)$  and its derivative determine how much shaft-axis angular momentum those same flyweights carry as their radial mass distribution changes. The axial and rotational balances are therefore two views of the same flyweight motion: one asks how rotation drives the movable sheave, while the other asks how that moving flyweight mass changes the rotational response of the pulley.

The secondary shows the same reciprocity through  $I_{s,M}$  and the helix map  $\theta_s(x_s)$ . In the axial balance, the helix reflects the movable assembly's rotational inertia into the sheave motion, so part of the available torque is required to change the movable side's rotation before the remainder appears as axial force. In the rotational balance, that same relative motion means the movable side carries angular momentum at a speed different from the shaft-side assembly, so the axial shift acceleration enters the shaft-axis balance. Again, the two equations are describing one constrained piece of hardware from different free bodies.

Mechanical actuation therefore creates a direct coupling between the axial and rotational dynamics inside each pulley assembly. This is distinct from the coupling carried by the belt reaction. Belt contact can connect the two kinds of balance because its normal action loads the sheaves axially while its tangential action transmits torque about the shafts. The present terms add a second path: the flyweight and helix mechanisms that generate the clamping response also carry inertia, so their own motion appears directly in the rotational equations. The actuator is therefore part of both balances, rather than acting only as a source of clamping force whose influence reaches shaft rotation through the belt.

This distinction gives a precise comparison with earlier transient models. In the rubber-belt formulation of [Ballew \(2015\)](#), belt-contact torque changes pulley speed through a fixed lumped pulley inertia, while pulley axial force is supplied to the belt model. Axial and rotational behaviour can still interact through the belt solution, but motion of the clamping hardware does not alter the pulley angular momentum itself. [Duan et al. \(2016\)](#) provide a stronger adjacent precedent in a chain CVT: rotational dynamics, movable-pulley axial dynamics, chain motion, and contact forces are solved as a coupled system. Their shaft balance nevertheless retains a fixed pulley inertia, while the movable-pulley mass appears in a separate axial equation. The coupling is therefore carried through the chain, contact forces, and changing radius rather than through actuator motion entering both balances.

A mechanically actuated rubber-belt comparison is provided by [da Silva et al. \(2023\)](#), who retain axial equations for both movable sheaves and reduce their translations to a shared shift coordinate. Their rotational treatment uses a fixed primary-pulley inertia. This establishes direct precedent for mechanical clamping, movable-sheave inertia, and coupled shift motion, but not for the additional

shaft-axis angular-momentum terms produced by the moving actuator hardware.

The helical model of [Sorge and Cammalleri \(2011\)](#) provides a complementary secondary comparison. Their formulation resolves the relative motion of the driven sheaves and the generally unequal belt torques acting on the two faces in substantially greater local detail. That work therefore captures secondary helix and face-contact mechanics that are reduced here. The distinction in the present balance is different: the helix-constrained movable assembly is retained as part of the complete secondary angular momentum, so the same relative motion that appears in its axial dynamics also appears in the shaft-axis equation.

Taken together, these comparisons show that axial-rotational coupling itself is not the new ingredient. Existing formulations can produce that coupling through belt or chain reactions, changing radii, and shared shift motion. Among the reviewed reduced-order formulations for mechanically actuated rubber-belt CVTs, however, no equivalent pair of pulley balances was identified in which the moving clamping hardware itself provides this additional inertial coupling on both sides: the primary flyweights change their shaft-axis mass distribution as they generate closing action, while the secondary movable assembly changes its shaft-axis angular momentum through the same helix motion that produces its axial response.

The result is a single mechanical picture of each pulley rather than separate axial and rotational descriptions joined only through transmitted load. The same mechanism maps that determine how the pulley clamps also determine how its internal motion changes the shaft dynamics. When those actuator motions cease, the additional terms vanish and the familiar fixed-configuration inertia balances are recovered.

### 3.4 Belt Transport and Pulley Contact Mechanics

The preceding sections established the axial and rotational equations of motion for the two pulley assemblies, but both stop at the same remaining mechanical body: the belt. The axial balances still contain the belt normal loads  $N_p$  and  $N_s$ , while the rotational balances contain the belt torques  $\tau_p$  and  $\tau_s$ . The belt transport speed  $v_b$  is also retained as a state, but its acceleration has not yet been established.

So far, each of these quantities has appeared only through the action of the belt on one of the pulley assemblies. Newton's third law gives the same interactions from the opposite side. The force and torque exerted by the belt on a pulley are accompanied by equal-and-opposite actions on the belt itself. The remaining mechanics can therefore be viewed from the belt rather than from either pulley.

This change of free body is important because the belt is not two independent contacts. It is one closed body connecting both pulley assemblies. Whatever forces and torques are exchanged at the primary must coexist with those at the secondary and with the motion of the belt between them. The belt mechanics must therefore be self-consistent around the complete loop.

#### 3.4.1 Complete-Belt Transport

The belt transport speed provides the natural starting point. The belt speed  $v_b$  can change only if something pushes or pulls the belt along its path. Those tangential actions come from the primary and secondary pulley contacts. The belt's transport acceleration must therefore come from the two pulley wraps.

At either wrap, the mechanism is simple. The pulley contact has both normal and tangential components. Normal contact acts perpendicular to the local belt path, so it cannot push or pull the belt in the direction that defines  $v_b$ . Frictional traction, by contrast, is spread over the contact and acts tangent to the belt at every point. Traction acting in the direction of belt travel does positive work on the belt, while traction acting against belt travel does negative work and removes energy from it. Moreover, when the complete closed belt is considered, the tensions between adjacent belt sections are internal and cancel. The primary and secondary tractions are therefore the only external actions that can change the belt's transport speed.

Each small traction force also acts at a radius from the shaft and therefore produces a small moment on the pulley. Adding these moments around the wrap gives the contact torque  $\tau_j$  already used in the pulley rotational balance. Thus,  $\tau_j$  is simply the accumulated turning effect of the same tangential contact forces that can change  $v_b$ .

Newton's third law places equal-and-opposite traction forces on the belt. Their total moment is therefore  $-\tau_j$ , and at the effective contact radius  $r_{j,\text{eff}}$ , they are equivalent to the net tangential pull  $-\tau_j/r_{j,\text{eff}}$ . If the pull at one wrap supplies exactly the power removed by the pull at the other, the belt simply carries that power between the pulleys. If the two actions do not balance, the resulting energy excess or deficit must be taken up by the belt.

The belt assumptions leave that imbalance only one place to go. The fixed length of **Assumption 6 (Fixed Belt Length)** excludes longitudinal extension and the strain energy it would store. The single transport coordinate of **Assumption 7 (Single Belt Transport Coordinate)** gives every belt region the same speed  $v_b$ , so one region cannot speed up independently of the rest. The belt's only retained longitudinal energy is therefore  $\frac{1}{2}m_b v_b^2$ . Net power into the belt increases  $v_b$ ; net power out decreases it.

The energy argument now explains why an imbalance between the pulley actions must change  $v_b$ . Newton's second law determines the corresponding signed acceleration directly, where the whole belt loop is treated as a single body. Place one boundary around the entire loop. From the force accounting above, the only external tangential actions on this free body are the two equivalent pulley pulls, acting on the complete belt mass  $m_b$ . The complete-belt transport balance is therefore

#### Complete-Belt Transport Balance

(3.4.1)

$$\begin{aligned}\sum F_{b,t}^{\text{ext}} &= -\frac{\tau_p}{r_{p,\text{eff}}} - \frac{\tau_s}{r_{s,\text{eff}}} \\ &= \frac{d}{dt}(m_b v_b) = m_b \dot{v}_b.\end{aligned}$$

This balance gives the basic longitudinal motion of the belt. If the two pulley contacts exert a net force in the positive belt direction, the belt speeds up; if they exert a net force in the opposite direction, it slows down. If the two actions balance, the belt can carry power between the pulleys without changing  $v_b$ .

### Interpreting the Transport Balance

As far as speeding up or slowing down the complete belt is concerned, this result is sufficient. The quantity  $v_b$ , however, describes only how quickly belt material travels along its prescribed path. Consider one marked piece of belt. It approaches a pulley along one straight span and leaves along

the other. Its direction of travel has changed. The complete-belt balance says how its speed changes, but contains nothing that explains this change in direction. Left to itself, the marked piece would continue straight past the pulley. Its motion around the wrap is therefore possible only because forces act on it throughout that turn.

The clearest way to separate these forces from the transport motion is to remove the motion altogether. Consider first the familiar case of a rope wrapped around a classical pulley. Pull equally on the two ends of the rope. If both pulls are increased together, the tension throughout the rope rises and the pulley becomes more heavily loaded, even though no tension difference has been introduced. The taut rope is redirected around the pulley, and in doing so it exerts a resultant load on it. The pulley supplies the corresponding equal-and-opposite contact reaction.

The same basic effect occurs when the CVT belt is wrapped around a sheave. Suppose the belt and pulley are stationary and the entry and exit tensions are both 500 N. Their difference is zero, so there is no transmitted torque or longitudinal acceleration, yet the tensioned belt still loads the pulley through the wrap. Raising both the entry and exit tensions to 1000 N changes neither the tension difference nor the belt's motion, but the belt now loads the pulley more strongly and the corresponding normal contact increases. The geometry and motion have not changed; the pulley loading has.

The loading described above is carried through the belt–sheave contact around the wrap. In the model, the resultant normal contact loads on the primary and secondary pulleys are denoted by  $N_p$  and  $N_s$ . Their axial components already appear in the primary and secondary movable-sheave balances; see [Equations \(3.2.14\)](#) and [\(3.2.33\)](#). The static example therefore has an important consequence: even with no transmitted torque or belt acceleration, changing the belt tension changes  $N_p$  and  $N_s$ , and can therefore change the shift balance. Determining these normal loads requires knowing the tension carried around each pulley. The remaining difficulty is that this tension is not generally constant around the pulley.

As established above, a pulley transfers torque through tangential traction distributed around its wrap. As the belt passes through the contact, this traction changes how strongly the belt is being pulled. The tension leaving the wrap therefore generally differs from the tension entering it. In the simplest steady, inertia-free picture, the transmitted torque is equal to the pulley radius multiplied by this tension difference. Retaining the inertia of the wrapped belt modifies that balance, but not the basic interpretation: torque transfer tells us how the tension changes through the contact.

The quantity needed to determine the normal loading is therefore itself changing as the belt moves through the wrap. Knowing the entry and exit tensions tells us only how that process begins and ends, not what tension is carried at each point between them. The normal contact must therefore be determined locally as well.

### 3.4.2 Local Equations of Motion on a Pulley Wrap

The previous subsection leaves one local question: how is the tension carried at a particular point in the wrap related to the contact force acting there? The next step is therefore to build the free-body diagram of an infinitesimal piece of belt within the wrap of either pulley  $j \in \{p, s\}$ .

Let  $\theta$  increase through the wrap in the positive belt-transport direction defined in [Section 2.3.1](#), and isolate the piece bounded by cuts at  $\theta$  and  $\theta + d\theta$ . Let  $T_j(\theta)$  denote the tension carried by the belt at angular position  $\theta$ . The tensions at the two cuts are therefore  $T_j(\theta)$  and  $T_j(\theta + d\theta)$ . To keep the labels in the free-body diagram compact, define

$$T_j \equiv T_j(\theta), \quad dT_j \equiv T_j(\theta + d\theta) - T_j(\theta). \quad (3.4.2)$$

The tension at the second cut can then be written exactly as  $T_j + dT_j$ . Thus,  $dT_j$  is the signed change in tension across the element: it is positive if the tension increases with  $\theta$  and negative if the tension decreases. No direction of change has yet been assumed.

Once the piece is cut free, the surrounding belt pulls on its two exposed ends. Under **Assumption 8 (Prescribed Planar Belt Path)**, the belt is perfectly flexible and follows the prescribed planar path. Each cut therefore carries only a longitudinal tension force directed along the local belt tangent; there is no transverse shear force or bending moment. At both cuts, the surrounding belt pulls away from the isolated piece.

Because the belt curves around the pulley, those two tension forces do not point in the same direction. The belt tangent turns through  $d\theta$  from one cut to the other. It is therefore convenient to draw the local directions at the midpoint of the element:  $\hat{\mathbf{e}}_r$  points radially outward from the pulley axis, while  $\hat{\mathbf{e}}_\theta$  points with increasing  $\theta$ . Each end tangent is inclined by  $d\theta/2$  from the midpoint tangent. The left-hand view in **Figure 3.11** shows this in-plane geometry.

The pulley also acts on the element through both sheave faces. To complete the free-body diagram, these contact forces must be added to the cut-end tensions. Their geometry cannot be represented clearly in the pulley plane, so the right-hand view in **Figure 3.11** shows the cross-section of the same element. Under **Assumption 4 (Fixed, Uniform Belt Cross-Section)**, its shape and mass centroid remain fixed. Under **Assumption 12 (Symmetric Sheave-Face Sharing)**, the two faces carry equal shares of the normal contact. Let  $dN_j$  denote the sum of the magnitudes of the two face-normal forces; each face therefore exerts a force of magnitude  $dN_j/2$ . Their axial components oppose one another and cancel, whereas their outward radial components add to  $dN_j \sin \beta$ .

The remaining contact action is tangential. Under **Assumption 10 (Gross Tangential Coulomb Traction)**, let  $dF_{t,j}$  denote the resultant tangential force exerted by both sheave faces on the element. It is defined as positive when it acts in the direction of increasing  $\theta$ ; its actual sign will be determined by the contact state. Because this force acts in the pulley plane, it appears in the left-hand view of **Figure 3.11**. The two views therefore complete a single free-body diagram: the left shows the forces acting around the wrap, while the right shows how the two face-normal forces produce the radial contact force appearing on the left.

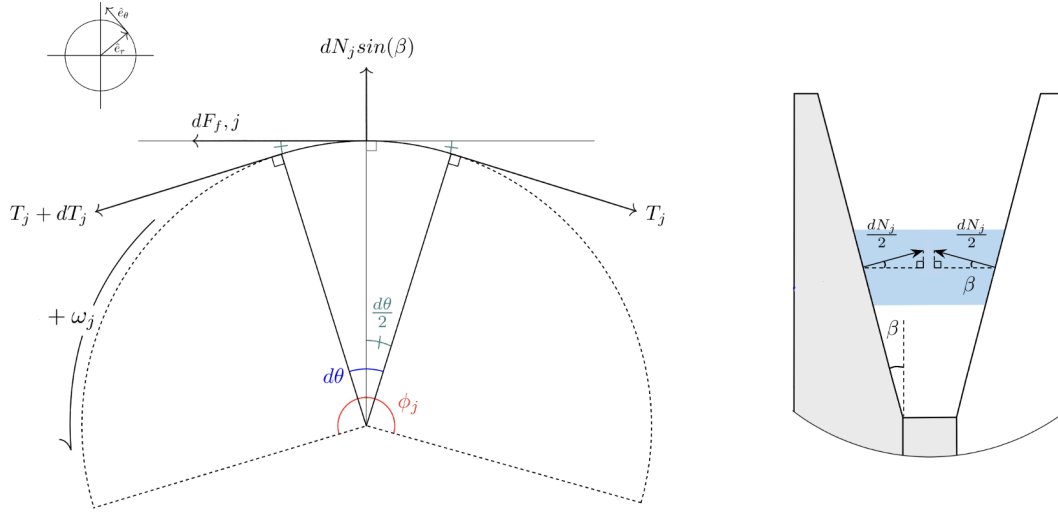


Figure 3.11: Two views of the same infinitesimal belt element, bounded by  $\theta$  and  $\theta + d\theta$  within the wrap of pulley  $j$ . The in-plane free-body diagram on the left shows the cut-end tensions  $T_j = T_j(\theta)$  and  $T_j + dT_j = T_j(\theta + d\theta)$ , the midpoint directions, the half-angle  $d\theta/2$ , the signed tangential contact force  $dF_{t,j}$ , and the outward radial contact force  $dN_j \sin \beta$ . The cross-section on the right shows the two face-normal forces of magnitude  $dN_j/2$ , whose axial components cancel and whose radial components add.

**Equations of motion for the infinitesimal element** With the free-body diagram complete, Newton's second law can now be assembled in the two local directions. The radial direction comes first because it contains the normal contact that motivated the local analysis.

In the radial direction, the two sheave-face forces supply the combined outward component  $dN_j \sin \beta$ . The surrounding belt pulls inward at both cuts. Since each end tangent is inclined by  $d\theta/2$  from the midpoint tangent, the corresponding inward components are  $T_j \sin(d\theta/2)$  and  $(T_j + dT_j) \sin(d\theta/2)$ . Taking outward as positive, Newton's second law therefore gives

$$dN_j \sin \beta - T_j \sin\left(\frac{d\theta}{2}\right) - (T_j + dT_j) \sin\left(\frac{d\theta}{2}\right) = dm_j a_{r,j}, \quad (3.4.3)$$

where  $dm_j$  is the mass of the element and  $a_{r,j}$  is its signed acceleration in the outward radial direction.

The same element must also satisfy Newton's second law in the tangential direction. The face-normal forces have no tangential component. At the cut at  $\theta$ , the tension pulls opposite increasing  $\theta$ , giving the component  $-T_j \cos(d\theta/2)$ . At the cut at  $\theta + d\theta$ , the tension pulls with increasing  $\theta$ , giving  $(T_j + dT_j) \cos(d\theta/2)$ . Including the signed tangential contact force  $dF_{t,j}$ , the tangential equation is therefore

$$(T_j + dT_j) \cos\left(\frac{d\theta}{2}\right) - T_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = dm_j a_{\theta,j}, \quad (3.4.4)$$

where  $a_{\theta,j}$  is the signed acceleration in the direction of increasing  $\theta$ .

The force side of each equation is now fully determined by the free-body diagram. Their right-hand sides still express the belt element's response only in the general form of mass times acceleration. Reducing them requires knowing how much belt mass lies within the angle  $d\theta$ , and how that material accelerates while it travels around a pulley whose wrap radius can simultaneously move as the CVT shifts. These quantities follow from the belt cross-section and the radius kinematics already established in [Section 3.1](#).

**Mass and motion of the belt element** The mass and acceleration of the element must be evaluated at the belt's mass centroid. As established in [Equations \(3.1.8\)](#) and [\(3.1.14\)](#),  $r_j$  locates the outer surface used to describe where the belt is seated, whereas  $r_{j,\text{cm}}$  locates the mass centroid. Under [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#), the centroid remains a fixed radial distance from the outer surface. Its radial velocity and acceleration are therefore the radius rate and acceleration already derived for the seated belt:

$$\dot{r}_{j,\text{cm}} = \dot{r}_j, \quad \ddot{r}_{j,\text{cm}} = \ddot{r}_j.$$

The element spans the angle  $d\theta$ , so its arc length at the mass centroid is  $r_{j,\text{cm}}d\theta$ . Multiplying this length by the belt cross-sectional area  $A_b$  from [Equation \(3.1.12\)](#) and the belt density  $\rho_b$  gives the element mass:

$$dm_j = \rho_b A_b r_{j,\text{cm}} d\theta. \quad (3.4.5)$$

The element can move in two directions at once. It moves radially with the shifting wrap at speed  $\dot{r}_j$ , while the belt's transport carries it tangentially at speed  $v_b$ . Its velocity is therefore

$$\mathbf{v}_{b,j} = \dot{r}_j \hat{\mathbf{e}}_r + v_b \hat{\mathbf{e}}_\theta. \quad (3.4.6)$$

Acceleration is the time derivative of this velocity. Both scalar speeds can change, but the local directions also change as the element travels around the pulley. Differentiating while retaining both effects gives

$$\begin{aligned} \mathbf{a}_{b,j} &= \frac{d}{dt} (\dot{r}_j \hat{\mathbf{e}}_r + v_b \hat{\mathbf{e}}_\theta) \\ &= \ddot{r}_j \hat{\mathbf{e}}_r + \dot{r}_j \dot{\hat{\mathbf{e}}}_r + \dot{v}_b \hat{\mathbf{e}}_\theta + v_b \dot{\hat{\mathbf{e}}}_\theta. \end{aligned} \quad (3.4.7)$$

The two basis-vector derivatives follow from how quickly the element moves around the wrap. At the centroidal radius, its tangential speed and angular rate are related by

$$v_b = r_{j,\text{cm}} \dot{\theta}, \quad \dot{\theta} = \frac{v_b}{r_{j,\text{cm}}}. \quad (3.4.8)$$

As  $\theta$  increases, the outward radial direction turns toward  $\hat{\mathbf{e}}_\theta$ , while the tangential direction turns toward  $-\hat{\mathbf{e}}_r$ . Multiplying these changes per unit angle by the angular rate in [Equation \(3.4.8\)](#) gives

$$\dot{\mathbf{e}}_r = \frac{v_b}{r_{j,\text{cm}}} \hat{\mathbf{e}}_\theta, \quad \dot{\mathbf{e}}_\theta = -\frac{v_b}{r_{j,\text{cm}}} \hat{\mathbf{e}}_r. \quad (3.4.9)$$

Substituting these basis rates into Equation (3.4.7) and collecting the radial and tangential terms gives

$$\mathbf{a}_{b,j} = \left( \ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}} \right) \hat{\mathbf{e}}_r + \left( \dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}} \right) \hat{\mathbf{e}}_\theta. \quad (3.4.10)$$

The acceleration components required by the two equations of motion are therefore

$$a_{r,j} = \ddot{r}_j - \frac{v_b^2}{r_{j,\text{cm}}}, \quad (3.4.11)$$

$$a_{\theta,j} = \dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}}. \quad (3.4.12)$$

The radial component contains the outward acceleration of the shifting wrap and the familiar inward centripetal acceleration produced as the belt turns around the pulley. The tangential component contains the change in belt speed and an additional contribution from the radial velocity changing direction as the element moves through the wrap. Together with Equation (3.4.5), these two components now provide the complete inertial sides of the radial and tangential balances.

**Radial equation of motion** We first return to the radial balance in Equation (3.4.3). Substituting the element mass from Equation (3.4.5) and the radial acceleration from Equation (3.4.11), then collecting the two inward tension components, gives

$$dN_j \sin \beta - (2T_j + dT_j) \sin\left(\frac{d\theta}{2}\right) = \rho_b A_b (r_{j,\text{cm}} \ddot{r}_j - v_b^2) d\theta. \quad (3.4.13)$$

The desired local quantity is the normal contact carried per unit wrap angle,  $dN_j/d\theta$ . Dividing Equation (3.4.13) by  $d\theta$  gives

$$\sin \beta \frac{dN_j}{d\theta} - (2T_j + dT_j) \frac{\sin(d\theta/2)}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \ddot{r}_j - v_b^2).$$

Only the tension term still depends on the finite angular width of the element. To put its sine factor into the standard small-angle form, write

$$\frac{\sin(d\theta/2)}{d\theta} = \frac{1}{2} \frac{\sin(d\theta/2)}{d\theta/2}.$$

Redistributing this factor of 1/2 through  $2T_j + dT_j$ , the radial balance can be written directly as

$$\sin \beta \frac{dN_j}{d\theta} - \left( T_j + \frac{dT_j}{2} \right) \frac{\sin(d\theta/2)}{d\theta/2} = \rho_b A_b (r_{j,\text{cm}} \ddot{r}_j - v_b^2).$$

The infinitesimal element is obtained by taking  $d\theta \rightarrow 0$ . Because  $dT_j = T_j(\theta + d\theta) - T_j(\theta)$ , continuity of the tension field gives  $dT_j \rightarrow 0$ . At the same time, the small-angle ratio tends to one:

$$T_j + \frac{dT_j}{2} \longrightarrow T_j, \quad \frac{\sin(d\theta/2)}{d\theta/2} \longrightarrow 1.$$

The complete local radial balance is consequently

$$\boxed{\sin \beta \frac{dN_j}{d\theta} = T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)}. \quad (3.4.14)$$

To see the physical meaning of this result most clearly, it is useful first to consider a stationary belt on a fixed wrap. Setting  $v_b = 0$  and  $\ddot{r}_j = 0$  gives

$$\sin \beta \frac{dN_j}{d\theta} = T_j. \quad (3.4.15)$$

We now recover the result anticipated from the physical picture: increasing the belt tension increases the normal loading carried per unit wrap angle. This remains true even when the tension is uniform through the contact, so there is no tension difference and no torque is transmitted. The normal loading therefore depends on the absolute tension level, not only on the tension difference across the wrap.

Returning to the full balance, motion modifies this stationary relationship. As the belt turns around the pulley, part of the inward tension is required to provide its centripetal acceleration, reducing the outward sheave reaction at a given tension. An outward wrap acceleration has the opposite effect and increases that reaction.

**Tangential equation of motion** Finally, we complete the tangential equation of motion introduced in Equation (3.4.4). Substituting the element mass from Equation (3.4.5) and the tangential acceleration from Equation (3.4.12) gives the complete balance

$$(T_j + dT_j) \cos\left(\frac{d\theta}{2}\right) - T_j \cos\left(\frac{d\theta}{2}\right) + dF_{t,j} = \rho_b A_b r_{j,\text{cm}} d\theta \left( \dot{v}_b + \frac{\dot{r}_j v_b}{r_{j,\text{cm}}} \right). \quad (3.4.16)$$

The desired result is again a local relation per unit wrap angle. Cancelling the common  $T_j$  contribution from the two end tensions, simplifying the acceleration term, and dividing through by  $d\theta$  gives

$$\frac{dT_j}{d\theta} \cos\left(\frac{d\theta}{2}\right) + \frac{dF_{t,j}}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b).$$

The remaining cosine records that the end tensions are not quite parallel on a finite element. In the infinitesimal limit, the two end tangents coincide and

$$\lim_{d\theta \rightarrow 0} \cos\left(\frac{d\theta}{2}\right) = \cos(0) = 1.$$

The cosine therefore becomes a factor of one; the tension-gradient term remains. The local tangential equation of motion is

$$\frac{dT_j}{d\theta} + \frac{dF_{t,j}}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \quad (3.4.17)$$

This balance identifies the two tangential forces acting per unit wrap angle: the change in belt tension and the traction supplied by the pulley. Their sum produces the tangential acceleration of the belt. What remains is to relate that tangential contact force to the normal contact already present in the radial balance.

Under **Assumption 10 (Gross Tangential Coulomb Traction)**, the sheave contact obeys Coulomb friction. While the belt sticks, the tangential force is a reaction that can act in either direction and take any magnitude up to the static limit. Once the belt slides, its magnitude is fixed by kinetic friction and its direction opposes the relative sliding. Both cases relate the tangential force on the element to the same normal-force increment  $dN_j$ , but with different rules for the allowable proportionality.

Rather than carrying a separate magnitude condition and direction sign through the remaining equations, define the signed traction utilization  $\lambda_j$  by

|                                                                                                                                                                                                                          |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Signed Traction Utilization</b>                                                                                                                                                                                       | <b>(3.4.18)</b> |
| $dF_{t,j} = \lambda_j dN_j, \quad \lambda_j = \begin{cases} \lambda_{j,\text{st}}, &  \lambda_{j,\text{st}}  \leq \mu_s, & \text{sticking,} \\ \sigma_j \mu_k, & \sigma_j \in \{-1, +1\}, & \text{sliding.} \end{cases}$ |                 |

The magnitude of  $\lambda_j$  gives the tangential force used per unit normal load, while its sign gives the direction of that force. Thus,  $\lambda_j > 0$  acts with increasing  $\theta$ ,  $\lambda_j < 0$  acts against it, and  $\lambda_j = 0$  represents normal contact with no tangential force. In sliding,  $\sigma_j$  selects the direction that opposes the relative motion, while  $\mu_k$  fixes the magnitude. Under **Assumption 11 (Wrap-Level Contact Resolution)**, one value of  $\lambda_j$  represents the complete wrap at a given instant.

Dividing  $dF_{t,j} = \lambda_j dN_j$  by  $d\theta$  and substituting it into **Equation (3.4.17)** gives the complete local tangential balance:

|                                                                                                                   |                 |
|-------------------------------------------------------------------------------------------------------------------|-----------------|
| $\frac{dT_j}{d\theta} + \lambda_j \frac{dN_j}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b).$ | <b>(3.4.19)</b> |
|-------------------------------------------------------------------------------------------------------------------|-----------------|

To expose the physical meaning of this result, it is useful again to remove the effects of motion. For a fixed wrap and steady belt speed,  $\dot{r}_j = 0$  and  $\dot{v}_b = 0$ , so the tangential inertia vanishes and the balance becomes

$$\frac{dT_j}{d\theta} = -\lambda_j \frac{dN_j}{d\theta}. \quad (3.4.20)$$

We now recover the expected local relationship: the pulley traction must be balanced by an equal and opposite tension gradient. Traction therefore forces the tension to evolve around the wrap, and integrating this local evolution produces the tension difference associated with transmitted torque.

Since the normal contact is compressive,  $dN_j/d\theta \geq 0$ : positive traction makes the tension fall as  $\theta$  increases, while traction in the opposite direction makes it rise. Retaining the full balance, some of the traction may be used to accelerate the belt, but the same underlying link between traction and tension evolution remains.

### 3.4.3 Tension Evolution Through a Pulley Wrap

The balances derived above describe the forces on one infinitesimal belt element at an arbitrary position within a pulley wrap. The next step is to carry that local description across the complete contact, so that the effect of every element from the wrap entrance to the wrap exit can be accumulated.

For pulley  $j$ , let  $\theta$  increase from 0 at the entrance to  $\phi_j$  at the exit. The belt enters with tension  $T_{j,\text{in}}$ , carries the tension  $T_j(\theta)$  at an intermediate position, and leaves with tension  $T_{j,\text{out}}$ :

$$0 \leq \theta \leq \phi_j, \quad T_j(0) = T_{j,\text{in}}, \quad T_j(\phi_j) = T_{j,\text{out}}.$$

Both  $T_j(\theta)$  and the normal loading  $dN_j/d\theta$  are needed throughout this interval. Tension provides the natural starting point:  $T_{j,\text{in}}$  is the entrance boundary value, and following its evolution through the wrap produces  $T_{j,\text{out}}$ . Once that tension field is known, the radial balance recovers the normal loading at every position.

The tangential balance, Equation (3.4.19), contains the desired tension gradient but remains coupled to  $dN_j/d\theta$ . Rearranging the radial balance, Equation (3.4.14), gives

$$\frac{dN_j}{d\theta} = \frac{1}{\sin \beta} [T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)]. \quad (3.4.21)$$

Substituting this expression into the tangential balance, then expanding and rearranging, gives

$$\begin{aligned} \frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin \beta} [T_j - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)] \\ = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \\ \frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin \beta} T_j = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) + \rho_b A_b \frac{\lambda_j}{\sin \beta} (v_b^2 - r_{j,\text{cm}} \ddot{r}_j). \end{aligned}$$

Factoring the common  $\rho_b A_b$  on the right gives the local equation that governs tension evolution through either pulley wrap:

|                                      |                 |
|--------------------------------------|-----------------|
| <b>Pulley-Wrap Tension Evolution</b> | <b>(3.4.22)</b> |
|--------------------------------------|-----------------|

|                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{dT_j}{d\theta} + \frac{\lambda_j}{\sin \beta} T_j = \rho_b A_b [r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b + \frac{\lambda_j}{\sin \beta} (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)], \quad j \in \{p, s\}.$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The two local balances have now been reduced to one first-order equation for  $T_j(\theta)$ .

**Solving the tension equation** Before Equation (3.4.22) can be integrated through the wrap, its coefficients must be considered along the spatial coordinate  $\theta$ . At a fixed instant, the circular wrap prescribed by Assumption 8 (Prescribed Planar Belt Path) gives one  $r_{j,\text{cm}}$ ,  $\dot{r}_j$ , and  $\ddot{r}_j$  for pulley  $j$ . Likewise, Assumption 7 (Single Belt Transport Coordinate) makes  $v_b$  and  $\dot{v}_b$  common throughout the belt, while Assumption 11 (Wrap-Level Contact Resolution) assigns one  $\lambda_j$  to the complete wrap. None of these quantities therefore varies with  $\theta$  during the integration.

To expose the structure of the equation, define the two constant coefficients

$$\begin{aligned} a_j &= \frac{\lambda_j}{\sin \beta}, \\ c_j &= \rho_b A_b [r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b + a_j (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)]. \end{aligned} \quad (3.4.23)$$

Then Equation (3.4.22) becomes

$$\frac{dT_j}{d\theta} + a_j T_j = c_j. \quad (3.4.24)$$

To integrate Equation (3.4.24), it is useful to combine the two terms on the left into a single derivative. The exponential  $e^{a_j \theta}$  is chosen because its derivative is  $a_j e^{a_j \theta}$ , matching the coefficient multiplying  $T_j$ . The product rule then gives

$$\frac{d}{d\theta} (e^{a_j \theta} T_j) = e^{a_j \theta} \left( \frac{dT_j}{d\theta} + a_j T_j \right). \quad (3.4.25)$$

Multiplying Equation (3.4.24) by  $e^{a_j \theta}$  therefore gives

$$\frac{d}{d\theta} (e^{a_j \theta} T_j) = c_j e^{a_j \theta}. \quad (3.4.26)$$

The entrance condition  $T_j(0) = T_{j,\text{in}}$  now supplies the lower boundary. Integrating from the wrap entrance to an arbitrary position  $\theta$  gives

$$\int_0^\theta \frac{d}{d\vartheta} (e^{a_j \vartheta} T_j(\vartheta)) d\vartheta = c_j \int_0^\theta e^{a_j \vartheta} d\vartheta. \quad (3.4.27)$$

For  $a_j \neq 0$ , evaluating the integrals yields

$$e^{a_j \theta} T_j(\theta) - T_{j,\text{in}} = \frac{c_j}{a_j} (e^{a_j \theta} - 1), \quad (3.4.28)$$

and isolating  $T_j(\theta)$  gives

$$T_j(\theta) = T_{j,\text{in}} e^{-a_j \theta} + \frac{c_j}{a_j} (1 - e^{-a_j \theta}). \quad (3.4.29)$$

Substituting the constant coefficients back into Equation (3.4.29) and rearranging gives the tension carried at every position through the wrap:

$$T_j(\theta) = T_{j,\text{in}} e^{-\frac{\lambda_j}{\sin \beta} \theta} + \rho_b A_b \left( v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin \beta}{\lambda_j} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \right) \left( 1 - e^{-\frac{\lambda_j}{\sin \beta} \theta} \right), \quad \lambda_j \neq 0. \quad (3.4.30)$$

At the wrap exit,  $\theta = \phi_j$  and  $T_j(\phi_j) = T_{j,\text{out}}$ . Evaluating Equation (3.4.30) at that endpoint gives

$$T_{j,\text{out}} = T_{j,\text{in}} e^{-\frac{\lambda_j}{\sin \beta} \phi_j} + \rho_b A_b \left( v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin \beta}{\lambda_j} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \right) \left( 1 - e^{-\frac{\lambda_j}{\sin \beta} \phi_j} \right), \quad \lambda_j \neq 0. \quad (3.4.31)$$

The closed form is written for  $\lambda_j \neq 0$  because its derivation divided by  $\lambda_j$ , but zero utilization presents no difficulty. For completeness, setting  $\lambda_j = 0$  directly in the tangential balance of Equation (3.4.19) gives

$$\frac{dT_j}{d\theta} = \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \quad (3.4.32)$$

Integrating from the wrap entrance to  $\theta$ , and then evaluating the result at  $\theta = \phi_j$ , gives

$$\begin{aligned} T_j(\theta) &= T_{j,\text{in}} + \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \theta, \\ T_{j,\text{out}} &= T_{j,\text{in}} + \rho_b A_b (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \phi_j. \end{aligned} \quad (3.4.33)$$

**Interpreting the wrap solution** The local balances have now been carried across the complete wrap. For a specified entrance tension and instantaneous wrap state, Equation (3.4.30) gives the tension at every position, while Equation (3.4.31) gives the resulting exit tension. Within the constant-wrap description established by Assumption 11 (Wrap-Level Contact Resolution), the solution has therefore retained every transient term admitted by the model: belt acceleration, radial migration, radial acceleration, and centripetal inertia. These instantaneous quantities are uniform with respect to  $\theta$ , even though the resulting tension and normal loading vary through the wrap. Improving the contact resolution further would therefore mean allowing local belt speed, radial motion, and traction state to vary within the wrap itself, for which the constant-coefficient solution would be replaced by a spatially varying model (Kong and Parker, 2006; Messick, 2018; Julió and Plante, 2011).

The full expression describes the wrap even while the belt speed and pulley radius are changing. When those changes stop, the transient terms disappear, leaving a direct connection to the familiar steady belt relation. Holding the pulley radius fixed and the belt transport speed steady gives  $\dot{r}_j = \ddot{r}_j = \dot{v}_b = 0$ . Substituting these conditions directly into the entry-to-exit result, Equation (3.4.31), and rearranging gives

$$\frac{T_{j,\text{out}} - \rho_b A_b v_b^2}{T_{j,\text{in}} - \rho_b A_b v_b^2} = e^{-\frac{\lambda_j \phi_j}{\sin \beta}}. \quad (3.4.34)$$

This is the inertia-corrected structure used by Gerbert (Gerbert, 1996), written in the same in/out convention as the full wrap solution. The correction has a direct physical meaning:  $\rho_b A_b v_b^2$  is the

portion of the tension needed simply to turn the moving belt, whereas the remainder produces normal contact and is therefore called the tractive tension. The ratio shows that it is this tractive portion, rather than the total tension, that friction causes to evolve across the steady wrap. At the traction limit,  $|\lambda_j| = \mu$ , while the sign of  $\lambda_j$  determines whether the tension rises or falls from the wrap entrance to the wrap exit.

The same comparison leads directly to the familiar classical limit. If belt inertia is neglected and the V-groove is removed by taking  $\beta = 90^\circ$ , then the centrifugal correction vanishes and  $\sin \beta = 1$ , so Equation (3.4.34) becomes

$$\frac{T_{j,\text{out}}}{T_{j,\text{in}}} = e^{-\lambda_j \phi_j}. \quad (3.4.35)$$

This is the Euler–Eytelwein equation in the same signed in/out convention (Euler, 1769; Eytelwein, 1808). No new law has been introduced: the classical result is simply what remains of the entry-to-exit balance after transient acceleration, centrifugal inertia, and V-groove amplification are removed. Read in reverse, the full solution restores each of those effects while preserving the same underlying idea that distributed traction changes the tension through the wrap. With that tension now known at every position, Equation (3.4.21) can recover the corresponding normal loading, allowing the pulley load and transmitted torque to be assembled from the same contact distribution.

#### 3.4.4 Pulley Contact Loads and Torque

With  $T_j(\theta)$  now known, the same distributed contact can supply the two belt actions left in the pulley equations: the normal resultant  $N_j$  and the contact torque  $\tau_j$ . Begin with the radial balance. Equation (3.4.21) gives the normal loading per unit wrap angle, so accumulating that distribution from the entrance at  $\theta = 0$  to the exit at  $\theta = \phi_j$  gives

$$N_j = \frac{1}{\sin \beta} \int_0^{\phi_j} [T_j(\theta) - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)] d\theta. \quad (3.4.36)$$

Here,  $N_j$  is the total normal load integrated over both sheave faces. Since  $T_j(\theta)$  is supplied by Equation (3.4.30), this integral now determines the normal resultant required in Equations (3.2.14) and (3.2.33).

It is important to note that, under Assumption 8 (Prescribed Planar Belt Path), this mathematical solution is physically admissible only while

$$T_j(\theta) \geq 0, \quad \frac{dN_j}{d\theta} \geq 0, \quad 0 \leq \theta \leq \phi_j. \quad (3.4.37)$$

A negative tension indicates slack, while a negative normal-loading gradient indicates local lift-off; neither represents a physical negative load.

Although Equation (3.4.36) determines  $N_j$ , evaluating it requires the complete tension field. The tangential balance gives the same resultant directly from the wrap boundary values. Under Assumption 11 (Wrap-Level Contact Resolution),  $\lambda_j$  is constant throughout the wrap, so integrating Equation (3.4.19) from  $\theta = 0$  to  $\theta = \phi_j$  gives

$$T_{j,\text{out}} - T_{j,\text{in}} + \lambda_j N_j = \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \quad (3.4.38)$$

For  $\lambda_j \neq 0$ , rearranging gives the normal resultant in its compact entry-to-exit form:

|                                                                                                                                                       |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Pulley-Wrap Normal Resultant</b>                                                                                                                   | <b>(3.4.39)</b> |
| $N_j = \frac{T_{j,\text{in}} - T_{j,\text{out}} + \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b)}{\lambda_j}, \quad \lambda_j \neq 0.$ |                 |

The division by  $\lambda_j$  only excludes zero traction from this compact form. When  $\lambda_j = 0$ , the tangential balance no longer determines  $N_j$ , but the radial integral remains valid. Substituting the zero-traction tension field from Equation (3.4.33) into Equation (3.4.36) gives

$$\begin{aligned} N_j &= \frac{\phi_j}{\sin \beta} [T_{j,\text{in}} - \rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j)] \\ &\quad + \frac{\rho_b A_b \phi_j^2}{2 \sin \beta} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b). \end{aligned} \quad (3.4.40)$$

Thus  $\lambda_j = 0$  removes the tangential contact force, not the normal contact: the pulley can remain clamped even though the modeled contact transmits no torque.

The axial derivation already established how the normal contact enters each movable-sheave free body. Under Assumption 12 (Symmetric Sheave-Face Sharing), the movable face carries  $N_j/2$ ; projecting that load onto the local shaft axis gives the previously introduced belt-opening term

$$F_{\text{belt},x_j} = \frac{N_j}{2} \cos \beta, \quad j \in \{p, s\}.$$

With  $N_j$  now known, this determines the belt-opening term left unknown in Equations (3.2.14) and (3.2.33).

The remaining unresolved belt term is the contact torque  $\tau_j$  in Equations (3.3.5) and (3.3.13). Recall from Section 3.4.1 that  $dF_{t,j} = \lambda_j dN_j$  acts on the belt, while the equal-and-opposite force acts on the pulley. Using the cord-line lever arm  $r_{j,\text{eff}}$  from Equation (3.1.9), and then using the integrated tangential balance in Equation (3.4.38), gives

$$\begin{aligned} \tau_j &= -r_{j,\text{eff}} \int_0^{\phi_j} \lambda_j \frac{dN_j}{d\theta} d\theta \\ &= -r_{j,\text{eff}} \lambda_j N_j \\ &= r_{j,\text{eff}} [T_{j,\text{out}} - T_{j,\text{in}} - \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b)], \quad j \in \{p, s\}. \end{aligned} \quad (3.4.41)$$

The final line makes the transient correction explicit: the pulley torque is set by the entry-to-exit tension change only when the tangential acceleration of the wrapped belt vanishes.

For a specified wrap state, the local mechanics are now complete, but the absolute tension level is not. Equation (3.4.30) maps the entrance tension  $T_{j,\text{in}}$  to the exit tension  $T_{j,\text{out}}$ , but an isolated wrap cannot determine either boundary value on its own. Those tensions continue into the adjoining

straight spans and, ultimately, around the same closed belt. The next step is therefore to reconnect the two pulley wraps through the straight spans and require the tension to be compatible after one complete circuit of the belt loop.

### 3.4.5 Reassembling the Closed Belt

Dividing the complete belt at the four in-out-of-wrap points in [Figure 3.12](#) produces four regions: the primary wrap, the secondary wrap, and the upper and lower straight spans. Newton's second law has already been applied throughout both wrapped regions. The two straight spans remain. They carry no distributed pulley contact, but they still contain belt mass, so their end tensions need not be equal while the belt accelerates.

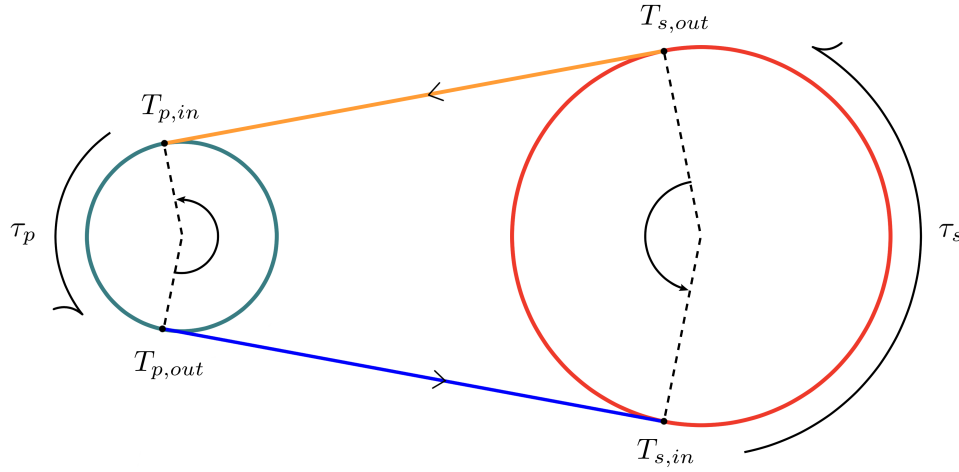


Figure 3.12: Complete belt divided into two wrapped regions and two straight spans. The arrows show the positive belt-transport direction. The upper span runs from the secondary wrap exit,  $T_{s,out}$ , to the primary wrap entrance,  $T_{p,in}$ ; the lower span runs from  $T_{p,out}$  to  $T_{s,in}$ . The torque arrows identify the contact actions on the wrapped regions already treated above.

As the in-out-of-wrap points move during a shift, each straight span also rotates. [Assumption 8 \(Prescribed Planar Belt Path\)](#) prescribes this changing path but explicitly omits the transverse inertia associated with its reorientation. Accordingly, Newton's second law is applied only along the instantaneous straight span.

Take this direction as positive with belt travel, and let  $p_u$  and  $p_l$  be the tangential momenta of the belt particles occupying the two spans at the instant considered; Newton's second law follows those same particles. The downstream tension acts positively and the upstream tension negatively, so

$$\begin{aligned} T_{p,in} - T_{s,out} &= \frac{dp_u}{dt}, \\ T_{s,in} - T_{p,out} &= \frac{dp_l}{dt}. \end{aligned} \tag{3.4.42}$$

At that instant, a straight span  $i \in \{u, l\}$  has mass  $\rho_b A_b l_i$ . Under [Assumption 7 \(Single Belt Transport Coordinate\)](#) and [Assumption 4 \(Fixed, Uniform Belt Cross-Section\)](#), its stored tangential momentum is  $p_i = \rho_b A_b l_i v_b$ , so the obvious first step is to differentiate:

$$\frac{d}{dt}(\rho_b A_b l_i v_b) = \rho_b A_b (l_i \dot{v}_b + \dot{l}_i v_b).$$

The term  $\rho_b A_b \dot{l}_i v_b$  exposes the complication. Each station marks where the belt enters or leaves the current pulley geometry; it is not fixed to a particular belt particle. As the stations move, belt particles pass between the wraps and straight span, so the region contains a changing mass. The derivative above therefore describes the momentum stored within a moving region, not yet the material momentum rate required by Newton's second law. The momentum crossing its boundaries must also be included.

Let  $v_i^-$  and  $v_i^+$  denote the velocities of the upstream and downstream stations, respectively, projected along the straight span. The belt crosses those boundaries at the relative speeds  $v_b - v_i^-$  and  $v_b - v_i^+$ . Adding the outgoing momentum flux and subtracting the incoming flux gives

$$\begin{aligned} \frac{dp_i}{dt} &= \frac{d}{dt}(\rho_b A_b l_i v_b) \\ &\quad + \rho_b A_b v_b [(v_b - v_i^+) - (v_b - v_i^-)]. \end{aligned} \tag{3.4.43}$$

These same station velocities determine the changing distance between them:

$$\dot{l}_i = v_i^+ - v_i^-. \tag{3.4.44}$$

Expanding the stored momentum and substituting [Equation \(3.4.44\)](#) gives

$$\begin{aligned} \frac{dp_i}{dt} &= \rho_b A_b (l_i \dot{v}_b + \dot{l}_i v_b) + \rho_b A_b v_b (v_i^- - v_i^+) \\ &= \rho_b A_b (l_i \dot{v}_b + \dot{l}_i v_b - \dot{l}_i v_b) \\ &= \rho_b A_b l_i \dot{v}_b. \end{aligned} \tag{3.4.45}$$

The cancellation is exact. The station motion changes the momentum stored within the geometric span by  $\rho_b A_b \dot{l}_i v_b$ , but the same motion produces an equal and opposite momentum flux through its boundaries. Once that flux is included, the moving stations leave no additional term in the tangential balance; only the acceleration of the belt transport remains.

The aligned-pulley geometry gives both straight spans the same length  $l$ , as shown in [Equation \(3.1.19\)](#). Substituting [Equation \(3.4.45\)](#) into the two Newton balances therefore yields

$$\begin{aligned} T_{p,\text{in}} - T_{s,\text{out}} &= \rho_b A_b l \dot{v}_b, \\ T_{s,\text{in}} - T_{p,\text{out}} &= \rho_b A_b l \dot{v}_b. \end{aligned} \tag{3.4.46}$$

Because the right-hand sides of the two straight-span balances are identical, their tension differences must be equal:

$$T_{p,\text{in}} - T_{s,\text{out}} = T_{s,\text{in}} - T_{p,\text{out}} \quad (3.4.47)$$

Rearranging this result places the two tensions belonging to each pulley on the same side:

$$T_{p,\text{in}} + T_{p,\text{out}} = T_{s,\text{in}} + T_{s,\text{out}}. \quad (3.4.48)$$

The closed belt therefore requires the sum of the primary boundary tensions to match the sum of the secondary boundary tensions. The remaining task is to evaluate that sum from the mechanics already obtained for a generic pulley wrap.

The integrated tangential balance in Equation (3.4.38) gives the boundary-tension difference as

$$T_{j,\text{out}} - T_{j,\text{in}} = \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) - \lambda_j N_j. \quad (3.4.49)$$

This and the entry-to-exit solution in Equation (3.4.31) are two relations for the same two boundary tensions. Solving them for the entrance tension gives

$$\begin{aligned} T_{j,\text{in}} = \rho_b A_b \left[ v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin \beta}{\lambda_j} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \right] \\ + \frac{\lambda_j N_j - \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b)}{1 - e^{-\frac{\lambda_j}{\sin \beta} \phi_j}}, \quad \lambda_j \neq 0. \end{aligned} \quad (3.4.50)$$

The required sum can now be written as  $2T_{j,\text{in}} + (T_{j,\text{out}} - T_{j,\text{in}})$ . Substituting Equations (3.4.49) and (3.4.50) and simplifying gives

$$\begin{aligned} T_{j,\text{in}} + T_{j,\text{out}} = 2\rho_b A_b \left[ v_b^2 - r_{j,\text{cm}} \ddot{r}_j + \frac{\sin \beta}{\lambda_j} (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b) \right] \\ + \frac{1 + e^{-\frac{\lambda_j}{\sin \beta} \phi_j}}{1 - e^{-\frac{\lambda_j}{\sin \beta} \phi_j}} [\lambda_j N_j - \rho_b A_b \phi_j (r_{j,\text{cm}} \dot{v}_b + \dot{r}_j v_b)], \quad \lambda_j \neq 0. \end{aligned} \quad (3.4.51)$$

Applying this expression to the primary and secondary sides of Equation (3.4.48) eliminates all four temporary boundary tensions:

$$\begin{aligned}
& 2\rho_b A_b \left[ v_b^2 - r_{p,\text{cm}} \ddot{r}_p + \frac{\sin \beta}{\lambda_p} (r_{p,\text{cm}} \dot{v}_b + \dot{r}_p v_b) \right] \\
& + \frac{1 + e^{-\frac{\lambda_p}{\sin \beta} \phi_p}}{1 - e^{-\frac{\lambda_p}{\sin \beta} \phi_p}} [\lambda_p N_p - \rho_b A_b \phi_p (r_{p,\text{cm}} \dot{v}_b + \dot{r}_p v_b)] \\
& = 2\rho_b A_b \left[ v_b^2 - r_{s,\text{cm}} \ddot{r}_s + \frac{\sin \beta}{\lambda_s} (r_{s,\text{cm}} \dot{v}_b + \dot{r}_s v_b) \right] \\
& + \frac{1 + e^{-\frac{\lambda_s}{\sin \beta} \phi_s}}{1 - e^{-\frac{\lambda_s}{\sin \beta} \phi_s}} [\lambda_s N_s - \rho_b A_b \phi_s (r_{s,\text{cm}} \dot{v}_b + \dot{r}_s v_b)], \quad \lambda_p \lambda_s \neq 0.
\end{aligned}$$

The displayed form assumes nonzero utilization only because the analytical wrap solution contains factors of  $1/\lambda_j$ . The zero-utilization limit remains regular. Using the zero-traction relations from [Equations \(3.4.33\) and \(3.4.40\)](#) gives

$$T_{j,\text{in}} + T_{j,\text{out}} = 2\rho_b A_b (v_b^2 - r_{j,\text{cm}} \ddot{r}_j) + \frac{2N_j \sin \beta}{\phi_j}, \quad \lambda_j = 0. \quad (3.4.53)$$

This expression replaces the corresponding side of [Equation \(3.4.52\)](#) whenever  $\lambda_p = 0$  or  $\lambda_s = 0$ . The four boundary tensions have therefore disappeared from the closed-loop equation: the straight spans require the two boundary-tension sums to match, and the local wrap mechanics express each sum in terms of the pulley normal load, traction utilization, and retained belt kinematics. Once the coupled mechanical equations determine those quantities, [Equations \(3.4.31\) and \(3.4.50\)](#) recover the boundary tensions and [Equation \(3.4.30\)](#) recovers the tension through each wrap.

### 3.4.6 Belt-Mechanics Closure and Scope

The closed-loop compatibility in [Equation \(3.4.52\)](#) is the final relation required from the belt itself. Rather than fixing the absolute tension level through a prescribed tight- or slack-side boundary condition, it makes that level emerge from the two pulley-wrap solutions together with the momentum balances of the two straight spans. This way of closing the loop also defines the level of belt resolution retained by the present model.

This treatment combines ideas that more commonly appear at different levels of resolution. Classical belt relations describe the tension change through a single wrap once an endpoint tension and contact state are supplied ([Gerbert, 1996](#)). Steady whole-belt boundary-value models couple the pulley contacts and free spans, but typically require operating-point boundary data such as a prescribed slack-side tension ([Kong and Parker, 2006](#); [Messick, 2018](#)). Transient discretized rubber-belt models instead enforce whole-loop compatibility through the states of many belt elements ([Julió and Plante, 2011](#); [Ballew, 2015](#)). The present formulation occupies a reduced middle ground: it retains one global belt-transport coordinate and an analytical closed-loop compatibility without introducing spatial or nodal belt states.

The chain-CVT model of [Duan et al. \(2016\)](#) provides the closest comparison for the retained transient physics. Both formulations account for centripetal inertia, tangential belt or chain acceleration, and the additional tangential acceleration produced as the contact radius changes. Duan also couples the

pulley axial and rotational dynamics through the contact solution, as is done here. One difference is that Duan neglects radial acceleration when reducing the final spatial wrap equations, whereas the present radial balance retains it.

Duan resolves considerably more physics within each pulley wrap. The chain trajectory may depart locally from the nominal pitch circle; radial and circumferential relative motion determine the sliding direction; friction varies continuously with sliding speed; and pulley deformation changes the contact geometry with angular position. Tension and normal loading are therefore obtained from coupled spatial equations. In the present model, [Assumption 8 \(Prescribed Planar Belt Path\)](#) fixes a circular contact path on rigid pulley geometry, [Assumption 7 \(Single Belt Transport Coordinate\)](#) assigns one transport speed to the belt, and [Assumption 11 \(Wrap-Level Contact Resolution\)](#) assigns one traction utilization  $\lambda_j$  to each wrap. The present formulation consequently retains the mechanism-level transient dynamics while sacrificing within-wrap variation of migration, microslip, traction reversal, and pulley deformation.

The complete belt loop is also closed differently. Duan concentrates the chain’s longitudinal deformation in the tight span, obtains its absolute tension from a lumped spring–damper law, and assumes that tension is equal at the driving-pulley entrance and driven-pulley exit. The tension difference across the slack span then accelerates the mass of the complete chain. Under [Assumption 6 \(Fixed Belt Length\)](#), the present formulation has no longitudinal stretch law from which to obtain an absolute tension. Instead, both straight spans are treated as massive belt regions, including the momentum carried through their moving in–out-of-wrap boundaries. Combining those balances with the two wrap solutions closes the loop without prescribing a tight- or slack-side tension.

The comparison is therefore best understood as a difference in where each model places its resolution. Duan resolves the local contact, friction, chain elasticity, and pulley deformation in greater detail. The present formulation uses a coarser contact description, but carries radial acceleration, both straight-span inertias, moving-boundary momentum transport, movable-sheave dynamics, and pulley rotation through the complete belt loop. The distributed tangential action within each wrap is reduced to the single signed utilization  $\lambda_j$ , allowing these mechanism-level transients to be retained without introducing a spatially discretized belt state.

### 3.5 Tangential Contact-State Closure

The fully engaged CVT has now been followed through every retained motion and free body. [Section 3.1](#) relates the sheave positions and pulley radii, [Section 3.2](#) balances the two movable assemblies, [Section 3.3](#) balances the two rotating assemblies, and [Section 3.4](#) balances the belt around both wraps and straight spans. No additional free body or degree of freedom remains to be introduced.

Looking across those relations, only two quantities remain without relations that fix their values at the current instant:  $\lambda_p$  and  $\lambda_s$ . The reason lies in how  $\lambda_j$  was introduced in [Equation \(3.4.18\)](#). There, the separate Coulomb conditions on traction magnitude, direction, and whether the contact is sticking or sliding were collected into the single signed utilization  $\lambda_j$ . That compression let the remainder of the belt derivation carry one contact quantity through the wrap tension, normal loading, pulley torque, and closed-loop belt relations without reopening the piecewise contact law at every step.

Importantly, this is different from adding one more balance. The equations derived so far keep the same form as the CVT moves. Changing a belt–pulley contact between sticking and sliding,

however, is a change of mechanical regime: the actual relation used to determine  $\lambda_j$  changes with the contact state. This is the type of regime change introduced in [Section 2.5](#), where a change in the active contact or constraint state changes the equations governing the mechanism. What remains is therefore not another motion or free-body balance, but the choice of which tangential contact relation governs each wrap.

### 3.5.1 Relative Motion at the Pulley Contacts

With the remaining ambiguity reduced to the two Coulomb branches, the first useful distinction is the contact motion itself. Sticking and sliding differ in whether the belt has relative tangential motion with respect to the pulley contact represented by a wrap. Before that relative motion can be written, the surface speed represented by each reduced contact must therefore be identified.

For the primary, this comparison is direct. Both sheave faces rotate with the primary shaft, so their angular speed is  $\omega_p$ . The secondary is different. Under [Assumption 11 \(Wrap-Level Contact Resolution\)](#), the complete wrap carries one traction state rather than separate states for the fixed and movable faces, while the helix kinematics already established that the two physical faces do not generally have the same speed during a shift. The fixed face rotates at  $\omega_s$ , whereas [Equation \(3.2.24\)](#) gives the movable-face speed  $\omega_{s,M} = \omega_s + \dot{\theta}_s$ .

One representative secondary surface speed is therefore required. Its value is fixed by the same face-torque reduction already used in [Assumption 12 \(Symmetric Sheave-Face Sharing\)](#). In particular, [Equation \(3.2.21\)](#) gives  $\tau_{s,M} = \tau_s/2$ , and therefore  $\tau_{s,F} = \tau_s/2$ . The instantaneous power delivered by the belt torque to the two faces is

$$\begin{aligned} P_{s,\text{faces}} &= \tau_{s,F}\omega_s + \tau_{s,M}\omega_{s,M} \\ &= \frac{\tau_s}{2}\omega_s + \frac{\tau_s}{2}(\omega_s + \dot{\theta}_s) \\ &= \tau_s\left(\omega_s + \frac{1}{2}\dot{\theta}_s\right). \end{aligned} \tag{3.5.1}$$

Let  $\bar{\omega}_s$  be the angular speed assigned to the single aggregate secondary contact. The same total torque represented by that contact carries power  $\tau_s\bar{\omega}_s$ . Equating this power with [Equation \(3.5.1\)](#) fixes the representative speed. The primary and secondary contact speeds can therefore be written together as

$$\boxed{\bar{\omega}_p = \omega_p, \quad \bar{\omega}_s = \omega_s + \frac{1}{2}\dot{\theta}_s = \frac{\omega_s + \omega_{s,M}}{2}.} \tag{3.5.2}$$

Thus the average in [Equation \(3.5.2\)](#) is not an additional empirical approximation. Once one secondary traction state and the equal face-torque split have been adopted, power consistency selects this representative surface speed. The arithmetic mean appears specifically because the two torque shares are equal. It remains an aggregate quantity: the model still does not determine the separate sliding states of the fixed and movable faces.

Returning now to relative motion, look at pulley  $j$ . The belt moves through the wrap with signed transport speed  $v_b$ , while the represented pulley surface moves tangentially at  $r_{j,\text{eff}}\bar{\omega}_j$ . Their difference defines the relative tangential speed,

$$v_{\text{rel},j} = v_b - r_{j,\text{eff}}\bar{\omega}_j, \quad j \in \{p, s\}. \quad (3.5.3)$$

The sign follows the common belt-transport and rotational directions defined in [Section 2.3.1](#). Positive  $v_{\text{rel},j}$  means the belt moves ahead of the represented pulley contact in the positive belt direction, while a negative value means it moves the other way.

If  $v_{\text{rel},j} \neq 0$ , the contact is already sliding. Its relative motion fixes the direction of friction, and kinetic Coulomb friction fixes the magnitude. The sliding branch of [Equation \(3.4.18\)](#) therefore gives

$$\begin{aligned} \sigma_j &= -\text{sgn}(v_{\text{rel},j}), \\ \lambda_j &= \sigma_j \mu_k, \quad v_{\text{rel},j} \neq 0. \end{aligned} \quad (3.5.4)$$

Sliding therefore closes  $\lambda_j$  directly from the current contact motion. Sticking begins at the opposite kinematic condition,

$$v_{\text{rel},j} = 0 \quad \Longleftrightarrow \quad v_b = r_{j,\text{eff}}\bar{\omega}_j. \quad (3.5.5)$$

The important point is that zero relative speed does not imply zero tangential force. It only says that the belt and represented pulley contact move together at that instant. Static friction may still supply a nonzero reaction, with  $\lambda_j$  taking whatever signed value is required to maintain sticking so long as the static limit in [Equation \(3.4.18\)](#) is not exceeded.

That speed match must also persist. If the belt speed and represented pulley-surface speed begin changing at different rates, relative motion appears immediately. Differentiating [Equation \(3.5.2\)](#) gives

$$\begin{aligned} \dot{\bar{\omega}}_p &= \dot{\omega}_p, \\ \dot{\bar{\omega}}_s &= \dot{\omega}_s + \frac{1}{2}\ddot{\theta}_s \\ &= \dot{\omega}_s + \frac{1}{2} \left[ \frac{d\theta_s}{dx_s} \ddot{x}_s + \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2 \right], \end{aligned} \quad (3.5.6)$$

where the final line follows directly from [Equation \(3.2.25\)](#). Differentiating [Equation \(3.5.3\)](#) then gives the relative tangential acceleration,

$$\begin{aligned} a_{\text{rel},j} &\equiv \dot{v}_{\text{rel},j} \\ &= \dot{v}_b - r_{j,\text{eff}}\dot{\bar{\omega}}_j - \dot{r}_{j,\text{eff}}\bar{\omega}_j. \end{aligned} \quad (3.5.7)$$

The first two terms are the direct comparison between belt acceleration and the acceleration of a represented pulley surface at fixed radius. The final term appears because the contact radius is not fixed during a shift. As established in [Equation \(3.1.58\)](#), the effective radius moves with the pulley radius, so  $r_{j,\text{eff}}\omega_j$  can change even when  $\omega_j$  is momentarily constant. A sticking belt must follow that change as well.

Established sticking therefore requires

$$a_{\text{rel},j} = 0. \quad (3.5.8)$$

The two Coulomb branches now leave different things to determine. Sliding already supplies  $\lambda_j$ . Sticking instead supplies the condition that the mechanics must satisfy,  $a_{\text{rel},j} = 0$ , while  $\lambda_j$  remains the static reaction required to make that condition hold within the available friction limit.

### 3.5.2 Determining the Sticking Traction

For a sticking contact, the previous subsection leaves a simple-looking pair:  $\lambda_j$  is unknown, while the contact must satisfy  $a_{\text{rel},j} = 0$ . At first, this seems no different from any other unknown and equation already added to the model. If  $\lambda_j$  entered the mechanics linearly, it could simply be solved algebraically alongside the other instantaneous unknowns.

Whether that is possible can be read directly from the equations already derived. The pulley torque in [Equation \(3.4.41\)](#), for example, contains

$$\tau_j = -r_{j,\text{eff}} \lambda_j N_j, \quad (3.5.9)$$

so an unknown  $\lambda_j$  multiplies the unknown normal resultant  $N_j$ . The equation is therefore no longer linear in the unknowns. If this product were the only complication, however, another rearrangement might still isolate  $\lambda_j$ . The stronger obstacle comes from the wrap relation carried into the closed-loop belt compatibility. Through [Equation \(3.4.51\)](#), the same  $\lambda_j$  also appears reciprocally and inside terms such as

$$\frac{\sin \beta}{\lambda_j}, \quad \frac{1 + e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}{1 - e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}. \quad (3.5.10)$$

With the same unknown appearing both in products with other unknowns and inside reciprocal and exponential terms, there is no useful algebraic rearrangement that isolates the required static utilization.

There is one important simplification. This nonlinear dependence comes from the unknown static  $\lambda_j$  itself; the other remaining accelerations and contact loads appear linearly. If the static utilization were known, those quantities could be obtained directly from the balances already derived.

That suggests turning the problem around. Rather than trying to isolate  $\lambda_j$  explicitly, choose a candidate static utilization and determine the response it would produce. The resulting accelerations give  $a_{\text{rel},j}$  through [Equation \(3.5.7\)](#). Since sticking requires the relative acceleration to vanish,  $a_{\text{rel},j}$  can be used directly as the residual for the static-traction solve,

$$a_{\text{rel},j} = 0. \quad (3.5.11)$$

This is the same basic idea used for the geometric compatibility roots in [Section 3.1](#). A nonzero residual means that the candidate traction would make the belt and pulley surface begin accelerating relative to one another. The required static utilization is the value for which the residual vanishes. Finding that value is therefore a root solve in the unknown static traction.

If only one wrap is sticking, the other utilization is already fixed by its sliding branch. One static  $\lambda_j$  and one sticking residual remain, so the root is one-dimensional. If both wraps are sticking, the two static utilizations influence the same mechanical response and must be varied together until both residuals vanish,

$$a_{\text{rel},p}(\lambda_p, \lambda_s) = 0, \quad a_{\text{rel},s}(\lambda_p, \lambda_s) = 0. \quad (3.5.12)$$

This leaves only a one- or two-dimensional nonlinear search in the static traction utilizations. For each candidate value or pair, everything else is obtained from the linear balances already derived.

### 3.5.3 The Common Mechanical Solve

The two contact branches now reduce to the same mechanical calculation. Sliding supplies  $\lambda_j$  directly, while sticking tests candidate values as its root is found. In either case, evaluating the instantaneous response begins with one traction utilization for each wrap. The remaining step is to solve the mechanics for that pair.

With  $\lambda_p$  and  $\lambda_s$  specified, the equations already derived can now be read simply for the quantities they still have to determine. The axial balances expose the first of these unknowns. The primary balance in Equation (3.2.38) relates the shared shift acceleration  $\ddot{s}$  to the primary normal resultant  $N_p$ . The secondary balance in Equation (3.2.39) contains the same  $\ddot{s}$ , the secondary normal resultant  $N_s$ , and also the secondary shaft acceleration  $\dot{\omega}_s$  and belt torque  $\tau_s$ .

The rotational balances add the shaft dynamics. The primary and secondary relations in Equations (3.3.5) and (3.3.13) contain  $\dot{\omega}_p$  and  $\dot{\omega}_s$ , together with the belt torques  $\tau_p$  and  $\tau_s$ ; the secondary relation also contains the same  $\ddot{s}$  through the helix kinematics. The torques may look like two more unknowns, but the specified traction pair removes them immediately. From Equation (3.4.41),

$$\tau_j = -r_{j,\text{eff}} \lambda_j N_j, \quad j \in \{p, s\}, \quad (3.5.13)$$

so each pulley torque is set by the same normal resultant already appearing in the axial balances.

The complete-belt transport balance in Equation (3.4.1) brings in the belt acceleration  $\dot{v}_b$ . Its two tangential pulley actions are again the belt torques, so after the relation above is substituted the transport equation involves  $\dot{v}_b$ ,  $N_p$ , and  $N_s$  for the specified traction pair.

Finally, the closed-loop tension compatibility in Equation (3.4.52) relates the same belt acceleration and normal resultants to the radial accelerations of the two wraps. Those radial accelerations introduce no new degree of freedom: Equations (3.1.66) and (3.1.83) already express them in terms of the current geometry,  $\dot{s}$ , and the same shift acceleration  $\ddot{s}$ . The four temporary boundary tensions have likewise already disappeared from the closed-loop relation.

No new unknown appears as these equations are brought together. The quantities left to determine are exactly

$$\mathbf{z} = [\dot{\omega}_p \quad \dot{\omega}_s \quad \dot{v}_b \quad \ddot{s} \quad N_p \quad N_s]^\top. \quad (3.5.14)$$

At this point, every other term in the six equations is fixed by the current state, the specified traction pair, and the component relations already derived. Each equation is also linear in the six entries of  $\mathbf{z}$ . Collecting them therefore gives

$$\mathbf{A}(\lambda_p, \lambda_s) \mathbf{z} = \mathbf{b}(\lambda_p, \lambda_s). \quad (3.5.15)$$

To write  $\mathbf{A}$  and  $\mathbf{b}$  out cleanly, a small amount of temporary notation is useful. For this matrix only, primes shorten derivatives:  $x'_s$ ,  $r'_j$ ,  $q'_f$ , and  $J'_{f,p}$  denote derivatives with respect to  $s$ , while  $\theta'_s$  denotes differentiation with respect to  $x_s$ ; double primes follow the same convention. The only additional symbols are the two analytical wrap factors repeated in the closed-loop row,

$$\begin{aligned} \mathcal{C}_{v,j} &\equiv \begin{cases} \frac{2 \sin \beta}{\lambda_j} - \phi_j \frac{1 + e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}{1 - e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}, & \lambda_j \neq 0, \\ 0, & \lambda_j = 0, \end{cases} \\ \mathcal{C}_{N,j} &\equiv \begin{cases} \lambda_j \frac{1 + e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}{1 - e^{-\frac{\lambda_j \phi_j}{\sin \beta}}}, & \lambda_j \neq 0, \\ \frac{2 \sin \beta}{\phi_j}, & \lambda_j = 0. \end{cases} \end{aligned} \quad (3.5.16)$$

These two symbols are temporary assembly notation only. Their  $\lambda_j = 0$  branches are the finite limits already established in [Equation \(3.4.53\)](#).

The coefficient matrix is then

$$\mathbf{A} = \begin{bmatrix} I_{B,p} + I_{p,\text{const}} + J_{f,p} & 0 & 0 & 0 & r_{p,\text{eff}} \lambda_p & 0 \\ 0 & I_{B,s} + I_{s,F} + I_{s,M} & 0 & I_{s,M} \theta'_s x'_s & 0 & r_{s,\text{eff}} \lambda_s \\ 0 & 0 & 0 & m_p + I_f (q'_f)^2 & \frac{1}{2} \cos \beta & 0 \\ 0 & I_{s,M} \theta'_s & 0 & [m_s + I_{s,M} (\theta'_s)^2] x'_s & 0 & \frac{1}{2} (\cos \beta + r_{s,\text{eff}} \lambda_s \theta'_s) \\ 0 & 0 & m_b & 0 & -\lambda_p & -\lambda_s \\ 0 & 0 & \rho_b A_b (r_{p,\text{cm}} C_{v,p} - r_{s,\text{cm}} C_{v,s}) & 2 \rho_b A_b (r_{s,\text{cm}} r'_s - r_{p,\text{cm}} r'_p) & C_{N,p} & -C_{N,s} \end{bmatrix}. \quad (3.5.17)$$

The corresponding known-side vector is

$$\mathbf{b} = \begin{bmatrix} \tau_{B,p} - J'_{f,p} \dot{s} \omega_p \\ \tau_{B,s} - I_{s,M} [\theta'_s x''_s + \theta''_s (x'_s)^2] \dot{s}^2 \\ \frac{1}{2} \omega_p^2 J'_{f,p} - I_f q'_f q''_f \dot{s}^2 - k_p (x_{p,\text{pre}} + s) \\ \theta'_s k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s) + k_{s,x} (x_{s,\text{pre}} - x_s) - \left\{ [m_s + I_{s,M} (\theta'_s)^2] x''_s + I_{s,M} \theta'_s \theta''_s (x'_s)^2 \right\} \dot{s}^2 \\ 0 \\ 2 \rho_b A_b (r_{p,\text{cm}} r''_p - r_{s,\text{cm}} r''_s) \dot{s}^2 + \rho_b A_b v_b (\dot{r}_s C_{v,s} - \dot{r}_p C_{v,p}) \end{bmatrix}. \quad (3.5.18)$$

The row order follows the derivation: the first two rows are the pulley rotational balances, the next two are the axial balances, the fifth is the complete-belt transport balance, and the sixth is the closed-loop tension compatibility.

For a specified traction pair, every entry of  $\mathbf{A}$  and  $\mathbf{b}$  is fixed by the current state and the component relations already derived. Solving [Equation \(3.5.15\)](#) then gives the two shaft accelerations, belt

acceleration, shift acceleration, and the two normal resultants. A sliding  $\lambda_j$  reaches this solve directly from kinetic friction; a sticking  $\lambda_j$  reaches the same solve as a candidate value whose resulting  $a_{\text{rel},j}$  is returned to the root described in [Section 3.5.2](#).

The four contact combinations can now be read off directly:

|                         | Secondary sliding                                                                                                                                                                                        | Secondary sticking                                                                                                                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary sliding</b>  | Both $\lambda_p$ and $\lambda_s$ are fixed by kinetic friction. One $6 \times 6$ solve gives the instantaneous response.                                                                                 | $\lambda_p$ is fixed by kinetic friction, while $\lambda_s$ is found from the one-dimensional root $a_{\text{rel},s} = 0$ . Each candidate $\lambda_s$ is evaluated through the same $6 \times 6$ solve. |
| <b>Primary sticking</b> | $\lambda_s$ is fixed by kinetic friction, while $\lambda_p$ is found from the one-dimensional root $a_{\text{rel},p} = 0$ . Each candidate $\lambda_p$ is evaluated through the same $6 \times 6$ solve. | $\lambda_p$ and $\lambda_s$ are found together from $a_{\text{rel},p} = 0$ and $a_{\text{rel},s} = 0$ . Each candidate pair is evaluated through the same $6 \times 6$ solve.                            |

The six mechanical balances are therefore common to all four cases. What changes with the contact regime is only how the traction pair supplied to those balances is obtained: directly from kinetic friction, from one static-traction root, or from the two sticking conditions together.

### 3.5.4 Contact Admissibility and Regime Transitions

The four contact combinations above are enough to obtain an instantaneous response for any assumed tangential regime. They do not yet tell us whether that regime can actually be maintained, or what should happen when it cannot.

For a sticking contact,  $\lambda_j$  is the static reaction required to keep the belt and pulley moving together, as introduced in [Equation \(3.4.18\)](#). The root in [Section 3.5.2](#) finds the value of that reaction that satisfies  $a_{\text{rel},j} = 0$ . Sticking is physically available only while the required reaction lies within the static Coulomb limit,

$$|\lambda_j| \leq \mu_s. \quad (3.5.19)$$

If this condition is satisfied, the contact can supply the traction needed to remain stuck. If the required  $|\lambda_j|$  would exceed  $\mu_s$ , no admissible static reaction can preserve zero relative motion and the sticking regime must end.

Sliding is simpler away from zero relative speed. While  $v_{\text{rel},j} \neq 0$ , its sign identifies the sliding direction and [Equation \(3.5.4\)](#) fixes the corresponding kinetic traction. The next ambiguity appears only when  $v_{\text{rel},j}$  reaches zero. At that instant the belt and pulley are momentarily moving together, so the first possibility to test is sticking. The required static  $\lambda_j$  is found from  $a_{\text{rel},j} = 0$  and checked against [Equation \(3.5.19\)](#). If it is admissible, the contact captures into sticking.

If sticking is not admissible, the contact must leave  $v_{\text{rel},j} = 0$  on a sliding branch. For a contact that arrived at zero while already sliding, two continuations are possible: the relative velocity may grow again in the same direction as before, or the sliding direction may reverse and grow in the opposite direction. At the event itself,  $v_{\text{rel},j} = 0$ , so the relative velocity cannot distinguish between those two possibilities. The relative acceleration does: its sign tells which way  $v_{\text{rel},j}$  begins to move away from zero.

For a candidate kinetic sign  $\sigma_j$ , consistency therefore requires

$$\sigma_j a_{\text{rel},j} < 0, \quad v_{\text{rel},j} = 0, \quad (3.5.20)$$

because  $\sigma_j$  is defined opposite to the direction of sliding. The valid kinetic branch is the one whose solved acceleration produces the relative-motion direction assumed by that branch. The same check selects the initial sliding direction when a sticking contact first releases.

Together, these checks also make the order of a regime update clear. Consider a sliding–sliding solution in which the primary relative velocity reaches zero. The secondary contact can initially be left on its current kinetic branch while the primary is tested as sticking. If the required static  $\lambda_p$  is admissible, the candidate regime becomes sticking–sliding. If it is not, the primary remains sliding and the kinetic branch consistent with Equation (3.5.20) is selected instead. The resulting traction pair changes the common mechanical response, so that response is solved again and the contact conditions at both wraps are checked once more. If both contacts reach a regime boundary at the same instant, the same logic is applied to the combined candidate regime rather than updating the two wraps independently.

These tangential contact changes are nonimpulsive regime transitions in the sense introduced in Section 2.5. The belt and pulley reach the event with continuous position and velocity; changing the active contact law changes the acceleration that follows rather than resetting the state instantaneously. The remaining structural regime changes associated with engagement and shift boundaries require a separate treatment.

### 3.6 Shift-Regime Dynamics and Boundary Transitions

The freely shifting engaged CVT is now mechanically closed. For any admissible tangential contact regime, Section 3.5 determines how the pulley speeds, belt transport, and shift coordinate evolve from the current state. But that solution does more than describe the mechanism at one instant: as it is integrated forward in time, it moves the hardware.

Consider what that motion can do. Continued backshift closes the secondary until it can move no farther; continued primary opening can then separate the primary sheave from the belt. In the opposite direction, continued upshift eventually carries the primary to the end of its closing travel. The equations developed so far can bring the CVT to each of these configurations, but once one is reached, the physical problem has changed. A sheave that was free to move may now be supported by a stop, or a belt–sheave interaction that previously transmitted force may no longer exist.

Reaching one of these boundaries therefore changes the shift regime of the CVT, in the same sense that a change between sticking and sliding changes a tangential contact regime in Section 3.5. That structural change has two separate consequences. First, the active contacts, supports, and permitted motions are different, so the continuous governing equations must change with the regime. Second, the state arriving at the boundary may contain a velocity that the new regime cannot support, so the incoming motion must be transferred into a compatible outgoing state.

These are different mechanical problems. The first determines how the CVT accelerates once the new regime is active. The second determines how its velocity changes at the instant the regime is entered. Both are required if the model is to pass through a shift-regime boundary without replacing the mechanics by an imposed reset.

### 3.6.1 Shift Domain and Structural Regimes

Before the governing equations can change with the shift regime, the boundaries between those regimes must first be located along the shift motion. The shift coordinate already traces the primary through its full travel, so the geometry derived in [Section 3.1](#) identifies where contact is gained or lost, where hardware begins to support the motion, and which arrangement exists on either side of each boundary. Following the primary from fully open to fully closed makes those changes explicit.

Consider first the primary closing during engaged upshift. Its groove narrows, moving the belt outward. The fixed belt length requires the secondary radius to decrease, so the secondary opens. This motion can continue until the primary reaches its closing stop at  $x_p = x_{p,\max}$ . Further upshift would require the primary to close farther, into the stop. The belt remains between both pairs of sheave faces, so the mechanism can be held at this ratio with both pulley contacts present.

Opening the primary reverses these motions. The belt moves inward on the primary and outward on the secondary, bringing the secondary toward its fully closed stop. The secondary reaches that stop when the primary is at  $x_p = x_{p,dz}$ , as given by [Equation \(3.1.32\)](#). Now try to continue the same engaged backshift: the belt would have to move farther outward on the secondary, which would require its sheave to close beyond the stop. With shift motion at rest and primary belt contact maintained, the secondary stop can therefore hold the mechanism at its minimum ratio. This is the *engaged low-ratio seat*.

Yet the primary has not reached its own opening stop. Its groove can open wider than the belt, leaving the clearance introduced in [Equation \(3.1.16\)](#). The primary can therefore open farther if it separates from the belt. That opening no longer moves the belt inward or requires the secondary to close farther. Under the deadzone mapping, the secondary stays fully closed and both prescribed belt radii remain fixed while the primary crosses its clearance. Only when the primary reaches  $x_p = 0$  does its own opening stop prevent further travel.

The secondary stop thus limits the engaged backshift before the primary exhausts its opening travel. Closing the primary through the same clearance first brings it back to the belt; further closure then moves the belt outward and opens the secondary away from its stop.

Since  $s \equiv x_p$  by [Equation \(3.1.31\)](#), the same travel and boundary positions in the generalized shift coordinate are

$$0 \leq s \leq s_{\max}, \quad s_{dz} \equiv x_{p,dz}, \quad s_{\max} \equiv x_{p,\max}. \quad (3.6.1)$$

These motions divide the full primary travel into the five structural arrangements summarized in [Table 3.1](#).

| Structural regime          | Primary belt contact | Axial motion or support                                                        |
|----------------------------|----------------------|--------------------------------------------------------------------------------|
| Fully open primary stop    | Absent               | Primary held at $s = 0$ ; secondary remains fully closed.                      |
| Free deadzone              | Absent               | Primary free for $0 < s < s_{dz}$ ; secondary remains fully closed.            |
| Engaged low-ratio seat     | Present              | Fully closed secondary stop holds the engaged mechanism at $s = s_{dz}$ .      |
| Free engaged shift         | Present              | Free shift for $s_{dz} < s < s_{max}$ ; both sheaves follow the belt geometry. |
| Engaged upper primary stop | Present              | Primary closing stop holds the engaged mechanism at $s = s_{max}$ .            |

Table 3.1: Structural arrangements over the primary travel. The secondary retains belt contact in all five arrangements.

A boundary position does not, by itself, determine which shift regime is active. At  $s = s_{dz}$ , the secondary may be held against its fully closed stop while the primary remains in contact with the belt. The primary may instead begin closing, opening the secondary away from that stop, or it may begin opening and separate from the belt into the clearance. All three continuations pass through the same value of  $s$ . What changes is which contacts and supports are active and which motions they permit. The shift regime must therefore be carried as part of the mechanical description; it cannot be inferred from the shift coordinate alone. This is the same regime-dependent logic introduced in [Section 2.5](#), now applied to the changing mechanical arrangement of the CVT.

The same boundary also shows that shift regime and tangential contact state describe different parts of the mechanics. While the secondary stop holds the engaged mechanism at  $s = s_{dz}$ , the two shafts and the belt can continue moving. The stop determines whether further axial motion is available, but it does not determine whether the belt sticks to either pulley or slides relative to it. Each engaged shift regime must therefore still be combined with the corresponding tangential contact regimes from [Section 3.5](#).

The three boundary positions and the five arrangements in [Table 3.1](#) therefore define where structural regime changes may occur and which arrangements are physically admissible there. They do not assign a unique regime to each value of  $s$ ; at a shared boundary, the active contacts, supports, and direction of continuation still matter. The active shift regime and tangential contact regimes together specify the mechanical regime that the model must track as the CVT evolves.

### 3.6.2 Governing Equations in Each Shift Regime

Knowing which shift regime is active tells us which contacts and supports exist, but it does not yet determine how the CVT moves within that regime. The continuous equations must reflect those physical conditions. The useful starting point is the mechanical system already assembled for free engaged motion in [Equation \(3.5.15\)](#). Its six rows are the primary and secondary rotational balances, the primary and secondary axial balances, the complete-belt transport balance, and the closed-loop tension compatibility. A different shift regime does not require these mechanics to be derived again. It changes which of those balances remain active, which contact terms disappear, and which motions are prescribed by the hardware.

## Disengaged Primary and Deadzone Motion

At the fully open end of travel, the primary sheaves do not touch the belt, and the same remains true while the primary crosses the clearance toward  $s = s_{\text{dz}}$ . The immediate consequence is

$$N_p = 0, \quad \tau_p = 0. \quad (3.6.2)$$

Putting this contact loss back into the assembled mechanics shows exactly what changes on the primary side. The rotational balance in Equation (3.3.5) and the axial balance in Equation (3.2.38) still describe the same primary hardware, but their belt terms vanish. The primary rotational row therefore becomes

$$\tau_{B,p} = [I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)] \dot{\omega}_p + \frac{dJ_{f,p}}{ds} \dot{s} \omega_p, \quad (3.6.3)$$

while the primary axial row becomes

$$\begin{aligned} \left[ m_p + I_f \left( \frac{dq_f}{ds} \right)^2 \right] \ddot{s} &= \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{ds} \\ &\quad - I_f \frac{dq_f}{ds} \frac{d^2 q_f}{ds^2} \dot{s}^2 - k_p (x_{p,\text{pre}} + s). \end{aligned} \quad (3.6.4)$$

The belt side changes more substantially. During engaged shift, the secondary axial balance in Equation (3.2.39) participates in the shared shift motion, while Equation (3.4.52) closes the taut belt path through both pulley contacts. Neither role remains in the deadzone. The secondary stays fully closed while  $s$  moves only the primary, so its axial balance no longer supplies an equation for  $\ddot{s}$ . Primary separation also breaks the two-contact tension loop, so the engaged closed-loop compatibility cannot be retained.

This is where the reduced deadzone belt model is needed. The fully closed secondary cannot close farther to take up the remaining belt slack and establish the taut path used by the engaged model. The belt model also has no longitudinal strain or pretension state from which a residual tension could be recovered while that slack remains. Although the belt may still rest on the minimum-radius primary support, that contact is not represented as a clamped belt contact. Resolving the loads produced by a slack or pretensioned belt as it is taken up would therefore require additional belt elasticity and slack-span geometry beyond the present reduced description.

Under Assumption 9 (Reduced Belt Motion While the Primary Is Disengaged), the disengaged belt is instead kept seated on the fully closed secondary and carried with its represented contact surface. Because the secondary has no axial or helix-relative motion,  $\dot{\theta}_s = 0$ , and Equation (3.5.2) reduces to  $\bar{\omega}_s = \omega_s$ . Hence,

$$v_b = r_{s,\text{eff,dz}} \bar{\omega}_s = r_{s,\text{eff,dz}} \omega_s, \quad r_{s,\text{eff,dz}} \equiv r_{s,\text{eff}}(s_{\text{dz}}). \quad (3.6.5)$$

The radius is fixed, so

$$\dot{v}_b = r_{s,\text{eff,dz}} \dot{\omega}_s. \quad (3.6.6)$$

The secondary rotational balance remains active, but at fixed secondary geometry it takes the form given by Equation (3.3.14). The complete-belt transport balance in Equation (3.4.1) also remains active. The deadzone lock ties these two motions together, so their internal secondary contact torque can be eliminated by combining the balances. This gives

$$\tau_{B,s} = [I_{B,s} + I_{s,F} + I_{s,M} + m_b r_{s,\text{eff,dz}}^2] \dot{\omega}_s. \quad (3.6.7)$$

The final term is the complete belt transport inertia reflected through the fixed secondary radius. Primary disengagement removes the primary belt load without removing the belt mass that the secondary must still accelerate.

The active free-deadzone system is now visible. Its unknown accelerations are

$$\mathbf{z}_{\text{dz}} = [\dot{\omega}_p \quad \dot{\omega}_s \quad \dot{v}_b \quad \ddot{s}]^T, \quad (3.6.8)$$

and the four active relations are Equations (3.6.3), (3.6.4), (3.6.6) and (3.6.7). They may therefore be assembled, just as in the engaged case, as

$$\mathbf{A}_{\text{dz}} \mathbf{z}_{\text{dz}} = \mathbf{b}_{\text{dz}}. \quad (3.6.9)$$

For  $0 < s < s_{\text{dz}}$ , this reduced system determines  $\dot{\omega}_p$ ,  $\dot{\omega}_s$ ,  $\dot{v}_b$ , and  $\ddot{s}$  while the primary moves freely through the clearance.

**Fully open primary stop.** The boundary at  $s = 0$  introduces a different kind of regime change. Primary belt contact is already absent there, so no additional interaction disappears. Instead, the hardware prevents the primary from moving beyond the end of its travel. A sustained stop condition requires

$$s = 0, \quad \dot{s} = 0, \quad \ddot{s} = 0. \quad (3.6.10)$$

Inside the free deadzone, the primary axial balance in Equation (3.6.4) determines  $\ddot{s}$ . At the stop, that acceleration is no longer unknown because the hardware prescribes it to be zero. The force balance must still hold, so a new quantity appears instead: the reaction exerted by the stop on the primary.

Let  $F_{\text{stop},0} \geq 0$  denote the magnitude of this reaction in the  $+s$  direction, the only direction in which the stop can push the primary. The primary axial balance becomes

$$0 = F_{\text{fw},x_p} - F_{\text{spring},x_p} + F_{\text{stop},0}, \quad (3.6.11)$$

and therefore

$$F_{\text{stop},0} = F_{\text{spring},x_p} - F_{\text{fw},x_p}. \quad (3.6.12)$$

The axial balance has not been removed; its role has changed. During free deadzone motion it determines the primary acceleration. At the stop the acceleration is known, so the same balance

determines the support force. Equivalently, the free-deadzone system replaces its primary axial row by  $\ddot{s} = 0$ , and the original axial balance is evaluated afterward to recover  $F_{\text{stop},0}$ .

The fully open stop can sustain the shift regime only while

$$F_{\text{stop},0} \geq 0. \quad (3.6.13)$$

A negative value would require the stop to pull the primary in the  $-s$  direction, which a unilateral contact cannot do. Once the required reaction reaches zero and the unconstrained primary tendency points into the deadzone, the stop no longer supports the motion.

### Engagement and the Low-Ratio Seat

Closing through the deadzone leaves the belt radii unchanged until the primary reaches  $s = s_{\text{dz}}$ . At that point the movable primary sheave reaches the belt, so the primary normal load and contact torque can reappear and the engaged belt mechanics become active again. The secondary is still at its fully closed position, so the first engaged shift regime is the low-ratio seat rather than free shift.

For a sustained low-ratio seat,

$$s = s_{\text{dz}}, \quad \dot{s} = 0, \quad \ddot{s} = 0. \quad (3.6.14)$$

The fixed shift also gives  $\dot{\theta}_s = 0$ , so [Equation \(3.5.2\)](#) again reduces to  $\bar{\omega}_s = \omega_s$ . The shafts and belt may nevertheless continue to rotate and accelerate, and the two pulley contacts retain the tangential contact regimes developed in [Section 3.5](#).

The stop mechanics are the same type just introduced at  $s = 0$ , but the supported member is now the secondary. The secondary axial balance in [Equation \(3.2.39\)](#) would determine the shared shift acceleration during free engagement. At the low-ratio seat,  $\ddot{s} = 0$  is prescribed instead, and the fully closed secondary stop supplies the reaction needed to satisfy the original secondary free body in [Figure 3.8](#).

Let  $F_{\text{stop,LR}} \geq 0$  denote the magnitude of that reaction in the secondary opening direction. Using [Equations \(3.2.30\)](#) and [\(3.2.32\)](#), the held secondary balance is

$$0 = F_{\text{helix},x_s} + F_{\text{spring},x_s} - F_{\text{belt},x_s} - F_{\text{stop,LR}}, \quad (3.6.15)$$

so

$$F_{\text{stop,LR}} = F_{\text{helix},x_s} + F_{\text{spring},x_s} - F_{\text{belt},x_s}. \quad (3.6.16)$$

As before, the stopped axial balance changes role rather than disappearing. The only additional complication is that the reaction cannot be evaluated before the coupled engaged response is known. The belt term contains the unknown  $N_s$ , while the helix term contains the transmitted secondary torque and  $\dot{\omega}_s$ .

For a trial traction pair  $(\lambda_p, \lambda_s)$ , the mechanical core in [Equation \(3.5.15\)](#) normally determines  $\ddot{s}$  together with  $\dot{\omega}_p$ ,  $\dot{\omega}_s$ ,  $\dot{v}_b$ ,  $N_p$ , and  $N_s$ . At the seat, the secondary axial row is replaced by  $\ddot{s} = 0$ . The primary axial row in [Equation \(3.2.38\)](#) remains active because the primary itself is not supported

by a stop. The rotational balances in [Equations \(3.3.5\) and \(3.3.13\)](#), the complete-belt transport balance in [Equation \(3.4.1\)](#), and the closed-loop tension compatibility in [Equation \(3.4.52\)](#) retain their engaged forms. Together these active rows determine the fixed-geometry response for the chosen tangential contact regimes.

The original secondary axial balance is then evaluated with the solved quantities to recover  $F_{\text{stop,LR}}$  from [Equation \(3.6.16\)](#). The low-ratio seat is supportable only while

$$F_{\text{stop,LR}} \geq 0. \quad (3.6.17)$$

The stop reaction is a hardware load path, not an additional belt normal force. Any spring or helix loading carried directly by the stop does not add to  $N_s$  and therefore does not increase the secondary tangential traction capacity.

Once the secondary moves away from its closing stop while primary belt contact is retained, its axial row becomes active again. All six rows of [Equation \(3.5.15\)](#) are then the free engaged system already developed in [Section 3.5](#). No additional continuous equation is required in the interior shift regime.

### Upper Primary Stop

Free upshift continues until the primary reaches  $s = s_{\text{max}}$ . Belt contact remains present, but the primary can no longer move farther in the  $+s$  direction. This is the same unilateral support mechanism as the previous two stops, now applied to the primary axial row of the engaged system. A sustained upper-stop state satisfies

$$s = s_{\text{max}}, \quad \dot{s} = 0, \quad \ddot{s} = 0. \quad (3.6.18)$$

Let  $F_{\text{stop,max}} \geq 0$  denote the magnitude of the primary stop reaction in the  $-s$  direction. Using the primary axial balance in [Equation \(3.2.38\)](#),

$$0 = F_{\text{fw},x_p} - F_{\text{spring},x_p} - F_{\text{belt},x_p} - F_{\text{stop,max}}, \quad (3.6.19)$$

so

$$F_{\text{stop,max}} = F_{\text{fw},x_p} - F_{\text{spring},x_p} - F_{\text{belt},x_p}. \quad (3.6.20)$$

The constrained solve follows the same pattern as the low-ratio seat. The primary axial row is replaced by  $\ddot{s} = 0$ , while the secondary axial, rotational, belt-transport, and closed-loop compatibility rows remain active. The solved engaged response is then substituted into [Equation \(3.6.20\)](#) to recover  $F_{\text{stop,max}}$ . The upper stop can sustain the shift regime only while

$$F_{\text{stop,max}} \geq 0. \quad (3.6.21)$$

As at low ratio, the recovered reaction is carried by the hardware rather than added to the belt normal load.

The deadzone and stop-supported regimes therefore modify the assembled mechanics in two distinct ways. Primary separation removes the primary belt interaction and replaces the engaged belt closure with the reduced secondary-carried belt dynamics of Equation (3.6.9). A travel stop instead preserves the remaining contacts while prescribing the stopped axial motion.

The three stop-supported shift regimes then follow the same rule: impose  $\ddot{s} = 0$  in place of the axial row belonging to the supported member, solve the remaining active mechanics, and evaluate the original axial balance to recover the unilateral stop reaction. The resulting cases are summarized in Table 3.2.

| Supported shift regime     | Stopped member | Reaction direction | Balance used to recover the reaction |
|----------------------------|----------------|--------------------|--------------------------------------|
| Fully open primary stop    | Primary        | $+s$               | Primary axial balance                |
| Engaged low-ratio seat     | Secondary      | Opening secondary  | Secondary axial balance              |
| Engaged upper primary stop | Primary        | $-s$               | Primary axial balance                |

Table 3.2: Hardware reactions for the three stop-supported shift regimes. In each case, the stopped member’s original axial balance is evaluated after the corresponding constrained continuous response has been solved.

Each stop is a unilateral contact, so all three supported shift regimes require

$$F_{\text{stop},0} \geq 0, \quad F_{\text{stop},\text{LR}} \geq 0, \quad F_{\text{stop},\text{max}} \geq 0. \quad (3.6.22)$$

The continuous mechanics are now defined for the fully open stop, free deadzone, low-ratio seat, free engaged shift, and upper primary stop. Within each shift regime, the active balances determine the instantaneous accelerations and any support reaction required by the hardware.

### 3.6.3 Shift-Regime Transitions and Momentum Transfer

The continuous equations now describe the motion within every shift regime, but they do not by themselves complete the trajectory through the boundaries between them. At such a boundary, the mechanics on either side may already be known, yet it remains to determine when the current constraints should be released and whether the existing velocity can be carried into the new regime unchanged.

A stop-supported regime exposes this missing boundary logic directly. While the stop is active, its continuous equations impose  $\ddot{s} = 0$ . As the surrounding loads evolve, that prescribed acceleration cannot indicate when the mechanism should begin moving away from the stop. The release condition must instead come from the quantity that measures whether the constraint is still mechanically required: the reaction needed to enforce  $\ddot{s} = 0$ .

The stop reactions derived in Section 3.6.2 provide that condition directly. For any of the three stops, the supported regime remains admissible while

$$F_{\text{stop}} \geq 0. \quad (3.6.23)$$

Release occurs when the required reaction reaches zero and crosses toward the inadmissible side. The reaction is precisely the force needed to hold the supported member at  $\ddot{s} = 0$ ; once that support vanishes, restoring the free axial balance gives the acceleration away from the stop. Thus the same quantity that closes the constrained mechanics also supplies its breakaway condition.

For the fully open primary stop, this occurs when  $F_{\text{stop},0}$  crosses zero, releasing the primary into the free deadzone. At the upper primary stop,  $F_{\text{stop},\text{max}}$  crossing zero similarly releases the mechanism into free engaged shift.

The low-ratio seat is different because two unilateral interactions are present at the same configuration: the secondary movable sheave is supported at its fully closed stop, while the primary remains in contact with the belt. There are therefore two physically distinct ways to leave the seat.

The first is the same stop-release mechanism just described. When

$$F_{\text{stop,LR}} = 0 \quad (3.6.24)$$

is crossed toward inadmissibility, the fully closed secondary stop no longer needs to carry load. The secondary is then free to move away from the stop, and the CVT returns to free engaged shifting.

The second possibility is for the primary to separate from the belt and move into the deadzone while the secondary remains fully closed. Here a vanishing primary normal resultant is only part of the release condition. Reaching  $N_p = 0$  establishes that the belt is no longer supplying a compressive load to the primary, but it does not tell us whether the primary would actually open away from the belt or immediately close back into contact.

While the low-ratio seat is active,  $\ddot{s} = 0$  is prescribed, so that direction cannot be read from the seated solution itself. Instead, evaluate the contact-free primary axial equation in [Equation \(3.6.4\)](#) at the same instantaneous state. Let  $\ddot{s}_{\text{cf}}$  denote the acceleration returned by this contact-free evaluation. It is not used to evolve the seated regime; it is only the acceleration the primary would have if belt contact were removed at that instant. Since positive  $s$  closes the primary, separation into the deadzone requires

$$N_p = 0, \quad \ddot{s}_{\text{cf}} < 0. \quad (3.6.25)$$

The contrast with a metal stop is important. A stop reaction crossing zero already identifies the direction of the free motion because that reaction is the force suppressing the corresponding free axial acceleration. Loss of primary normal load does not provide the same information, so the contact-free acceleration must be checked separately before primary separation is admitted.

At the instant a stop releases, the mechanism is already stationary in the constrained shift direction. The stop has held  $\dot{s} = 0$  up to the moment its reaction vanishes; removing that support does not require the hardware to acquire a new velocity instantaneously. It only allows the restored free axial balance to determine what acceleration follows. The same is true when the primary separates from an established low-ratio seat. The seated state already has  $\dot{s} = 0$ , and the deadzone can begin from that same instantaneous velocity once belt contact is lost.

Position and velocity therefore remain continuous through these releases, as in [Equation \(2.5.2\)](#). Their effect is only to change which continuous equations govern the next acceleration. In this

sense the release is nonimpulsive: the outgoing regime already admits the velocity with which the constrained regime ends.

The reverse crossing need not have this property. A free regime can reach a new boundary carrying a velocity that the newly active constraint or kinematic mapping does not admit.

### Finite-Speed Arrival at a Boundary

The release conditions determine when a stop can cease supporting the mechanism. Shift is then already at rest, so the free axial balance determines the acceleration away from the stop. For the reverse crossing, arrival is detected simply by reaching the stop position. The difficulty is the shift velocity with which the sheaves arrive. Bringing that motion to rest requires an impulse, and the motion immediately after the event must be determined before the continuous solution can proceed.

At the upper primary stop, for example, free upshift can reach  $s = s_{\max}$  while the primary is still closing. The movable sheave strikes the fixed sheave, metal on metal. Under the perfectly inelastic impact model of [Assumption 13 \(Instantaneous, Perfectly Inelastic Constraint Activation\)](#), this collision arrests its axial motion without rebound. With superscripts  $-$  and  $+$  denoting values immediately before and after the event, an arrival with  $\dot{s}^- > 0$  therefore gives

$$\dot{x}_p^+ = \dot{s}^+ = 0. \quad (3.6.26)$$

The belt remains seated between both pairs of sheaves. The secondary cannot keep opening while the primary holds its radius fixed: the two radii must still fit the same belt length. The belt therefore brings the secondary's axial travel to rest with the primary's.

During this travel, however, the movable secondary has also been turning relative to its shaft as its pin follows the helix. With axial travel blocked, the pin can no longer move along the inclined track. Continued relative rotation would drive it into the side of the track, which pushes back on the pin. The force exerts a torque on the movable sheave; the pin exerts an equal and opposite force on the track attached to the shaft. Unlike the axially fixed stop, the shaft and fixed sheave remain free to change their angular speed. The reaction therefore changes their rotation too, until the movable sheave and shaft turn together.

To see how this exchange changes the shaft speed, follow the angular momentum through the collision. During this upshift, the movable sheave arrives rotating more slowly than the shaft, as established in [Section 3.3.3](#). If the belt supplies no angular impulse during the collision, the faster shaft must supply the angular momentum that brings the movable sheave up to their common speed. The movable assembly gains exactly what the shaft and its attached rotating hardware lose. The shaft slows while the movable assembly speeds up, conserving their combined angular momentum. Their common speed lies between the two incoming speeds.

The incoming state supplies the shaft speed  $\omega_s^-$  and shift rate  $\dot{s}^-$ . The helix relation then gives the known angular speed of the movable assembly before impact:

$$\omega_{s,M}^- = \omega_s^- + \frac{d\theta_s}{ds} \dot{s}^-. \quad (3.6.27)$$

After impact the relative rotation has ended, and the two assemblies share the unknown speed  $\omega_s^+$ :

$$\omega_{s,M}^+ = \omega_s^+. \quad (3.6.28)$$

The stop condition has fixed the axial velocities, but this common rotation speed still has to be found from the incoming angular momentum and the inertias that now turn together.

The belt can take part in the same exchange. If an existing secondary sticking contact remains stuck, changing the pulley speed while keeping the belt speed unchanged would create slip. A tangential contact impulse must change their momenta together so that  $v_b^+ = r_{s,\text{eff}}\omega_s^+$ . If the primary contact remains stuck too, the same belt speed must also satisfy  $v_b^+ = r_{p,\text{eff}}\omega_p^+$ , extending the exchange to the primary. These are the retained sticking conditions of [Assumption 14 \(Tangential Contact Through Boundary Events\)](#); sliding interfaces and newly formed contacts receive no automatic impulsive sticking condition. The shaft and belt speeds must therefore be found together, with the stop and every retained contact condition satisfied by the same outgoing state.

Arrival at the engaged low-ratio seat or the fully open primary stop requires the same accounting. The stop position locates the event, and the stop condition specifies the motion that must end. The remaining velocities depend on which bodies exchange momentum as that motion ends. A complete stop transition must supply those velocities before the continuous balances can be used again.

**Changing the coupling** At these stops, the connections between the bodies remain in place. Primary separation introduces a related difficulty: the primary can keep moving while its connection to the secondary is lost. At the low-ratio boundary  $s = s_{\text{dz}}$ , the secondary is fully closed, but the primary still has room to open. If the primary separates from the belt, it can continue through the deadzone while the secondary remains at its seat. The shift rate still gives  $\dot{x}_p = \dot{s}$ , but no longer produces secondary translation, helix-relative rotation, or changing belt radii.

For a backshift that reaches this boundary at finite speed, simply carrying all the state velocities across unchanged would keep the primary moving while secondary translation and radial belt motion ceased. The movable secondary would also lose the helix contribution to its angular speed. The same state-velocity values would thus describe different physical motions on the two sides. Those changes still involve momentum: the secondary seat arrests its translation, while the helix can transfer angular momentum into the continuing shaft rotation. The event balance must use the actual body velocities produced by the connections on each side of the boundary.

The same separation can impose a velocity restriction as well. In the deadzone, the belt and secondary must satisfy the lock  $v_b^+ = r_{s,\text{eff,dz}}\omega_s^+$  from [Equation \(3.6.5\)](#). If the incoming belt and secondary surface speeds differ, bringing them into agreement requires an impulse even when the shift rate is zero. Entry into the deadzone can change the coupling between bodies and impose a tangential restriction at the same instant.

**The general boundary problem** A boundary can thus change how the state velocities move the bodies, restrict which velocities are allowed, or do both. These two effects account for all five structural regimes. Free engaged shift, the low-ratio seat, and the upper primary stop share the engaged relations between state velocities and body motion. Free deadzone motion and the fully open primary stop share the deadzone relations. There are only two sets of these relations; the stops impose  $\dot{s} = 0$  within them, and tangential locks impose conditions on the shaft and belt speeds.

These examples leave two separate pieces of mechanics that a general boundary law must carry together. The state velocities do not by themselves specify the physical momentum; the current connections determine which masses and inertias those velocities move. The outgoing regime can also restrict the velocity state beyond those connections. A finite-speed transition must therefore

carry the physical momentum from the incoming connection to the outgoing one and, when the outgoing state is more restricted, account for the impulse that establishes those restrictions.

The two effects identified above enter the boundary problem differently. Momentum is carried by the physical hardware in motion, so its contribution can be made explicit before any additional stop or contact restriction is imposed. That accounting begins with the velocities already retained by the model.

**Momentum carried through the boundary** The first part of the boundary problem is to identify the momentum carried by the moving hardware. For each retained translation or rotation, that requires its physical velocity together with the mass or inertia moving at that speed. The model describes these motions through two shaft speeds, a belt transport speed, and the shift rate, collected as

$$\boldsymbol{\nu} = [\omega_p \quad \omega_s \quad v_b \quad \dot{s}]^T. \quad (3.6.29)$$

The upper-stop example already showed that the movable secondary's angular speed contains both shaft rotation and helix-relative rotation, while its axial travel carries a separate linear momentum. The separation example established the complementary point: the physical motion produced by a given  $\boldsymbol{\nu}$  depends on which kinematic connection is active. Applying the established kinematics to the remaining translations and rotations gives the physical velocities in [Table 3.3](#), with the corresponding mass or inertia beside each one.

| Physical motion                 | Mass or inertia                             | Engaged velocity                          | Deadzone velocity         |
|---------------------------------|---------------------------------------------|-------------------------------------------|---------------------------|
| Primary shaft-carried rotation  | $I_{B,p} + I_{p,\text{const}} + J_{f,p}(s)$ | $\omega_p$                                | $\omega_p$                |
| Primary movable axial motion    | $m_p$                                       | $\dot{s}$                                 | $\dot{s}$                 |
| Primary flyweight pivot motion  | $I_f$                                       | $\frac{dq_f}{ds} \dot{s}$                 | $\frac{dq_f}{ds} \dot{s}$ |
| Secondary shaft-side rotation   | $I_{B,s} + I_{s,F}$                         | $\omega_s$                                | $\omega_s$                |
| Secondary movable-side rotation | $I_{s,M}$                                   | $\omega_s + \frac{d\theta_s}{ds} \dot{s}$ | $\omega_s$                |
| Secondary movable axial motion  | $m_s$                                       | $\frac{dx_s}{ds} \dot{s}$                 | 0                         |
| Complete-belt transport         | $m_b$                                       | $v_b$                                     | $v_b$                     |
| Primary-wrap radial motion      | $\rho_b A_b r_{p,\text{cm}} \phi_p$         | $\frac{dr_p}{ds} \dot{s}$                 | 0                         |
| Secondary-wrap radial motion    | $\rho_b A_b r_{s,\text{cm}} \phi_s$         | $\frac{dr_s}{ds} \dot{s}$                 | 0                         |

Table 3.3: Retained physical motions and their momentum-carrying masses and inertias. Each row identifies one physical motion, the mass or inertia carried with it, and the velocity produced by the engaged and deadzone kinematics.

Each velocity in the table is a linear combination of the same four state entries. For the engaged movable secondary, for example,

$$\omega_{s,M} = \omega_s + \frac{d\theta_s}{ds}\dot{s} = \begin{bmatrix} 0 & 1 & 0 & \frac{d\theta_s}{ds} \end{bmatrix} \boldsymbol{\nu}. \quad (3.6.30)$$

In the deadzone, the same row becomes  $\begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}$ : the shaft still turns the movable assembly, but primary travel no longer adds helix-relative rotation. Stacking the nine rows in table order gives the engaged and deadzone velocity maps

$$J_{\text{eng}}, J_{\text{dz}} \in \mathbb{R}^{9 \times 4}. \quad (3.6.31)$$

For either topology,  $J\boldsymbol{\nu}$  therefore contains the nine physical velocities produced by the four state velocities. Read by columns, the same matrix shows where each retained velocity appears through the mechanism; in particular, the fourth column gives the coefficients multiplying  $\dot{s}$  in each retained physical velocity.

The linear or angular momentum carried by each motion is obtained by multiplying its velocity by the corresponding mass or inertia. Placing the masses and inertias from [Table 3.3](#) on the diagonal of  $W \in \mathbb{R}^{9 \times 9}$ , in the same row order, performs these multiplications together:

$$\mathbf{p} = WJ\boldsymbol{\nu} \in \mathbb{R}^9, \quad J \in \{J_{\text{eng}}, J_{\text{dz}}\}. \quad (3.6.32)$$

The entries are the linear and angular momenta of the individual translations and rotations in the table; they are not the components of one spatial momentum vector.

The same construction gives the momentum immediately before and after the boundary event. Positions remain continuous through the instantaneous transition, so the masses and inertias are evaluated at the same configuration and  $W$  is common to both sides. The incoming and outgoing connections determine  $J_-$  and  $J_+$ , respectively, so both maps are known for the transition being considered:

$$J_- \in \{J_{\text{eng}}, J_{\text{dz}}\}, \quad J_+ \in \{J_{\text{eng}}, J_{\text{dz}}\}. \quad (3.6.33)$$

A crossing between engaged and deadzone motion uses different maps, whereas arrival at the upper primary stop has  $J_- = J_+ = J_{\text{eng}}$ . At the fully open primary stop, both sides instead use  $J_{\text{dz}}$ .

The incoming state supplies the known velocity  $\boldsymbol{\nu}^-$ , while  $\boldsymbol{\nu}^+$  remains to be found. The corresponding physical momenta are therefore

$$\mathbf{p}^- = WJ_- \boldsymbol{\nu}^-, \quad \mathbf{p}^+ = WJ_+ \boldsymbol{\nu}^+. \quad (3.6.34)$$

Because each row follows the same physical translation or rotation on both sides, subtracting the two gives its momentum change through the event:

$$\Delta \mathbf{p} = WJ_+ \boldsymbol{\nu}^+ - WJ_- \boldsymbol{\nu}^- \in \mathbb{R}^9. \quad (3.6.35)$$

If  $J_- \neq J_+$ , even identical state-velocity vectors can correspond to different physical motions and therefore different momenta.

**Velocities allowed after the event** With the momentum on both sides now expressed through  $J_-$  and  $J_+$ , the effect of changing the kinematic connection is accounted for in the momentum inventory. The boundary examples established a second possibility as well: even after the appropriate outgoing map  $J_+$  is known, the new regime can impose additional relations on the outgoing velocities  $\boldsymbol{\nu}^+$  themselves. At the upper primary stop, for example, the engaged connection remains valid, so  $J_- = J_+ = J_{\text{eng}}$ , while the already-established stop condition is

$$\dot{s}^+ = 0. \quad (3.6.36)$$

The shift rate  $\dot{s}^+$  is the fourth entry of  $\boldsymbol{\nu}^+$ . Writing the stop condition's coefficients in that state order gives a row that selects this entry, so the required relation becomes

$$c_{\text{stop}} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}, \quad c_{\text{stop}} \boldsymbol{\nu}^+ = 0. \quad (3.6.37)$$

The engaged low-ratio seat and fully open primary stop arrest the same shift coordinate and therefore use the same row. The engaged low-ratio seat uses  $J_{\text{eng}}$ , while the fully open primary stop uses  $J_{\text{dz}}$ .

A belt–pulley contact can restrict the outgoing velocities for a different physical reason. If a contact that was already sticking remains stuck through the collision, the two represented surface speeds must still agree afterward. At the primary, both sheave faces rotate with the shaft, so retained sticking requires

$$v_b^+ - r_{p,\text{eff}} \omega_p^+ = 0. \quad (3.6.38)$$

In the state ordering of [Equation \(3.6.29\)](#), this condition is

$$c_{p,\text{stick}} = \begin{bmatrix} -r_{p,\text{eff}} & 0 & 1 & 0 \end{bmatrix}, \quad c_{p,\text{stick}} \boldsymbol{\nu}^+ = 0. \quad (3.6.39)$$

The same no-slip requirement applies at the secondary, but the represented surface speed there also contains the movable sheave's helix-relative rotation. Using the representative contact speed from [Equation \(3.5.2\)](#), retained secondary sticking requires

$$v_b^+ - r_{s,\text{eff}} \left( \omega_s^+ + \frac{1}{2} \frac{d\theta_s}{ds} \dot{s}^+ \right) = 0, \quad (3.6.40)$$

which gives

$$c_{s,\text{stick}} = \begin{bmatrix} 0 & -r_{s,\text{eff}} & 1 & -\frac{r_{s,\text{eff}}}{2} \frac{d\theta_s}{ds} \end{bmatrix}, \quad c_{s,\text{stick}} \boldsymbol{\nu}^+ = 0. \quad (3.6.41)$$

These rows apply only to sticking contacts retained through the collision. An incoming sliding contact contributes no sticking relation, and a contact created by the event is not assigned one automatically.

Retained sticking through the event is treated here as an ideal impulsive constraint under [Assumption 14 \(Tangential Contact Through Boundary Events\)](#). The resulting tangential impulse is therefore not re-tested against a separate impulsive Coulomb bound; doing so would require the normal and tangential contact impulses to be resolved together during the event.

For outgoing deadzone motion, the reduced model's prescribed belt–secondary lock replaces the engaged secondary sticking condition. The outgoing map is  $J_+ = J_{\text{dz}}$ , and the belt remains locked to the fully closed secondary after primary separation. The closure already derived in [Equation \(3.6.5\)](#) requires

$$v_b^+ - r_{s,\text{eff,dz}} \omega_s^+ = 0, \quad (3.6.42)$$

so an outgoing deadzone state must satisfy

$$c_{s,dz} = \begin{bmatrix} 0 & -r_{s,\text{eff},dz} & 1 & 0 \end{bmatrix}, \quad c_{s,dz}\boldsymbol{\nu}^+ = 0. \quad (3.6.43)$$

The lock applies even when entry into the deadzone must first bring unequal incoming belt and secondary surface speeds into agreement.

At the upper primary stop, if both belt contacts remain sticking, the stop condition and the two sticking conditions must all hold for the same  $\boldsymbol{\nu}^+$ . Their rows can therefore be imposed together by stacking them into a single matrix. More generally, for the  $n_c$  restrictions present in a given event,

$$C_+ = \begin{bmatrix} c_1 \\ \vdots \\ c_{n_c} \end{bmatrix} \in \mathbb{R}^{n_c \times 4}, \quad C_+\boldsymbol{\nu}^+ = 0. \quad (3.6.44)$$

**Constraint impulses** These relations determine which combinations of outgoing velocities are allowed, but they do not themselves supply the momentum change required to establish them. At the upper primary stop, that distinction has a direct physical source: the movable sheave strikes the axially fixed sheave, and the metal contact develops whatever axial reaction is required to prevent further closing. Over the instantaneous event, that reaction appears as a finite impulse associated with the stop constraint.

Let  $\Gamma_{\text{stop}}$  denote this impulse, signed positive toward increasing  $s$ . Since  $c_{\text{stop}}$  identifies the retained direction in which the stop acts, its impulse is

$$c_{\text{stop}}^\top \Gamma_{\text{stop}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \Gamma_{\text{stop}} \end{bmatrix}. \quad (3.6.45)$$

A retained sticking condition has the same structure. The condition  $c_{p,\text{stick}}\boldsymbol{\nu}^+ = 0$  requires the belt and primary surface to leave with zero relative velocity; the tangential contact can supply the impulse required to maintain that relation through the collision. Let  $\Gamma_p$  be positive when the impulse on the belt acts in the positive  $v_b$  direction. The belt then receives  $+\Gamma_p$ , while the equal and opposite action on the primary gives angular impulse  $-r_{p,\text{eff}}\Gamma_p$ :

$$\begin{bmatrix} -r_{p,\text{eff}}\Gamma_p \\ 0 \\ \Gamma_p \\ 0 \end{bmatrix} = c_{p,\text{stick}}^\top \Gamma_p. \quad (3.6.46)$$

When both contacts remain sticking, both tangential impulses contribute to the belt's momentum change. Contributions from the active restrictions therefore add wherever they act on the same retained motion. Associate each row  $c_i$  of  $C_+$  with an unknown signed impulse amplitude  $\Gamma_i$ , and collect these amplitudes in

$$\boldsymbol{\Gamma} = [\Gamma_1, \dots, \Gamma_{n_c}]^\top \in \mathbb{R}^{n_c}, \quad (3.6.47)$$

Their combined generalized impulse is then

$$C_+^T \mathbf{\Gamma} \in \mathbb{R}^4. \quad (3.6.48)$$

Each row therefore carries both the velocity restriction it imposes and the unknown impulse required to enforce it:  $c_i \mathbf{v}^+ = 0$  specifies the outgoing velocity relation, while  $c_i^T \mathbf{\Gamma}_i$  gives the corresponding generalized impulse.

**Momentum balance in the retained velocity directions** The impulses have known places to act, but their magnitudes and the resulting speeds remain unknown. Determining them requires balancing those impulses against the momentum changes of the moving hardware. The difficulty is that  $\Delta \mathbf{p} \in \mathbb{R}^9$  still lists the nine physical translations and rotations separately, whereas  $C_+^T \mathbf{\Gamma} \in \mathbb{R}^4$  already acts in the four retained velocity directions. The body momentum changes must therefore be gathered into those same four directions before the two can be balanced.

That gathering is already encoded in the outgoing velocity map. Each column of  $J_+$  states how one retained velocity appears among the nine outgoing body velocities; the same coefficients therefore state how the corresponding body momentum changes contribute back to that retained direction. For secondary rotation, for example, the  $\omega_s$  column has coefficient one in both the shaft-side and movable-side rotation rows. Denote their angular-momentum changes by  $\Delta H_{s,F}$  and  $\Delta H_{s,M}$ , with the shaft-side term including the boundary inertia  $I_{B,s}$ . These changes therefore enter together:

$$\left[ J_+^T \Delta \mathbf{p} \right]_{\omega_s} = \Delta H_{s,F} + \Delta H_{s,M}. \quad (3.6.49)$$

The upper-stop exchange described at the outset reappears here: the pin-track reaction transfers angular momentum between the two secondary assemblies, and its equal and opposite angular impulses cancel from their combined rotational balance. That transfer remains present in the individual momentum changes. The shift balance, however, also contains the reaction's axial contribution, so the angular and axial impulses must be expressed in the same shift direction before they can be combined.

The compatible-work relation already derived for the helix in [Equation \(3.2.28\)](#) supplies this conversion. A torque contributes to the shift balance through the relative rotation produced per unit shift,  $d\theta_s/ds$ . Because the geometry is fixed through the instantaneous event, integrating the balance preserves this factor: the movable secondary's angular-momentum change contributes

$$\frac{d\theta_s}{ds} \Delta H_{s,M}$$

to the shift-direction balance. The other rotational and translational momentum changes enter with their corresponding motion ratios, which are precisely the entries of the shift column.

For the ideal pin-track reaction, the same compatible-work relation makes its weighted angular and axial impulse contributions equal and opposite in this balance. Their cancellation here, together with the angular cancellation in the secondary rotational balance, explains why the helix reaction requires no additional component of  $\mathbf{\Gamma}$ , even though its momentum transfer remains present in  $\Delta \mathbf{p}$ .

Applying this operation to all four columns collects the nine physical momentum changes into

$$J_+^T \Delta \mathbf{p} \in \mathbb{R}^4, \quad (3.6.50)$$

with one balance associated with each of  $\omega_p$ ,  $\omega_s$ ,  $v_b$ , and  $\dot{s}$ .

Taken together, the gathered balance has a useful interpretation. Motion that exists only on the incoming side is still retained in  $\Delta \mathbf{p}$  through  $W J_- \boldsymbol{\nu}^-$ , while reactions internal to the ideal kinematic connections cancel when the body balances are combined, as the helix exchange does above. The stop and contact impulses enforce additional restrictions rather than those kinematic connections, so they remain explicitly as  $C_+^T \boldsymbol{\Gamma}$ .

**Coupled event solve** The physical momentum changes and the explicit constraint impulses are now expressed in the same four retained directions. They can therefore be balanced directly, while the outgoing velocities must simultaneously satisfy the restrictions already collected in  $C_+$ :

$$\begin{aligned} J_+^T \Delta \mathbf{p} &= C_+^T \boldsymbol{\Gamma}, \\ C_+ \boldsymbol{\nu}^+ &= 0. \end{aligned} \quad (3.6.51)$$

The first relation determines the velocity response to an impulse exchange, but the amplitudes remain unknown until the second relation is satisfied. The second fixes the required speed relationships, but in the upper-stop example their common value still depends on the incoming momentum in the first.

These two relations contain the complete event mechanics, but they are not yet written directly in the unknowns. Substituting the momentum difference from Equation (3.6.35) and placing the known incoming term on the right collects both sets of unknowns:

$$\begin{bmatrix} J_+^T W J_+ & -C_+^T \\ C_+ & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\nu}^+ \\ \boldsymbol{\Gamma} \end{bmatrix} = \begin{bmatrix} J_+^T W J_- \boldsymbol{\nu}^- \\ 0 \end{bmatrix}. \quad (3.6.52)$$

At this point the event can be solved in one step. For a known incoming state and a selected outgoing regime, every block in Equation (3.6.52) is known; solving the system returns the outgoing velocities  $\boldsymbol{\nu}^+$  and the constraint impulses  $\boldsymbol{\Gamma}$  together.

The block matrix has the standard saddle-point, or KKT, structure of a constrained mechanical system (?). Here the constraint multipliers are impulses rather than continuous reaction forces, because the balance is taken across an instantaneous transition (?). The block  $J_+^T W J_+$  is the generalized inertia seen in the outgoing velocity directions,  $C_+^T$  places the unknown constraint impulses into those directions, and  $C_+$  imposes the required outgoing velocity relations. The right-hand side carries the known incoming momentum into the same directions.

The resulting event state must still respect the unilateral character of any active stop. With  $\Gamma_{\text{stop}}$  signed positive toward increasing  $s$ , an impulse at the fully open stop or engaged low-ratio seat must satisfy  $\Gamma_{\text{stop}} \geq 0$ , whereas the upper primary stop can act only toward decreasing  $s$  and therefore requires  $\Gamma_{\text{stop}} \leq 0$ . A solution requiring the opposite sign cannot be supplied by that stop, so the proposed outgoing constraint is not mechanically admissible.

**Completing the upper primary stop** The final system can now resolve the secondary-speed question posed at the outset. Take  $J_- = J_+ = J_{\text{eng}}$  and first consider the case in which the belt supplies no angular impulse to the secondary. The shaft-side and movable-side secondary rotations contribute the following  $(\omega_s, \dot{s})$  block to the generalized inertia matrix:

$$\left[ J_{\text{eng}}^T W J_{\text{eng}} \right]_{\omega_s, \dot{s}}^{\text{secondary rotation}} = \begin{bmatrix} I_{B,s} + I_{s,F} + I_{s,M} & I_{s,M} \frac{d\theta_s}{ds} \\ I_{s,M} \frac{d\theta_s}{ds} & I_{s,M} \left( \frac{d\theta_s}{ds} \right)^2 \end{bmatrix}. \quad (3.6.53)$$

The rows identify the retained direction in which momentum is counted, while the columns identify the retained velocity contributing that momentum. The upper-left entry therefore describes the contribution from  $\omega_s$  to momentum in the  $\omega_s$  direction: both secondary assemblies rotate with the shaft. The upper-right entry describes the contribution from  $\dot{s}$  to that same momentum, because shift rotates the movable assembly through the helix. The lower-left entry is the reciprocal coupling, in which shaft rotation contributes to the shift-direction momentum. The lower-right entry describes the contribution from  $\dot{s}$  to the shift-direction momentum, giving the movable secondary's rotational inertia reflected through the helix into shift.

At the upper stop, the constraint row in [Equation \(3.6.52\)](#) imposes  $\dot{s}^+ = 0$ . The outgoing terms multiplied by  $\dot{s}^+$  therefore vanish, while the incoming shift-induced rotation remains in the known right-hand side. With no secondary belt impulse, the  $\omega_s$  row of the same system becomes

$$(I_{B,s} + I_{s,F} + I_{s,M}) \omega_s^+ = (I_{B,s} + I_{s,F} + I_{s,M}) \omega_s^- + I_{s,M} \frac{d\theta_s}{ds} \dot{s}^-. \quad (3.6.54)$$

The final term on the right is the angular momentum carried into the event by the movable secondary's helix-relative rotation. Using

$$\omega_{s,M}^- = \omega_s^- + \frac{d\theta_s}{ds} \dot{s}^-, \quad (3.6.55)$$

the same row gives

$$\omega_s^+ = \frac{(I_{B,s} + I_{s,F}) \omega_s^- + I_{s,M} \omega_{s,M}^-}{I_{B,s} + I_{s,F} + I_{s,M}}. \quad (3.6.56)$$

The common secondary speed introduced at the beginning of the boundary example has therefore emerged directly from [Equation \(3.6.52\)](#). The incoming helix-relative motion is carried into the solve through the off-diagonal coupling, while the stop condition removes that relative motion from the outgoing state. The pin-track impulse that transfers angular momentum between the shaft-side and movable-side assemblies is internal to the engaged connection; its equal-and-opposite actions cancel from the combined  $\omega_s$  balance rather than appearing as another component of  $\mathbf{\Gamma}$ .

### Remark (Energy consistency check)

The completed event solution also provides a useful consistency check. Because the physical velocities on both sides are already known from  $J_- \boldsymbol{\nu}^-$  and  $J_+ \boldsymbol{\nu}^+$ , the kinetic energy before and after the transition follows directly:

$$T^- = \frac{1}{2}(J_- \boldsymbol{\nu}^-)^\top W(J_- \boldsymbol{\nu}^-), \quad T^+ = \frac{1}{2}(J_+ \boldsymbol{\nu}^+)^\top W(J_+ \boldsymbol{\nu}^+). \quad (3.6.57)$$

For the perfectly inelastic event model, the solved transition should not increase the total kinetic energy. This follows directly from the event balance. Premultiplying the first relation in [Equation \(3.6.51\)](#) by  $(\boldsymbol{\nu}^+)^\top$  gives

$$(J_+ \boldsymbol{\nu}^+)^\top \Delta \mathbf{p} = (C_+ \boldsymbol{\nu}^+)^\top \boldsymbol{\Gamma} = 0,$$

because the outgoing solution satisfies  $C_+ \boldsymbol{\nu}^+ = 0$ . Substituting the momentum change from [Equation \(3.6.35\)](#) then gives

$$T^- - T^+ = \frac{1}{2}(J_+ \boldsymbol{\nu}^+ - J_- \boldsymbol{\nu}^-)^\top W(J_+ \boldsymbol{\nu}^+ - J_- \boldsymbol{\nu}^-) \geq 0. \quad (3.6.58)$$

The right-hand side is a sum of one half of each retained mass or inertia times the square of its physical velocity change. The result therefore provides a direct check that the event solution is dissipative even when the kinematic connection changes. Individual bodies can still gain kinetic energy through the momentum exchange, as the movable secondary does at the upper stop; only the total must not increase.

Together, these rules determine how the state passes from one mechanical regime to the next. While the current regime remains admissible, its continuous balances advance the state. Loss of a support or contact admissibility condition ends that interval and determines which neighboring mechanics can become active. If the new regime already admits the incoming velocity, the state passes into it continuously; if a newly active constraint does not, [Equation \(3.6.52\)](#) supplies the compatible outgoing velocity and the impulse required to establish it. That outgoing state then provides the initial condition for the next interval of continuous motion.

## Chapter 4

# Results and Discussion

The formulation in [Chapter 3](#) determines the CVT response from its internal mechanics and the loading at its shafts. The numerical investigation begins with verification of this coupled model and comparison with an independent transient formulation. It then examines when the retained actuator and belt dynamics change the predicted behaviour. Full transient simulations finally bring these mechanisms together, following how the CVT engages, shifts, and responds to changing loads.

### 4.1 Computational Implementation and Reproducibility

CINDER is the simulator developed in this work to implement the formulation and carry out these numerical studies. Its numerical methods are documented in [Appendix D](#).

The model was developed around a Baja SAE CVT, whose hardware forms the basis of the reference configuration in [Appendix E](#). Much of the investigation therefore concerns the operating range of a small off-road vehicle. The quantitative importance of the effects examined here requires separate assessment for substantially different machines.

Complete study definitions, including the physical configurations and numerical settings, are collected in [Appendix F](#). The external systems that drive and load the CVT, including simplified engine and vehicle models, are defined separately in [Appendix E.2](#).

The results reported here use a frozen CINDER release and a separate frozen study record containing the case definitions, analysis scripts, and outputs. Both records are identified in [Table 4.1](#). The correspondence between the archived outputs and the manuscript figures and tables is given in ??.

Table 4.1: Frozen software and study records used for the results in this chapter.

| Record                | Release / source revision                                                           | Source / archive                                                             |
|-----------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| CINDER implementation | <code>cinder-v1.1.2</code><br><code>7637a38b4fb9ec21dfb953c1c80a27ec5f389654</code> | <a href="#">Tagged source</a>                                                |
| Study record          | <code>develop (working)</code><br><code>TBD-at-results-freeze</code>                | <a href="#">Working Results source</a><br><code>TBD-at-results-freeze</code> |

## 4.2 Internal Verification

The same belt–pulley contact forces enter the belt transport, shaft rotation, and axial sheave balances. Their numerical evaluation must preserve these coupled effects as contacts stick or slide and travel constraints engage or release. Verification therefore covers both the individual mechanical conditions and their combined response: the operating-domain audit and work–energy balance check local and global consistency, while refinement of the time solution and examination of the instantaneous contact closure assess the numerical reliability of that response.

### 4.2.1 Mechanical Invariants and Operating-Domain Audit

The audit combines the ten-second, flat-road Baja launch with constructed constant torque–inertia cases that exercise additional contact and structural regimes. The launch retains the engine and vehicle boundaries of [Appendices E.2.2](#) and [E.2.3](#); the constructed cases instead prescribe the torque and attached inertia at each shaft using [Appendix E.2.1](#), while allowing both shaft speeds to evolve. All use the reference CVT and its zero-clearance bilateral secondary helix, defined in [Appendices E.1](#) and [E.1.1](#). Case definitions and audit thresholds are collected in [Appendix F.1](#).

All 25 required coverage classes were exercised. These include rest and deadzone motion, engagement, free shifting in both directions, the travel stops, forward and reverse rotation, either slipping contact with either slip sign, and all four both-slip sign combinations. Across 6,257 audited states and 81 hybrid transitions, the recorded mechanical checks passed, including those applied to the exact post-transition states. The largest row-scaled mechanical-closure residual was  $1.11 \times 10^{-14}$ , using the normalization in [Equation \(F.1.1\)](#). The coverage record is given in [Appendix F.1](#); [Figure 4.1](#) shows how differently the canonical contact cases behave within that successful audit.

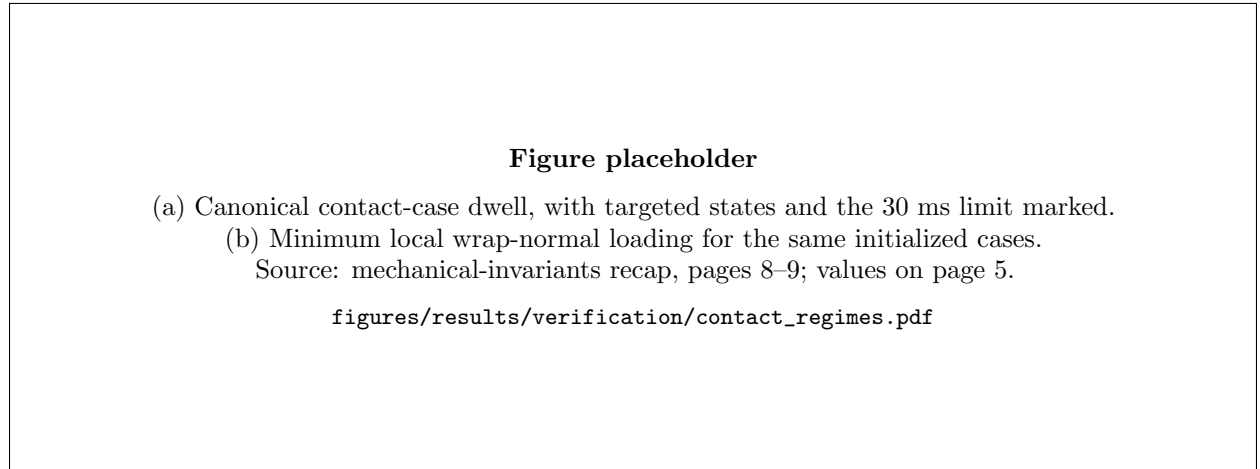


Figure 4.1: Contact-regime coverage and persistence in the reference CVT. (a) Time to the first transition from each requested initial contact state; a case with no transition in the 30 ms interval has a dwell of at least 30 ms. Slip signs refer to  $v_{\text{rel},j}$  in [Equation \(3.5.3\)](#). (b) Minimum distributed normal loading on each pulley over the corresponding audited case. The contact cases satisfy the retained constraints, but differ substantially in persistence and local contact margin.

Several requested branches persist throughout the 30 ms interval, whereas secondary-slip (+) and both-slip (–, +) leave their initial branches through secondary resticking after approximately 414  $\mu\text{s}$  and 9.82  $\mu\text{s}$ , respectively. Both required targeted initial states in this reference configuration. The

readily reproduced both-slip (+, +) case also resticks quickly, after 31.7  $\mu\text{s}$ . Thus short dwell and difficulty finding an initial state are distinct observations. In these cases the coupled dynamics rapidly bring the secondary relative speed back to zero; a brief initial branch is a resolved part of the trajectory rather than a failure to continue the simulation. These observations describe the selected hardware, loads, and initial states, not the frequency of these regimes in vehicle operation generally.

The local contact margins in [Figure 4.1\(b\)](#) also show why integrated clamping alone is an incomplete admissibility check. In the case initialized in both-slip (−, −), the minimum primary distributed normal loading is 0.0482 N/rad, while the integrated primary normal resultant remains at least 374 N over that case. The small local margin is therefore not accompanied by disappearance of the overall normal load. The retained full-wrap description requires nonnegative contact loading everywhere along the wrap as well as a nonnegative resultant. Together with the exact successor checks, this audit demonstrates consistent operation across the tested regime set, without implying that every admissible initial state or every possible transition has been exhausted.

#### 4.2.2 Energy Consistency

The energy audit follows the reference launch for ten seconds on flat road, with the numerical settings in [Appendix F.2](#). Its inventory includes the retained kinetic stores and conservative actuator-spring energy. The boundary work is the signed net work supplied at the two shafts,

$$W_B(t) = \int_0^t (\tau_{B,p}\omega_p + \tau_{B,s}\omega_s) dt, \quad (4.2.1)$$

so a resisting vehicle load contributes negative work. The audited balance is

$$W_B(t) = \Delta E_{\text{stored}}(t) + E_{\text{slip}}(t) + E_{\text{capture}}(t) + R_E(t), \quad (4.2.2)$$

where  $R_E$  is the unexplained numerical residual, not an additional loss mechanism.

[Figure 4.2](#) compares this accounting on the throughput scale and then resolves the much smaller balance defect. The final residual is 0.0254 J against 52.17 kJ of net boundary work, giving  $\epsilon_E = 4.88 \times 10^{-7}$  under [Equation \(F.2.1\)](#). The final energy ledger in [Table 4.2](#) distinguishes the two retained irreversible channels from stored energy: kinetic belt–pulley slip accounts for approximately 1.588 kJ, whereas the cumulative inelastic capture loss is only 0.254 J in this trajectory. Spring storage and inertial storage are not counted as dissipation. The omitted mechanisms remain those stated in [Assumption 15 \(Negligible Unmodeled Parasitic Losses\)](#); this ledger quantifies the losses represented by the model.

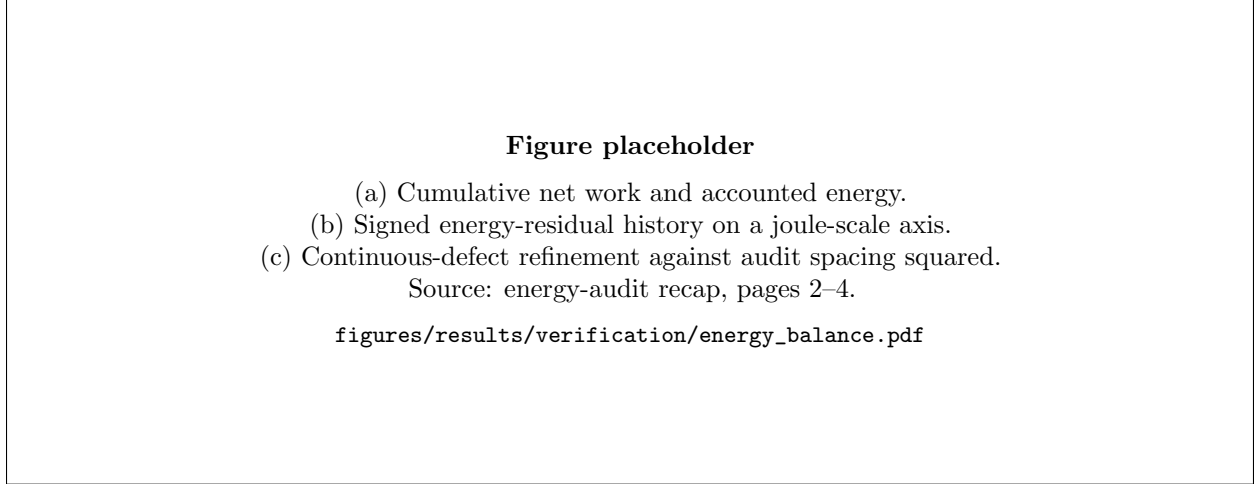


Figure 4.2: Mechanical-energy accounting for the reference launch. (a) Net boundary work and stored-energy change plus modeled dissipation. (b) Their signed residual, shown separately from the kilojoule-scale throughput. (c) Refinement of the summed continuous-segment defect using audit spacings of 5, 2.5, 1.25, and 0.625 ms; the horizontal line is the Richardson zero-spacing estimate. Capture losses are accounted for at the discrete events rather than integrated across the velocity resets.

Table 4.2: Final energy ledger for the reference audit. The residual is evaluated from unrounded quantities; displayed entries are rounded.

| Quantity                      | Energy [J] |
|-------------------------------|------------|
| Net boundary work             | 52 168.893 |
| Stored-energy change          | 50 580.274 |
| Kinetic-slip dissipation      | 1 588.340  |
| Inelastic capture dissipation | 0.253859   |
| Balance residual              | 0.025444   |

The refinement panel is important because coincident cumulative curves alone would conceal a small systematic defect. Reducing the quadrature spacing approaches a finite continuous-defect estimate of 0.024510 J: post-processing refinement alone does not drive the fixed-trajectory residual to zero. A tighter trajectory integration gives a final residual of  $-0.022843$  J, changing its sign while retaining the same small scale. The maximum normalized final-state change is  $1.30 \times 10^{-7}$ , using [Equation \(F.2.2\)](#). Discrete captures separately satisfy the recorded energy-drop relation to within  $2.27 \times 10^{-13}$  J. These checks support energy consistency at the observed numerical-error scale across both the continuous motion and the discrete updates. The residual fraction measures accounting accuracy; it is not a transmission-efficiency estimate.

### 4.2.3 Solver Convergence

A hybrid trajectory includes the contact history as well as the shaft speeds and shift motion. Its numerical convergence therefore requires both parts to agree. The convergence benchmark uses the ten-second reference launch, with a tight LSODA solution at  $\text{rtol} = 10^{-7}$ ,  $\text{atol} = 10^{-10}$ , and a 2.5

ms maximum step as the numerical reference. The physical case and tested settings are specified in [Appendix F.3](#).

The transition signature records the ordered events, successor modes, and occurrence of velocity resets. State differences are evaluated between corresponding smooth segments after matching that signature; a different signature is itself a convergence failure. Each segment is sampled at the same fractional progress between its endpoint events rather than at the same absolute time, so a small event-time offset does not compare a pre-reset state with a post-reset state. Event-time error is reported separately, together with the fraction of time spent in different regimes. The normalization and duration-weighted root-mean-square (RMS) error are defined by [Equations \(F.3.2\) to \(F.3.4\)](#). The acceptance criteria in [Table F.11](#) require an exact transition signature as well as bounds on the state and timing errors.

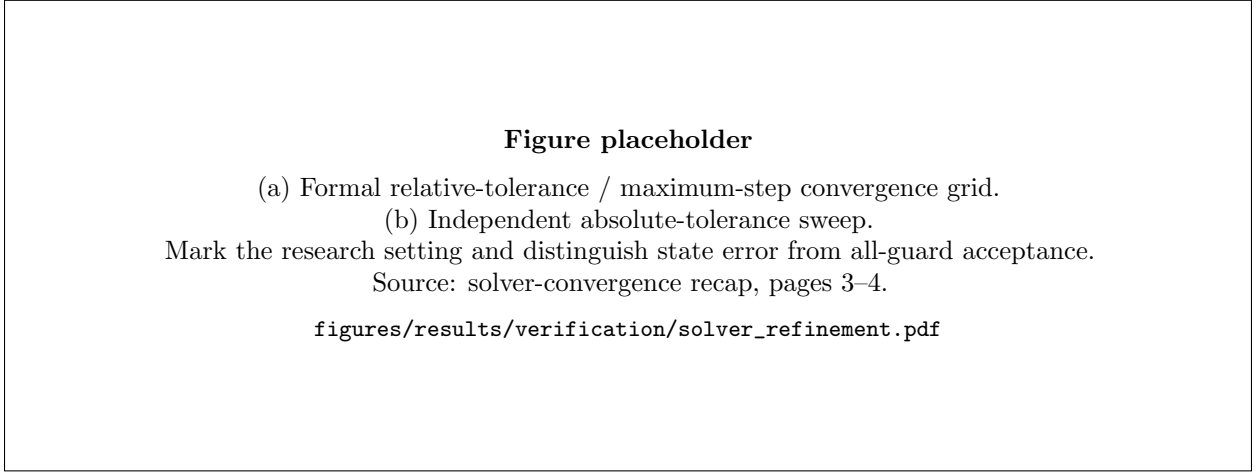


Figure 4.3: Convergence of the reference hybrid trajectory. (a) Event-aligned normalized state error over the formal relative-tolerance and maximum-step grid. (b) Independent absolute-tolerance sensitivity at  $\text{rtol} = 10^{-4}$  and a 10 ms maximum step. The research setting is  $\text{rtol} = 10^{-4}$ ,  $\text{atol} = 10^{-7}$ , and a 10 ms maximum step. The maximum step is an upper bound on the adaptive integration, not a fixed time increment.

[Figure 4.3\(a\)](#) shows a contiguous passing region: every tested maximum step from 5 to 100 ms satisfies all criteria when  $\text{rtol} \leq 10^{-4}$ , with  $\text{atol} = 10^{-3}\text{rtol}$  in that sweep. At the research setting, the event-aligned trajectory RMS is  $4.27 \times 10^{-7}$ , the maximum normalized state error is  $6.96 \times 10^{-5}$ , and the largest corresponding event-time difference is  $7.99 \mu\text{s}$ . The complete 16-transition signature is reproduced, and the regime-mismatch fraction is  $8.46 \times 10^{-6}$ .

The independent absolute-tolerance sweep in [Figure 4.3\(b\)](#) matters because the angular speeds, shift position, and shift speed have different numerical scales. Loose absolute tolerances alter the transition history; the tighter tested range from  $3 \times 10^{-7}$  to  $10^{-8}$  recovers the reference history, with  $10^{-7}$  satisfying all criteria. The one failed formal integration uses the coarsest combination,  $\text{rtol} = 10^{-2}$  and a 100 ms maximum step, and reaches an inadmissible primary flyweight-contact state. This is a numerical/domain failure of that calculation, not evidence of physical lift-off on the converged trajectory.

### Figure placeholder

Gross trajectory error versus native solver points, grouped by transition count.

Source: solver-convergence recap, page 9, Figure 8.

This existing population plot supports the current text without selecting a new case.

`figures/results/verification/gross_motion_hybrid_history.pdf`

Figure 4.4: Macroscopic agreement and hybrid-history agreement are distinct. The dense exploratory sweep is shown using the absolute-time gross-state error for  $(\omega_p, \omega_s, v_b, s)$ , excluding  $\dot{s}$ , and grouped by transition count. A low gross error need not reproduce the reference contact history. Native solver points are a work proxy; they are not right-hand-side evaluation counts.

The dense sweep makes the distinction visible in [Figure 4.4](#). Among 609 completed runs with a different transition signature, the median absolute-time gross-state RMS error is only  $1.23 \times 10^{-4}$ . This diagnostic includes shaft speeds, belt speed, and shift position, but excludes shift speed; it is not the event-aligned five-state metric used for acceptance. Gross motion can remain close while brief contact transitions differ. Resolving internal traction and shift mechanisms therefore requires the hybrid checks, not only a visually converged speed or ratio curve. The demonstrated tolerance range applies to this benchmark; settings for other cases remain part of their individual study definitions.

#### 4.2.4 Closure Conditioning and Root Structure

The engaged solve separates into a linear mechanical response at fixed  $(\lambda_p, \lambda_s)$  and a nonlinear selection of the traction required by sticking contact. The eight-variable form, including the retained pulley torques, is given in [Equation \(D.2.1\)](#). These two levels are assessed separately: uniqueness of the fixed-traction response does not imply uniqueness of the traction pair that enforces stick.

At the frozen states drawn from the reference launch and constructed operating cases, no rank-deficient mechanical system is found in the sampled static-capacity or expanded traction maps. Each sampled fixed-traction system therefore has a unique linear response. The condition numbers in [Figure 4.5\(a\)](#) use the equilibrated matrix of [Equation \(D.2.3\)](#); the largest reported value among the four representative static-capacity maps is  $3.38 \times 10^5$ . Full rank does not imply uniform sensitivity, and some trial points violate contact or support admissibility even within the static-capacity box. The map domains and mechanical-admissibility checks are distinguished in [Appendix F.4](#).

**Figure placeholder**

- (a) Four representative static-capacity-map maxima of equilibrated  $\kappa(\mathbf{A}_s)$ , replotted from the closure recap, page 3.
- (b) Mid-shift multistart outcomes, closure recap, page 5.  
Distinguish accepted roots from unsuccessful starts.
- `figures/results/verification/closure_robustness.pdf`

Figure 4.5: Representative fixed-traction and sticking-root behaviour. (a) Maximum equilibrated mechanical-matrix condition number in each of four sampled static-capacity maps; these are map maxima rather than values at the accepted sticking root. (b) Starting guesses and acceptance outcomes for the mid-shift sticking solve. Accepted solutions belong to one root cluster; unsuccessful starts are shown separately. Mechanical admissibility is assessed independently of membership in the static-capacity box.

The representative nonlinear solves are also reproducible across starting estimates. Across 15 states and 375 multistart attempts in total, all accepted solutions for each state belong to one physical root cluster. **Figure 4.5(b)** illustrates this result at mid-shift. The statement concerns the accepted roots: it neither assumes that every initial guess succeeds nor proves nonlinear uniqueness outside the tested states.

Targeted continuation reveals a different structure elsewhere in the reduced contact equations. In the selected stick–stick example of **Figure 4.6**, a continuous solution branch turns back in loading space. A fixed loading then intersects that branch twice, producing two traction pairs that satisfy both sticking residuals and the physical admissibility checks. The smaller static-capacity reserve of the selected pair is 0.098, so the multiplicity is not explained simply by including an above-capacity mathematical root. It occurs in the nonlinear traction selection while the fixed-traction mechanics still give a definite response at either root.

### Figure placeholder

- (a) Focused traction-plane view with two admissible stick-stick roots.
- (b) Continuation curve and fixed-load line intersecting both sides of the fold.

Source: closure recap, page 6; selected late-shift opening case.

`figures/results/verification/sticking_closure_fold.pdf`

Figure 4.6: Two admissible sticking solutions in a deliberately selected frozen-state example. The traction-plane view identifies the two roots; the continuation view shows that they lie on opposite sides of one folded branch at the same applied loading. Multiple nonlinear sticking roots can therefore coexist with a unique mechanical response for each prescribed traction pair.

Multiplicity also occurs in a mixed branch with primary slip at  $-\mu_k$  and secondary stick. The targeted 62-state investigation found one distinct scalar-fold geometry, with secondary sticking roots near 0.434 and 0.546 on the selected loading slice. One approaches local wrap-contact loss. The examples establish that neither the two-contact nor the one-contact sticking problem is globally single-rooted; their frequency in this targeted search is not an operating probability.

These folded branches were exposed by deliberate frozen-state searches. No spontaneous ambiguity of this type was observed in the maintained transient studies. During smooth evolution, the continuation strategy in [Appendix D.4.2](#) reuses the previous accepted traction solution, encouraging the solver to remain on the nearby branch. Fresh mechanical-residual and admissibility checks still apply, as described in [Appendix D.4.1](#). Continuation reduces arbitrary root changes but cannot remove a fold or guarantee a unique continuation when a branch ends. The positive sampled solvability results and the global multiplicity finding therefore describe different, compatible properties of the reduced closure.

## 4.3 Independent Transient-Model Benchmark

Ballew’s discretized rubber-belt model provides a five-second ATV acceleration case with published shaft-speed histories, primary clamping force, and controller gains ([Ballew, 2015](#)). It permits a comparison at two different levels: whether CINDER reproduces the response under reconstructed speed feedback, and whether it reproduces the response when the published clamping force itself is imposed. The distinction is consequential. In the first comparison the controller can adjust the force in response to CINDER’s motion; in the second, the force history is fixed and the transmission must respond to it directly.

The two comparisons give markedly different impressions of agreement. Reconstructed feedback keeps the shaft-speed errors relatively small despite a large difference in clamp demand. Force replay then exposes a much larger difference in primary speed. Together, they identify a limit on what agreement in controlled shaft speeds can establish about the underlying belt and shift mechanics.

Table 4.3: RMSE relative to the mean absolute reference value at the comparison samples, expressed as a percentage. The shaft-speed ratio is  $\omega_p/\omega_s$ . Primary clamp is prescribed in force replay and is therefore not an independently predicted output in that column.

| Quantity          | Closed loop | Force replay     |
|-------------------|-------------|------------------|
| Primary speed     | 4.42%       | 77.89%           |
| Secondary speed   | 2.71%       | 1.09%            |
| Shaft-speed ratio | 4.81%       | 78.31%           |
| Primary clamp     | 46.66%      | Prescribed input |

### 4.3.1 Benchmark Reconstruction

The benchmark is assembled from Ballew’s published case rather than from the Baja reference. Both protocols use the same reconstructed belt geometry, rotating inertias, constant primary torque of 18 N m, and ATV road-load boundary. The secondary closing force is fixed at 2000 N. The initial primary and secondary speeds are the explicitly tabulated 2500 and 1136 rpm; their compatible belt speed and shift position are obtained from the reconstructed geometry. The boundary and protocol parameters are collected in [Appendix F.5](#), and the interchangeable force laws are defined in [Appendix E.3.2](#).

Several translations are necessary to make those inputs comparable. The published belt mass and combined shaft inertias are preserved, and the friction coefficients are converted between the two normal-force conventions so that the gross tangential capacity corresponds to the same applied clamp. Both literal moving-sheave masses are set to zero because Ballew determines radial position from an algebraic axial-force balance rather than integrating a movable-sheave axial equation. CINDER’s belt inertia and reduced shift dynamics remain active. Thus the comparison does not introduce the Baja sheave masses as an additional difference between the two models.

The published model still contains internal belt deformation states that cannot be initialized identically in CINDER. Those differences are retained as part of the model-to-model comparison. No physical parameter or controller gain is fitted to the reference traces. The published information, the necessary reconstruction choices, and the digitization procedure are recorded separately in the study’s [reconstruction record](#).

### 4.3.2 Reconstructed Closed-Loop Response

In the closed-loop protocol, primary speed is regulated toward 2500 rpm by the reconstructed PI and feed-forward law of [Equation \(E.3.7\)](#). The published gains are used with an integrated controller-error state. Ballew’s primary force history is an output reference in this protocol, not an input to the controller.

The left-hand comparison in [Figure 4.7](#) shows why the shaft-speed agreement is useful but incomplete. Primary and secondary speed RMSE are approximately 110 and 33 rpm, respectively. Yet the primary force differs substantially, and CINDER’s controlled response is visibly more oscillatory. The percentage errors in [Table 4.3](#) quantify the contrast: errors of only a few percent in the shaft histories coexist with a primary-clamp error of approximately 47%.

The controller is therefore generating a different force history while keeping the primary speed near the same objective. Agreement in the regulated quantity does not imply agreement in the force

### Existing results: paired protocol comparison

Left: reconstructed feedback. Right: published force replay. Compare the same shaft outputs and show how primary clamp is supplied in each protocol. Include a resolved closed-loop time-window inset.

`figures/results/ballew/protocol_comparison.pdf`

Figure 4.7: Ballew model-to-model comparison under reconstructed speed feedback and prescribed primary-force replay. The secondary closing force and the reconstructed shaft boundaries are common to both protocols. In the closed-loop case, primary clamp is generated by feedback; in force replay, the digitized force from Ballew’s Figure 45 is imposed. The original shaft-speed references are from Ballew’s Figure 41 (Ballew, 2015).

required to produce it. This is particularly important here because clamping controls both the shift balance and the available traction.

#### 4.3.3 Force Replay and the Internal Response

Force replay removes that adjustment. The prepared Figure 45 history is applied through the tabulated primary actuator of Equation (E.3.5), while the other reconstructed inputs are unchanged. The right-hand comparison in Figure 4.7 then shows a substantial reduction in CINDER’s primary speed relative to Ballew’s history. Its primary-speed and shaft-ratio errors approach 78%, even though secondary speed remains close to the reference. Agreement at the output shaft alone would consequently conceal a very different transmission response.

The internal kinematic comparison in Figure 4.8 makes the limitation more direct. For this benchmark, which has no relative sheave-rotation mechanism, the shaft-speed ratio  $\omega_p/\omega_s$  and geometric ratio  $r_{s,\text{eff}}/r_{p,\text{eff}}$  would coincide if both contacts rolled without slip. In CINDER’s reconstructed closed loop they instead follow substantially different histories. A relatively close shaft-speed ratio is therefore not evidence that the model has reproduced the same belt position or contact state. The accompanying slip-energy history supplies an energetic view of that distinction within CINDER’s retained loss model.

#### 4.3.4 Interpretation of the Model-Form Difference

The leading physical difference is the treatment of the belt itself. Ballew resolves internal deformation and local nodal motion, whereas CINDER represents the belt with a fixed length, one transport speed, and a prescribed path controlled by the shift coordinate. Local deformation, force redistribution, and contact evolution can therefore change the response to the same clamp history. The unmatched distributed initial-tension state and the reconstructed controller equation also bound how directly individual differences can be attributed to one belt mechanism.

Rubber compliance and dissipation are plausible contributors to a less abrupt response, but they

### Existing results: kinematics and slip loss

Show shaft-speed and effective-radius ratios separately, with a resolved oscillation window. Add cumulative modeled slip dissipation. Do not use a smoothed geometric ratio as a substitute for the actual trajectory.

`figures/results/ballew/internal_response.pdf`

Figure 4.8: Internal response of the reconstructed closed-loop CINDER case. Shaft-speed ratio, effective-radius ratio, and modeled slip dissipation describe different aspects of the response. Their comparison shows why macroscopic speed agreement alone does not establish agreement of the internal belt and shift mechanics.

are not interchangeable explanations. Compliance allows energy to be stored and motion to be redistributed; damping dissipates energy. Moreover, Figures 41 and 45 are Ballew’s case *without the added linear inter-node damping*, rather than his separate linearly damped case (Ballew, 2015, Figures 40–45). The present mismatch cannot therefore be assigned specifically to that added damper. Separating the remaining causes would require controlled changes to the belt descriptions, beyond the comparison available here.

### Evidence to complete: B1 — benchmark-specific numerical evidence

Recover the existing four-case closed-loop refinement outputs before using the oscillatory waveform to make a stronger physical claim. Compare the nominal 1 ms limit with 0.5 and 0.25 ms limits and the tighter-tolerance 0.5 ms case specified in the reconstruction’s numerical-stability record. The supplied recap supports the aggregate comparison above, but does not provide the quantitative refinement table. No new convergence result is asserted here.

The useful result is the contrast between the two protocols. CINDER can approach the controlled shaft histories while requiring a different clamp history, and the discrepancy becomes much more pronounced when that freedom is removed. The benchmark consequently tests more than whether two speed curves look similar: it reveals a difference in the reconstructed clamp-to-shift/contact response. It is an independent model comparison, not experimental validation of either internal belt description.

## 4.4 Dynamic Mechanics Retained by CINDER

At a given mechanism position, the same centrifugal or torque loading need not produce the same clamp during a transient. Part of that loading may be required to accelerate the actuator itself. A quasi-static actuator law omits that inertial demand. The resulting approximation may reproduce a smooth response while changing the reaction to a rapid load disturbance. The relevant question is therefore which retained inertial contributions are actually excited, and when their omission changes the solved CVT motion.

The comparisons begin with the primary and secondary mechanisms under this common question,

then examine the transient terms retained in the reduced belt. The actuator variants use the interchangeable interface in [Appendix D.3](#); their exact reductions and redistribution of rotating inertia are given in [Appendix E.3.1](#). Evaluating an omitted term on the full-model trajectory isolates its local contribution. Integrating the corresponding reduced model answers the different question of whether that omission changes the trajectory itself.

#### 4.4.1 Primary Actuation Dynamics

##### Centrifugal drive and the motion it must accelerate

The flyweights both generate the centrifugal closing action and move about their pivots as the primary sheave shifts. These roles involve different inertias. The shaft-axis map  $J_{f,p}(x_p)$  sets the centrifugal drive, whereas the pivot inertia  $I_f$  sets the torque required to accelerate the flyweight motion. Writing [Equation \(3.2.11\)](#) as its quasi-static part plus its omitted contribution gives

$$\begin{aligned} F_{fw,x_p} &= F_{fw,x_p}^{QS} + \Delta F_{fw,x_p}, \\ F_{fw,x_p}^{QS} &= \frac{1}{2} \omega_p^2 \frac{dJ_{f,p}}{dx_p}, \\ \Delta F_{fw,x_p} &= -I_f \left( \frac{dq_f}{dx_p} \right)^2 \ddot{x}_p - I_f \frac{dq_f}{dx_p} \frac{d^2 q_f}{dx_p^2} \dot{x}_p^2. \end{aligned} \tag{4.4.1}$$

The acceleration term has a positive inertial coefficient

$$m_{f,\text{ref}} = I_f \left( \frac{dq_f}{ds} \right)^2, \tag{4.4.2}$$

because  $x_p = s$ . Thus even a modest pivot inertia can resist shifting strongly if the mechanism requires a large flyweight rotation per unit sheave travel. The final term in [Equation \(4.4.1\)](#) instead depends on how that motion ratio changes along the ramp. A large centrifugal force alone does not determine either correction.

This distinction gives a useful first sensitivity test before selecting a transient. For a sheave excursion of fixed shape and amplitude executed over a time scale  $T$ , both  $\ddot{x}_p$  and  $\dot{x}_p^2$  scale as  $T^{-2}$ . Shortening the same prescribed excursion can therefore magnify the dynamic correction without changing the centrifugal force at the same position and shaft speed. This is a kinematic scaling argument; in an actual CVT the excursion and its duration are themselves outcomes of the coupled solve.

##### Small force corrections do not make flyweight inertia irrelevant

The reference actuator study identifies ordinary continuous Baja operation as close to quasi-static. A small primary correction means that little of the centrifugal action is consumed by accelerating the flyweights about their pivots. The comparison in [Figure 4.9](#) separates this force-level statement from its consequence for shaft and shift motion. A local measure is

$$\Pi_p(t) = \frac{|\Delta F_{fw,x_p}(t)|}{|F_{fw,x_p}^{QS}(t)|}, \quad F_{fw,x_p}^{QS} \neq 0. \tag{4.4.3}$$

The signed force terms must accompany this ratio so that a small denominator or cancellation between corrections is not mistaken for a large change in physical loading.

**New figure: primary geometry, excitation, and response**

Map the motion-ratio-weighted pivot inertia over travel. Below it, show signed centrifugal/inertial force terms and a matched full-versus-QS primary comparison. Keep the shaft-torque correction separate from the axial-force correction. Quantitative primary evidence is pending.

`figures/results/dynamics/primary_geometry_and_response.pdf`

Figure 4.9: Primary-actuator comparison to be assembled from the fixed-pivot mechanism map and its matched dynamic and quasi-static runs. The upper display locates geometric sensitivity; the lower displays test whether the realized motion excites that sensitivity and changes shaft or shift response. All variants use the same tested primary mechanism and the inertia treatment in [Appendix E.3.1](#).

**Evidence to complete: P1 — recover the primary comparison**

The supplied commercial-actuator recap gives the qualitative ordinary-Baja conclusion, but not the primary’s force peaks, shaft-torque corrections, or paired trajectory errors. Insert those from the archived fixed-pivot study. Use at least the smooth reference interval and the strongest admissible primary transient already tested; report the interval, denominator, and full/QS comparison. If those outputs do not support the qualitative statement for the primary separately, revise that statement. No numerical primary bound or universal negligible-inertia claim is assumed.

There is a second distinction to preserve. The same flyweight mass also changes the primary’s shaft-axis angular momentum. Its contribution to the rotational balance is

$$\frac{d}{dt}[J_{f,p}(s)\omega_p] = J_{f,p}(s)\dot{\omega}_p + \frac{dJ_{f,p}}{ds}\dot{s}\omega_p. \quad (4.4.4)$$

Even when the pivot-acceleration force is small, outward flyweight motion changes the rotating inertia. During upshift, angular momentum is required to move that mass outward; during backshift, the redistribution reverses. The axial-force comparison and the shaft-torque comparison therefore answer different questions. In particular, the trajectory ablation changes the mechanism’s inertial coupling as specified in the appendix; it is not a test of deleting a single axial term while everything else remains identical.

The primary result is consequently specific to the fixed-pivot roller mechanism examined here. It can justify the quasi-static force approximation for that mechanism over the supported operating intervals, while the maps in [Equation \(4.4.1\)](#) identify what must be checked for another design. Neither greater flyweight mass nor a similar static clamp curve, by itself, establishes the size of the dynamic error.

#### 4.4.2 Secondary Helix Dynamics and Transient Sensitivity

##### The helix converts rotational inertia into axial response

The secondary adds a direct route by which shaft acceleration changes clamp. Its movable assembly rotates with the shaft and relative to it through the helix. The torque available for axial reaction must therefore supply both motions. Define the quasi-static reacted torque, with the formulation's symmetric face share, as

$$\tau_{s,\text{helix}}^{\text{QS}} = \frac{\tau_s}{2} + k_{s,\theta}(\theta_{s,\text{pre}} - \theta_s). \quad (4.4.5)$$

Then Equation (3.2.30) separates into

$$\begin{aligned} F_{\text{helix},x_s} &= \frac{d\theta_s}{dx_s} \tau_{s,\text{helix}}^{\text{QS}} + \Delta F_{\text{helix},x_s}, \\ \Delta F_{\text{helix},x_s} &= -I_{s,M} \frac{d\theta_s}{dx_s} \dot{\omega}_s - I_{s,M} \left( \frac{d\theta_s}{dx_s} \right)^2 \ddot{x}_s \\ &\quad - I_{s,M} \frac{d\theta_s}{dx_s} \frac{d^2\theta_s}{dx_s^2} \dot{x}_s^2. \end{aligned} \quad (4.4.6)$$

The three omitted contributions are driven by shaft acceleration, axial acceleration, and changing helix motion ratio. The first can remain active even while the sheave is instantaneously stationary. The second changes the inertia multiplying axial acceleration in Equation (3.2.33). Its contribution to the shared shift coordinate is

$$m_{h,\text{ref}} = I_{s,M} \left( \frac{d\theta_s}{ds} \right)^2. \quad (4.4.7)$$

The force correction and the reflected inertia are thus two appearances of the same mechanism, not separate effects that can be assessed independently once the trajectory changes.

The equation also distinguishes excitation time scales. A change in shaft speed over a shorter interval increases the first term through  $\dot{\omega}_s$ ; a fixed-shape axial excursion executed more rapidly increases the acceleration and squared-velocity contributions. A constant-angle helix removes the local profile-curvature term, but retains both acceleration terms. Nonlinear mapping from  $x_s$  to  $s$  may still introduce  $\dot{s}^2$  contributions in the shared-coordinate equations.

Geometry can amplify these contributions strongly because the motion ratio appears squared in the reflected axial inertia. The commercial-secondary estimate in Figure 4.10 illustrates why physical rotational inertia and motion ratio must be considered together. Its motion ratio is weaker than the Baja reference, yet the larger movable-member inertia raises the nominal reflected local axial inertia to approximately 1.49 times the reference at  $31^\circ$ . The low and high estimates span approximately 0.45 to 2.92 times the reference. The  $28^\circ$  high-estimate sensitivity reaches approximately 3.72. These are component estimates, not measurements of a complete commercial CVT response.

##### A continuous transient with a measurable response change

To connect that scaling to motion, the nominal commercial helix geometry and movable-member rotational inertia are inserted into the otherwise reference Baja system. A secondary torque increment of  $-120 \text{ N m}$  is applied over 25 ms from a conditioned state near the middle of active shift. The selected case remains in clean continuous operation and compares the full helix with its quasi-static counterpart. The unchanged host makes this a mechanism-transplant experiment, not a reconstructed prediction for a Sidewinder vehicle.

### Existing estimates + new equation-based display

Compare movable rotational inertia, helix motion ratio, and their squared product. Mark Baja and the low/nominal/high commercial estimates; separate component-level prescribed acceleration sensitivity from simulated motion.

`figures/results/dynamics/secondary_inertia_scaling.pdf`

Figure 4.10: Secondary-actuator sensitivity to movable-member inertia and helix geometry. The coefficient  $I_{s,M}(d\theta_s/dx_s)^2$  is local axial reflected inertia; conversion to the shared shift coordinate also includes  $(dx_s/ds)^2$ . The provisional commercial estimates show that a less aggressive motion ratio need not imply a smaller dynamic contribution.

The total relative helix-reaction correction is evaluated on the full-model trajectory as

$$\Pi_{s,\text{total}}(t) = \frac{I_{s,M}|\dot{\omega}_s + \ddot{\theta}_s|}{|\tau_{s,\text{helix}}^{\text{QS}}|}, \quad \tau_{s,\text{helix}}^{\text{QS}} \neq 0. \quad (4.4.8)$$

It reaches 7.87% in the nominal case, with an approximately 89.3 N peak dynamic helix-force correction. The quasi-static torque magnitude remains above 92.2% of its onset value, so the peak relative correction is not generated by a denominator approaching zero. The shaft-acceleration term dominates this example. The component maxima are not added to obtain the total: the signed contributions are combined at each instant before their magnitude is taken.

A local correction is not yet a trajectory error. Each model is therefore also run with a zero-disturbance control. For any output  $y$ , the compared perturbation responses are

$$\delta y^M(t) = y_{\text{loaded}}^M(t) - y_{\text{control}}^M(t), \quad M \in \{\text{full}, \text{QS}\}. \quad (4.4.9)$$

**Figure 4.11** compares these responses alongside the actual stressed trajectories. In the nominal case,  $\max |\delta y^{\text{full}} - \delta y^{\text{QS}}|$  is 0.274 mm for shift, 40.4 rpm for primary speed, and 109.1 N for secondary clamp. Under the same disturbance, the high estimate reaches a 13.6% reaction correction and a 68.5 rpm primary-speed response difference.

This example establishes a secondary dynamic effect without relying on a contact loss or an almost-vanishing static reaction. The gross trajectories remain close, but the disturbance-induced response changes measurably. The low, nominal, and high hardware estimates provide a sensitivity range; they are not statistical confidence bounds on a measured machine.

### Rapid backshift couples reaction, inertia, and traction

A more sensitive response occurs when the transient approaches a traction transition. The severe reversal case uses the reference secondary with 270° torsional preload and a conditioned state near 30% active shift. Primary applied torque is blended toward −28 N·m over 50 ms. The full and quasi-static helix systems receive the same initial state and external disturbance. Both use the zero-clearance bilateral slot, so the comparison changes the movable-member dynamics rather than permitting detachment in only one model.

### Existing results: continuous secondary response

Show signed dynamic contributions and quasi-static torque, then actual stressed motion and full/QS perturbation-response differences. Include the low/nominal/high sensitivity without mixing unlike response units.

`figures/results/dynamics/secondary_continuous_response.pdf`

Figure 4.11: Continuous-transient consequence of the commercial-secondary mechanism transplant. The nominal case uses a  $-120$  N m added secondary torque over 25 ms. Force diagnostics are evaluated on the full-model motion. Perturbation-response differences subtract each model’s own zero-disturbance control as defined in Equation (4.4.9); they are not raw differences between independently conditioned trajectories.

Figure 4.12 shows that the trajectories remain close until the late contact event, after which the quasi-static model backshifts more rapidly. It leaves stick approximately 4.66 ms earlier and accumulates approximately 4.66 ms more non-stick duration within the reported window. The maximum shift separation is 2.026 mm. Near the event,  $I_{s,M}(d\theta_s/ds)^2$  contributes approximately 7.48 kg to the full model’s shift-inertia accounting. This is a reflected coefficient, not additional physical mass attached to the sheave.

The reduced model returns the movable rotational inertia to rigid shaft rotation. It consequently retains that rotating hardware but loses its resistance to relative helix motion. The resulting shift acceleration feeds back into the reaction terms in Equation (4.4.6), changing the clamp and traction solution. This explains why the difference is not captured by subtracting an inertial force from a fixed quasi-static trajectory. The trajectory changes as well. The extra resistance to rapid backshift is inertial; it is not an added dissipative damper.

The reaction chronology adds an important distinction. In the selected full-dynamic reversal, the quasi-static reacted torque is still approximately  $+1.811$  N m when the dynamic correction cancels it. The required slot reaction then changes sign. Yet secondary normal load remains positive: across the thirty crossing cases in the chronology analysis, the smallest reported secondary resultant at a helix crossing is approximately 827 N. Those crossings did not coincide with loss of the integrated secondary normal load. Slot-flank reversal and belt-contact admissibility must therefore be assessed separately. The microsecond separation between the recorded helix and traction crossings is not used to establish a resolved causal ordering.

Conversely, the stock high-margin case develops a visible shift-acceleration reaction contribution while sticking, but the paired shift trajectories differ by only about 0.051 mm. Nonzero dynamic terms are thus neither confined to slipping operation nor sufficient, by themselves, to produce a large trajectory difference. Their importance depends on the available reaction margin and on the coupled response being excited.

### Existing results: severe transient and high-margin contrast

Severe pair: shift response, reflected inertia, helix reaction, and exact contact histories through the event.

Adjacent contrast: the stock high-margin case on explicitly stated axes. Preserve absolute event-time differences rather than aligning away the change being measured.

`figures/results/dynamics/secondary_sensitive_transient.pdf`

Figure 4.12: Secondary-helix dynamics near a sensitive traction event, contrasted with a high-margin stock case. In the severe case, full and quasi-static models start from the same conditioned state under the same forcing. The stock contrast uses  $300^\circ$  preload and a different disturbance, an added secondary torque of  $-120$  N m over 2 ms; it is not a preload-only comparison. Contact histories and shift motion are displayed with their original timing.

### Evidence to complete: S1 — archive-level ratio and event checks

The recap reports a maximum ratio separation of 0.231, but the inspected plot routine labels a field sourced from an effective-radius ratio as a shaft-speed ratio. Confirm the archived field and convention before adding that number here. The 4.66 ms timing difference is retained at the recap’s precision; retain the underlying event records and observation window when preparing the contact strips. Do not convert the reported non-stick duration into a claim about eventual reattachment outside that window.

### What the comparison changes for prediction and design

The continuous and sensitive-transient cases answer complementary questions. The commercial estimate shows that ordinary helix kinematics and plausible movable-member inertia can produce a measurable dynamic correction during continuous motion. The reversal comparison shows how the same type of coupling can change the timing and magnitude of a rapid shift/contact response. A quasi-static model may therefore describe a smooth acceleration well while missing a consequential response to a harsher disturbance.

This has a direct design implication. A static torque-to-clamp relationship contains the spring, transmitted torque, and helix motion ratio, but not the acceleration demand of the movable assembly. Matching that static relationship alone does not fix  $I_{s,M}(d\theta_s/ds)^2$  or the  $I_{s,M}\dot{\omega}_s$  reaction contribution. Inertia and mechanism geometry must therefore be assessed alongside preload when transient behavior matters. The evidence does not select an optimal inertia or helix angle; it establishes why a static clamp comparison alone is insufficient for that design decision.

### 4.4.3 Reduced-Belt Transient Mechanics

#### Judge each term in the equation it must close

The belt retains two different balances. The global transport equation [Equation \(3.4.1\)](#) relates the net pulley traction to  $m_b \dot{v}_b$ . The closed tension compatibility [Equation \(3.4.52\)](#) instead requires the

two wraps and straight spans to form one consistent tension field. A small global inertia force need not imply that acceleration is unimportant to that second balance.

For the term comparison, the final tension compatibility is collected as

$$F_{\dot{s}} + F_{\dot{s}^2} + F_{\dot{v}_b} + F_{\dot{s}v_b} + F_N = 0. \quad (4.4.10)$$

Here the terms are the surviving contributions in [Table 4.4](#); the common  $2\rho_b A_b v_b^2$  contribution has already cancelled between the two wrap-boundary sums. That cancellation is specific to this compatibility equation and does not remove centrifugal loading from the local contact conditions.

To state the coefficients compactly, let  $r'_j = dr_{j,\text{cm}}/ds$  and  $r''_j = d^2 r_{j,\text{cm}}/ds^2$ . For nonzero  $\lambda_j$ , define

$$\begin{aligned} G_j &= \lambda_j \frac{1 + e^{-\lambda_j \phi_j / \sin \beta}}{1 - e^{-\lambda_j \phi_j / \sin \beta}}, \\ H_j &= \frac{2 \sin \beta}{\lambda_j} - \phi_j \frac{1 + e^{-\lambda_j \phi_j / \sin \beta}}{1 - e^{-\lambda_j \phi_j / \sin \beta}}. \end{aligned} \quad (4.4.11)$$

The continuous limits are  $G_j = 2 \sin \beta / \phi_j$  and  $H_j = 0$  at  $\lambda_j = 0$ . These factors collect the existing wrap solution; they do not add a contact law.

Table 4.4: Surviving contributions to the reduced tension compatibility. Radius derivatives are with respect to the shared shift coordinate. The contact term is the difference of the two mapped normal-load contributions, not either gross pulley load.

| Contribution       | Final-equation term                                                                      |
|--------------------|------------------------------------------------------------------------------------------|
| Shift acceleration | $F_{\dot{s}} = -2\rho_b A_b (r_{p,\text{cm}} r'_p - r_{s,\text{cm}} r'_s) \dot{s}$       |
| Path curvature     | $F_{\dot{s}^2} = -2\rho_b A_b (r_{p,\text{cm}} r''_p - r_{s,\text{cm}} r''_s) \dot{s}^2$ |
| Belt acceleration  | $F_{\dot{v}_b} = \rho_b A_b (H_p r'_{p,\text{cm}} - H_s r'_{s,\text{cm}}) \dot{v}_b$     |
| Moving radius      | $F_{\dot{s}v_b} = \rho_b A_b (H_p r'_p - H_s r'_s) \dot{s} v_b$                          |
| Net contact        | $F_N = G_p N_p - G_s N_s$                                                                |

The distinction between gross and net loading is central to the result.  $G_p N_p$  and  $G_s N_s$  can each be hundreds or thousands of newtons while nearly cancelling. A sub-newton transient term can then be important to the remaining compatibility. Its scale should not be dismissed by comparing it only with gross clamp.

The reported activity share uses the absolute contributions to this final balance. For a time window  $\mathcal{W}$ ,

$$a_i(t) = \frac{|F_i(t)|}{\sum_k |F_k(t)|}, \quad A_i(\mathcal{W}) = \frac{\int_{\mathcal{W}} |F_i(t)| dt}{\int_{\mathcal{W}} \sum_k |F_k(t)| dt}, \quad (4.4.12)$$

where the sum contains the five terms in [Equation \(4.4.10\)](#). Windows with vanishing total activity carry no such percentage interpretation. Signed terms and dimensional magnitudes remain necessary: activity describes participation in the equation, not an energy fraction or the trajectory error caused by deleting a term.

### Existing term data: two distinct belt balances

Compare the global transport row with the closed tension balance on the same reference interval. Show gross contact cancellation, signed surviving terms, and equation-activity shares using separate denominators.

`figures/results/dynamics/belt_balance_hierarchy.pdf`

Figure 4.13: Term hierarchy in the global transport and closed tension compatibility equations. The global inertia term is compared with the two opposing pulley reactions; the local acceleration terms are compared within the surviving tension compatibility. These are distinct force balances, so their relative contributions need not have the same hierarchy.

### Rapid changes in shift motion activate radial belt inertia

During smooth free shifting, belt acceleration and net contact provide the main opposing contributions to the tension compatibility. The shift-acceleration contribution grows when sheave motion is strongly accelerated, arrested, or reversed. This is the distinction tested by applying the same  $+18^\circ$  road-load change over different rise times. The finite-rise cases produce peak  $|F_s|$  values of approximately 0.0360, 0.00941, and 0.00269 N for 50, 200, and 800 ms, respectively.

Figure 4.14 separates this rate effect from load magnitude. Across the three smooth rises, the fitted peak varies approximately as  $T_{\text{rise}}^{-0.94}$ , close to inverse rise time over that range. This is a response of the coupled system to those forcing programmes, not a universal power law or a prescribed scaling of sheave motion. The instantaneous-load case is shown separately; only the boundary load is stepped, not a CVT state.

The wider twenty-eight-case envelope connects the controlled experiment to other operating states. A rapid uphill case near 64% active shift, with about  $+20.7^\circ$  loading applied over 56 ms, reaches approximately 24% shift-acceleration activity over the load-rise window. Increased road load does not always produce backshift: the tested responses include continued upshift, arrest, and reversal. The better predictor of this term's importance is the realized shift acceleration, not the name assigned to the maneuver.

The same mechanism remains active when power flow is reversed. A separate overrun case combines downhill loading with a resisting primary torque and verifies secondary-to-primary power flow. Its shift-acceleration contribution accounts for approximately 17% of integrated tension-equation activity in the reported transition window. The result is therefore not limited to uphill backshift: rapid changes in the shift motion also matter during controlled overrun. The ordinary downhill cases, by contrast, retain full-throttle drive and represent unloading rather than this reverse-power condition.

### A sensitive coefficient need not produce an important term

The ordinary load envelope reaches only about 19% of the static traction limit, so it does not alone test whether contact asymmetry could amplify the moving-radius contribution. Independent

### Existing results: loading rate and realized shift acceleration

Pair the controlled rise-time curve with the envelope's activity versus actual shift acceleration. Add a short trace of the rapid-shift exemplar to connect the percentage to arrest/reversal of sheave motion.

`figures/results/dynamics/belt_rate_dependence.pdf`

Figure 4.14: Activation of the shift-acceleration term by rapid loading. The controlled  $+18^\circ$  family isolates boundary rise time, while the wider envelope links the contribution to the solved sheave acceleration. Activity is evaluated over the stated load-rise window using Equation (4.4.12); it is not a percentage change in vehicle acceleration or shift position.

evaluation of the coefficients toward 95% utilization shows that it can. This motivates the high-contact comparisons rather than treating small terms in the nominal launch as sufficient evidence for omission.

The subsequent clamp-stress cases reach sticking states near 40, 60, 80, and 95% utilization without altering the friction coefficients. More severe cases enter primary-slip/secondary-stick, and one also enters both-slip. The largest realized moving-radius coefficient is approximately  $0.928 \text{ N}/(\text{m}^2 \text{ s}^{-2})$ , yet the largest realized moving-radius force is only 0.0182 N. Its greatest instantaneous activity in these stress cases is approximately 0.13%.

Figure 4.15 should be read as a product, not as three independent rankings. The large coefficient tends to appear near engagement, when belt speed is still low, or in heavily utilized seated states, when shift speed is near zero. Either condition suppresses  $\dot{s}v_b$ . Multiplying the largest coefficient from one instant by the largest driver from another would describe a state the trajectory did not visit. The mixed-slip example retains the same distinction: high contact asymmetry does not by itself supply the missing kinematic driver.

The path-curvature term remains very small for a related but distinct reason: the realized  $\dot{s}^2$  is too small to activate its finite coefficient strongly. Belt acceleration does not share that suppression and remains a substantial partner to net contact, including at high traction demand and in overrun. These observations support a hierarchy of complete terms, not a claim that the small coefficients vanish or that every contact branch has been sampled. In particular, the opposite mixed branch, primary-stick/secondary-slip, did not arise naturally in this belt-study set.

### What can be reduced without breaking the coupled description

The global transport inertia presents a different reduction question. Across the investigated envelope,  $m_b \dot{v}_b$  is small relative to the two opposing pulley reactions. Scaling only that coefficient to 10% or 3%, however, prevents admissible engagement in the flat, fast-backshift, and fast-unloading tests. This isolated modification leaves the local wrap inertia unchanged and therefore does not represent a consistent change in belt mass.

### New composition from existing high-contact and overrun data

Show the moving-radius coefficient, its simultaneous driver, and their product through high utilization and mixed slip. Contrast with the significant shift-acceleration contribution in true overrun.

`figures/results/dynamics/belt_coefficient_driver.pdf`

Figure 4.15: Why coefficient sensitivity alone overstates the realized moving-radius effect. The complete term uses coefficient and driver at the same state. High utilization can enlarge the coefficient while small belt speed or nearly stationary shift suppresses its contribution. Overrun supplies an additional test with verified reverse power flow.

Scaling belt density instead changes the global mass and local wrap inertia together. Both the 10% and 3% density continuations complete all three cases and preserve their respective hybrid transition counts. [Figure 4.16](#) compares the trajectory differences from this coherent modification with the isolated-row outcomes. This is evidence for investigating a coupled low-inertia reduction, rather than deleting a small coefficient from one equation on its own. It does not solve the exact zero-density limit.

#### Evidence to complete: R1 — reconcile the continuation error scale

The source recap’s graph and narrative report approximately 0.81–0.86% worst five-state normalized RMSE at 3% density; its page-10 table prints approximately 81–86%. Match the archived `inertia_continuation_comparison.csv` to the normalization and plot before inserting the quantitative error bound. The completion and unchanged transition-count statements above do not resolve this discrepancy.

#### Evidence to complete: R2 — test the proposed small-term omission

Re-integrate a consistently assembled model omitting  $F_{s^2}$  and  $F_{svb}$ , first jointly and separately only if attribution is needed. Use the flat reference, rapid backshift, rapid unloading, high-contact mixed-slip, and true-overrun cases. Compare state errors, event histories, and admissibility at resolved numerical settings. This is the evidence needed to turn small observed equation activity into a quantitative bound on the reduced model’s response; no passing outcome is presumed.

The resulting approximation guidance is summarized in [Table 4.5](#). Belt acceleration belongs in the present finite-inertia tension compatibility. Shift-acceleration inertia must also be retained when rapid changes in sheave motion are part of the intended use. Curvature and moving-radius terms are the strongest candidates for omission over the investigated Baja envelope, because their complete realized contributions remain small even in the targeted high-contact and overrun cases. Their actual omission error must still be established by the reduced comparison. Global low-belt-inertia modeling is a separate, coupled limit.

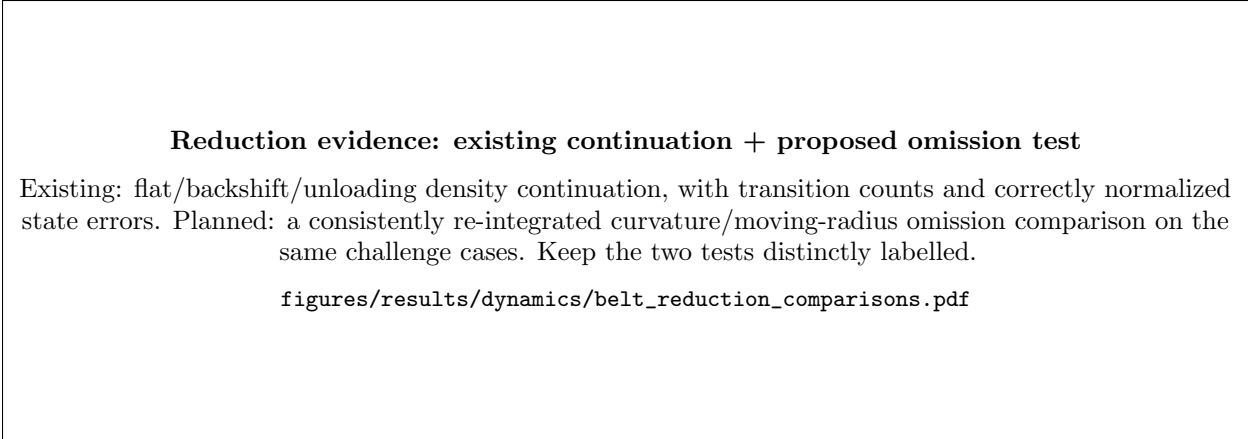


Figure 4.16: Tests of further belt-model reduction. The completed continuation changes density in all retained belt-inertia contributions; isolated transport-row changes are shown separately. A second, explicitly planned comparison tests omission of the two small kinematic terms. Its results remain to be generated, and the density-continuation error scale requires the source check identified in the draft.

Table 4.5: Reduction guidance for the investigated belt model and operating envelope. Equation-level smallness, a completed continuation, and a quantitatively verified reduced trajectory are different levels of evidence.

| Contribution                         | Disposition                 | Condition / evidence                                                                                                             |
|--------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Belt acceleration in tension closure | Retain                      | Substantial contribution throughout the tested finite-inertia balance.                                                           |
| Shift acceleration                   | Retain for rapid transients | Appreciable during arrest/reversal and overrun; smaller in smooth shifting.                                                      |
| Path curvature                       | Candidate omission          | Small realized squared-speed contribution; re-integrated omission error pending.                                                 |
| Moving radius                        | Candidate omission          | Small simultaneous coefficient–driver product, including tested high utilization and slip; re-integrated omission error pending. |
| Global transport inertia             | Reduce coherently           | Density continuation completes; local and global belt inertia must change together. Exact massless limit not evaluated.          |

The distinction across these results is not simply between a dynamic model and a static one. Some retained mechanisms participate throughout ordinary motion, some become important only when the corresponding acceleration is excited, and others remain small because their coefficient and driver do not become large together. That separation explains both why a relatively simple approximation can succeed in smooth operation and why it can miss the response to a rapid disturbance. It also identifies which simplifications should be tested first, with the remaining coupled balances preserved.

## 4.5 Coupled Operation Across Regimes

### 4.5.1 Launch and Upshift

### 4.5.2 Load-Induced Backshift

### 4.5.3 Reverse-Power Operation and Engine Braking

## 4.6 Design and Tuning Consequences

### 4.6.1 Representative Tune Comparison

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# Appendix A

## Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

### A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies  $\lambda \approx 0$  and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint ?? reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \quad (\text{A.1.1})$$

### A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute  $r_p = r_{p,\text{outer,min}}$  and  $r_s = r_{s,\text{outer,max}}$  into Equation (A.1.1) and solve for  $C$ .

**Step 1: Isolate the square root.** Rearranging Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \quad (\text{A.2.1})$$

**Step 2: Square both sides.**

$$[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})]^2 = 4[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2]. \quad (\text{A.2.2})$$

**Step 3: Solve for  $C^2$ .**

$$C^2 = \frac{[L_b - \pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})]^2}{4} + (r_{s,\text{outer},\text{max}} - r_{p,\text{outer},\text{min}})^2. \quad (\text{A.2.3})$$

**Step 4: Take the square root.** Expanding the squared term in the numerator:

$$[L_b - \pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}}) + \pi^2(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})^2,$$

the approximate center distance becomes

$$C_{\text{approx}} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}}) + \pi^2(r_{p,\text{outer},\text{min}} + r_{s,\text{outer},\text{max}})^2}{4} + (r_{s,\text{outer},\text{max}} - r_{p,\text{outer},\text{min}})^2}. \quad (\text{A.2.4})$$

This closed-form expression avoids the need for numerical root finding but neglects the  $\sin^{-1}$  term in the exact constraint.

### A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

#### System Parameters

| Parameter                                              | Value     | SI Value      |
|--------------------------------------------------------|-----------|---------------|
| Belt length $L_b$                                      | 37.538 in | 0.9535 m      |
| Belt height $h_b$                                      | 0.613 in  | 0.015 57 m    |
| Groove half-angle $\beta$                              | 11.5°     | 0.2007 rad    |
| Inner primary radius                                   | 0.8125 in | 0.020 637 5 m |
| Primary outer radius $r_{p,\text{outer},\text{min}}$   | 1.4255 in | 0.036 21 m    |
| Secondary outer radius $r_{s,\text{outer},\text{max}}$ | 4.0 in    | 0.1016 m      |

The primary outer radius is computed as the inner radius plus belt height:  $r_{p,\text{outer},\text{min}} = 0.8125 + 0.613 = 1.4255$  in.

**Computed Center Distances** Solving the exact belt-length constraint ?? numerically yields

$$C_{\text{exact}} = 0.251\,74\text{ m} \approx 9.911\text{ in.}$$

Evaluating the approximate formula Equation (A.2.4) yields

$$C_{\text{approx}} = 0.268\,37\text{ m} \approx 10.566\text{ in.}$$

**Error Metrics** The absolute and relative errors are:

$$\Delta C = C_{\text{approx}} - C_{\text{exact}} = 0.016\,63\text{ m} = 0.655\text{ in,} \quad (\text{A.3.1})$$

$$\varepsilon_{\text{rel}} = \frac{\Delta C}{C_{\text{exact}}} = \frac{0.01663}{0.25174} = 6.61\%. \quad (\text{A.3.2})$$

**Assessment** The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03621}{0.25174} = 0.260$$

is not small. The assumption  $\lambda \approx 0$  is therefore not justified for this CVT geometry.

Since the center distance  $C$  propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in  $C$  can compound into larger errors in predicted system behavior.

## A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

### A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius  $r_{p,\text{outer}}$  and center distance  $C$ , the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation [Equation \(A.1.1\)](#), this yields a closed-form expression.

**Step 1: Isolate the square root.** From [Equation \(A.1.1\)](#):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \quad (\text{A.4.1})$$

**Step 2: Square both sides.**

$$[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}]^2 = 4[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2]. \quad (\text{A.4.2})$$

**Step 3: Expand and collect terms.** Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of  $r_{s,\text{outer}}$  yields a quadratic equation:

$$(\pi^2 + 4)r_{s,\text{outer}}^2 - 2[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}]r_{s,\text{outer}} + [(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2] = 0. \quad (\text{A.4.3})$$

**Step 4: Apply the quadratic formula.** The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$r_{s,\text{approx}} = \frac{\pi L_b + (4 - \pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2 + 4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2 + 4}. \quad (\text{A.4.4})$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

### A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance  $C_{\text{approx}}$ , so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

**Primary Radius at Zero Shift** At  $s = 0$ , the primary is within the deadzone, so from ??:

$$r_{p,\text{outer}}(0) = r_{p,\text{outer},\text{min}} = 0.036\,21\text{ m}.$$

**Secondary Radius Comparison** Solving the exact belt-length constraint with  $C_{\text{exact}}$  yields

$$r_{s,\text{exact}} = 0.101\,60\text{ m}.$$

Evaluating the approximate formula [Equation \(A.4.4\)](#) with  $C_{\text{approx}} = 0.268\,37\text{ m}$  yields

$$r_{s,\text{approx}} = 0.101\,60\text{ m}.$$

**Transmission Ratio Error** Computing effective radii with  $h_b/2 = 0.007\,79\text{ m}$ :

| Quantity                                | Exact    | Approximate |
|-----------------------------------------|----------|-------------|
| $r_{p,\text{eff}}$                      | 0.0284 m | 0.0284 m    |
| $r_{s,\text{eff}}$                      | 0.0938 m | 0.0938 m    |
| $R = r_{s,\text{eff}}/r_{p,\text{eff}}$ | 3.300    | 3.300       |

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.300 - 3.300|}{3.300} \approx 0.000\%.$$

Thus, although  $C_{\text{approx}}$  is already 6.61% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure A.1: Scaled diagram of the CVT at zero shift ( $s = 0$ ). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

### A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

**Primary Radius at Maximum Shift** Using ?? with  $s = s_{\max} = 0.75 \text{ in} = 0.01905 \text{ m}$  and deadzone  $s_{\text{dz}} = 0.0024892 \text{ m}$ :

$$\begin{aligned}
r_{p,\text{outer}}(s_{\max}) &= r_{p,\text{outer},\min} + \frac{s_{\max} - s_{\text{dz}}}{2 \tan \beta} \\
&= 0.03621 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5^\circ)} \\
&= 0.03621 + \frac{0.01656}{0.4070} \\
&= 0.0769 \text{ m.}
\end{aligned} \tag{A.4.5}$$

**Secondary Radius Comparison** Solving the exact belt-length constraint ?? numerically with  $C_{\text{exact}} = 0.25174 \text{ m}$  yields

$$r_{s,\text{exact}} = 0.06673 \text{ m.}$$

Evaluating the approximate formula Equation (A.4.4) with  $C_{\text{approx}} = 0.26837 \text{ m}$  yields

$$r_{s,\text{approx}} = 0.05626 \text{ m.}$$

**Transmission Ratio Error** The CVT ratio is defined using effective radii, which account for belt seating depth (??):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With  $h_b/2 = 0.00779 \text{ m}$ :

| Quantity                                | Exact    | Approximate |
|-----------------------------------------|----------|-------------|
| $r_{p,\text{eff}}$                      | 0.0691 m | 0.0691 m    |
| $r_{s,\text{eff}}$                      | 0.0589 m | 0.0485 m    |
| $R = r_{s,\text{eff}}/r_{p,\text{eff}}$ | 0.853    | 0.701       |

Both models predict overdrive ( $R < 1$ ), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.701 - 0.853|}{0.853} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure A.2: Scaled diagram of the CVT at maximum shift ( $s = s_{\max}$ ). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

## A.5 Summary of Approximation Errors

| Configuration                          | $R_{\text{exact}}$ | $R_{\text{approx}}$ | Ratio Error |
|----------------------------------------|--------------------|---------------------|-------------|
| Zero shift ( $s = 0$ )                 | 3.300              | 3.300               | 0.000%      |
| Maximum shift ( $s = s_{\text{max}}$ ) | 0.853              | 0.701               | 17.8%       |

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.61% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (??), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.

## Appendix B

# Rate and Acceleration of the Geometric CVT Ratio

The main derivation uses the individual pulley-radius rates and accelerations from [Section 3.1.4](#). Those same results can be combined to show how quickly the available geometric CVT ratio changes as the sheaves move.

Recall the geometric CVT ratio from [??](#):

$$R_g = \frac{r_s}{r_p}, \quad (\text{B.0.1})$$

The individual primary and secondary radius rates and accelerations have already been established in [Section 3.1.4](#). This appendix only combines those results to show how the available geometric ratio itself changes as the sheaves move.

### B.0.1 Rate of Change of the Geometric CVT Ratio

Both the numerator and denominator in [Equation \(B.0.1\)](#) can change with time. Using the quotient rule,

$$\begin{aligned} \dot{R}_g &= \frac{d}{dt} \left( \frac{r_s}{r_p} \right) \\ &= \frac{\dot{r}_s r_p - r_s \dot{r}_p}{r_p^2} \\ &= \frac{\dot{r}_s}{r_p} - \frac{r_s}{r_p^2} \dot{r}_p. \end{aligned} \quad (\text{B.0.2})$$

The secondary-radius result in [Equation \(3.1.57\)](#) supplies the coupled secondary motion. Within the active-shift branch,

$$\dot{r}_s = -\frac{\phi_p}{\phi_s} \dot{r}_p. \quad (\text{B.0.3})$$

Substituting this into Equation (B.0.2) gives

$$\begin{aligned}
\dot{R}_g &= \frac{1}{r_p} \left( -\frac{\phi_p}{\phi_s} \dot{r}_p \right) - \frac{r_s}{r_p^2} \dot{r}_p \\
&= -\frac{\dot{r}_p}{r_p} \left( \frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) \\
&= -\frac{\dot{r}_p}{r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right).
\end{aligned} \tag{B.0.4}$$

Finally, Equation (3.1.36) gives  $\dot{r}_p = \dot{s}/(2 \tan \beta)$  in the active-shift branch. Therefore,

|                                                                                                                                                                               |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Rate of Change of Geometric CVT Ratio</b>                                                                                                                                  | <b>(B.0.5)</b> |
| $ \dot{R}_g = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ -\frac{\dot{s}}{2 \tan \beta r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right), & s_{dz} < s \leq s_{max}. \end{cases} $ |                |

For an upshift,  $\dot{s} > 0$ , so the available geometric reduction decreases as the belt moves outward on the primary pulley.

## B.0.2 Geometry Families in the Radius Plane

The preceding result gives the local rate at which the geometric ratio changes once the pulley geometry is known. To interpret that result in terms of actual CVT design choices, it is useful to first ask which pulley-radius combinations are available in the first place.

Recall once more that the geometric CVT ratio is

$$R_g = \frac{r_s}{r_p}. \tag{B.0.6}$$

Thus, every point in the  $(r_p, r_s)$  plane corresponds to one geometric ratio, and every constant-ratio contour is a line of geometrically similar pulley states. This view is helpful because it separates three different design questions that are easy to blur together:

1. What geometric ratio does a given radius pair provide?
2. Which radius pairs are compatible with a selected belt and installed geometry?
3. How does the chosen compatible path influence the way ratio changes as the CVT shifts?

The first question is answered by the constant- $R_g$  contours. The other two come from the belt-length compatibility relation. Once one of the global geometric quantities—belt length  $L_b$  or center-to-center distance  $C$ —is fixed, the compatibility relation defines a family of allowable radius pairs in the same plane.

Figure B.1 compares these two views directly. The left panel fixes the center distance and varies the belt length, while the right panel fixes the belt length and varies the center distance. In both

panels, the solid contours are constant geometric ratio. The dashed families are the geometrically compatible radius pairs selected by the remaining design choice.

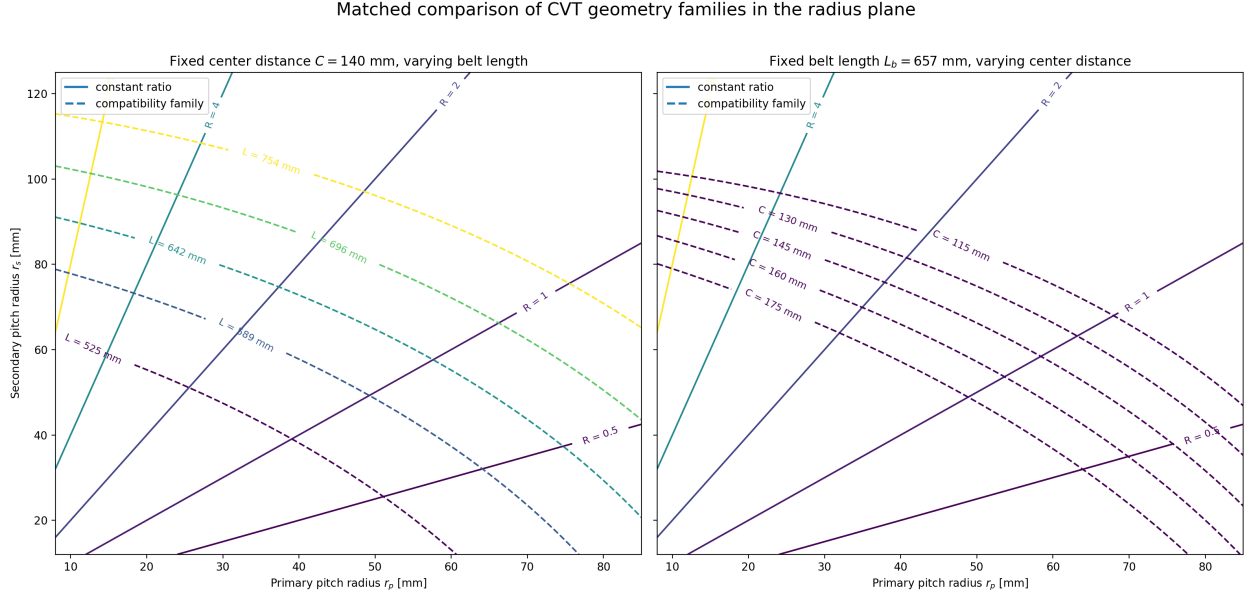


Figure B.1: Matched comparison of geometry families in the pulley-radius plane. In both panels, the solid contours are constant geometric ratio  $R_g = r_s/r_p$ . In (a), the center-to-center distance is fixed and the dashed families correspond to different belt lengths. In (b), the belt length is fixed and the dashed families correspond to different center-to-center distances.

This figure makes two points especially clear.

First, meeting a target ratio aperture is not only a matter of choosing two endpoint ratios. The selected belt and installed geometry determine which radius pairs are even reachable. A longer or shorter belt shifts the compatible family through the plane, and so does a change in center distance. Since the constant-ratio lines are not parallel to those compatibility families, changing  $L_b$  or  $C$  changes not only the reachable endpoints, but also the geometric route between them.

Second, the figure makes the roles of belt selection and packaging easier to distinguish. The belt length is largely a component-selection choice, while the center distance is largely a packaging choice. Both appear in the same compatibility relation, but they move the admissible family in different ways. This is why a target ratio range cannot be discussed cleanly without also specifying either the chosen belt or the available center-to-center distance.

In this sense, [Figure B.1](#) is the static geometric view of the design problem. It shows what ratio states are available. The next question is dynamic in a geometric sense: once the CVT moves along one of those compatible paths, how much ratio change does a small increment of axial travel produce?

### B.0.3 Exploring the Local Ratio-Change Map

The expression for  $\dot{R}_g$  gives the instantaneous rate of change of the geometric ratio. To compare different parts of the geometry without committing to a particular time history, it is useful to divide

out the shift speed and look instead at the local ratio change produced by a small increment of axial travel:

$$\frac{dR_g}{ds} = -\frac{1}{2 \tan \beta r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right). \quad (\text{B.0.7})$$

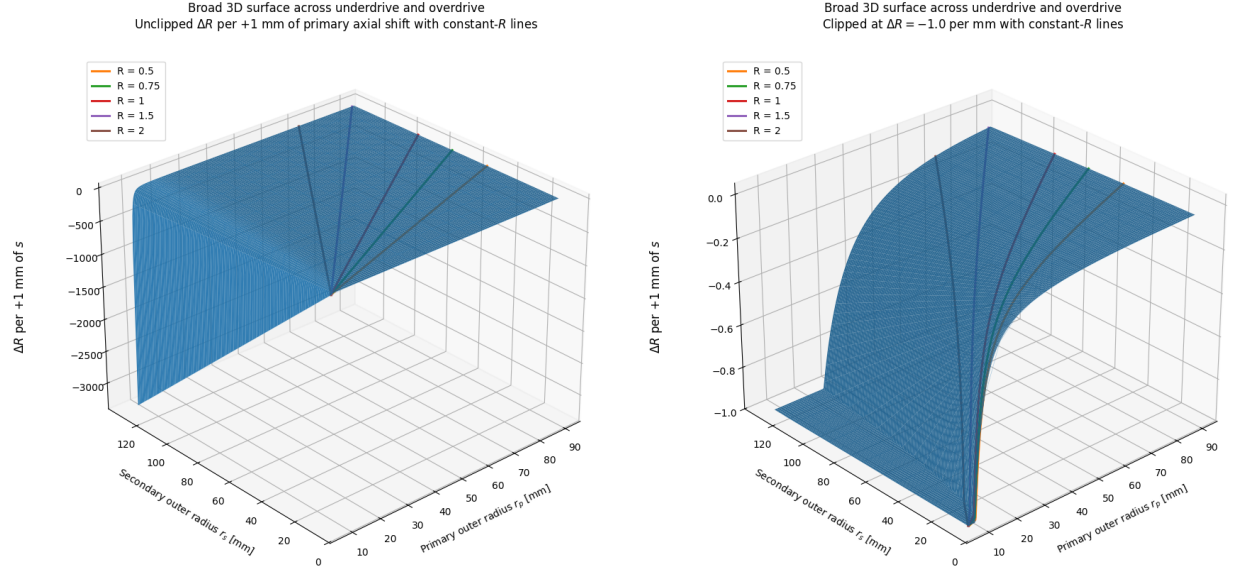
For a small positive increment  $\Delta s$ ,

$$\Delta R_g \approx \frac{dR_g}{ds} \Delta s. \quad (\text{B.0.8})$$

The plots below take  $\Delta s = 1 \text{ mm}$ , so the displayed quantity may be read as the local change in geometric ratio caused by an additional 1 mm of primary closure. The negative sign simply reflects the usual upshift direction: increasing  $s$  reduces the geometric reduction ratio.

**A sensitivity field over the radius plane** The first view is a broad surface over the radius plane. Here,  $r_p$  and  $r_s$  are treated as independent coordinates so that every part of the geometry can be inspected. A real CVT does not wander arbitrarily across this surface; its fixed belt length and center distance constrain it to one belt-length-compatible path. Even so, the full surface is valuable because it shows where the geometry itself makes ratio change especially strong or weak.

**Figure B.2** shows two versions of this surface. The unclipped view exposes the dominant trend: the local ratio change becomes much larger in magnitude as the primary radius becomes small. This follows directly from the explicit  $1/r_p$  factor in **Equation (B.0.7)**, and is reinforced by the fact that the geometric ratio  $R_g = r_s/r_p$  is also largest in that same region. The clipped view hides the most extreme values so that the otherwise compressed mid-range curvature can be seen more clearly. The orange curve marks the  $R_g = 1$  locus. It is a convenient reference, but not a special boundary in the local ratio-change law itself.



(a) Broad radius domain. The full scale exposes the rapidly increasing magnitude of local ratio change near small  $r_p$ . (b) The same domain with values below  $-1$  per  $1$  mm clipped, exposing the curvature through the mid-range.

Figure B.2: Local geometric-ratio change for a  $1$  mm increase in primary axial closure across a broad pulley-radius domain. The orange curve marks the  $R_g = 1$  locus.

**How belt choice and installed geometry select a path** The broad surface is useful, but a particular CVT design only experiences the part of that field lying along its own compatible geometry path. This is where the earlier radius-plane view becomes directly relevant. Once a belt length and center distance are fixed, the CVT is constrained to one compatibility curve in the  $(r_p, r_s)$  plane. As the primary sheave closes, the operating point moves along that curve and samples the local ratio-change field point by point.

Figure B.3 overlays three representative belt-length-compatible paths on the local ratio-change surface. Each path corresponds to a different belt-length/reference-geometry choice. The figure shows that two designs with similar overall ratio aperture can still distribute their ratio change very differently over the available shift travel. One path may enter the steep small- $r_p$  region early, concentrating a large amount of ratio change into a short portion of the stroke. Another may spend more of the stroke in the flatter region, distributing the same overall change more gradually.

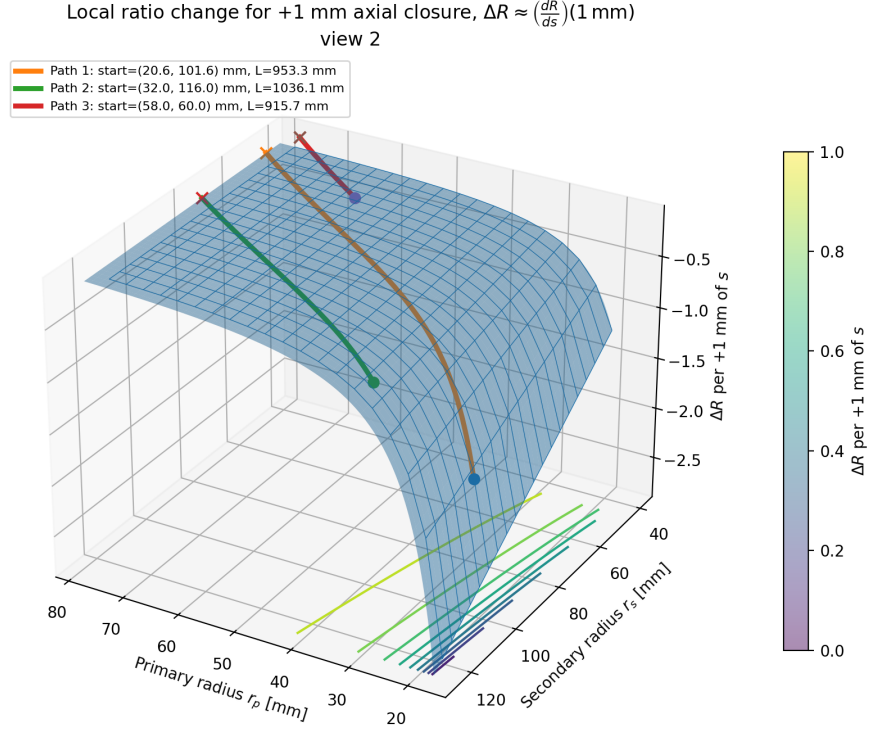
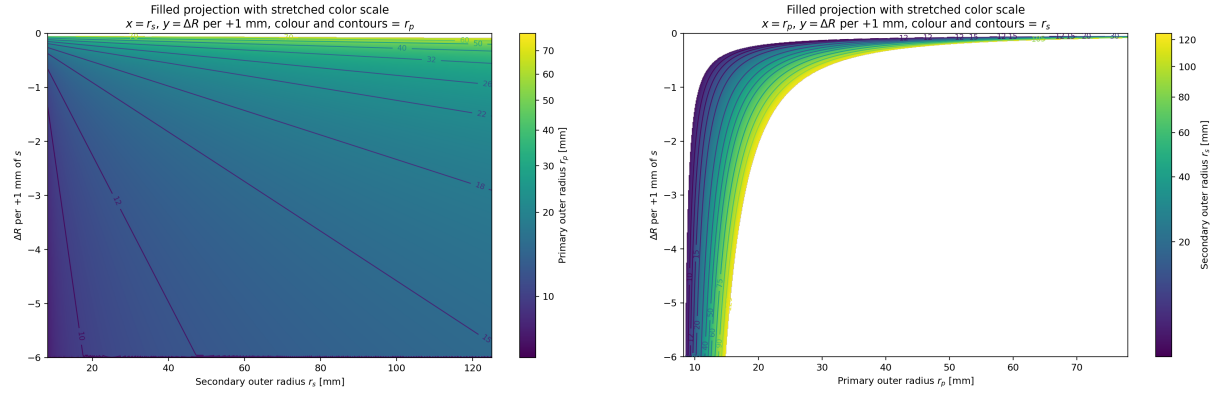


Figure B.3: Representative fixed-belt-length paths through the local ratio-change surface. Each coloured curve is a geometrically compatible pulley-radius path for a different belt-length/reference-geometry choice. The surface value gives the local change in geometric ratio associated with an additional 1 mm of primary closure at that point.

This is the key design value of the map: it does not merely say what overall ratio range is available. It shows *where along the shift stroke* that ratio change is spent.

**Separating the roles of the two pulley radii** The two projections in [Figure B.4](#) help isolate the individual effects of  $r_p$  and  $r_s$ . In [Figure B.4\(a\)](#), the horizontal axis is the secondary radius and the contours identify the primary radius. For a fixed primary radius, increasing  $r_s$  increases the geometric ratio and makes the local ratio change more negative. The effect is visible, but the contour spacing shows that it remains secondary to the influence of the primary radius.

[Figure B.4\(b\)](#) makes the primary-radius effect more explicit. Every secondary-radius contour bends sharply downward as  $r_p$  approaches its lower limit, then flattens at larger radius. This is the visual form of the  $1/r_p$  dependence in [Equation \(B.0.7\)](#). Thus, the small-primary-radius region is not only where the geometric reduction is large; it is also where the ratio becomes most sensitive to additional axial motion.



(a) Local ratio change against secondary radius. (b) Local ratio change against primary radius. Colour and contours identify the primary radius. Colour and contours identify the secondary radius.

Figure B.4: Two projections of the local ratio-change surface for a 1 mm primary closure. The projections separate the contributions of the two radii and show the dominant sensitivity to small primary radius.

**Design interpretation** Taken together, [Figure B.1](#) and [Figure B.2](#)–[Figure B.4](#) give a more complete geometric design picture than endpoint ratios alone.

The radius-plane view answers the question of feasibility. It makes clear how a target ratio aperture depends on the selected belt length and the available center-to-center distance. A longer or shorter belt can move the admissible family enough to change both the reachable ratio range and the portion of the plane through which the CVT shifts. Likewise, a packaging change in center distance can move the same belt into a very different family of compatible radius pairs.

The ratio-change view answers the question of distribution. Once a design selects its belt and installed geometry, the compatible path through the plane determines where ratio change is concentrated. A path that reaches very small  $r_p$  can produce large ratio change in little axial travel, which may be attractive when launch or climbing performance demands strong reduction with limited shift stroke. A path that stays farther from that corner distributes ratio change more gently and may be easier to control or to package within a desired sheave travel.

This does *not* mean that the smallest possible  $r_p$  is always best. The present plots are geometric only. They show the leverage of the geometry, but not its cost. Experimental work by [Chen et al. \(1998\)](#) showed that, for their rubber-belt CVT, the highest transmission efficiency occurred near a 1:1 speed ratio, reminding us that the most attractive geometric reduction state is not automatically the most efficient operating state. More directly, [Childs and Cowburn \(1987\)](#) identified additional loss penalties at small pulley radii in V-belt drives, with belt-carcass warping and shearing proposed as important contributors. In other words, the same small- $r_p$  region that offers strong geometric leverage is also the region where the belt is bent and seated most severely.

For that reason, these plots are best used as an early design lens rather than a final design verdict. They help screen combinations of belt length, center distance, and pulley radii by showing not only how much ratio range is available, but how that range is distributed over the stroke. Later traction, loss, and dynamic analyses can then be used to judge whether the geometrically attractive part of the design space is physically supportable ([Gerbert, 1996](#); [Bertini et al., 2014](#); [Messick, 2018](#)).

#### B.0.4 Acceleration of the Geometric CVT Ratio

The ratio rate can change for two reasons. The primary sheave may accelerate, and the pulley geometry continues to change as the belt moves. Differentiate Equation (B.0.2) term by term to capture both effects.

For the first term,  $\dot{r}_s/r_p$ , use the quotient rule:

$$\frac{d}{dt} \left( \frac{\dot{r}_s}{r_p} \right) = \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s \dot{r}_p}{r_p^2}. \quad (\text{B.0.9})$$

For the second term,  $r_s \dot{r}_p / r_p^2$ , all three factors may change. Using the product rule,

$$\frac{d}{dt} \left( \frac{r_s \dot{r}_p}{r_p^2} \right) = \frac{\dot{r}_s \dot{r}_p}{r_p^2} + \frac{r_s \ddot{r}_p}{r_p^2} - \frac{2r_s \dot{r}_p^2}{r_p^3}. \quad (\text{B.0.10})$$

Subtracting Equation (B.0.10) from Equation (B.0.9) gives

$$\begin{aligned} \ddot{R}_g &= \frac{\ddot{r}_s}{r_p} - \frac{\dot{r}_s \dot{r}_p}{r_p^2} - \left[ \frac{\dot{r}_s \dot{r}_p}{r_p^2} + \frac{r_s \ddot{r}_p}{r_p^2} - \frac{2r_s \dot{r}_p^2}{r_p^3} \right] \\ &= \frac{\ddot{r}_s}{r_p} - \frac{2\dot{r}_s \dot{r}_p}{r_p^2} - \frac{r_s \ddot{r}_p}{r_p^2} + \frac{2r_s \dot{r}_p^2}{r_p^3}. \end{aligned} \quad (\text{B.0.11})$$

The main derivation gives the required coupled secondary motion in ????:

$$\dot{r}_s = -\frac{\phi_p}{\phi_s} \dot{r}_p, \quad (\text{B.0.12})$$

$$\ddot{r}_s = -\frac{\phi_p}{\phi_s} \ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}} \dot{r}_p^2. \quad (\text{B.0.13})$$

Substituting these into Equation (B.0.11) gives

$$\begin{aligned} \ddot{R}_g &= \frac{1}{r_p} \left[ -\frac{\phi_p}{\phi_s} \ddot{r}_p - \frac{8\pi^2}{\phi_s^3 \sqrt{C^2 - (r_s - r_p)^2}} \dot{r}_p^2 \right] - \frac{2}{r_p^2} \left[ -\frac{\phi_p}{\phi_s} \dot{r}_p \right] \dot{r}_p \\ &\quad - \frac{r_s}{r_p^2} \ddot{r}_p + \frac{2r_s}{r_p^3} \dot{r}_p^2 \\ &= -\frac{1}{r_p} \left( \frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) \ddot{r}_p + \left[ \frac{2}{r_p^2} \left( \frac{\phi_p}{\phi_s} + \frac{r_s}{r_p} \right) - \frac{8\pi^2}{\phi_s^3 r_p \sqrt{C^2 - (r_s - r_p)^2}} \right] \dot{r}_p^2 \\ &= -\frac{1}{r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right) \ddot{r}_p + \left[ \frac{2}{r_p^2} \left( \frac{\phi_p}{\phi_s} + R_g \right) - \frac{8\pi^2}{\phi_s^3 r_p \sqrt{C^2 - (r_s - r_p)^2}} \right] \dot{r}_p^2. \end{aligned} \quad (\text{B.0.14})$$

Finally, use the primary radius rate and acceleration from [Equations \(3.1.36\)](#) and [\(3.1.66\)](#):

#### Acceleration of Geometric CVT Ratio

(B.0.15)

$$\ddot{R}_g = \begin{cases} 0, & 0 \leq s < s_{\text{dz}}, \\ -\frac{\ddot{s}}{2 \tan \beta r_p} \left( \frac{\phi_p}{\phi_s} + R_g \right) + \frac{\dot{s}^2}{2 \tan^2 \beta r_p^2} \left( \frac{\phi_p}{\phi_s} + R_g \right) - \frac{2\pi^2 \dot{s}^2}{\phi_s^3 r_p \tan^2 \beta \sqrt{C^2 - (r_s - r_p)^2}}, & s_{\text{dz}} < s \leq s_{\text{max}}. \end{cases}$$

The ratio acceleration has two sources. The term proportional to  $\ddot{s}$  is the direct conversion of primary-sheave acceleration into ratio acceleration. Its magnitude is scaled by the local ratio sensitivity  $dR_g/ds$ , so the same axial acceleration produces a larger ratio response where the geometric ratio changes rapidly with shift.

The terms proportional to  $\dot{s}^2$  arise even when the primary sheave moves at constant axial speed. They reflect the curvature of the geometric relation  $R_g(s)$ : as the belt moves through the sheaves, the ratio change produced by each additional increment of axial travel need not remain constant. In the usual upshift direction, the large ratio sensitivity associated with small primary radius progressively weakens as the primary radius grows. The ratio therefore often changes rapidly at the beginning of the shift and then eases off, even without a change in axial shift speed.

## Appendix C

# Construction of Primary Flyweight Geometry Maps

The primary actuator dynamics derived in the main text do not require a particular flyweight shape directly. They require the mechanism maps

$$q_f = q_f(x_p), \quad \frac{dq_f}{dx_p}, \quad \frac{d^2q_f}{dx_p^2}, \quad J_{f,p} = J_{f,p}(x_p), \quad \frac{dJ_{f,p}}{dx_p}, \quad (\text{C.0.1})$$

together with the constant pivot-axis inertia  $I_f$ . This appendix constructs those quantities for the fixed-pivot roller flyweight used by the present primary mechanism.

The construction follows the physical mechanism directly. A ramp attached to the movable sheave defines the contact surface. The finite roller radius offsets that surface to a roller-center locus. Intersecting that locus with the rigid flyweight-arm circle determines the assembled flyweight pose and therefore  $q_f(x_p)$ . Differentiating the same geometric constraint supplies the motion ratios required by the acceleration-level actuator model. Finally, the flyweight mass distribution is evaluated at the resulting pose to obtain  $I_f$ ,  $J_{f,p}(x_p)$ , and  $dJ_{f,p}/dx_p$ .

No independent initial flyweight angle is introduced. The assembled pose is a consequence of the pivot, ramp, roller, and arm geometry.

### C.1 Fixed-Pivot Flyweight Geometry

Consider the axial–radial plane of one primary flyweight. Let  $O$  denote the shaft centre, let  $+\mathbf{e}_x$  point in the positive primary closing direction, and let  $+\mathbf{e}_r$  point radially outward. The flyweight pivots about the fixed point

$$\mathbf{r}_P = \begin{bmatrix} x_P \\ r_P \end{bmatrix}, \quad (\text{C.1.1})$$

which is fixed to the pulley body. The movable sheave carries the physical ramp. The roller centre is denoted by  $C$ , the physical roller–ramp contact point by  $K$ , and the reference point at the start of the active ramp by  $A$ .

The rigid distance from the pivot to the roller centre is

$$\boxed{L_f = |PC|}, \quad (\text{C.1.2})$$

and the roller radius is  $R_{\text{roll}}$ . The flyweight angle  $q_f$  is measured from the positive axial reference to the line from  $P$  to  $C$ , with increasing  $q_f$  corresponding to outward flyweight rotation. Thus

$$\mathbf{r}_C(q_f) = \mathbf{r}_P + L_f \begin{bmatrix} \cos q_f \\ \sin q_f \end{bmatrix}. \quad (\text{C.1.3})$$

The arm is therefore constrained to a circle of radius  $L_f$  about  $P$ .

### C.1.1 Physical Ramp Representation

Introduce a scalar ramp coordinate  $\xi$ , with  $A$  corresponding to  $\xi = 0$ . Let  $h(\xi)$  be the radial profile measured from  $A$ , and let  $d_x \in \{-1, +1\}$  specify whether increasing  $\xi$  proceeds in the positive or negative physical axial direction. For movable-sheave closure  $x_p$ , a point on the physical ramp is

$$\mathbf{r}_K(\xi, x_p) = \begin{bmatrix} x_A + x_p + d_x \xi \\ r_A + h(\xi) \end{bmatrix}. \quad (\text{C.1.4})$$

The present Baja parameterization uses  $d_x = -1$ : increasing profile coordinate proceeds axially toward the pivot while the ramp rises radially outward.

Ramp angle is measured from the axial reference line, so  $0^\circ$  denotes an axial ramp. For a graph of the form in [Equation \(C.1.4\)](#), the local unsigned ramp angle satisfies

$$\alpha(\xi) = \tan^{-1} |h'(\xi)|. \quad (\text{C.1.5})$$

Accordingly, a  $20^\circ$  ramp is nearly axial rather than nearly radial.

The unnormalized tangent is

$$\mathbf{t}(\xi) = \begin{bmatrix} d_x \\ h'(\xi) \end{bmatrix}, \quad \|\mathbf{t}\| = \sqrt{1 + [h'(\xi)]^2}. \quad (\text{C.1.6})$$

With  $\sigma_n \in \{-1, +1\}$  selecting the physical side of the ramp on which the roller lies, a compatible unit normal is

$$\mathbf{n}(\xi) = \sigma_n \frac{\begin{bmatrix} -h'(\xi) \\ d_x \end{bmatrix}}{\sqrt{1 + [h'(\xi)]^2}}. \quad (\text{C.1.7})$$

The sign  $\sigma_n$  is fixed once from the assembled mechanism and is then used throughout the active ramp range.

The profile  $h$  may be assembled from straight, circular, spline, or other smooth segments. Those shapes define only the physical contact surface; they do not themselves prescribe the flyweight motion or force.

### C.1.2 Finite-Radius Roller-Centre Locus

The roller centre does not lie on the physical ramp. At contact it is offset by one roller radius along the selected ramp normal:

$$\mathbf{r}_C(\xi, x_p) = \mathbf{r}_K(\xi, x_p) + R_{\text{roll}} \mathbf{n}(\xi). \quad (\text{C.1.8})$$

This finite-radius offset is the roller-centre path used by the mechanism. The point  $K$  remains the physical contact point;  $C$  is the centre of the roller bearing.

The contact problem can now be viewed geometrically as the intersection of the roller-centre offset curve with the rigid circle of radius  $L_f$  about  $P$ . A point-contact or zero-radius follower approximation is not used for the present mechanism.

### C.1.3 Rigid-Arm Contact Constraint and Assembled Branch

A roller-centre location is compatible with the rigid flyweight when

$$\boxed{g(\xi, x_p) = \|\mathbf{r}_C(\xi, x_p) - \mathbf{r}_P\|^2 - L_f^2 = 0.} \quad (\text{C.1.9})$$

For any fixed  $x_p$ , every root of [Equation \(C.1.9\)](#) is a mathematical flyweight configuration that satisfies the rigid arm and roller contact geometry.

Define

$$a = x_C - x_P, \quad b = r_C - r_P. \quad (\text{C.1.10})$$

Then

$$a^2 + b^2 = L_f^2, \quad (\text{C.1.11})$$

and the corresponding flyweight angle is

$$\boxed{q_f(x_p) = \text{atan2}(b, a).} \quad (\text{C.1.12})$$

Thus the starting angle is recovered from the assembled geometry rather than specified as an independent parameter.

Multiple mathematical roots can exist at the same  $x_p$ . At the beginning of the declared operating interval,  $x_p = x_{p,\min}$ , enumerate the admissible contact configurations and initialize the assembled branch with

$$\boxed{q_f(x_{p,\min}) = \min_i q_{f,i}(x_{p,\min}).} \quad (\text{C.1.13})$$

This smallest- $q_f$  rule is an *initial assembly rule only*. For increasing or decreasing closure thereafter, the same solution branch is followed continuously. The flyweight is not permitted to jump to a disconnected arm orientation simply because another root has a smaller angle at a later position.

Alternate arm orientations that separately satisfy [Equation \(C.1.9\)](#) are not themselves a contact failure. They are mathematically distinct assembled configurations. A different condition occurs when the *same selected roller pose* contacts two distinct locations of the ramp simultaneously; that true two-point contact lies outside the present single-contact mechanism.

For the selected branch the geometry chain is therefore

$$\boxed{h(\xi) \longrightarrow \mathbf{r}_C(\xi, x_p) \longrightarrow g(\xi, x_p) = 0 \longrightarrow q_f(x_p).} \quad (\text{C.1.14})$$

## C.2 Contact-Branch Kinematics

The main actuator dynamics require  $q'_f(x_p)$  and  $q''_f(x_p)$ . These quantities follow from the same physical contact constraint rather than from an independent fitted motion law.

Along the selected branch,

$$g(\xi(x_p), x_p) = 0. \quad (\text{C.2.1})$$

Differentiating with respect to  $x_p$  gives

$$g_\xi \xi' + g_x = 0, \quad (\text{C.2.2})$$

where  $g_x \equiv \partial g / \partial x_p$ , and therefore

$$\boxed{\xi' = -\frac{g_x}{g_\xi}.} \quad (\text{C.2.3})$$

Differentiating once more gives

$$g_{xx} + 2g_{x\xi}\xi' + g_{\xi\xi}(\xi')^2 + g_\xi\xi'' = 0, \quad (\text{C.2.4})$$

and hence

$$\boxed{\xi'' = -\frac{g_{xx} + 2g_{x\xi}\xi' + g_{\xi\xi}(\xi')^2}{g_\xi}.} \quad (\text{C.2.5})$$

These relations require

$$\boxed{g_\xi \neq 0} \quad (\text{C.2.6})$$

along the selected branch. A zero value corresponds to a fold or dead-centre at which the local contact map cannot be continued smoothly in  $x_p$ .

For compactness write

$$\mathbf{c}(\xi, x_p) \equiv \mathbf{r}_C(\xi, x_p). \quad (\text{C.2.7})$$

Its total first and second derivatives along the selected branch are

$$\boxed{\mathbf{c}' = \mathbf{c}_x + \mathbf{c}_\xi \xi',} \quad (\text{C.2.8})$$

and

$$\boxed{\mathbf{c}'' = \mathbf{c}_{xx} + 2\mathbf{c}_{x\xi}\xi' + \mathbf{c}_{\xi\xi}(\xi')^2 + \mathbf{c}_\xi\xi''}. \quad (\text{C.2.9})$$

The explicit  $x_p$ -dependence is particularly simple because the ramp carrier undergoes rigid axial translation, but the chain-rule form above keeps the construction general and makes the geometric origin of every derivative clear.

Using  $a$  and  $b$  from [Equation \(C.1.10\)](#), the first angle derivative is

$$\boxed{q'_f = \frac{ab' - ba'}{L_f^2}.} \quad (\text{C.2.10})$$

Because  $a^2 + b^2 = L_f^2$  is constant, the second derivative reduces to

$$\boxed{q''_f = \frac{ab'' - ba''}{L_f^2}.} \quad (\text{C.2.11})$$

The angle, motion ratio, and change of motion ratio are therefore all direct consequences of the ramp, roller, pivot, and arm geometry.

### C.2.1 Smoothness of the Dynamic Ramp Representation

The finite roller-centre curve contains the normal  $\mathbf{n}(h')$ . Its first profile derivative therefore contains  $h''$ , while its second profile derivative contains  $h'''$ . Since  $q_f''(x_p)$  depends on  $\mathbf{c}_{\xi\xi}$ , the acceleration-level mechanism map requires the active ramp representation to satisfy

$$\boxed{h \in C^3} \quad (\text{C.2.12})$$

through piecewise junctions used by the production dynamic map. A discontinuity in slope, curvature, or the third derivative can contaminate the acceleration terms even when the physical profile itself appears visually continuous.

When idealized or measured ramp pieces do not meet this requirement directly, a short artificial smoothing segment may be introduced explicitly as a model approximation. The provisional Baja ramp used during development consists of

$$20^\circ \text{ linear over } 5 \text{ mm} \longrightarrow 3 \text{ mm } C^3 \text{ blend} \longrightarrow 35^\circ \rightarrow 10^\circ \text{ circular over approximately } 30 \text{ mm}. \quad (\text{C.2.13})$$

The blend is represented by a sixth-order polynomial transition selected so that derivatives through third order match the neighbouring segments. The 3 mm blend length is a modelling choice, not a measured hardware dimension. A sharp-corner finite-roller construction may still be useful for geometry diagnostics, but it is not admitted directly into the production map while  $q_f''$  is retained.

## C.3 Flyweight Mass Properties

Once  $q_f(x_p)$  is known, the remaining mechanism quantities follow from the flyweight mass distribution. Introduce body-fixed coordinates relative to the pivot  $P$ :  $u$  along the nominal arm from the pivot toward the roller centre,  $v$  perpendicular to  $u$  in the axial–radial mechanism plane, and  $z$  circumferentially along the pivot axis.

For one flyweight define

$$m_f = \int dm, \quad (\text{C.3.1})$$

$$M_u = \int u dm, \quad M_v = \int v dm, \quad (\text{C.3.2})$$

$$S_u = \int u^2 dm, \quad S_v = \int v^2 dm, \quad (\text{C.3.3})$$

$$S_{uv} = \int uv dm, \quad S_z = \int z^2 dm. \quad (\text{C.3.4})$$

These first and second moments may be obtained from a simplified analytic mass model, direct measurement, or CAD.

For  $n_f$  identical flyweights, the inertia about their individual pivot axes is

$$\boxed{I_f = n_f(S_u + S_v)}. \quad (\text{C.3.5})$$

This is the inertia multiplying the relative pivot acceleration in the primary actuator balance.

At flyweight angle  $q_f$ , a mass element has radial shaft coordinate

$$r = r_P + u \sin q_f + v \cos q_f. \quad (\text{C.3.6})$$

Including its circumferential offset  $z$ , integration gives the total flyweight inertia about the primary shaft:

$$\boxed{J_{f,p}(q_f) = n_f \left[ m_f r_P^2 + 2r_P(M_u \sin q_f + M_v \cos q_f) + S_u \sin^2 q_f + S_v \cos^2 q_f + 2S_{uv} \sin q_f \cos q_f + S_z \right].} \quad (\text{C.3.7})$$

Composition with the geometry map gives

$$\boxed{J_{f,p}(x_p) = J_{f,p}(q_f(x_p)),} \quad (\text{C.3.8})$$

and

$$\boxed{\frac{dJ_{f,p}}{dx_p} = \frac{dJ_{f,p}}{dq_f} q'_f.} \quad (\text{C.3.9})$$

The same mass description therefore supplies both the pivot inertia and the configuration-dependent shaft-axis inertia required by the dynamic actuator.

### C.3.1 Present Baja Mass Approximation

The present parameterization does not attempt to reproduce the exact outline and centroidal inertia of every flyweight component. For each flyweight, the arm/body mass  $m_a$  is represented as a uniform slender member extending from  $u = 0$  to  $u = L_f$ , while the hardware carried at the outer end is collected into a concentrated tip mass  $m_t$  at  $u = L_f$ ,  $v = z = 0$ . The tip package is

$$m_t = m_{\text{roller/bearing}} + m_{\text{bolt}} + m_{\text{nut/washer}} + m_{\text{fixed hardware}} + m_{\text{tuning}}. \quad (\text{C.3.10})$$

For the uniform arm,

$$M_{u,a} = \frac{m_a L_f}{2}, \quad S_{u,a} = \frac{m_a L_f^2}{3}, \quad (\text{C.3.11})$$

and for the concentrated tip package,

$$M_{u,t} = m_t L_f, \quad S_{u,t} = m_t L_f^2. \quad (\text{C.3.12})$$

Hence

$$\boxed{M_u = \frac{m_a L_f}{2} + m_t L_f,} \quad (\text{C.3.13})$$

and

$$\boxed{S_u = \frac{m_a L_f^2}{3} + m_t L_f^2.} \quad (\text{C.3.14})$$

with

$$M_v = S_v = S_{uv} = S_z = 0, \quad m_f = m_a + m_t. \quad (\text{C.3.15})$$

The resulting pivot inertia is

$$\boxed{I_f = n_f \left( \frac{1}{3} m_a L_f^2 + m_t L_f^2 \right).} \quad (\text{C.3.16})$$

The corresponding shaft-axis inertia map is

$$J_{f,p}(q_f) = n_f [(m_a + m_t)r_P^2 + 2r_P M_u \sin q_f + S_u \sin^2 q_f]. \quad (\text{C.3.17})$$

Its configuration derivative is

$$\frac{dJ_{f,p}}{dq_f} = 2n_f \cos q_f (r_P M_u + S_u \sin q_f), \quad (\text{C.3.18})$$

so the required axial derivative becomes

$$\frac{dJ_{f,p}}{dx_p} = 2n_f \cos q_f (r_P M_u + S_u \sin q_f) q'_f. \quad (\text{C.3.19})$$

This reduction is an explicit approximation to the hardware mass distribution. It neglects the centroidal inertia associated with the finite roller/bearing radius, bolt diameter and length, nut and washer dimensions, tuning-weight dimensions, arm thickness and nonuniform outline, holes and reliefs, and any off-centre placement of tip hardware not represented by the roller-centre station. These features are not assumed physically absent; they are omitted when reducing the real flyweight to the mass moments above.

For the present assembly, the tuning mass is of order  $> 100$  g, while the measured arm/body mass is approximately 13.646 g per flyweight. The parallel-axis contribution of the tip package is therefore expected to dominate many small centroidal corrections, although this observation does not make the omitted terms identically zero. If greater fidelity is needed, the same governing equations accept CAD- or measurement-derived  $M_u, M_v, S_u, S_v, S_{uv}, S_z$  without changing the actuator dynamics.

### C.3.2 Present Geometry Values

The provisional geometry used during development is summarized below. These values identify the mechanism currently being parameterized; the ramp blend and unverified tip masses should not be interpreted as final metrology.

| Quantity                             | Present value                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------|
| Number of flyweights, $n_f$          | 3                                                                                                    |
| Pivot radius, $r_P$                  | 1.675 in = 42.545 mm                                                                                 |
| Pivot-to-roller-centre length, $L_f$ | 1.241 in = 31.5214 mm                                                                                |
| Roller radius, $R_{\text{roll}}$     | 6.5 mm                                                                                               |
| Ramp-reference offset from pivot     | $\Delta x_A = 1.5$ in, $\Delta r_A = 0.2776$ in                                                      |
| Measured arm/body mass, $m_a$        | 13.646 g per flyweight                                                                               |
| Profile direction                    | $d_x = -1$                                                                                           |
| Provisional active ramp              | 20° linear over 5 mm, 3 mm $C^3$ blend, then circular 35° $\rightarrow$ 10° over approximately 30 mm |

Table C.1: Present fixed-pivot flyweight geometry and provisional ramp data. The 3 mm smoothing length is a modelling choice rather than a measured hardware dimension.

The roller/bearing, bolt, nut/washer, fixed tip-hardware, and tuning masses are entered individually when measured and then collected into  $m_t$  through [Equation \(C.3.10\)](#).

## C.4 Resulting Mechanism Interface

The construction above supplies exactly the quantities required by the primary actuator dynamics:

$$\boxed{q_f(x_p), \quad q'_f(x_p), \quad q''_f(x_p), \quad I_f, \quad J_{f,p}(x_p), \quad J'_{f,p}(x_p).} \quad (\text{C.4.1})$$

For the fixed-pivot mechanism, the complete chain is

$$\boxed{\text{ramp geometry} \rightarrow \text{roller-centre locus} \rightarrow \text{rigid-arm branch} \rightarrow q_f, q'_f, q''_f \rightarrow \text{mass moments} \rightarrow I_f, J_{f,p}, J'_{f,p}.} \quad (\text{C.4.2})$$

Substitution into the main actuator result in [Equation \(3.2.11\)](#) gives

$$\boxed{F_{p,\text{fly}} = \frac{1}{2}\omega_p^2 J'_{f,p} - I_f(q'_f)^2 \ddot{x}_p - I_f q'_f q''_f \dot{x}_p^2.} \quad (\text{C.4.3})$$

The three terms are respectively the centrifugal generalized closing drive, the pivot inertia reflected into the axial coordinate, and the inertial term created by variation of the mechanism motion ratio.

The same configuration-dependent mass distribution contributes to the owning shaft through

$$\boxed{J_{f,p}\dot{\omega}_p + J'_{f,p}\dot{x}_p\omega_p,} \quad (\text{C.4.4})$$

which is the local  $x_p$ -coordinate form of the primary angular-momentum term used in [Equation \(3.3.4\)](#). Under the assumed circumferential symmetry, the corresponding flyweight kinetic energy may be written

$$\boxed{T_f = \frac{1}{2}J_{f,p}\omega_p^2 + \frac{1}{2}I_f(q'_f\dot{x}_p)^2.} \quad (\text{C.4.5})$$

Replacing the simplified mass moments with higher-fidelity CAD or measured moments changes only the supplied maps, not these governing relations.

## C.5 Mechanism Admissibility and Assumptions

The maps in [Equation \(C.4.1\)](#) are valid only while the geometric, mass, and unilateral-contact assumptions of the mechanism remain satisfied. The conditions below define the admissible mechanism used by the present model.

### C.5.1 Geometry and Branch Admissibility

All physical dimensions and masses must be finite, with

$$L_f > 0, \quad R_{\text{roll}} > 0, \quad r_P > 0, \quad n_f \geq 1, \quad (\text{C.5.1})$$

and nonnegative component masses. The declared ramp domain and primary travel interval must both be nonempty.

At  $x_p = x_{p,\text{min}}$ , at least one contact solution of [Equation \(C.1.9\)](#) must exist. The physical branch is selected there using [Equation \(C.1.13\)](#) and must then continue continuously through the entire declared travel. Loss of that branch or a jump to a disconnected arm orientation is inadmissible.

The selected branch must remain regular:

$$\boxed{g_\xi \neq 0, \quad 0 < q'_f(x_p) < \infty, \quad |q''_f(x_p)| < \infty.} \quad (\text{C.5.2})$$

The sign requirement corresponds to the present fixed-pivot mechanism, for which primary closure rotates the flyweight outward. A zero or negative motion ratio indicates a dead-centre, reversal, or incompatible branch for this mechanism definition.

The finite roller offset must also remain regular. Let  $\kappa_n$  denote the signed ramp curvature consistent with the selected normal. The normal-offset regularity factor must satisfy

$$\boxed{1 - R_{\text{roll}}\kappa_n \neq 0} \quad (\text{C.5.3})$$

through the selected contact range. Operation close to this singular condition should be treated as geometrically sensitive even before equality is reached.

The selected contact coordinate must remain within the physical ramp extent for the full primary travel. In practice, the active ramp should extend beyond the selected contact path with an appropriate geometric and manufacturing margin.

The rigid arm condition itself is exact up to numerical tolerance:

$$\boxed{\|\mathbf{r}_C - \mathbf{r}_P\| = L_f.} \quad (\text{C.5.4})$$

It is not an approximate fitted arm length.

### C.5.2 No Penetration or Simultaneous Double Contact

For the selected roller pose, no second portion of the physical ramp may penetrate the roller. Conceptually, for every distinct ramp coordinate  $\tilde{\xi}$  away from the selected contact,

$$\boxed{\|\mathbf{r}_C - \mathbf{r}_K(\tilde{\xi}, x_p)\| \geq R_{\text{roll}}.} \quad (\text{C.5.5})$$

A strict violation indicates interference. Equality at a second distinct ramp location indicates true simultaneous two-point contact and lies outside the present single-contact model. This condition is different from the existence of multiple arm orientations that separately solve the contact equation.

### C.5.3 Mass and Inertia Admissibility

The supplied mass description must satisfy

$$\boxed{I_f \geq 0, \quad J_{f,p}(x_p) > 0} \quad (\text{C.5.6})$$

throughout the selected branch. These conditions follow naturally for the nonnegative arm-plus-tip approximation. If arbitrary CAD moments are supplied, they must correspond to a physically realizable nonnegative mass distribution.

Flyweight mass and inertia represented here must not be included again in the movable-sheave translating mass, the constant rigid pulley rotational inertia, or another actuator inertia term. This separation prevents double counting of the same kinetic energy.

#### C.5.4 Unilateral Contact-Force Admissibility

Geometric compatibility alone does not guarantee that the ramp can maintain the constrained roller contact during every dynamic state. Under the sign convention used in the main text, compressive ramp contact requires

$$\boxed{F_{p,\text{fly}} \geq 0.} \quad (\text{C.5.7})$$

If  $F_{p,\text{fly}} < 0$ , the constrained solution would require the ramp to pull on the roller. That state is physically inadmissible. The result is not replaced by  $\max(0, F_{p,\text{fly}})$ , since clipping the force would hide a change in mechanism topology.

A complete treatment of such a state would introduce flyweight lift-off, free-flight, re-contact, and any associated impact dynamics. Those modes are not retained here. Until such an extension is introduced, negative contact force is treated as an explicit admissibility failure. The condition  $F_{p,\text{fly}} = 0$  marks the unilateral-contact boundary.

#### C.5.5 Mechanism and Contact Assumptions

The fixed-pivot construction uses the following assumptions:

- the flyweight pivot is fixed relative to the pulley body;
- the ramp translates rigidly with the movable sheave;
- the pivot-to-roller-centre length is rigid and constant;
- the roller has finite radius and the flyweight motion is planar in the axial–radial plane;
- repeated flyweights are identical and arranged with sufficient circumferential symmetry that omitted cross and gyroscopic terms cancel;
- structural deformation of the arm, pivot, roller, ramp, and carrier is neglected;
- backlash and manufacturing clearance in the kinematic constraint are neglected;
- the dynamic ramp representation is  $C^3$  over the active contact range;
- ramp contact is an ideal geometric, frictionless constraint in the flyweight force law;
- roller-bearing loss, roller spin inertia, local Hertzian compliance, and wear are neglected;
- only one active roller–ramp contact is retained; and
- the selected contact must remain compressive.

For the present Baja mass approximation, the arm/body is additionally treated as a uniform slender member and all outer hardware is concentrated at the roller-centre station. The finite dimensions and centroidal inertias listed in [Appendix C.3.1](#) are consequently omitted. A higher fidelity mass description may replace this reduction without changing the geometry or dynamics developed above.

### C.6 Relation to Existing Actuator Models

The geometry and mass reductions used here are not intended as novelty claims by themselves. Detailed CVT actuator models already use measured geometry, centre-of-mass locations, contact construction, and mechanical advantage to recover flyweight thrust. For example, [Messick \(2018\)](#) derives primary thrust from measured/CAD flyweight geometry under quasi-equilibrium, [Kim et al. \(2002\)](#) retain detailed roller/contact geometry and contact friction in an equilibrium actuator treatment, and [Skinner \(2020\)](#) resolves the flyweight, supporting link, and ramp reactions with explicit free bodies while representing the flyweight and link centrifugal loading through their mass and working radii. These approaches establish the use of reduced mass and geometry information for detailed actuator force prediction.

If the pivot inertia is neglected and the mass distribution is reduced to a point or centre-of-mass description at radius  $r_f(x_p)$ , then

$$J_{f,p} = m_f r_f^2 \quad (\text{C.6.1})$$

and the centrifugal term reduces to

$$\boxed{\frac{1}{2}\omega_p^2 \frac{dJ_{f,p}}{dx_p} = m_f r_f \omega_p^2 \frac{dr_f}{dx_p}.} \quad (\text{C.6.2})$$

This is the familiar centrifugal force-times-motion-ratio form used by quasi-static actuator descriptions. The extension developed here is that the same geometry and mass maps are consumed by the transient balance while retaining the flyweight's pivoting inertia, changing mechanism motion ratio, and configuration-dependent shaft angular momentum. The distinction is therefore dynamic coupling of the actuator to the shift and rotational equations, not the existence of curved ramps, finite rollers, or detailed flyweight geometry.

## Appendix D

# Numerical Implementation

*Implementation version: `cinder-v1.1.2` (commit `7637a38b4fb9ec21dfb953c1c80a27ec5f389654`).*

This appendix records the parts of the CINDER implementation that require a numerical procedure beyond the analytical equations themselves, together with cases where the implemented algebra differs from the form used in the formulation. The discussion therefore focuses on root solves, conditioning, contact closure, event handling, and the equivalent equation forms used in the code. Code references are included so that these numerical choices can be traced to the tagged implementation.

### D.1 Numerical Solution of Belt Geometry

The exact belt geometry derived in [Sections 3.1.2](#) and [3.1.3](#) is retained in the implementation. The quantities that remain implicit in the belt-length closure are found with SciPy's `scipy.optimize.brentq` routine (?).

The first such solve is performed once, during construction, to determine the fixed pulley centre distance  $C$  from the specified zero-shift radii  $r_{p,0}$  and  $r_{s,0}$ . A physical bracket can be obtained directly from the same belt geometry. The centre distance cannot be smaller than the sum of the two radii,

$$C_{\text{lo}} = r_{p,0} + r_{s,0}, \tag{D.1.1}$$

because the two pulley circles would otherwise overlap. The upper bound comes from the finite belt length. The total wrapped length is

$$\begin{aligned} r_p \phi_p + r_s \phi_s &= \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1} \left( \frac{r_s - r_p}{C} \right) \\ &\geq \pi(r_p + r_s). \end{aligned} \tag{D.1.2}$$

At the zero-shift geometry, let  $\ell_0$  denote the length of either straight span. The two straight spans can then use no more than

$$2\ell_0 \leq L_b - \pi(r_{p,0} + r_{s,0}). \tag{D.1.3}$$

From the exact span relation in [Equation \(3.1.19\)](#),

$$\ell_0^2 = C^2 - \Delta r_0^2, \quad \Delta r_0 = r_{s,0} - r_{p,0}. \quad (\text{D.1.4})$$

Combining these relations gives the upper bracket

$$C_{\text{hi}} = \sqrt{\Delta r_0^2 + \left[ \frac{L_b - \pi(r_{p,0} + r_{s,0})}{2} \right]^2}. \quad (\text{D.1.5})$$

The belt-length residual is checked at both endpoints before iteration. An endpoint is accepted when its absolute residual is no greater than  $10^{-12}$  m; otherwise a sign change is required and Brent iteration uses an absolute root-location tolerance of  $10^{-12}$  m. The bracket is therefore set by the geometry itself rather than by an arbitrary numerical search interval.

A second broad geometry solve is performed once at construction to make the repeated runtime solve cheaper. At  $s = s_{\text{max}}$ , the primary radius is known explicitly,

$$r_{p,\text{max}} = r_{p,0} + \frac{s_{\text{max}} - s_{\text{dz}}}{2 \tan \beta}, \quad (\text{D.1.6})$$

while the compatible secondary radius is still obtained from belt-length closure. This one setup solve uses the broad physical bracket

$$\begin{aligned} r_{s,\text{lo}}^{\text{setup}} &= h_b, \\ r_{s,\text{hi}}^{\text{setup}} &= \min \left( C - r_{p,\text{max}}, \frac{L_b}{\pi} - r_{p,\text{max}} \right). \end{aligned} \quad (\text{D.1.7})$$

The lower limit keeps the full belt section outside the pulley axis. The first upper limit leaves enough centre spacing for the primary and secondary radii to fit without intersecting, while the second follows from [Equation \(D.1.2\)](#): the two wrapped portions cannot by themselves require more belt than is available.

Once the secondary radius at maximum shift has been found, the two travel endpoints are stored as

$$r_s(s) \in [r_s(s_{\text{max}}), r_s(0)]. \quad (\text{D.1.8})$$

Every ordinary geometry evaluation during integration can then solve for  $r_s(s)$  on this much tighter interval instead of repeating the broad setup search. The extra work is paid once at construction so that the geometry root solved throughout the run uses the smallest precomputed physical bracket. The other radius definitions used by the model follow from fixed offsets from these solved radii. First and second radius derivatives are obtained directly from the differentiated belt-length closure, which avoids adding finite-difference error to quantities used by the dynamic equations. At  $s = s_{\text{dz}}$ , the radius itself is common to both sides of the boundary, but its governing branch changes; separate deadzone-side and engaged-side evaluations therefore provide the appropriate one-sided derivatives.

**Code reference** The geometry implementation is contained in [belt\\_length.py](#), [spec.py](#), and [belt\\_pulley.py](#).

## D.2 Fixed- $\lambda$ Engaged Mechanical Closure

For a proposed signed traction pair  $(\lambda_p, \lambda_s)$ , [Section 3.5.3](#) presents the freely shifting engaged mechanics after eliminating the two pulley torques  $\tau_p$  and  $\tau_s$ . The implementation instead retains those torques as simultaneous unknowns,

$$\mathbf{z}_{\text{impl}} = [\dot{\omega}_p \quad \dot{\omega}_s \quad \dot{v}_b \quad \ddot{s} \quad \tau_p \quad \tau_s \quad N_p \quad N_s]^\top, \quad (\text{D.2.1})$$

and keeps the two traction relations as explicit rows,

$$\frac{\tau_p}{r_{p,\text{eff}}} + \lambda_p N_p = 0, \quad \frac{\tau_s}{r_{s,\text{eff}}} + \lambda_s N_s = 0. \quad (\text{D.2.2})$$

Eliminating  $\tau_p$  and  $\tau_s$  from these rows recovers the reduced six-variable system used in the formulation, so the implementation does not add a physical degree of freedom. This form has two practical advantages: torque-reactive mechanisms remain coupled directly to the same simultaneous solve, and the normal resultants never have to be recovered by dividing a solved torque by  $\lambda_j$ . In particular, when  $\lambda_j = 0$ , the corresponding traction row simply gives  $\tau_j = 0$  while the remaining balances continue to determine  $N_j$ .

### D.2.1 Equilibrated linear solve

The closure combines accelerations, torques, forces, and normal resultants in one linear system. These quantities have different units and can produce very different coefficient magnitudes even when the underlying equations are well posed. Before the dense solve, CINDER therefore equilibrates the algebraically identical system. For  $\mathbf{A}\mathbf{z} = \mathbf{b}$ ,

$$\mathbf{A}_s = \mathbf{D}_r \mathbf{A} \mathbf{D}_c, \quad \mathbf{b}_s = \mathbf{D}_r \mathbf{b}, \quad \mathbf{z} = \mathbf{D}_c \mathbf{y}. \quad (\text{D.2.3})$$

Each row scale is the reciprocal of the largest absolute magnitude in that row or its right-hand side; column scales are then formed from the row-scaled matrix. The scaled system is solved for  $\mathbf{y}$  and mapped back to the physical unknowns. Residuals are still evaluated against the original, unscaled equations. Diagnostic solves additionally expose the rank and both raw and equilibrated condition numbers. The scaling therefore changes only the numerical conditioning of the solve, not the mechanical equations being enforced.

### D.2.2 Regular evaluation through zero traction

The wrap solution is analytically regular at  $\lambda_j = 0$ , as shown in [Equation \(3.4.33\)](#). A clamped contact can therefore pass continuously through zero signed traction while retaining  $N_j > 0$ ; zero traction is not a separate mechanical regime. The direct coefficient formulas are nevertheless numerically awkward near this limit because they subtract nearly equal quantities.

With  $z = \lambda_j \phi_j / \sin \beta$ ,

$$\Phi_-(z) = \frac{1 - e^{-z}}{z}, \quad \Psi_-(z) = \frac{z - 1 + e^{-z}}{z^2}. \quad (\text{D.2.4})$$

For  $|z| < 10^{-4}$ , CINDER evaluates the local series

$$\begin{aligned}\Phi_-(z) &= 1 - \frac{z}{2} + \frac{z^2}{6} - \frac{z^3}{24} + \frac{z^4}{120}, \\ \Psi_-(z) &= \frac{1}{2} - \frac{z}{6} + \frac{z^2}{24} - \frac{z^3}{120} + \frac{z^4}{720},\end{aligned}\tag{D.2.5}$$

and uses `expm1`-based expressions outside that neighbourhood. The change is only in how the same smooth analytical functions are evaluated; no artificial topology change is introduced at  $\lambda_j = 0$ .

**Code reference** The closure basis and linear solve are implemented in `closure.py` and `trial_system.py`; row assembly is in `equations.py`, and the regular wrap functions are evaluated in `equation_context.py`.

### D.3 Actuator Contributions

The actuator forces and torques derived in [Sections 3.2](#) and [3.3](#) enter the same coupled mechanical closure as the belt and pulley dynamics. The implementation preserves that coupling, but separates the actuator law from the closure assembly. A pulley actuator therefore supplies its contribution to the common force and torque balances through a shared interface rather than being built directly into those balances.

This separation allows the same CVT mechanics to be used with different clamping mechanisms without changing the closure itself. The derived flyweight, primary spring, secondary helix, and secondary spring models all use this interface, but so can a different mechanical actuator, a prescribed axial-force law, or a feedback-controlled actuator. This also allows study configurations to replace the native mechanical clamping without defining a separate CVT model. Only the actuator contribution changes; the belt, pulley, and contact equations continue to be solved as the same coupled system.

The interface places one numerical restriction on those contributions. Once the current state, geometry, and trial traction pair are fixed, each contribution must be affine in the instantaneous closure unknowns so that the common mechanical system remains linear. Thus a contribution is represented as

$$q(\mathbf{z}_{\text{impl}}) = q_0 + \mathbf{g}^T \mathbf{z}_{\text{impl}}.\tag{D.3.1}$$

The affine requirement applies to the closure unknowns, not to the state on which the actuator law depends. Both  $q_0$  and  $\mathbf{g}$  may therefore depend nonlinearly on the current position and speed. Centrifugal flyweight action, for example, may retain its  $\omega_p^2$  dependence because  $\omega_p$  is known when the instantaneous closure is assembled.

Dependence on an unknown being solved at that instant is carried by  $\mathbf{g}$ . If a mounted mechanism uses a local pulley coordinate  $x_j(s)$ , for example,

$$\ddot{x}_j = x_j''(s)\dot{s}^2 + x_j'(s)\ddot{s}.\tag{D.3.2}$$

At the current state,  $x_j''(s)\dot{s}^2$  is known, while  $x_j'(s)\ddot{s}$  contributes a coefficient on the unknown global shift acceleration. The same interface can carry dependence on  $\dot{\omega}_j$ ,  $\tau_j$ , or  $N_j$  when those quantities appear in an actuator law, while preserving the linear simultaneous closure.

**Code reference** The affine closure objects are defined in `closure.py`; the pulley element contract and local context are in `actuation/types.py`, and mounted contributions are combined by `pulley_actuator.py`.

## D.4 Numerical Solution of Tangential Contact

Sticking introduces the nonlinear traction-root conditions in [Equations \(3.5.11\) and \(3.5.12\)](#). CINDER solves these roots numerically around the fixed- $\lambda$  mechanical closure of [Appendix D.2](#).

### D.4.1 Search domain, Jacobian, and acceptance

The numerical root search is deliberately allowed to extend beyond the physical static-friction interval. This separates two questions that are physically distinct: first, what traction would be required to enforce stick; second, whether the available static friction can supply it. An above-capacity root can therefore be found numerically and then rejected by the static limit in [Equation \(3.5.19\)](#), rather than being hidden by the search bounds themselves.

When no continuation estimate is available, the initial guess is zero. CINDER uses SciPy’s bounded `scipy.optimize.least_squares` routine (?) with the trust-region reflective method, a three-point finite-difference residual Jacobian, and Jacobian-based scaling. The tagged defaults use `ftol=xtol=gtol=10-12` and at most 100 residual evaluations. A returned candidate is not accepted from the optimizer status alone: the mechanical system is evaluated once more at that traction, and the resulting sticking residual must satisfy the declared contact tolerance.

### D.4.2 Continuation between consecutive contact solves

Successive right-hand-side evaluations within one smooth contact topology are normally close in state space, so the required sticking traction is normally close as well. CINDER uses this local continuity to avoid restarting the full nonlinear solve from zero at every evaluation. The previous accepted traction root and its local residual Jacobian are stored, and one bounded Newton correction is attempted first. After a successful correction, the Jacobian is updated with a rank-one good-Broyden step.

If the continuation estimate is unavailable, reaches a search bound, cannot be evaluated mechanically, or fails the sticking-residual tolerance, the code falls back to the full bounded least-squares solve. Besides reducing repeated nonlinear work, following the nearby accepted solution helps the simulation remain on the locally connected traction branch if other roots exist elsewhere in the residual map. The stored numerical information is only a solver aid; it does not represent physical friction memory.

### D.4.3 Numerical hysteresis at reattachment

The physical static-capacity limit remains [Equation \(3.5.19\)](#). At finite numerical precision, however, using exactly the same zero-margin capacity for both loss and recovery of stick can cause repeated stick-slip switching at nearly the same state. CINDER therefore leaves stick loss at the physical zero-margin boundary but normally requires a positive restick reserve of  $10^{-3}$  in traction-utilization margin.

This reserve is a numerical switching margin, not a change to the friction law. If applying it would leave the event resolver with no admissible successor state, the sticking candidate is retried at the

physical static boundary. The extra margin therefore suppresses numerical chatter where it can do so without preventing an otherwise valid transition.

**Code reference** The nonlinear solve and continuation are implemented in [engaged\\_contact.py](#); contact residuals and tolerances are in [relative\\_motion.py](#) and [tolerances.py](#); hybrid switching is implemented in [cvt\\_contact\\_switching.py](#).

## D.5 Time Integration and Hybrid Boundary Events

### D.5.1 Continuous integration and host coupling

CINDER is implemented as a CVT component within a host simulation. The standard study configurations attach the surrounding engine, load, vehicle, or other required system through the shaft interfaces defined in [Section 2.1](#). At each interface the host supplies the applied torque and equivalent referred inertia seen by the CVT, while the shaft speed remains part of the coupled dynamic response.

The host layer is modular rather than fixed to those standard configurations. A different host can be attached without changing the internal CVT equations, and it may introduce additional states and boundary laws outside the transmission. Those states can be advanced together with the CVT in the same initial-value problem, allowing the transmission to be embedded in a different engine, vehicle, dynamometer, flywheel, controller, or larger drivetrain model while retaining the same shaft-port contract.

Within any fixed mechanical regime, the combined first-order initial-value problem is advanced through SciPy’s `scipy.integrate.solve_ivp` interface (?). The generic runner is method-configurable. The reported transient study drivers select LSODA, whose automatic selection between stiff and nonstiff multistep formulas follows the strategy of ?. Relative and absolute tolerances and maximum-step limits are study settings and are therefore reported with the Results configuration rather than treated as constants of the mechanical model.

The integration mesh is adaptive. The right-hand side may be evaluated at several internal trial states before a single step is accepted, so a plotted or reported sampling interval is not an integration time step.

### D.5.2 Hybrid segmentation and event localization

The regime structure derived in [Section 3.6](#) is integrated one smooth regime at a time. Each active regime exposes only the boundary events that it can reach, together with the allowed crossing direction. When an event is localized, the current `solve_ivp` segment ends, the transition resolver determines the successor regime and any required state update, and integration restarts at the same physical time.

Events localized to the same endpoint within the configured association tolerance are resolved together. The adaptive integrator can also probe a trial state slightly beyond a valid boundary while attempting or rejecting a step. Roundoff-scale boundary snapping and protected evaluation of these trial states keep such numerical probes inside valid geometry; they do not move the accepted physical event surface.

### D.5.3 Integration mesh and reported output

The native adaptive trajectory is retained separately from the user-facing reporting grid. Uniform reports are generated with SciPy dense output within each smooth segment, and exact segment endpoints are kept explicitly. Dense output is therefore never used to interpolate across a discrete state reset.

**Code reference** The shaft ports are implemented in [ports.py](#); segmented integration is in [hybrid.py](#); and reporting materialization is in [reporting.py](#).

## D.6 Formulation-to-Implementation Correspondence

| Analytical presentation                         | cinder-v1.1.2 implementation                                                                              | Numerical purpose                                                                                                      |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Implicit exact belt geometry                    | SciPy <code>brentq</code> ; physical construction bounds and a cached runtime radius bracket              | Retain the exact belt-length closure while keeping the repeated radius solve tightly bounded.                          |
| Six-variable fixed- $\lambda$ closure           | Eight-variable closure retaining pulley torques and explicit traction rows                                | Keep torque-dependent mechanisms simultaneous and remain regular through zero traction.                                |
| $Az = b$                                        | Row/column equilibration followed by a dense direct solve                                                 | Reduce avoidable mixed-scale conditioning problems without changing the physical equations.                            |
| Continuous $\lambda_j \rightarrow 0$ wrap limit | Regular $\Phi_-/\Psi_-$ evaluation using local series and <code>expm1</code>                              | Avoid cancellation while preserving the same smooth contact solution.                                                  |
| Actuator laws entering the coupled balances     | Shared actuator interface with affine bias-plus-gain dependence on the closure unknowns                   | Allow actuator models to be exchanged without changing the CVT closure while preserving its linear simultaneous solve. |
| Static-traction root                            | Bounded trust-region <code>least_squares</code> , local continuation, and independent residual acceptance | Find the traction required for stick efficiently, then apply physical admissibility separately.                        |
| Continuous regime equations                     | Adaptive <code>solve_ivp</code> segments with explicit hybrid transitions                                 | Integrate smooth dynamics separately from changes in the active mechanical regime.                                     |

Table D.1: Correspondence between the analytical formulation and the numerical choices documented in this appendix.

# Appendix E

## Reference Inputs, Shaft Boundaries, and Study Actuators

This appendix records the shared Baja reference inputs used by the Results campaign and the shaft-boundary models attached to CINDER in the different simulation cases. Case-specific changes are listed with the corresponding study in [Appendix F](#).

The executable reference input is [defaults/baja/simulation\\_case.json](#), with the human-readable tuning quantities collected in [defaults/baja/tuning.json](#).

### E.1 Reference CVT Configuration

| Group                           | Quantity                                     | Symbol                          | Value                  |
|---------------------------------|----------------------------------------------|---------------------------------|------------------------|
| <b>Belt and pulley geometry</b> |                                              |                                 |                        |
| Belt                            | Outer-path length                            | $L_b$                           | 0.953 262 m            |
| Belt                            | Radial thickness                             | $h_b$                           | 0.015 570 2 m          |
| Belt                            | Outer-face width                             | $b_{\text{out}}$                | 0.021 336 m            |
| Belt                            | Inner-face width                             | $b_{\text{in}}$                 | 0.016 814 8 m          |
| Belt                            | Cord-layer depth from the outer surface      | $d_{\text{cord}}$               | 0.002 54 m             |
| Belt                            | Material density                             | $\rho_b$                        | 1100 kg/m <sup>3</sup> |
| Pulley geometry                 | Sheave half-angle                            | $\beta$                         | 0.200 712 864 rad      |
| Pulley geometry                 | Primary outer-surface radius at zero shift   | $r_{p,\text{outer},\text{min}}$ | 0.036 207 7 m          |
| Pulley geometry                 | Secondary outer-surface radius at zero shift | $r_{s,\text{outer},\text{max}}$ | 0.1016 m               |
| Pulley geometry                 | Primary deadzone travel                      | $s_{\text{dz}}$                 | 0.002 489 2 m          |
| Pulley geometry                 | Maximum shift coordinate                     | $s_{\text{max}}$                | 0.019 05 m             |
| <b>Belt–pulley contact</b>      |                                              |                                 |                        |
| Contact                         | Static traction coefficient                  | $\mu_s$                         | 0.65                   |
| Contact                         | Kinetic traction coefficient                 | $\mu_k$                         | 0.55                   |

Table E.1: Geometry, belt, and contact parameters of the shared Baja reference configuration.

| Group            | Quantity                                       | Symbol                  | Value                                               |
|------------------|------------------------------------------------|-------------------------|-----------------------------------------------------|
| <b>Primary</b>   |                                                |                         |                                                     |
| Primary          | Number of flyweights                           | $n_f$                   | 3                                                   |
| Primary          | Mass per flyweight                             | $m_f$                   | 0.263 646 kg                                        |
| Primary          | Flyweight pivot radius                         | —                       | 0.042 545 m                                         |
| Primary          | Pivot-to-roller-center distance                | —                       | 0.031 521 4 m                                       |
| Primary          | Roller radius                                  | —                       | 0.0065 m                                            |
| Primary          | Linear ramp angle                              | —                       | 35°                                                 |
| Primary          | Linear ramp span                               | —                       | 0.005 m                                             |
| Primary          | $C^3$ transition span                          | —                       | 0.003 m                                             |
| Primary          | Circular ramp angle range                      | —                       | 35° → 20°                                           |
| Primary          | Circular ramp span                             | —                       | 0.03 m                                              |
| Primary          | Fixed rotating hardware inertia                | —                       | 3.220 283 8 ×<br>10 <sup>-3</sup> kg m <sup>2</sup> |
| Primary          | Moving-sheave translational mass               | $m_{p,m}$               | 1.0681 kg                                           |
| Primary          | Axial spring stiffness                         | $k_p$                   | 12 784 N/m                                          |
| Primary          | Axial spring initial compression               | $x_{p,0}$               | 0.094 090 49 m                                      |
| <b>Secondary</b> |                                                |                         |                                                     |
| Secondary        | Fixed rotating hardware inertia                | $I_{s,F}$               | 2.326 26 ×<br>10 <sup>-3</sup> kg m <sup>2</sup>    |
| Secondary        | Movable-member rotational inertia              | $I_{s,M}$               | 0.002 513 9 kg m <sup>2</sup>                       |
| Secondary        | Moving-sheave translational mass               | $m_{s,m}$               | 0.705 141 kg                                        |
| Secondary        | Axial spring stiffness                         | $k_{s,x}$               | 3532 N/m                                            |
| Secondary        | Axial spring initial compression               | $x_{s,0}$               | 0.11 m                                              |
| Secondary        | Torsional stiffness                            | $k_{s,\theta}$          | 3.476 N m/rad                                       |
| Secondary        | Initial torsional preload                      | $\theta_{s,\text{pre}}$ | 300°                                                |
| Secondary        | Helix reference radius                         | $r_h$                   | 0.044 45 m                                          |
| Secondary        | Helix angle from the circumferential direction | $\alpha_s$              | 20°                                                 |
| Secondary        | Movable-face share of belt torque              | $\tau_{s,M}/\tau_s$     | 0.5                                                 |

Table E.2: Actuation and inertia parameters of the shared Baja reference configuration.

### E.1.1 Reference secondary-helix contact topology

The Results reference CVT uses a zero-clearance bilateral slot at the secondary helix. The signed torque and axial reaction are those derived in [Section 3.2.2](#); a positive or negative required reaction is carried by the corresponding slot flank. The slot is idealized as having no backlash or free-play, so changing reaction sign does not introduce a separate kinematic state.

This is a physical reference-model assumption, not a change to the helix equations derived in the main text. The stock `cinder-v1.1.2` helix element additionally applies a one-sided contact-admissibility

check that requires the reaction on its implemented helix flank to remain compressive. For the reference Results model, the loader replaces that element by a bilateral subclass that preserves the same spring, torque-sharing, inertia, and helix kinematics while removing only that one-sided flank check. The implementation is in `defaults/reference_model/slotted_helix.py` and is applied by `defaults/reference_model/reference_case.py`.

The shared reference launch begins from

$$\omega_p(0) = 1800 \text{ rpm}, \quad \omega_s(0) = 0, \quad v_b(0) = 0, \quad s(0) = 0, \quad \dot{s}(0) = 0. \quad (\text{E.1.1})$$

### E.1.2 Basis of the reference parameters

The reference set is an engineering baseline representative of the custom CVT and vehicle used by McMaster Baja Racing. Its inputs combine direct hardware measurements, CAD- or geometry-derived quantities, tuning choices, manufacturer or published data, and engineering estimates. The principal basis of each parameter family is summarized in [Table E.3](#).

Table E.3: Basis of the shared Baja reference parameters.

| Parameter family                              | Basis used for the reference set                                                                                                                                                                                          |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CVT and pulley dimensions                     | Based on the team’s custom Baja CVT hardware, using direct measurements and derived geometry where available.                                                                                                             |
| Primary and secondary actuation               | Based on the team’s hardware and reference tune; preload and profile quantities are the values used by the frozen simulation input.                                                                                       |
| Secondary helix contact topology              | Zero-clearance bilateral slot used as the Results reference mechanism; this is a modelling assumption for the ideal signed helix relation, not a measured backlash/contact-compliance model.                              |
| Masses and inertias                           | Combination of known component masses, CAD/geometry-derived quantities, and engineering estimates where direct inertia measurements were unavailable.                                                                     |
| Belt density and traction coefficients        | Engineering estimates based on generic rubber/V-belt material and contact data; these quantities were not measured for the exact belt used by the team.                                                                   |
| Engine torque curve                           | Based on the Kohler CH440 torque/power data used in <a href="#">Figure 1.1 (Baja SAE, 2022)</a> ; interpolation and the out-of-range braking tails are numerical extensions described in <a href="#">Appendix E.2.2</a> . |
| Vehicle mass, wheel radius, and final drive   | Team vehicle/reference-drivetrain values.                                                                                                                                                                                 |
| Rolling resistance and aerodynamic parameters | Engineering estimates used to provide a representative longitudinal road load rather than an experimentally identified coast-down model.                                                                                  |

## E.2 Shaft-Boundary Interface

At each shaft, the host boundary supplies the signed external torque and the rotational inertia referred to that shaft. Using the shaft-interface notation introduced in [Section 2.1](#), a boundary evaluation returns

$$\mathcal{B}_j(t, \mathbf{x}, \mathbf{x}_h) \longmapsto (\tau_{B,j}, I_{B,j}), \quad j \in \{p, s\}, \quad (\text{E.2.1})$$

where  $\mathbf{x}_h$  denotes any additional host state required by the boundary. The shaft speed remains a CINDER state unless a separate kinematic constraint is introduced by the host.

Appendix E.2 distinguishes boundary components provided by CINDER v1.1.2 from adapters defined in the Results code and attached through the same interface.

| Boundary used in the Results campaign   | Origin        | Implementation                                                         | Boundary quantity                                                                                               |
|-----------------------------------------|---------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Full-throttle engine                    | CINDER v1.1.2 | <code>engine boundary</code> ; <code>torque curve</code>               | Speed-dependent engine torque and constant referred engine-side inertia.                                        |
| Locked final-drive vehicle              | CINDER v1.1.2 | <code>vehicle boundary</code> ; <code>road load</code>                 | Vehicle road-load torque and reflected vehicle/wheel inertia.                                                   |
| Constant torque–inertia boundary        | CINDER v1.1.2 | <code>constant boundary</code>                                         | Prescribed constant torque and referred inertia; shaft speed is not fixed.                                      |
| Time-programmed final-drive vehicle     | Results code  | <code>time-programmed boundary</code>                                  | Same vehicle and inertia as the decoded reference boundary, with grade supplied as $\gamma(t)$ .                |
| Additive torque wrapper                 | Results code  | <code>actuator-study wrapper</code> ; <code>helix-study wrapper</code> | Adds a prescribed smooth torque perturbation while retaining the wrapped boundary inertia.                      |
| Blend-to-target / torque-scale wrappers | Results code  | <code>torque-blend wrappers</code>                                     | Smoothly replaces or scales the torque returned by another boundary while retaining its inertia.                |
| Prescribed torque bench boundary        | Results code  | <code>prescribed-torque boundary</code>                                | Smooth transition between two prescribed torque values with a specified constant referred inertia.              |
| Controlled overrun boundary pair        | Results code  | <code>overrun boundary pair</code>                                     | Blends primary torque toward a prescribed resisting value while independently programming secondary road grade. |

Table E.4: Shaft-boundary implementations used by the Results studies.

### E.2.1 CINDER constant torque–inertia boundary

The CINDER class named `FixedShaftBoundary` fixes the *boundary inputs*, not the shaft speed. It returns

$$\tau_{B,j} = \tau_{B,j}^{(0)}, \quad I_{B,j} = I_{B,j}^{(0)}, \quad (\text{E.2.2})$$

while  $\omega_j$  remains part of the dynamically integrated CVT state. This boundary is used for constructed bench states in the verification studies and for source-constrained external cases such as the independent Ballew benchmark; the selected  $\tau_{B,j}^{(0)}$  and  $I_{B,j}^{(0)}$  values are listed with those cases in Appendix F.

### E.2.2 Primary full-throttle engine boundary

The reference primary boundary uses the CINDER v1.1.2 full-throttle engine component,

$$\tau_{B,p} = \tau_{\text{eng}}(\omega_p), \quad I_{B,p} = I_{\text{eng}}, \quad (\text{E.2.3})$$

with  $I_{\text{eng}} = 0.05 \text{ kg m}^2$ . The positive-drive points are based on the Kohler CH440 data used in [Figure 1.1](#) ([Baja SAE, 2022](#)) and are interpolated by CINDER using a shape-preserving piecewise cubic Hermite interpolant. The numerical boundary adds bounded low- and high-speed braking tails so that externally driven states remain finite outside the positive-drive range.

Table E.5: Reference positive-drive engine torque points.

| Engine speed [rpm] | Torque [N m] |
|--------------------|--------------|
| 1000               | 0.000        |
| 1800               | 24.405       |
| 2400               | 25.083       |
| 2600               | 24.540       |
| 2800               | 23.591       |
| 3000               | 22.507       |
| 3200               | 20.880       |
| 3400               | 19.659       |
| 3600               | 18.304       |
| 4000               | 0.000        |

The low-speed extension passes through  $(0, 0)$ ,  $(500 \text{ rpm}, -5 \text{ N m})$ , and  $(1000 \text{ rpm}, 0)$ . The high-speed extension begins at the final positive-drive point and transitions over 1500 rpm to a  $-28 \text{ N m}$  braking plateau; a second 1500 rpm interval is included in the PCHIP support before the plateau is held constant. These tails are extrapolated boundary behaviour rather than CH440 measurements.

### E.2.3 Secondary locked-final-drive vehicle boundary

The reference secondary boundary rigidly maps secondary-shaft speed to wheel speed through the final-drive reduction

$$G = \frac{\omega_s}{\omega_w}, \quad v = \frac{r_w}{G} \omega_s. \quad (\text{E.2.4})$$

For grade angle  $\gamma$ , the longitudinal forces are

$$F_{\text{grade}} = -m_v g \sin \gamma, \quad (\text{E.2.5})$$

$$F_{\text{roll}} = -C_{rr} m_v g \cos \gamma \frac{v}{\sqrt{v^2 + v_c^2}}, \quad (\text{E.2.6})$$

$$F_{\text{aero}} = -\frac{1}{2} \rho C_d A_f |v| v, \quad (\text{E.2.7})$$

and the external secondary torque is

$$\tau_{B,s} = \frac{r_w}{G} (F_{\text{grade}} + F_{\text{roll}} + F_{\text{aero}}). \quad (\text{E.2.8})$$

The vehicle translation and wheel rotation are reflected to the secondary as

$$I_{B,s} = I_{s,\text{dir}} + \frac{I_w}{G^2} + m_v \left( \frac{r_w}{G} \right)^2. \quad (\text{E.2.9})$$

Table E.6: Parameters of the reference vehicle boundary.

| Quantity                                                | Value                             |
|---------------------------------------------------------|-----------------------------------|
| Vehicle mass $m_v$                                      | 300 kg                            |
| Wheel rotational inertia $I_w$                          | 1.107 817 44 kg m <sup>2</sup>    |
| Wheel radius $r_w$                                      | 0.2794 m                          |
| Final-drive reduction $G$                               | 7.556                             |
| Direct secondary-shaft inertia $I_{s,\text{dir}}$       | 0.002 011 338 4 kg m <sup>2</sup> |
| Rolling-resistance coefficient $C_{rr}$                 | 0.015                             |
| Aerodynamic drag coefficient $C_d$                      | 0.6                               |
| Frontal area $A_f$                                      | 1.114 84 m <sup>2</sup>           |
| Air density $\rho$                                      | 1.225 kg/m <sup>3</sup>           |
| Gravity $g$                                             | 9.806 65 m/s <sup>2</sup>         |
| Rolling-speed regularization $v_\epsilon$               | 0.01 m/s                          |
| Total reflected secondary inertia from Equation (E.2.9) | 0.431 610 kg m <sup>2</sup>       |

### E.2.4 Results-side time-programmed and torque-modified boundaries

Several studies retain a CINDER boundary’s inertia and constitutive law while changing only the imposed external loading history. The Results adapters use a smooth transition coordinate

$$u(t) = \begin{cases} 0, & t \leq t_0, \\ 3\xi^2 - 2\xi^3, & t_0 < t < t_0 + T_r, \\ 1, & t \geq t_0 + T_r, \end{cases} \quad \xi = \frac{t - t_0}{T_r}. \quad (\text{E.2.10})$$

For a shaft  $j$ , the boundary forms used in the Results campaign include

$$\tau_{B,j}^{\text{add}}(t) = \tau_{B,j}^{\text{base}}(t) + u(t) \Delta\tau_j, \quad (\text{E.2.11})$$

$$\tau_{B,j}^{\text{target}}(t) = [1 - u(t)]\tau_{B,j}^{\text{base}}(t) + u(t)\tau_{j,\text{target}}, \quad (\text{E.2.12})$$

$$\tau_{B,j}^{\text{scale}}(t) = [1 + u(t)(k_\tau - 1)]\tau_{B,j}^{\text{base}}(t), \quad (\text{E.2.13})$$

$$\tau_{B,j}^{\text{prescribed}}(t) = [1 - u(t)]\tau_{j,i} + u(t)\tau_{j,f}. \quad (\text{E.2.14})$$

The additive, target, and scale forms retain the equivalent inertia returned by the wrapped boundary. The standalone prescribed-torque form uses a case-specified constant inertia. The time-programmed vehicle boundary retains Equations (E.2.8) and (E.2.9) and replaces the road profile by a prescribed  $\gamma(t)$ . Exact amplitudes, inertias, grades, and transition times are listed with the study cases that use them.

## E.3 Study-Specific Actuator Implementations

The implementation described in Appendix D.3 allows the pulley actuator attached to either side to be exchanged without changing the belt, pulley, or contact closure. Each actuator contribution is returned through the same affine closure interface. The Results studies use that interface both for the full mechanisms derived in the main text and for the study-specific alternatives summarized below.

| Actuator law                       | Used in                                  | Mechanical change relative to the main-text actuator                                                                                                                                                                                   | Origin                    |
|------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Bilateral dynamic helix            | Shared Baja-reference studies            | Same signed dynamic helix relation as <a href="#">Equation (3.2.27)</a> ; the physical contact is the zero-clearance bilateral slot of <a href="#">Appendix E.1.1</a> , rather than the stock v1.1.2 one-sided flank check.            | <a href="#">Results</a>   |
| Quasi-static fixed-pivot flyweight | <a href="#">Section 4.4.1</a>            | Retains the centrifugal term of <a href="#">Equation (3.2.11)</a> and removes the two pivot-acceleration terms. The associated rotating inertia is reassigned to the fixed primary inertia rather than deleted.                        | <a href="#">Results</a>   |
| Quasi-static helix                 | <a href="#">Sections 4.4.1 and 4.4.2</a> | Retains spring torque and reacted movable-face belt torque, but removes the movable-member inertial term from <a href="#">Equation (3.2.27)</a> . The removed movable rotational inertia is reassigned to the fixed secondary inertia. | <a href="#">Results</a>   |
| Constant axial force               | <a href="#">Section 4.3</a>              | Replaces the native mechanical actuator by a prescribed axial closing force independent of state and closure unknowns.                                                                                                                 | <a href="#">Benchmark</a> |
| Tabulated axial force              | <a href="#">Section 4.3.3</a>            | Replaces the primary mechanical actuator by a prescribed time history, evaluated by piecewise-linear interpolation of the prepared force data.                                                                                         | <a href="#">Benchmark</a> |
| PI-controlled axial force          | <a href="#">Section 4.3.2</a>            | Replaces the primary mechanical actuator by a feedback force law driven by primary speed and an integrated controller-error state carried by the benchmark host.                                                                       | <a href="#">Benchmark</a> |

Table E.7: Study-specific actuator laws used through the common CINDER pulley-actuator interface.

### E.3.1 Quasi-static actuator reductions

The quasi-static primary-flyweight law used for the actuator-dynamics comparison keeps only the first term of [Equation \(3.2.11\)](#),

$$F_{\text{fw},x_p}^{\text{QS}} = \frac{1}{2}\omega_p^2 \frac{dJ_{f,p}}{dx_p}. \quad (\text{E.3.1})$$

The terms proportional to  $\ddot{x}_p$  and  $\dot{x}_p^2$  are omitted from the actuator force. To avoid deleting physical rotating mass, the ablation moves the corresponding flyweight rotational inertia into the constant primary rotating inertia used by that comparison. The implementation is in [ablation\\_core.py](#).

For the secondary, define the quasi-static reacted torque

$$\tau_{s,\text{helix}}^{\text{QS}} = \tau_{s,M} + k_{s,\theta} (\theta_{s,\text{pre}} - \theta_s). \quad (\text{E.3.2})$$

The quasi-static helix closing force is then

$$F_{\text{helix},x_s}^{\text{QS}} = \frac{d\theta_s}{dx_s} \tau_{s,\text{helix}}^{\text{QS}}. \quad (\text{E.3.3})$$

Relative to [Equation \(3.2.27\)](#), the movable-member inertial contribution  $I_{s,M}\dot{\omega}_{s,M}$  is omitted. Its rotational inertia is moved to the fixed secondary inertia for the quasi-static comparison. The same reduction is used by the actuator-dynamics and helix-topology studies through [ablation\\_core.py](#) and [helix\\_mechanics.py](#).

### E.3.2 Ballew benchmark axial-force actuators

The Ballew benchmark uses prescribed clamping-force actuators rather than the mechanical flyweight, spring, and helix combination of the Baja reference CVT. The secondary actuator is constant in both benchmark protocols,

$$F_{s,\text{act}} = 2000 \text{ N}. \quad (\text{E.3.4})$$

The force-replay protocol sets the primary actuator equal to the prepared Figure 45 force history, with piecewise-linear interpolation between adjacent samples,

$$F_{p,\text{act}}(t) = \text{interp}_{\text{lin}}[F_{45}(t)]. \quad (\text{E.3.5})$$

The benchmark-local constant and tabulated force laws are implemented in [actuation.py](#).

For the reconstructed closed-loop protocol, let  $n_p = 60\omega_p/(2\pi)$  be primary speed in rpm and  $e = n_p - n^*$ , with  $n^* = 2500$  rpm. The host carries the controller integral  $\eta$ ,

$$\dot{\eta} = e, \quad (\text{E.3.6})$$

and the commanded primary closing force is

$$F_{p,\text{act}} = F_{\text{ff}} + K_p e + K_i \eta, \quad F_{\text{ff}} = K_{\text{ff}} F_{s,\text{act}}. \quad (\text{E.3.7})$$

The source-constrained reconstruction uses  $K_{\text{ff}} = 1.2$ ,  $K_p = 5$ ,  $K_i = 75$ , so  $F_{\text{ff}} = 2400 \text{ N}$ , with  $\eta(0) = 0$ . No saturation or anti-windup term is added. The controller force law and host state are implemented in [controller.py](#).

## Appendix F

# Results Configuration and Reproducibility

This appendix records exact case inputs, numerical settings, normalization definitions, and other reproduction details that are referenced by the Results chapter but are not needed in its main discussion. The working paper references the Results tree on `develop` (`working`); the single ref used by `\ResultsCode` and `\ResultsTree` will be replaced by the frozen Results revision before publication.

### F.1 Mechanical Invariants and Operating-Domain Audit

Exact configuration data supporting [Section 4.2.1](#) are defined by [study.json](#) and the shared accepted-case library [operating\\_cases.json](#).

| Audited case family              | Shaft-boundary environment                                                                                        | Case role                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Reference Baja trajectory        | Full-throttle engine + locked final-drive vehicle from <a href="#">Appendices E.2.2</a> and <a href="#">E.2.3</a> | Ten-second reference launch/shift and nominal trajectory coverage.                          |
| Static rest and deadzone states  | Constant torque–inertia boundaries from <a href="#">Appendix E.2.1</a>                                            | Zero-speed rest, deadzone lock, and classifier checks.                                      |
| Canonical engaged contact states | Constant torque–inertia boundaries on both shafts; no vehicle host                                                | Stick–stick, both single-slip branches, and all four both-slip sign combinations.           |
| Free-shift states                | Constant torque–inertia boundaries on both shafts; no vehicle host                                                | Positive and negative engaged shift speed.                                                  |
| Structural arrivals              | Constant torque–inertia bench environment                                                                         | Engagement, low-ratio-seat, upper-stop, and lower-stop arrivals with exact successor audit. |

| Audited case family   | Shaft-boundary environment         | Case role                                                                                             |
|-----------------------|------------------------------------|-------------------------------------------------------------------------------------------------------|
| Geometry-domain audit | No dynamic shaft boundary required | Belt-length closure and radius derivatives over the physical shift domain.                            |
| Negative controls     | Initial-state/classifier checks    | Deliberately invalid states required to be rejected rather than projected into the admissible domain. |

Table F.1: Case families used by the mechanical-invariants audit.

The standard constructed bench cases use [Appendix E.2.1](#) on both shafts with  $I_{B,p} = 0.03 \text{ kg m}^2$  and  $I_{B,s} = 0.05 \text{ kg m}^2$ . Their shaft speeds remain dynamic. The two targeted anchors use the explicit boundary values in [Table F.5](#).

| Required coverage class           | Case family              | Boundary environment                         |
|-----------------------------------|--------------------------|----------------------------------------------|
| Static zero-speed lower-stop rest | Static rest              | Constant torque–inertia, zero applied torque |
| Deadzone free                     | Deadzone state           | Constant torque–inertia bench                |
| Engaged stick–stick               | Canonical contact state  | Constant torque–inertia bench                |
| Primary slip / secondary stick    | Canonical contact state  | Constant torque–inertia bench                |
| Primary stick / secondary slip    | Canonical contact state  | Constant torque–inertia bench                |
| Both slip                         | Canonical contact state  | Constant torque–inertia bench                |
| Both-slip (+, +)                  | Canonical contact state  | Constant torque–inertia bench                |
| Both-slip (+, −)                  | Canonical contact state  | Constant torque–inertia bench                |
| Both-slip (−, +)                  | Canonical contact state  | Constant torque–inertia bench                |
| Both-slip (−, −)                  | Canonical contact state  | Constant torque–inertia bench                |
| Primary slip in both directions   | Canonical contact states | Constant torque–inertia bench                |
| Secondary slip in both directions | Canonical contact states | Constant torque–inertia bench                |
| Forward and reverse rotation      | Stick/contact states     | Constant torque–inertia bench                |
| Free shift closing                | Free-shift state         | Constant torque–inertia bench                |
| Free shift opening                | Free-shift state         | Constant torque–inertia bench                |
| Low-ratio seat                    | Structural state         | Constant torque–inertia bench                |
| Upper stop                        | Structural state         | Constant torque–inertia bench                |
| Deadzone lower stop               | Structural state         | Constant torque–inertia bench                |
| Engagement transition             | Structural arrival       | Constant torque–inertia bench                |
| Low-ratio arrival                 | Structural arrival       | Constant torque–inertia bench                |
| Upper-stop arrival                | Structural arrival       | Constant torque–inertia bench                |
| Lower-stop arrival                | Structural arrival       | Constant torque–inertia bench                |
| Negative controls                 | Invalid-state checks     | No accepted dynamic case                     |
| Engagement-side classifier        | Classifier control       | Constructed boundary state                   |
| Nominal Baja reference            | Reference trajectory     | Full-throttle engine + locked final drive    |

Table F.2: Required mechanical-invariant coverage classes and the case family used to exercise them.

Table F.3: Canonical engaged contact cases retained by the mechanical-invariants audit. Dwell is the time spent on the requested initial branch before the first transition or the end of the 30 ms audit window.

| Contact regime       | Reproduction class  | Initial dwell  | First exit                   |
|----------------------|---------------------|----------------|------------------------------|
| Stick–stick, forward | standard            | 30.000 ms      | none within 30 ms            |
| Stick–stick, reverse | standard            | 13.114 ms      | primary static-capacity exit |
| Primary slip (+)     | standard            | 26.225 ms      | primary restick              |
| Primary slip (–)     | standard            | 5.624 ms       | primary restick              |
| Secondary slip (+)   | targeted anchor     | 414.44 $\mu$ s | secondary restick            |
| Secondary slip (–)   | standard            | 30.000 ms      | none within 30 ms            |
| Both slip (+, +)     | standard, transient | 31.71 $\mu$ s  | secondary restick            |
| Both slip (+, –)     | standard            | 30.000 ms      | none within 30 ms            |
| Both slip (–, +)     | targeted anchor     | 9.82 $\mu$ s   | secondary restick            |
| Both slip (–, –)     | standard            | 1.757 ms       | primary restick              |

The two targeted reproduction anchors are frozen explicitly in the shared case library. Their initial CVT states are listed in [Table F.4](#); their constant shaft-boundary values are listed separately in [Table F.5](#).

Table F.4: Initial CVT states of the two targeted mechanical-invariant reproduction anchors.

| Quantity                | Secondary slip (+) | Both slip (–, +) |
|-------------------------|--------------------|------------------|
| Primary speed [rad/s]   | 272.786402         | 536.870017       |
| Secondary speed [rad/s] | 223.014459         | 173.988007       |
| Belt speed [m/s]        | 17.211387          | 17.245403        |
| Shift position [m]      | 0.01446320         | 0.00292236       |
| Shift speed [m/s]       | 0                  | 0                |

Table F.5: Constant torque–inertia boundary values of the two targeted mechanical-invariant reproduction anchors.

| Boundary quantity                               | Secondary slip (+) | Both slip (–, +) |
|-------------------------------------------------|--------------------|------------------|
| Primary external torque [N m]                   | -38.596922         | -105.580961      |
| Primary referred inertia [kg m <sup>2</sup> ]   | 0.01279093         | 0.06468014       |
| Secondary external torque [N m]                 | -17.984371         | -282.073589      |
| Secondary referred inertia [kg m <sup>2</sup> ] | 0.04127251         | 0.00250569       |

Table F.6: Numerical settings used by the mechanical-invariants case families.

| Case family                                | rtol               | atol               | Max. step | Duration |
|--------------------------------------------|--------------------|--------------------|-----------|----------|
| Reference Baja trajectory                  | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 10 ms     | 10 s     |
| Selected contact-case audit                | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 1 ms      | 30 ms    |
| Targeted-anchor screening before selection | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 0.5 ms    | 30 ms    |
| Structural arrivals                        | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 0.5 ms    | 30 ms    |
| Free-shift cases                           | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 1 ms      | 20 ms    |
| Static rest                                | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 2 ms      | 50 ms    |

The geometry-domain audit evaluates 401 points over the engaged shift range and 100 points through the primary deadzone, for 501 geometry samples in total. A  $1 \times 10^{-6}$  m finite-difference displacement is used for the independent derivative checks away from one-sided boundaries. The two targeted contact anchors are first screened with the 0.5 ms maximum step shown in [Table F.6](#); once accepted, they are rerun by the main contact audit with the same 1 ms maximum step used for every selected contact case.

For the engaged closure, let  $\mathbf{r} = \mathbf{A}\mathbf{z} - \mathbf{b}$ . Each row is normalized by

$$s_i = \max\left(1, |b_i|, \sum_k |A_{ik}| |z_k|\right), \quad \epsilon_{\text{cl}} = \max_i \frac{|r_i|}{s_i}. \quad (\text{F.1.1})$$

The “scaled closure-equation residual” in [Appendix F.1](#) is  $\epsilon_{\text{cl}}$ . The other residual thresholds retain their physical units as shown.

| Audit threshold                                       | Value                              |
|-------------------------------------------------------|------------------------------------|
| Belt-length residual                                  | $10^{-9}$ m                        |
| Scaled closure-equation residual                      | $10^{-10}$                         |
| Sticking velocity drift                               | $10^{-3}$ m/s                      |
| Sticking acceleration residual                        | $10^{-7}$ m/s <sup>2</sup>         |
| Allowed negative static-margin roundoff               | $10^{-9}$                          |
| Allowed positive kinetic pair power                   | $10^{-3}$ W                        |
| Allowed negative integrated normal load               | $10^{-6}$ N                        |
| Allowed negative belt tension                         | $10^{-6}$ N                        |
| Allowed negative distributed normal load              | $10^{-6}$ N/rad                    |
| Wrap-field normal-integral residual                   | $10^{-6}$ N                        |
| Allowed negative stop reaction                        | $10^{-6}$ N                        |
| Fixed-shift speed residual                            | $10^{-8}$ m/s                      |
| Fixed-shift acceleration residual                     | $10^{-7}$ m/s <sup>2</sup>         |
| Deadzone secondary-lock speed residual                | $10^{-7}$ m/s                      |
| Deadzone secondary-lock acceleration residual         | $10^{-7}$ m/s <sup>2</sup>         |
| Static-rest state drift                               | $10^{-10}$                         |
| Deadzone radius-derivative residual                   | $10^{-12}$                         |
| Geometry first-derivative finite-difference residual  | $10^{-7}$                          |
| Geometry second-derivative finite-difference residual | $5 \times 10^{-3}$ m <sup>-1</sup> |

Table F.7: Numerical review thresholds used by the mechanical-invariants audit.

## F.2 Mechanical-Energy Consistency

Exact settings supporting [Section 4.2.2](#) are defined in [energy-consistency/study.json](#). The audited trajectory uses the physical reference configuration of [Appendix E.1](#) and the reference shaft boundaries of [Appendices E.2.2](#) and [E.2.3](#), with  $\gamma = 0$  throughout the 10 s interval.

Table F.8: Case and numerical settings used by the mechanical-energy audit.

| Configuration     | rtol               | atol               | Max. step | Duration |
|-------------------|--------------------|--------------------|-----------|----------|
| Nominal audit     | $3 \times 10^{-5}$ | $3 \times 10^{-8}$ | 10 ms     | 10 s     |
| Tight integration | $9 \times 10^{-6}$ | $9 \times 10^{-9}$ | 5 ms      | 10 s     |

The final balance fraction is

$$\epsilon_E = \frac{|R_E(t_f)|}{|W_B(t_f)|}, \quad (\text{F.2.1})$$

where  $R_E$  is the raw mechanical-energy residual and  $W_B$  is the net work supplied through the shaft boundaries. The ODE-refinement final-state metric compares the nominal and tight solutions componentwise. For  $x_k \in \{\omega_p, \omega_s, v_b, s, \dot{s}\}$ ,

$$S_k = \max(|x_{k,\text{nom}}(t_f)|, |x_{k,\text{tight}}(t_f)|, S_{k,\text{min}}), \quad \epsilon_{x,f} = \max_k \frac{|x_{k,\text{tight}}(t_f) - x_{k,\text{nom}}(t_f)|}{S_k}, \quad (\text{F.2.2})$$

with floors (1 rad/s, 1 rad/s, 1 m/s,  $10^{-3}$  m,  $10^{-3}$  m/s). The impact momentum and active-constraint residuals are the residuals returned by the finite-speed boundary projection described in [Section 3.6.3](#); they are audited independently of the energy defect.

Table F.9: Quadrature and review settings used by the mechanical-energy audit.

| Quantity                                           | Value                  |
|----------------------------------------------------|------------------------|
| Continuous-work quadrature steps                   | 5, 2.5, 1.25, 0.625 ms |
| Maximum final $ R_E / W_B $                        | $10^{-4}$              |
| Maximum absolute event-energy defect               | $10^{-8}$ J            |
| Maximum impact momentum residual                   | $10^{-7}$              |
| Maximum impact constraint residual                 | $10^{-8}$              |
| Maximum normalized final-state difference          | $10^{-2}$              |
| Allowed negative computed capture loss             | $10^{-9}$ J            |
| Maximum finest-to-previous continuous-defect ratio | 1.25                   |

### F.3 Solver Convergence

Exact settings supporting [Section 4.2.3](#) are defined in [solver-convergence/study.json](#). The benchmark trajectory uses the reference configuration and shaft boundaries of [Appendices E](#) and [E.2](#).

Table F.10: Numerical reference, formal sweep, and retained comparison point of the solver-convergence study.

| Component                  | Relative tolerance                         | Absolute tolerance                | Maximum step          |
|----------------------------|--------------------------------------------|-----------------------------------|-----------------------|
| Tight numerical reference  | $10^{-7}$                                  | $10^{-10}$                        | 2.5 ms                |
| Canonical comparison point | $10^{-4}$                                  | $10^{-7}$                         | 10 ms                 |
| Formal main sweep          | $10^{-2}$ to $3 \times 10^{-6}$ , 8 values | $10^{-3}$ rtol                    | 5, 10, 20, 50, 100 ms |
| Independent atol sweep     | $10^{-4}$                                  | $10^{-5}$ to $10^{-8}$ , 7 values | 10 ms                 |

Continuous-state error is evaluated only after matching the complete hybrid transition signature. Corresponding candidate and reference segments are each parameterized by normalized within-segment phase

$$u = \frac{t - t_m}{t_{m+1} - t_m} \in [0, 1], \quad (\text{F.3.1})$$

and each retained dense interpolant is evaluated directly on the same 2049-point phase grid. No interpolation is performed across a reset. For each state component  $x_k$ , the normalization scale is

$$S_k = \max\left(\max_t |x_{k,\text{ref}}|, \text{range}(x_{k,\text{ref}}), S_{k,\text{min}}\right), \quad (\text{F.3.2})$$

with floors (100 rad/s, 100 rad/s, 5 m/s, 0.02 m, 0.05 m/s) for  $(\omega_p, \omega_s, v_b, s, \dot{s})$ , respectively. The normalized error on segment  $m$  is

$$e_{k,m}(u) = \frac{x_{k,m}^{\text{cand}}(u) - x_{k,m}^{\text{ref}}(u)}{S_k}. \quad (\text{F.3.3})$$

Using the reference segment duration  $\Delta t_m$  and total reference duration  $T$ , the reported aggregate RMS is

$$\epsilon_{\text{RMS}} = \left[ \frac{1}{5T} \sum_k \sum_m \Delta t_m \int_0^1 e_{k,m}^2(u) du \right]^{1/2}, \quad (\text{F.3.4})$$

while the maximum metric is  $\epsilon_{\text{max}} = \max_{k,m,u} |e_{k,m}(u)|$ . Event-time error is compared separately between corresponding transitions. The regime-mismatch fraction is the exact fraction of total time for which the two mode labels differ after the union of all candidate and reference event times partitions the interval; it is therefore not evaluated on an arbitrary reporting grid.

Table F.11: Formal review guards used in the solver-convergence study.

| Metric                                  | Required value |
|-----------------------------------------|----------------|
| Event-aligned normalized trajectory RMS | $\leq 10^{-4}$ |
| Event-aligned normalized maximum error  | $\leq 10^{-3}$ |
| Maximum corresponding event-time error  | $\leq 1$ ms    |
| Exact regime-history mismatch fraction  | $\leq 10^{-3}$ |
| Hybrid transition signature             | exact match    |

## F.4 Closure Conditioning and Root Structure

Configuration details for [Section 4.2.4](#) are defined in [closure-conditioning/study.json](#). The accepted operating states are drawn from the reference launch and the shared verification case library used by [Appendix F.1](#).

Table F.12: Reference-trajectory and multistart settings used by the closure-conditioning study.

| Quantity                           | Value               |
|------------------------------------|---------------------|
| Reference launch duration          | 10 s                |
| Reference launch sampling interval | 2 ms                |
| Reference launch rtol / atol       | $10^{-4} / 10^{-7}$ |
| Reference launch maximum step      | 10 ms               |
| Physical-box multistart grid       | $7 \times 7$        |
| Expanded-box multistart grid       | $9 \times 9$        |
| Root-clustering tolerance          | $10^{-6}$           |

Table F.13: Named frozen states selected for closure maps.

| Case ID                            | Role                                                     |
|------------------------------------|----------------------------------------------------------|
| <code>low_ratio_seat</code>        | Engaged low-ratio support state                          |
| <code>mid_shift</code>             | Interior shifting state and broad-map showcase           |
| <code>upper_stop</code>            | Engaged upper-travel support state                       |
| <code>free_shift_opening_70</code> | Late-ratio opening/backshift state                       |
| Worst actual $A$ state             | Largest encountered mechanical-closure condition number  |
| Worst actual $J_R$ state           | Largest encountered stick-root Jacobian condition number |

Table F.14: Traction domains used by the closure-conditioning maps.

| Domain                | Grid             | Definition                                                                                             |
|-----------------------|------------------|--------------------------------------------------------------------------------------------------------|
| Static Coulomb box    | $241 \times 241$ | Exact static-capacity intervals at the frozen state; mechanical admissibility is evaluated separately. |
| Expanded continuation | $321 \times 321$ | Signed half-width $\max(2.5, 4\lambda_{\max, \text{phys}})$ ; normally about $[-2.6, 2.6]^2$ .         |
| Broad diagnostic      | $501 \times 501$ | $[-10, 10]^2$ , generated for the configured showcase state.                                           |

## F.5 Independent Ballew Benchmark

Exact benchmark inputs supporting [Section 4.3](#) are defined in [study.json](#). The benchmark assembly and shaft boundaries are constructed in [case.py](#), with published and reconstructed constants collected in [constants.py](#).

### F.5.1 Shaft boundaries

Table F.15: Shaft-boundary quantities used by the independent Ballew benchmark.

| Quantity                                             | CINDER bound-ary        | Value                          |
|------------------------------------------------------|-------------------------|--------------------------------|
| Primary applied torque $\tau_{B,p}$                  | Constant torque–inertia | 18 N m                         |
| Primary referred inertia $I_{B,p}$                   | Constant torque–inertia | 0.008 kg m <sup>2</sup>        |
| Final-drive reduction $G$                            | Locked final drive      | 8.93                           |
| Tire radius $r_w$                                    | Locked final drive      | 0.317 m                        |
| ATV mass $m_v$                                       | Locked final drive      | 226 kg                         |
| Wheel rotational inertia $I_w$                       | Locked final drive      | 0 kg m <sup>2</sup>            |
| Rolling-resistance coefficient $C_{rr}$              | Road load               | 0.048                          |
| Aerodynamic drag coefficient $C_d$                   | Road load               | 1.0                            |
| Frontal area $A_f$                                   | Road load               | 1.39 m <sup>2</sup>            |
| Road grade $\gamma$                                  | Constant grade          | 0                              |
| Air density $\rho$                                   | Reconstructed road load | 1.225 kg/m <sup>3</sup>        |
| Gravity $g$                                          | Reconstructed road load | 9.806 65 m/s <sup>2</sup>      |
| Rolling-speed regularization $v_\epsilon$            | Reconstructed road load | 0.01 m/s                       |
| Reflected ATV translation $m_v(r_w/G)^2$             | Locked final drive      | 0.284 789 55 kg m <sup>2</sup> |
| Direct secondary-boundary inertia $I_{s,\text{dir}}$ | Locked final drive      | 0.988 210 45 kg m <sup>2</sup> |
| Total secondary boundary inertia $I_{B,s}$           | Locked final drive      | 1.273 kg m <sup>2</sup>        |

The Ballew assembly assigns a further 0.002 kg m<sup>2</sup> to the output pulley inside the CVT. Thus the internal output-pulley inertia plus the secondary boundary inertia is  $0.002 + 1.273 = 1.275$  kg m<sup>2</sup>, matching the published combined output-pulley–ATV inertia used by the reconstruction.

### F.5.2 Actuation and host state

The benchmark-specific axial-force laws are defined in [Appendix E.3.2](#). Their use in the two comparison protocols is fixed as follows.

Table F.16: Actuator and host configuration of the two Ballew benchmark protocols.

| Protocol     | Primary actuator                                                 | Secondary actuator            | Host state                                           |
|--------------|------------------------------------------------------------------|-------------------------------|------------------------------------------------------|
| Force replay | Prepared Figure 45 force history; piecewise-linear interpolation | Constant 2000 N closing force | Secondary-shaft angle                                |
| Closed loop  | PI force law of <a href="#">Equation (E.3.7)</a>                 | Constant 2000 N closing force | Secondary-shaft angle and controller integral $\eta$ |

The common initial shaft speeds are

$$n_p(0) = 2500 \text{ rpm}, \quad n_s(0) = 1136 \text{ rpm}, \quad \dot{s}(0) = 0. \quad (\text{F.5.1})$$

The initial shift coordinate is the engaged geometry that makes those two shaft speeds compatible with one belt speed. In the closed-loop protocol,  $\eta(0) = 0$ . The exact construction is implemented in [simulation.py](#).

### F.5.3 Numerical settings

Table F.17: Numerical settings used by the two Ballew benchmark protocols.

| Protocol     | rtol      | atol      | Max. step | Report step | Duration |
|--------------|-----------|-----------|-----------|-------------|----------|
| Force replay | $10^{-7}$ | $10^{-9}$ | 1 ms      | 5 ms        | 5 s      |
| Closed loop  | $10^{-7}$ | $10^{-9}$ | 1 ms      | 0.2 ms      | 5 s      |

Both protocols permit up to 2000 hybrid transitions and retain dense output for the benchmark comparison.